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Preface

Regression is taught under at least five different descriptions, and each one is correct. In one, regression is least squares together with the distribution theory of quadratic forms. In another, it is orthogonal projection onto a subspace, and everything else follows as a corollary. A third starts from likelihood and treats the normal linear model as the Gaussian member of the generalized linear family. A fourth defines regression as estimating a conditional distribution, with linearity one structural assumption among many. A fifth treats it as a data-analytic procedure whose assumptions must be checked, one at a time, against the data.

Most textbooks pick one of these descriptions and stay with it. This book uses all of them, in order, as rungs of a single ladder. Parts I–III set up the classical linear model and its exact theory. After that, *each part relaxes exactly one assumption*, and asks what survives, what must be rebuilt, and what new tools the relaxation requires.

Part	What changes	What we study
I	—	The apparatus: matrices, random vectors, the normal distribution, quadratic forms
II	—	$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$, $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$, as a projection problem
III	adds $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$	Exact finite-sample inference
IV	drops $\text{rank}(\mathbf{X}) = p$	Estimability, analysis of variance, designed structure
V	audits every assumption	What breaks, how to see it, what inference survives
VI	drops unbiasedness	Shrinkage, selection, regularization, $p > n$
VII	drops $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$	Generalized least squares, general Gauss–Markov, mixed models
VIII	drops normality of $\mathbf{Y}$	Exponential families and generalized linear models
IX	drops the linear predictor	Polynomials, splines, smoothing, additive models
X	drops modeling only the mean	Quantile regression and distributional models

Choices this book makes

Geometry before distribution theory. Least squares is introduced as a nearest-point problem ([Chapter 6](#)) before any sampling theory. Fitted values, residuals and sums of squares are projections. Coefficients are coordinates, and coordinates depend on the basis you choose. This is the only framing in which rank-deficient models need no special treatment.

Estimability is an estimation idea. It appears in Part II, not as a preliminary to the analysis of variance. Putting it there is what allows Part IV to treat ANOVA as ordinary subspace comparison.

Computation is part of the theory. Conditioning, QR and the singular value decomposition come immediately after least squares (Chapter 10), because numerical stability and statistical stability are closely linked.

The Bayesian treatment is a thread, not a quarantined chapter. Boxed “Bayesian thread” sections close Parts II, III, VII and VIII. Each develops the Bayesian counterpart of the result just proved.

Departures have a theory. Part V opens by deriving what omitted variables, wrong covariance, non-normality, outliers and collinearity actually do to the estimator. Diagnostics and remedies come after that.

Material most regression texts omit. This book adds chapters on bootstrap and permutation inference (Chapter 23), errors in variables (Chapter 24), the causal interpretation of coefficients (Chapter 25), high-dimensional regression (Chapter 28) and missing data (Chapter 41).

Code

Every figure and every number obtained by computation is produced by a Python script in the book’s `code/` directory, and the scripts check their own claims with assertions. The code in the text is taken from those scripts. On this website each code block is a live cell: press **Run** (or Shift+Enter) to execute it in your browser, edit it, and run it again. Nothing needs to be installed; the first run downloads a Python environment with numpy, scipy, pandas and statsmodels. Cells on a page share their variables, and running a cell first runs the cells above it that have not run yet.

Sources

The coverage of this book was planned against six standard texts: Rencher and Schaalje (2008), Seber and Lee (2003), Christensen (2020), Sen and Srivastava (1990), Agresti (2015) and Fahrmeir et al. (2021). Their chapters were mapped onto a single prerequisite graph, and the order of this book respects that graph. Each chapter’s *Notes and sources* section names the treatments it drew on and the primary literature. The exposition, proofs, examples and exercises are original to this book.

How to Use This Book

Parts and chapters

Parts I–VI (Chapters 1–30) are the classical linear model, complete in itself: estimation, exact inference, designed experiments, diagnostics, and the move from unbiased estimation to regularization. Parts VII–X (Chapters 31–45) climb the rest of the ladder: correlated errors, non-normal responses, nonlinear predictors, and models for entire conditional distributions. Chapters are published here as they are written; the ones not yet on the site are named in plain text.

Reading paths

- **A first course on linear models.** Chapters 5–9, 11–12, 15, 19–20 and 29. Consult Part I as needed.
- **The theory spine.** Part I, then Chapters 6, 7, 8, 9, 11 and 31. This is the shortest route to a full structural understanding of the linear model.
- **For implementers.** Chapters 5, 6, 10 and 20, Part VI and Chapter 34.
- **A course on generalized models.** Chapters 5, 6 and 11, then Parts VII–X.

Prerequisites

Each chapter opens with a list of the earlier chapters it uses, taken from the prerequisite graph the book was planned against. Part I is written as reference material. Read it quickly the first time and come back to it when a later chapter sends you there.

Exercises

Exercises come at the end of each section in three groups: **A. Check your understanding** (routine: a definition or computation has been understood), **B. Practice** (core: the result is used later in the book) and **C. Going deeper** (hard: may need an idea not in the text). Many have a worked solution, folded under the exercise; try the exercise before opening it.

Notation

Symbol	Meaning
$\mathbf{Y}, \mathbf{y}$	response vector (random, observed), $n \times 1$
$\mathbf{X}$	model matrix, $n \times p$; columns $\mathbf{x}_1, \dots, \mathbf{x}_p$
$\boldsymbol{\beta}, \hat{\boldsymbol{\beta}}$	coefficient vector and a least squares estimate
$\boldsymbol{\varepsilon}$	error vector, $\boldsymbol{\varepsilon} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}$
$\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}}$	fitted values $\mathbf{X}\hat{\boldsymbol{\beta}}$ and residuals $\mathbf{Y} - \hat{\mathbf{Y}}$
σ^2	error variance
$\mathbb{R}^n$	n -dimensional real space with $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{v}$
$\mathbb{C}(\mathbf{A}), \mathcal{N}(\mathbf{A})$	column space and null space of $\mathbf{A}$
$\mathcal{S}^\perp$	orthogonal complement of a subspace $\mathcal{S}$
$\mathbf{A}^T, \mathbf{A}^-$	transpose; a generalized inverse ($\mathbf{A}\mathbf{A}^- \mathbf{A} = \mathbf{A}$)
$\mathbf{M}$	orthogonal projection onto $\mathbb{C}(\mathbf{X})$
$\mathbf{M}_0$	orthogonal projection onto the column space of a reduced model

Symbol	Meaning
rank, tr, diag	rank, trace, diagonal matrix
1 , 0 , I	vector of ones, zero vector, identity
$\ \cdot\ $	Euclidean norm unless stated otherwise
E , Var, Cov	expectation, variance, covariance (matrix)
$N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	n -variate normal distribution
$\chi^2(r, \gamma)$, $F(r, s, \gamma)$	noncentral distributions, noncentrality γ

Vectors are columns. Matrices and vectors are set in bold upright type and scalars in italic. A bare “projection” always means an orthogonal projection with respect to the Euclidean inner product. [Section 6.9](#) covers the other kinds.

Chapter 1

Matrix Algebra for Linear Models

A linear model is a statement about vectors and matrices, and nearly every argument in this book is an argument about them. This chapter collects the matrix algebra the book relies on: rank and the fundamental subspaces, inverses and generalized inverses, consistent systems of equations, orthogonal matrices, determinants and partitioned matrices, the spectral theorem and positive definiteness, the singular value decomposition, idempotent matrices, Kronecker products and the calculus of quadratic forms. It is a reference chapter, meant to be skimmed first and consulted later, but every central result is proved.

What you need. None beyond the Notation chapter. A first course in linear algebra helps, but is not assumed.

1.0.1 Roadmap

- [Vectors, matrices and the operations we use](#)
- [Rank, column space and null space](#)
- [Inverses and generalized inverses](#)
- [Systems of equations and consistency](#)
- [Orthogonality and orthogonal matrices](#)
- [Trace, determinants and partitioned matrices](#)
- [Eigenvalues, the spectral theorem and positive definiteness](#)
- [The singular value decomposition and conditioning](#)
- [Idempotent matrices](#)
- [Kronecker products and the vec operator](#)
- [Vector and matrix calculus](#)
- [Summary and notes](#)

1.1 Vectors, matrices and the operations we use

This chapter works in $\mathbb{R}^n$ and with real matrices only. It fixes the conventions, states the results later chapters cite, and proves them. A reader who already knows linear algebra can go straight

to the results that are specific to statistics: generalized inverses (Section 1.3 and Section 1.4), Schur complements and the Woodbury identity (Section 1.6), nonnegative definite matrices (Section 1.7), conditioning (Section 1.8) and idempotent matrices (Section 1.9). Orthogonal projections, the geometric heart of least squares, get only the minimum here. Chapter 6 develops them.

1.1.1 Conventions

An $m \times n$ matrix $\mathbf{A} = (a_{ij})$ has m rows and n columns. We write $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ for its columns and $\mathbf{a}_{(i)}\mathbf{T}$ for its i th row. A **vector** is always a column, so $\mathbf{x} \in \mathbb{R}^n$ is an $n \times 1$ matrix and $\mathbf{x}\mathbf{T}$ is a row. Bold upright letters denote vectors and matrices, and italic letters denote scalars. The special matrices we use constantly are the identity $\mathbf{I}_n$, the zero matrix $\mathbf{0}$, the vector of ones $\mathbf{1}_n$, the matrix of ones $\mathbf{J}_n = \mathbf{1}_n\mathbf{1}_n\mathbf{T}$, and the coordinate vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$, the columns of $\mathbf{I}_n$. Subscripts giving sizes are dropped when the size is clear. $\text{diag}(d_1, \dots, d_n)$ is the diagonal matrix with the given diagonal entries. A square matrix is **symmetric** if $\mathbf{A}\mathbf{T} = \mathbf{A}$, *upper triangular* if $a_{ij} = 0$ for $i > j$, and *lower triangular* if $\mathbf{A}\mathbf{T}$ is upper triangular.

The **transpose** satisfies $(\mathbf{A}\mathbf{T})\mathbf{T} = \mathbf{A}$, $(\mathbf{A} + \mathbf{B})\mathbf{T} = \mathbf{A}\mathbf{T} + \mathbf{B}\mathbf{T}$ and, most usefully,

$$(\mathbf{A}\mathbf{B})\mathbf{T} = \mathbf{B}\mathbf{T}\mathbf{A}\mathbf{T}. \quad (1.1.1)$$

1.1.2 Products as linear combinations

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times q$, the (i, j) entry of $\mathbf{A}\mathbf{B}$ is $\sum_k a_{ik}b_{kj}$. Three equivalent readings of the product carry most of the arguments in this book:

$$\mathbf{A}\mathbf{b} = \sum_{j=1}^n b_j \mathbf{a}_j, \quad \mathbf{A}\mathbf{B} = [\mathbf{A}\mathbf{b}_1, \dots, \mathbf{A}\mathbf{b}_q], \quad \mathbf{A}\mathbf{B} = \sum_{k=1}^n \mathbf{a}_k \mathbf{b}_{(k)}\mathbf{T}. \quad (1.1.2)$$

The first says that $\mathbf{A}\mathbf{b}$ combines the columns of $\mathbf{A}$, with weights taken from $\mathbf{b}$. The second says that each column of $\mathbf{A}\mathbf{B}$ is such a combination. The third writes a product as a sum of n rank-one matrices (outer products), and is the form in which spectral and singular value decompositions are usually used. Transposing the first reading shows that each row of $\mathbf{A}\mathbf{B}$ is a combination of the rows of $\mathbf{B}$.

Matrices partitioned into conformable blocks multiply as if the blocks were scalars, as long as the order of factors is kept:

$$\begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix} \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11}\mathbf{B}_{11} + \mathbf{A}_{12}\mathbf{B}_{21} & \mathbf{A}_{11}\mathbf{B}_{12} + \mathbf{A}_{12}\mathbf{B}_{22} \\ \mathbf{A}_{21}\mathbf{B}_{11} + \mathbf{A}_{22}\mathbf{B}_{21} & \mathbf{A}_{21}\mathbf{B}_{12} + \mathbf{A}_{22}\mathbf{B}_{22} \end{pmatrix}.$$

Multiplication is associative and distributive but not commutative, and $\mathbf{A}\mathbf{B} = \mathbf{0}$ does not force $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$. The **Hadamard product** $\mathbf{A} \circ \mathbf{B} = (a_{ij}b_{ij})$ multiplies entrywise. It appears only occasionally.

1.1.3 Inner products, lengths and the Cauchy–Schwarz inequality

The **inner product** of $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{v} = \sum_i u_i v_i$, and the **Euclidean norm** is $\|\mathbf{u}\| = (\mathbf{u}^T \mathbf{u})^{1/2}$. The entries of $\mathbf{A}^T \mathbf{A}$ are the inner products of the columns of $\mathbf{A}$, and those of $\mathbf{A} \mathbf{A}^T$ are the inner products of its rows. In particular

$$\sum_{i,j} a_{ij}^2 = \sum_j \|\mathbf{a}_j\|^2, \quad (1.1.3)$$

so $\mathbf{A}^T \mathbf{A} = \mathbf{0}$ forces every column to be zero, that is, $\mathbf{A} = \mathbf{0}$. The square root of Eq. 1.1.3 is the **Frobenius norm** $\|\mathbf{A}\|_F$.

Proposition 1.1.1 — Cauchy–Schwarz

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, $(\mathbf{u}^T \mathbf{v})^2 \leq (\mathbf{u}^T \mathbf{u})(\mathbf{v}^T \mathbf{v})$, with equality iff one vector is a multiple of the other. Consequently $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

Proof. If $\mathbf{v} = \mathbf{0}$ there is nothing to prove. Otherwise, the quadratic $q(t) = \|\mathbf{u} - t\mathbf{v}\|^2 = \mathbf{u}^T \mathbf{u} - 2t \mathbf{u}^T \mathbf{v} + t^2 \mathbf{v}^T \mathbf{v}$ is nonnegative for all t , so its discriminant $4(\mathbf{u}^T \mathbf{v})^2 - 4(\mathbf{u}^T \mathbf{u})(\mathbf{v}^T \mathbf{v})$ is at most zero. Equality holds iff q has a real root, that is, iff $\mathbf{u} = t\mathbf{v}$ for some t . The triangle inequality follows by expanding $\|\mathbf{u} + \mathbf{v}\|^2$ and bounding the cross term. ■

1.1.4 Quadratic forms

A **quadratic form** in $\mathbf{x} \in \mathbb{R}^n$ is a function $\mathbf{x}^T \mathbf{A} \mathbf{x} = \sum_{i,j} a_{ij} x_i x_j$. Since $\mathbf{x}^T \mathbf{A} \mathbf{x}$ is a scalar it equals its own transpose $\mathbf{x}^T \mathbf{A}^T \mathbf{x}$, so

$$\mathbf{x}^T \mathbf{A} \mathbf{x} = \mathbf{x}^T \left(\frac{1}{2} (\mathbf{A} + \mathbf{A}^T) \right) \mathbf{x}. \quad (1.1.4)$$

Every quadratic form therefore has a symmetric matrix, and that matrix is unique: if $\mathbf{x}^T \mathbf{S} \mathbf{x} = 0$ for all $\mathbf{x}$ with $\mathbf{S}$ symmetric, then taking $\mathbf{x} = \mathbf{e}_i$ gives $s_{ii} = 0$, and $\mathbf{x} = \mathbf{e}_i + \mathbf{e}_j$ gives $2s_{ij} = 0$. From now on we always take the matrix in a quadratic form to be symmetric.

Example. (Sums of squares as quadratic forms)

The sum of squared deviations of $\mathbf{y} \in \mathbb{R}^n$ from its mean is

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \mathbf{y}^T \mathbf{T} \mathbf{y} - n\bar{y}^2 = \mathbf{y}^T \left(\mathbf{I}_n - \frac{1}{n} \mathbf{J}_n \right) \mathbf{y},$$

because $n\bar{y}^2 = (\mathbf{1}^T \mathbf{y})^2 / n = \mathbf{y}^T \mathbf{J}_n \mathbf{y} / n$. The **centring matrix** $\mathbf{I}_n - n^{-1} \mathbf{J}_n$ is symmetric and satisfies $(\mathbf{I}_n - n^{-1} \mathbf{J}_n)^2 = \mathbf{I}_n - n^{-1} \mathbf{J}_n$, because $\mathbf{J}_n^2 = n \mathbf{J}_n$. Every sum of squares

in an analysis of variance is a quadratic form whose matrix has these two properties (Section 1.9).

1.2 Rank, column space and null space

1.2.1 Subspaces and dimension

A nonempty set $\mathcal{S} \subseteq \mathbb{R}^n$ is a **subspace** if $\mathbf{u}, \mathbf{v} \in \mathcal{S}$ and $a, b \in \mathbb{R}$ imply $a\mathbf{u} + b\mathbf{v} \in \mathcal{S}$. Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are **linearly independent** if $\sum_j c_j \mathbf{v}_j = \mathbf{0}$ only for $c_1 = \dots = c_k = 0$, that is, if the matrix $[\mathbf{v}_1, \dots, \mathbf{v}_k]$ sends only the zero vector to zero. Their **span** is the set of all their linear combinations. A **basis** of $\mathcal{S}$ is an independent set that spans it.

We use the following facts of elementary linear algebra without proof (Halmos 1958, chapter 1). Every subspace of $\mathbb{R}^n$ has a basis. All bases of a subspace have the same number of elements, its **dimension** $\dim \mathcal{S}$. In an r -dimensional subspace, any r independent vectors form a basis, and any independent set can be extended to a basis. If $\mathcal{S} \subseteq \mathcal{T}$ and $\dim \mathcal{S} = \dim \mathcal{T}$, then $\mathcal{S} = \mathcal{T}$. Finally, if $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$, the union of a basis of $\mathcal{S}$ and a basis of $\mathcal{T}$ is independent, so the sum $\mathcal{S} + \mathcal{T} = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in \mathcal{S}, \mathbf{v} \in \mathcal{T}\}$ has dimension $\dim \mathcal{S} + \dim \mathcal{T}$. Such a sum is called **direct** and written $\mathcal{S} \oplus \mathcal{T}$.

1.2.2 The four subspaces of a matrix

An $m \times n$ matrix $\mathbf{A}$ carries two subspaces of $\mathbb{R}^m$ and two of $\mathbb{R}^n$:

$$\begin{aligned} \mathbb{C}(\mathbf{A}) &= \{\mathbf{A}\mathbf{b} : \mathbf{b} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, & \mathcal{N}(\mathbf{A}) &= \{\mathbf{b} \in \mathbb{R}^n : \mathbf{A}\mathbf{b} = \mathbf{0}\} \subseteq \mathbb{R}^n, \\ \mathbb{C}(\mathbf{A}\mathbf{T}) &\subseteq \mathbb{R}^n, & \mathcal{N}(\mathbf{A}\mathbf{T}) &\subseteq \mathbb{R}^m. \end{aligned}$$

By Eq. 1.1.2, $\mathbb{C}(\mathbf{A})$ is the span of the columns, the **column space**, and $\mathbb{C}(\mathbf{A}\mathbf{T})$ is the span of the rows, the **row space**. $\mathcal{N}(\mathbf{A})$ is the **null space**. The **rank** of $\mathbf{A}$ is $\text{rank}(\mathbf{A}) = \dim \mathbb{C}(\mathbf{A})$, the largest number of independent columns. The number of independent rows, $\dim \mathbb{C}(\mathbf{A}\mathbf{T})$, looks like a different quantity. It is not.

Theorem 1.2.1 — Row rank equals column rank

For every matrix $\mathbf{A}$, $\dim \mathbb{C}(\mathbf{A}\mathbf{T}) = \dim \mathbb{C}(\mathbf{A})$. That is, $\text{rank}(\mathbf{A}\mathbf{T}) = \text{rank}(\mathbf{A})$. Moreover, if $\text{rank}(\mathbf{A}) = r \geq 1$ there are an $m \times r$ matrix $\mathbf{B}$ and an $r \times n$ matrix $\mathbf{F}$, both of rank r , with $\mathbf{A} = \mathbf{B}\mathbf{F}$.

Proof. Let $r = \text{rank}(\mathbf{A})$ and let the columns of the $m \times r$ matrix $\mathbf{B}$ be a basis of $\mathbb{C}(\mathbf{A})$. Each column $\mathbf{a}_j$ is a combination $\mathbf{B}\mathbf{f}_j$ of these, so $\mathbf{A} = \mathbf{B}\mathbf{F}$ with $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_n]$. By the transposed reading of Eq. 1.1.2, every row of $\mathbf{A} = \mathbf{B}\mathbf{F}$ is a combination of the r rows of $\mathbf{F}$, so $\dim \mathbb{C}(\mathbf{A}\mathbf{T}) \leq r = \dim \mathbb{C}(\mathbf{A})$. Applying this to $\mathbf{A}\mathbf{T}$ gives the reverse inequality. Finally $\text{rank}(\mathbf{F}) \leq r$ because $\mathbf{F}$ has r rows, and $r = \dim \mathbb{C}(\mathbf{A}\mathbf{T}) \leq \dim \mathbb{C}(\mathbf{F}\mathbf{T}) = \text{rank}(\mathbf{F})$, so $\text{rank}(\mathbf{F}) = r$. ■

The factorization $\mathbf{A} = \mathbf{B}\mathbf{F}$ is a **rank factorization**: $\mathbf{B}$ has **full column rank** (its columns are independent) and $\mathbf{F}$ has **full row rank**. It is a convenient tool for proofs, as [Proposition 1.9.1](#) shows.

Theorem 1.2.2 — Rank-nullity

For an $m \times n$ matrix $\mathbf{A}$, $\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = n$.

Proof. Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be a basis of $\mathcal{N}(\mathbf{A})$ and extend it to a basis $\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}_1, \dots, \mathbf{w}_{n-k}$ of $\mathbb{R}^n$. Every $\mathbf{A}\mathbf{b}$ is a combination of $\mathbf{A}\mathbf{w}_1, \dots, \mathbf{A}\mathbf{w}_{n-k}$, since $\mathbf{A}\mathbf{v}_i = \mathbf{0}$. These vectors are independent: if $\sum_j c_j \mathbf{A}\mathbf{w}_j = \mathbf{0}$, then $\sum_j c_j \mathbf{w}_j \in \mathcal{N}(\mathbf{A})$ is a combination of the $\mathbf{v}_i$, and independence of the full basis forces every $c_j = 0$. So $\text{rank}(\mathbf{A}) = n - k$. ■

In a linear model with $n \times p$ model matrix $\mathbf{X}$, $\dim \mathcal{N}(\mathbf{X}) = p - \text{rank}(\mathbf{X})$ counts the directions in which the coefficient vector can move without changing the mean $\mathbf{X}\boldsymbol{\beta}$. $\mathbf{X}$ has full column rank iff $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$.

1.2.3 Rank of products and of $\mathbf{A}\mathbf{T}\mathbf{A}$

Proposition 1.2.3

Let $\mathbf{A}$ be $m \times n$ and $\mathbf{B}$ be $n \times q$.

- (a) $\mathbb{C}(\mathbf{AB}) \subseteq \mathbb{C}(\mathbf{A})$ and $\mathcal{N}(\mathbf{B}) \subseteq \mathcal{N}(\mathbf{AB})$, so $\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$.
- (b) If $\mathbf{B}$ has full row rank, then $\mathbb{C}(\mathbf{AB}) = \mathbb{C}(\mathbf{A})$. If $\mathbf{A}$ has full column rank, then $\mathcal{N}(\mathbf{AB}) = \mathcal{N}(\mathbf{B})$. So $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{A})$ in the first case and $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B})$ in the second. In particular, multiplying by a nonsingular matrix does not change rank.
- (c) $\mathcal{N}(\mathbf{ATA}) = \mathcal{N}(\mathbf{A})$ and $\mathbb{C}(\mathbf{ATA}) = \mathbb{C}(\mathbf{AT})$, so $\text{rank}(\mathbf{ATA}) = \text{rank}(\mathbf{AAT}) = \text{rank}(\mathbf{A})$.
- (d) $\text{rank}(\mathbf{A} + \mathbf{C}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{C})$ for $\mathbf{C}$ of the same size as $\mathbf{A}$.

Proof.

- (a) is [Eq. 1.1.2](#), together with $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{BTAT}) \leq \text{rank}(\mathbf{BT})$ by the first inclusion and [Theorem 1.2.1](#). (b) If $\mathbf{B}$ has full row rank, $\mathbb{C}(\mathbf{B}) = \mathbb{R}^n$ because it is an n -dimensional subspace of $\mathbb{R}^n$, so every $\mathbf{A}\mathbf{u}$ equals $\mathbf{A}\mathbf{B}\mathbf{v}$ for some $\mathbf{v}$. If $\mathbf{A}$ has full column rank, $\mathbf{A}\mathbf{B}\mathbf{v} = \mathbf{0}$ forces $\mathbf{B}\mathbf{v} = \mathbf{0}$. The rank statements follow from rank-nullity. (c) If $\mathbf{ATA}\mathbf{v} = \mathbf{0}$ then $\|\mathbf{A}\mathbf{v}\|^2 = \mathbf{vTATA}\mathbf{v} = 0$, so $\mathbf{A}\mathbf{v} = \mathbf{0}$. Thus the null spaces agree, and by [Theorem 1.2.2](#) $\text{rank}(\mathbf{ATA}) = \text{rank}(\mathbf{A})$. Since $\mathbb{C}(\mathbf{ATA}) \subseteq \mathbb{C}(\mathbf{AT})$ and both have dimension $\text{rank}(\mathbf{A})$, they are equal. Applying this to $\mathbf{AT}$ gives $\text{rank}(\mathbf{AAT}) = \text{rank}(\mathbf{A})$. (d) $\mathbb{C}(\mathbf{A} + \mathbf{C}) \subseteq \mathbb{C}(\mathbf{A}) + \mathbb{C}(\mathbf{C})$, and the dimension of a sum is at most the sum of the dimensions. ■

Part (c) is the reason the normal equations $\mathbf{X}\mathbf{T}\mathbf{X}\mathbf{b} = \mathbf{X}\mathbf{T}\mathbf{y}$ always have a solution: the right-hand side lies in $\mathbb{C}(\mathbf{X}\mathbf{T}) = \mathbb{C}(\mathbf{X}\mathbf{T}\mathbf{X})$ (Section 1.4).

1.2.4 Nonsingular submatrices

Proposition 1.2.4

Let $\mathbf{A}$ have rank $r \geq 1$. Choose any r linearly independent rows, indexed by I , and any r linearly independent columns, indexed by J . Then the $r \times r$ submatrix $\mathbf{A}_{IJ}$ at their intersection is nonsingular, and

$$\mathbf{A} = \mathbf{A}_{\cdot J} \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I\cdot}, \quad (1.2.1)$$

where $\mathbf{A}_{\cdot J}$ holds the columns in J and $\mathbf{A}_{I\cdot}$ the rows in I .

Proof. The columns in J are a basis of $\mathbb{C}(\mathbf{A})$, so $\mathbf{A} = \mathbf{A}_{\cdot J} \mathbf{F}$ for some $r \times n$ matrix $\mathbf{F}$. Keeping only the rows in I gives $\mathbf{A}_{I\cdot} = \mathbf{A}_{IJ} \mathbf{F}$. The left side has rank r , so by Proposition 1.2.3(a) the $r \times r$ matrix $\mathbf{A}_{IJ}$ has rank r and is nonsingular. Then $\mathbf{F} = \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I\cdot}$, which gives Eq. 1.2.1. ■

Conversely, a nonsingular $k \times k$ submatrix makes the k columns of $\mathbf{A}$ that contain it linearly independent, so $\text{rank}(\mathbf{A}) \geq k$. In particular, a matrix has rank r iff it has a nonsingular $r \times r$ submatrix and no nonsingular larger one. This proposition gives an explicit generalized inverse in Theorem 1.3.2.

1.2.5 Exercises

1.2.5.1 A. Check your understanding

Exercise A1.

Show that $\sum_{i < j} (y_i - y_j)^2 = n \sum_i (y_i - \bar{y})^2$, and write both sides as $\mathbf{y}^T \mathbf{A} \mathbf{y}$ with $\mathbf{A}$ symmetric. Find the symmetric matrix of the quadratic form $2x_1^2 - x_2^2 + 6x_1x_2 - 4x_2x_3$.

Exercise A2.

Show that $\mathbf{X}\mathbf{T}\mathbf{X}\mathbf{B} = \mathbf{X}\mathbf{T}\mathbf{X}\mathbf{C}$ implies $\mathbf{X}\mathbf{B} = \mathbf{X}\mathbf{C}$. Deduce that $\mathbf{X}\mathbf{T}\mathbf{X}\mathbf{b} = \mathbf{0}$ iff $\mathbf{X}\mathbf{b} = \mathbf{0}$.

Solution. Put $\mathbf{D} = \mathbf{B} - \mathbf{C}$, so that $\mathbf{X}\mathbf{T}\mathbf{X}\mathbf{D} = \mathbf{0}$. Then $(\mathbf{X}\mathbf{D})^T(\mathbf{X}\mathbf{D}) = \mathbf{D}^T \mathbf{X}^T \mathbf{X} \mathbf{D} = \mathbf{0}$, and by Eq. 1.1.3 a matrix $\mathbf{Z}$ with $\mathbf{Z}^T \mathbf{Z} = \mathbf{0}$ is zero. So $\mathbf{X}\mathbf{D} = \mathbf{0}$. The second statement is the case $\mathbf{C} = \mathbf{0}$, with the converse obvious.

Exercise A3.

For the additive two-way layout with two rows, three columns and one observation per cell, write the 6×6 model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{R}, \mathbf{C}]$, where $\mathbf{R}$ and $\mathbf{C}$ hold the row and column indicators. Find $\text{rank}(\mathbf{X})$ and a basis of $\mathcal{N}(\mathbf{X})$. Compare $[\mathbf{R}, \mathbf{C}]\mathbf{T}$ with the matrix $\mathbf{A}$ of [Example \(Filling a table from its margins\)](#).

1.2.5.2 B. Practice**Exercise B1.**

Let $\mathbf{A}$ be $m \times n$ and $\mathbf{B}$ be $n \times q$. Prove $\text{rank}(\mathbf{AB}) \geq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}) - n$. *Hint:* apply rank-nullity to $\mathbf{AB}_0$, where the columns of $\mathbf{B}_0$ are a basis of $\mathbb{C}(\mathbf{B})$.

Solution. Let the $n \times s$ matrix $\mathbf{B}_0$, $s = \text{rank}(\mathbf{B})$, have columns forming a basis of $\mathbb{C}(\mathbf{B})$. Then $\mathbb{C}(\mathbf{AB}) = \mathbb{C}(\mathbf{AB}_0)$, so by rank-nullity $\text{rank}(\mathbf{AB}) = s - \dim \mathcal{N}(\mathbf{AB}_0)$. The map $\mathbf{c} \mapsto \mathbf{B}_0\mathbf{c}$ is one-to-one and sends $\mathcal{N}(\mathbf{AB}_0)$ into $\mathcal{N}(\mathbf{A})$, so $\dim \mathcal{N}(\mathbf{AB}_0) \leq \dim \mathcal{N}(\mathbf{A}) = n - \text{rank}(\mathbf{A})$. Hence $\text{rank}(\mathbf{AB}) \geq \text{rank}(\mathbf{B}) - n + \text{rank}(\mathbf{A})$.

Exercise B2.

Show that if $\mathbf{A} = \mathbf{B}_1\mathbf{F}_1 = \mathbf{B}_2\mathbf{F}_2$ are two rank factorizations, then $\mathbf{B}_2 = \mathbf{B}_1\mathbf{T}$ and $\mathbf{F}_2 = \mathbf{T}^{-1}\mathbf{F}_1$ for a nonsingular $\mathbf{T}$.

1.3 Inverses and generalized inverses

1.3.1 The inverse of a square matrix

An $n \times n$ matrix $\mathbf{A}$ is **nonsingular** if there is a matrix $\mathbf{A}^{-1}$ with $\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$. The inverse is unique: if $\mathbf{B}$ and $\mathbf{C}$ both qualify, then $\mathbf{B} = \mathbf{B}(\mathbf{AC}) = (\mathbf{BA})\mathbf{C} = \mathbf{C}$.

Proposition 1.3.1

For an $n \times n$ matrix $\mathbf{A}$ the following are equivalent: (a) $\mathbf{A}$ is nonsingular; (b) $\text{rank}(\mathbf{A}) = n$; (c) $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$; (d) $\mathbf{BA} = \mathbf{I}$ for some $\mathbf{B}$; (e) $\mathbf{AC} = \mathbf{I}$ for some $\mathbf{C}$. When they hold, $\mathbf{B} = \mathbf{C} = \mathbf{A}^{-1}$, and

$$(\mathbf{AT})^{-1} = (\mathbf{A}^{-1})\mathbf{T}, \quad (\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

for nonsingular $\mathbf{A}, \mathbf{B}$ of the same size. A nonsingular symmetric matrix has a symmetric inverse.

Proof. (b) $\Leftrightarrow$ (c) is rank-nullity. (a) $\Rightarrow$ (d) and (a) $\Rightarrow$ (e) are trivial. (d) $\Rightarrow$ (c): $\mathbf{Ax} = \mathbf{0}$ gives $\mathbf{x} = \mathbf{BAx} = \mathbf{0}$. (e) $\Rightarrow$ (b): $\mathbf{I} = \mathbf{AC}$ has rank n , and $\text{rank}(\mathbf{AC}) \leq \text{rank}(\mathbf{A})$. (b) $\Rightarrow$ (a): since $\mathbb{C}(\mathbf{A}) = \mathbb{R}^n$, we can solve $\mathbf{Ac}_j = \mathbf{e}_j$ for each j , giving $\mathbf{AC} = \mathbf{I}$. Then

$\mathbf{A}(\mathbf{CA} - \mathbf{I}) = \mathbf{0}$, and because $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$ each column of $\mathbf{CA} - \mathbf{I}$ is zero. So $\mathbf{CA} = \mathbf{I}$ as well. If $\mathbf{BA} = \mathbf{I}$ then $\mathbf{B} = \mathbf{BAA}^{-1} = \mathbf{A}^{-1}$, and similarly for $\mathbf{C}$. The formulas are verified by multiplication: $\mathbf{AT}(\mathbf{A}^{-1})\mathbf{T} = (\mathbf{A}^{-1}\mathbf{A})\mathbf{T} = \mathbf{I}$ and $(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{I}$. If $\mathbf{A} = \mathbf{AT}$, then $\mathbf{A}^{-1} = (\mathbf{AT})^{-1} = (\mathbf{A}^{-1})\mathbf{T}$.

■

A rectangular matrix has no two-sided inverse. If $\mathbf{X}$ is $n \times p$ with full column rank, then $\mathbf{XTX}$ is nonsingular by Proposition 1.2.3(c), and $(\mathbf{XTX})^{-1}\mathbf{XT}$ is a *left inverse*: it multiplies $\mathbf{X}$ on the left to give $\mathbf{I}_p$. This is the matrix that computes least squares coefficients.

1.3.2 Generalized inverses

When $\mathbf{A}$ is singular or rectangular we still want a matrix that inverts $\mathbf{A}$ on the vectors where inversion makes sense, namely those in $\mathbb{C}(\mathbf{A})$. The following weak requirement turns out to be exactly what solving equations needs.

Definition 1.3.1 — Generalized inverse

A **generalized inverse** of an $m \times n$ matrix $\mathbf{A}$ is any $n \times m$ matrix $\mathbf{A}^-$ with

$$\mathbf{AA}^-\mathbf{A} = \mathbf{A}.$$

Read columnwise, the definition says that $\mathbf{AA}^-$ acts as the identity on every column of $\mathbf{A}$, hence on all of $\mathbb{C}(\mathbf{A})$: if $\mathbf{u} = \mathbf{Ab}$ then $\mathbf{A}(\mathbf{A}^-\mathbf{u}) = \mathbf{AA}^-\mathbf{Ab} = \mathbf{u}$. So $\mathbf{A}^-\mathbf{u}$ is a solution of $\mathbf{Ax} = \mathbf{u}$ whenever one exists. If $\mathbf{A}$ is nonsingular, multiplying the definition by $\mathbf{A}^{-1}$ on both sides gives $\mathbf{A}^- = \mathbf{A}^{-1}$. In general there are many generalized inverses. Chapter 6 uses them to write projections, and Chapter 8 uses them to describe estimable functions in models of less than full rank.

Theorem 1.3.2 — Existence

Every matrix has a generalized inverse. Explicitly, if $\text{rank}(\mathbf{A}) = r \geq 1$ and $\mathbf{A}_{IJ}$ is the nonsingular $r \times r$ submatrix of Proposition 1.2.4, the $n \times m$ matrix $\mathbf{G}$ that holds $\mathbf{A}_{IJ}^{-1}$ in the rows indexed by J and the columns indexed by I , and zeros elsewhere, is a generalized inverse of $\mathbf{A}$.

Proof. For $\mathbf{A} = \mathbf{0}$ any matrix of the right size works. Otherwise let $\mathbf{E}_I$ and $\mathbf{E}_J$ be the columns of $\mathbf{I}_m$ and $\mathbf{I}_n$ indexed by I and J . Then $\mathbf{G} = \mathbf{E}_J\mathbf{A}_{IJ}^{-1}\mathbf{E}_I\mathbf{T}$, $\mathbf{AE}_J = \mathbf{A}_{.J}$ and $\mathbf{E}_I\mathbf{TA} = \mathbf{A}_{I.}$. So $\mathbf{AGA} = \mathbf{A}_{.J}\mathbf{A}_{IJ}^{-1}\mathbf{A}_{I.} = \mathbf{A}$ by Eq. 1.2.1.

■

Proposition 1.3.3 — Properties of generalized inverses

Let $\mathbf{A}$ be $m \times n$ of rank r and let $\mathbf{G}$ be a generalized inverse of $\mathbf{A}$.

- (a) $\mathbf{AG}$ and $\mathbf{GA}$ are idempotent. Moreover $\mathbb{C}(\mathbf{AG}) = \mathbb{C}(\mathbf{A})$ and $\mathcal{N}(\mathbf{GA}) = \mathcal{N}(\mathbf{A})$,

and $\text{rank}(\mathbf{AG}) = \text{rank}(\mathbf{GA}) = r \leq \text{rank}(\mathbf{G})$.

- (b) $\mathbf{GT}$ is a generalized inverse of $\mathbf{AT}$.
- (c) If $\mathbf{G}_1, \mathbf{G}_2$ are generalized inverses of $\mathbf{A}$, so is $\mathbf{G}_1\mathbf{AG}_2$. In particular $\mathbf{GAG}$ is a generalized inverse which also satisfies $(\mathbf{GAG})\mathbf{A}(\mathbf{GAG}) = \mathbf{GAG}$.
- (d) If $\mathbf{A}$ is symmetric, then $\mathbf{GT}$ and $\frac{1}{2}(\mathbf{G} + \mathbf{GT})$ are generalized inverses of $\mathbf{A}$, so a symmetric generalized inverse always exists.
- (e) $\mathbf{G} + (\mathbf{I} - \mathbf{GA})\mathbf{U} + \mathbf{V}(\mathbf{I} - \mathbf{AG})$ is a generalized inverse of $\mathbf{A}$ for every $\mathbf{U}$ and $\mathbf{V}$ of size $n \times m$.

Proof.

- (a) $(\mathbf{AG})^2 = (\mathbf{AGA})\mathbf{G} = \mathbf{AG}$, and likewise for $\mathbf{GA}$. Next, $\mathbb{C}(\mathbf{A}) = \mathbb{C}(\mathbf{AGA}) \subseteq \mathbb{C}(\mathbf{AG}) \subseteq \mathbb{C}(\mathbf{A})$, and $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{GA}) \subseteq \mathcal{N}(\mathbf{AGA}) = \mathcal{N}(\mathbf{A})$. The ranks follow from these and rank-nullity, and $r = \text{rank}(\mathbf{AG}) \leq \text{rank}(\mathbf{G})$.
- (b) Transpose $\mathbf{AGA} = \mathbf{A}$. (c) $\mathbf{AG}_1\mathbf{AG}_2\mathbf{A} = \mathbf{AG}_2\mathbf{A} = \mathbf{A}$. The second statement is $\mathbf{GAGAGAG} = \mathbf{G}(\mathbf{AGA})\mathbf{GAG} = \mathbf{GAGAG} = \mathbf{GAG}$.
- (c) By (b), $\mathbf{GT}$ is a generalized inverse of $\mathbf{AT} = \mathbf{A}$. The condition $\mathbf{AGA} = \mathbf{A}$ is linear in $\mathbf{G}$, so it is preserved by averaging.
- (d) Multiply out, using $\mathbf{A}(\mathbf{I} - \mathbf{GA}) = \mathbf{0}$ and $(\mathbf{I} - \mathbf{AG})\mathbf{A} = \mathbf{0}$.

■

A generalized inverse with the extra property in (c), $\mathbf{GAG} = \mathbf{G}$, is called **reflexive**. [Exercise \(B1\)](#) shows that the reflexive ones are exactly those of rank r . Part (e) in fact produces *every* generalized inverse from a single one as $\mathbf{U}$ and $\mathbf{V}$ vary. Idempotence in (a) means that $\mathbf{AG}$ and $\mathbf{GA}$ are projections, generally not orthogonal ones. [Chapter 6](#) shows that the particular combination $\mathbf{X}(\mathbf{XTX})^{-1}\mathbf{XT}$ is symmetric, does not depend on the choice of generalized inverse, and is the orthogonal projection onto $\mathbb{C}(\mathbf{X})$.

1.3.3 The Moore–Penrose inverse

One generalized inverse is singled out by symmetry requirements.

Definition 1.3.2 — Moore–Penrose inverse

The **Moore–Penrose inverse** of $\mathbf{A}$ is the matrix $\mathbf{A}^+$ satisfying

$$\mathbf{AA}^+\mathbf{A} = \mathbf{A}, \quad \mathbf{A}^+\mathbf{AA}^+ = \mathbf{A}^+, \quad (\mathbf{AA}^+)^T = \mathbf{AA}^+, \quad (\mathbf{A}^+\mathbf{A})^T = \mathbf{A}^+\mathbf{A}.$$

There is at most one such matrix. If $\mathbf{G}$ and $\mathbf{H}$ both satisfy the four conditions, then

$$\mathbf{G} = \mathbf{GAG} = \mathbf{G}(\mathbf{AG})^T = \mathbf{GGT}(\mathbf{AHA})^T = \mathbf{G}(\mathbf{AG})^T(\mathbf{AH})^T = \mathbf{GAGAH} = \mathbf{GAH},$$

where the steps use the second and third conditions for $\mathbf{G}$, the first for $\mathbf{H}$, and the third for both. The same argument with the fourth condition in place of the third gives

$$\mathbf{H} = (\mathbf{HA})^T\mathbf{H} = \mathbf{ATHTH} = (\mathbf{AGA})^T\mathbf{H}^T\mathbf{H} = (\mathbf{GA})^T(\mathbf{HA})^T\mathbf{H} = \mathbf{GAHAH} = \mathbf{GAH},$$

so $\mathbf{G} = \mathbf{H}$. Existence follows from the singular value decomposition, which gives $\mathbf{A}^+$ explicitly (Theorem 1.8.1).

1.4 Systems of equations and consistency

A system $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{A}$ of size $m \times n$ is **consistent** if it has at least one solution. Least squares leads to such systems in which $\mathbf{A} = \mathbf{XTX}$ may be singular, so we need to know when solutions exist, what all of them look like, and which features are common to all of them.

Theorem 1.4.1 — Consistency and the general solution

Let $\mathbf{G}$ be any generalized inverse of $\mathbf{A}$. The following are equivalent:

- (a) $\mathbf{Ax} = \mathbf{b}$ is consistent;
- (b) $\mathbf{b} \in \mathbb{C}(\mathbf{A})$;
- (c) $\text{rank}[\mathbf{A}, \mathbf{b}] = \text{rank}(\mathbf{A})$;
- (d) $\mathbf{AGb} = \mathbf{b}$.

If the system is consistent, $\mathbf{x}_0 = \mathbf{Gb}$ is a solution, and the set of all solutions is

$$\{\mathbf{Gb} + (\mathbf{I} - \mathbf{GA})\mathbf{z} : \mathbf{z} \in \mathbb{R}^n\} = \mathbf{x}_0 + \mathcal{N}(\mathbf{A}). \quad (1.4.1)$$

Proof. (a) $\Leftrightarrow$ (b) is the first reading in Eq. 1.1.2. (b) $\Leftrightarrow$ (c): appending $\mathbf{b}$ as a column leaves the column space unchanged iff $\mathbf{b}$ is already in it, and a subspace of $\mathbb{C}([\mathbf{A}, \mathbf{b}])$ with the same dimension equals it. (d) $\Rightarrow$ (a): $\mathbf{Gb}$ is a solution. (b) $\Rightarrow$ (d): if $\mathbf{b} = \mathbf{Au}$, then $\mathbf{AGb} = \mathbf{AGA}u = \mathbf{Au} = \mathbf{b}$.

The solutions of a consistent system are $\mathbf{x}_0 + \mathcal{N}(\mathbf{A})$, since $\mathbf{Ax} = \mathbf{b}$ iff $\mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{0}$. It remains to show $\mathcal{N}(\mathbf{A}) = \mathbb{C}(\mathbf{I} - \mathbf{GA})$. The inclusion $\supseteq$ holds because $\mathbf{A}(\mathbf{I} - \mathbf{GA}) = \mathbf{0}$. Conversely, if $\mathbf{Av} = \mathbf{0}$, then $\mathbf{v} = (\mathbf{I} - \mathbf{GA})\mathbf{v}$.

■

Criterion (d) is useful in practice: it tests consistency with a single matrix product, for any generalized inverse. By Proposition 1.3.3(a) and rank-nullity, $\text{rank}(\mathbf{I} - \mathbf{GA}) = n - r$, so the solution set is an affine subspace of dimension $n - r$. The solution is unique iff $\text{rank}(\mathbf{A}) = n$, and then $\mathbf{Gb}$ is the same for every $\mathbf{G}$.

Idea. Any generalized inverse does three jobs for $\mathbf{Ax} = \mathbf{b}$. It tests consistency ($\mathbf{AGb} = \mathbf{b}$), produces one solution ($\mathbf{Gb}$), and parametrizes all of them ($\mathbf{Gb} + (\mathbf{I} - \mathbf{GA})\mathbf{z}$). Which $\mathbf{G}$ is used changes the particular solution but never the solution set, and a linear function $\mathbf{qTx}$ is the same across solutions exactly when $\mathbf{q} \in \mathbb{C}(\mathbf{AT})$.

Corollary 1.4.2 — Invariant linear functions

Let $\mathbf{Ax} = \mathbf{b}$ be consistent and let $\mathbf{q} \in \mathbb{R}^n$. The value $\mathbf{qTx}$ is the same for every solution

$\mathbf{x}$ iff $\mathbf{q} \in \mathbb{C}(\mathbf{AT})$, that is, iff $\mathbf{qT} = \mathbf{qTGA}$ for one (and then every) generalized inverse $\mathbf{G}$.

Proof. By Eq. 1.4.1, $\mathbf{qTx}$ is constant over solutions iff $\mathbf{qT}(\mathbf{I} - \mathbf{GA})\mathbf{z} = 0$ for all $\mathbf{z}$, that is, iff $\mathbf{qT} = \mathbf{qTGA}$. That equation puts $\mathbf{q}$ in $\mathbb{C}(\mathbf{AT})$. Conversely, if $\mathbf{q} = \mathbf{ATw}$, then $\mathbf{qTGA} = \mathbf{wTAGA} = \mathbf{wTA} = \mathbf{qT}$. ■

Applied to the normal equations $\mathbf{XTXb} = \mathbf{XTy}$, which are consistent by Proposition 1.2.3(c), the corollary says that $\mathbf{qT}\hat{\mathbf{b}}$ is the same for all least squares solutions iff $\mathbf{q} \in \mathbb{C}(\mathbf{XTX}) = \mathbb{C}(\mathbf{XT})$. This is the algebra behind estimable functions (Chapter 8).

Proposition 1.4.3 — Minimum-norm solution

If $\mathbf{Ax} = \mathbf{b}$ is consistent, then $\mathbf{A}^+\mathbf{b}$ is its unique solution lying in $\mathbb{C}(\mathbf{AT})$, and $\|\mathbf{A}^+\mathbf{b}\| < \|\mathbf{x}\|$ for every other solution $\mathbf{x}$.

Proof. $\mathbf{A}^+\mathbf{b}$ is a solution because $\mathbf{A}^+$ is a generalized inverse. From the second and fourth Penrose conditions, $\mathbf{A}^+ = (\mathbf{A}^+\mathbf{A})\mathbf{A}^+ = (\mathbf{A}^+\mathbf{A})\mathbf{TA}^+ = \mathbf{AT}(\mathbf{A}^+)\mathbf{TA}^+$, so $\mathbf{A}^+\mathbf{b} \in \mathbb{C}(\mathbf{AT})$. Every vector of $\mathbb{C}(\mathbf{AT})$ is orthogonal to every $\mathbf{v} \in \mathcal{N}(\mathbf{A})$, since $(\mathbf{ATw})\mathbf{Tv} = \mathbf{wTA}^+\mathbf{v} = 0$. Any other solution is $\mathbf{x} = \mathbf{A}^+\mathbf{b} + \mathbf{v}$ with $\mathbf{0} \neq \mathbf{v} \in \mathcal{N}(\mathbf{A})$, and then $\|\mathbf{x}\|^2 = \|\mathbf{A}^+\mathbf{b}\|^2 + \|\mathbf{v}\|^2$. The same computation shows that $\mathbf{x} \notin \mathbb{C}(\mathbf{AT})$, because $\mathbf{xTv} = \|\mathbf{v}\|^2 \neq 0$. ■

Example. (Filling a table from its margins)

Suppose the six entries of a 2×3 table are unknown, but its row totals (9, 6) and column totals (4, 5, 6) are known. With the cells ordered $x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}$, the equations are $\mathbf{Ax} = \mathbf{b}$ with

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 9 \\ 6 \\ 4 \\ 5 \\ 6 \end{pmatrix}.$$

The rows of $\mathbf{A}$ satisfy one relation: the two row-total rows and the three column-total rows both add to $\mathbf{1}_6\mathbf{T}$. So $\text{rank}(\mathbf{A}) = 4$ and there is a $6 - 4 = 2$ -dimensional family of solutions. The same relation must hold for $\mathbf{b}$. By Theorem 1.4.1(c), the system is consistent iff the row totals and the column totals have the same sum, and here both sums are 15.

The listing builds three generalized inverses: $\mathbf{G}_1$ from the nonsingular submatrix in rows 1–4 and columns 1–4 (Theorem 1.3.2), the Moore–Penrose inverse $\mathbf{G}_2 = \mathbf{A}^+$, and $\mathbf{G}_3 = \mathbf{G}_1 + (\mathbf{I} - \mathbf{G}_1\mathbf{A})\mathbf{Z}$ for an integer matrix $\mathbf{Z}$ (Proposition 1.3.3(e)). They give the

solutions

$$\begin{aligned}\mathbf{G}_1\mathbf{b} &= (-2, 5, 6, 6, 0, 0)\mathbf{T}, & \|\mathbf{G}_1\mathbf{b}\| &= 10.050, \\ \mathbf{G}_2\mathbf{b} &= (5/2, 3, 7/2, 3/2, 2, 5/2)\mathbf{T}, & \|\mathbf{G}_2\mathbf{b}\| &= 6.325, \\ \mathbf{G}_3\mathbf{b} &= (21, -39, 27, -17, 44, -21)\mathbf{T}, & \|\mathbf{G}_3\mathbf{b}\| &= 73.192.\end{aligned}$$

The individual cell x_{11} takes the values -2 , $5/2$ and 21 : margins do not determine cells. But $x_{12} + x_{13} - x_{21}$ equals 5 for all three, as [Corollary 1.4.2](#) predicts, because its coefficient vector is the first row of $\mathbf{A}$ minus the third. The minimum-norm solution has the additive form $x_{ij} = r_i/3 + c_j/2 - 15/6$ in terms of the row totals r_i and column totals c_j ([Exercise \(A2\)](#)). If the last column total is changed to 7 , then $\mathbf{AGb} \neq \mathbf{b}$ for each of the three matrices, and the script confirms that the test detects the inconsistency. The script also checks all 60 submatrices formed by 4 independent rows and 4 independent columns, and finds each one nonsingular, as [Proposition 1.2.4](#) requires.

```
from fractions import Fraction
from itertools import combinations
import numpy as np
m, n = A.shape
# cells ordered (1,1), (1,2), (1,3), (2,1), (2,2), (2,3)
A = np.array([[1, 1, 1, 0, 0, 0],      # row 1 total
              [0, 0, 0, 1, 1, 1],      # row 2 total
              [1, 0, 0, 1, 0, 0],      # column 1 total
              [0, 1, 0, 0, 1, 0],      # column 2 total
              [0, 0, 1, 0, 0, 1]],      # column 3 total
             dtype=float)
b = np.array([9, 6, 4, 5, 6], dtype=float) # both sets of totals add to 15
r = np.linalg.matrix_rank(A)
print("rank(A) =", r, " rank([A, b]) =", np.linalg.matrix_rank(np.column_stack([A, b]
)))

def ginverse_from_submatrix(A, rows, cols):
    """Place the inverse of the nonsingular block A[rows, cols] at [cols, rows]."""
    G = np.zeros(A.T.shape)
    G[np.ix_(cols, rows)] = np.linalg.inv(A[np.ix_(rows, cols)])
    return G

rows, cols = [0, 1, 2, 3], [0, 1, 2, 3] # r independent rows and
# columns
G1 = ginverse_from_submatrix(A, rows, cols)
G2 = np.linalg.pinv(A) # Moore-Penrose inverse
rng = np.random.default_rng(1)
Z = rng.integers(-2, 3, size=A.T.shape).astype(float)
G3 = G1 + (np.eye(n) - G1 @ A) @ Z # another, non-reflexive
# one

for G in (G1, G2, G3):
    assert np.allclose(A @ G @ A, A) # each is a g-inverse
    assert np.allclose(A @ G @ b, b) # consistency test passes
    x = G @ b
    print(np.round(x, 4), " A x - b =", np.abs(A @ x - b).max())
```

1.4.1 Exercises

1.4.1.1 A. Check your understanding

Exercise A1.

Find all generalized inverses of a nonzero vector $\mathbf{a} \in \mathbb{R}^n$, regarded as an $n \times 1$ matrix. Which one is $\mathbf{a}^+$?

Exercise A2.

Generalize [Example \(Filling a table from its margins\)](#) to an $a \times b$ table with row totals r_i , column totals c_j and grand total $T = \sum_i r_i = \sum_j c_j$. Show that $x_{ij} = r_i/b + c_j/a - T/(ab)$ is a solution lying in $\mathbb{C}(\mathbf{AT})$, and hence is the minimum-norm solution.

Solution. Row i of the proposed table sums to $r_i + \sum_j c_j/a - bT/(ab) = r_i$, and column j sums to $\sum_i r_i/b + c_j - T/b = c_j$. The rows of $\mathbf{A}$ are the row indicators and the column indicators, so $\mathbb{C}(\mathbf{AT})$ consists of the tables of the form $x_{ij} = \alpha_i + \beta_j$. The proposed solution has this form. By [Proposition 1.4.3](#), it is $\mathbf{A}^+\mathbf{b}$. For $a = 2$, $b = 3$ and $T = 15$ this is the additive formula quoted in [Example \(Filling a table from its margins\)](#).

Exercise A3.

For which c and d is the system $x_1 + x_2 + x_3 = 2$, $x_1 - x_2 = c$, $3x_1 + x_2 + 2x_3 = d$ consistent? When it is, give its general solution in the form [Eq. 1.4.1](#).

1.4.1.2 B. Practice

Exercise B1.

Let $\mathbf{G}$ be a generalized inverse of $\mathbf{A}$. Show that $\mathbf{GAG} = \mathbf{G}$ iff $\text{rank}(\mathbf{G}) = \text{rank}(\mathbf{A})$.

Solution. If $\mathbf{GAG} = \mathbf{G}$, then $\text{rank}(\mathbf{G}) \leq \text{rank}(\mathbf{A})$ by [Proposition 1.2.3\(a\)](#), and $\text{rank}(\mathbf{A}) \leq \text{rank}(\mathbf{G})$ by [Proposition 1.3.3\(a\)](#). Conversely, suppose $\text{rank}(\mathbf{G}) = \text{rank}(\mathbf{A}) = r$. Since $\mathbb{C}(\mathbf{GA}) \subseteq \mathbb{C}(\mathbf{G})$ and $\text{rank}(\mathbf{GA}) = r$, the two column spaces coincide, so $\mathbf{G} = \mathbf{GAK}$ for some $\mathbf{K}$. Then $\mathbf{GAG} = \mathbf{GAGAK} = \mathbf{GAK} = \mathbf{G}$, using $\mathbf{AGA} = \mathbf{A}$.

Exercise B2.

Let $\mathbf{A} = \mathbf{BF}$ be a rank factorization. Show that $\mathbf{A}^+ = \mathbf{FT}(\mathbf{FFT})^{-1}(\mathbf{BTB})^{-1}\mathbf{BT}$. Deduce that $\mathbf{A}^+ = (\mathbf{ATA})^{-1}\mathbf{AT}$ when $\mathbf{A}$ has full column rank.

Solution. Write $\mathbf{G}$ for the proposed matrix. Then $\mathbf{AG} = \mathbf{B}(\mathbf{BTB})^{-1}\mathbf{BT}$ and $\mathbf{GA} = \mathbf{FT}(\mathbf{FFT})^{-1}\mathbf{F}$, both symmetric. Also $\mathbf{AGA} = \mathbf{B}(\mathbf{BTB})^{-1}\mathbf{BTBF} = \mathbf{A}$ and $\mathbf{GAG} = \mathbf{FT}(\mathbf{FFT})^{-1}\mathbf{FFT}(\mathbf{FFT})^{-1}(\mathbf{BTB})^{-1}\mathbf{BT} = \mathbf{G}$. By uniqueness $\mathbf{G} = \mathbf{A}^+$. If $\mathbf{A}$ has full column rank, take $\mathbf{B} = \mathbf{A}$ and $\mathbf{F} = \mathbf{I}$.

Exercise B3.

Let $\mathbf{A}$ be $m \times n$ of rank r and let $\mathbf{Ax} = \mathbf{b}$ be consistent with $\mathbf{b} \neq \mathbf{0}$. Show that the system has $n - r + 1$ linearly independent solutions and no more, and that every solution is an affine combination of them.

1.4.1.3 C. Going deeper**Exercise C1.**

Show that $(\mathbf{A}^+)^+ = \mathbf{A}$, $(\mathbf{AT})^+ = (\mathbf{A}^+)\mathbf{T}$, $(\mathbf{ATA})^+ = \mathbf{A}^+(\mathbf{A}^+)\mathbf{T}$ and $\mathbf{A}^+ = (\mathbf{ATA})^+\mathbf{AT}$. Give an example with $(\mathbf{AB})^+ \neq \mathbf{B}^+\mathbf{A}^+$.

1.5 Orthogonality and orthogonal matrices

1.5.1 Orthogonal vectors and complements

Vectors $\mathbf{u}, \mathbf{v}$ are **orthogonal**, written $\mathbf{u} \perp \mathbf{v}$, if $\mathbf{u}^T\mathbf{v} = 0$. For orthogonal vectors the Pythagorean identity $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ holds. A set of nonzero, mutually orthogonal vectors is independent: if $\sum_j c_j \mathbf{v}_j = \mathbf{0}$, taking the inner product with $\mathbf{v}_k$ leaves $c_k \|\mathbf{v}_k\|^2 = 0$. A set is **orthonormal** if, in addition, each vector has unit length. The columns of an $n \times r$ matrix $\mathbf{Q}$ are orthonormal iff $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_r$.

The **orthogonal complement** of a subspace $\mathcal{S} \subseteq \mathbb{R}^n$ is $\mathcal{S}^\perp = \{\mathbf{v} : \mathbf{v}^T\mathbf{u} = 0 \text{ for all } \mathbf{u} \in \mathcal{S}\}$. The next proposition is all that this chapter needs about complements. Chapter 6 builds the theory of orthogonal projections on it.

Proposition 1.5.1

Let $\mathbf{A}$ be $m \times n$ and let $\mathcal{S} \subseteq \mathbb{R}^n$ be a subspace.

- (a) $\mathcal{N}(\mathbf{A}) = \mathbb{C}(\mathbf{AT})^\perp$ and $\mathcal{N}(\mathbf{AT}) = \mathbb{C}(\mathbf{A})^\perp$.
- (b) $\dim \mathcal{S}^\perp = n - \dim \mathcal{S}$, and $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp$. In particular $\mathbb{R}^n = \mathbb{C}(\mathbf{AT}) \oplus \mathcal{N}(\mathbf{A})$ and $\mathbb{R}^m = \mathbb{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{AT})$.
- (c) $(\mathcal{S}^\perp)^\perp = \mathcal{S}$.

Proof.

- (a) $\mathbf{Av} = \mathbf{0}$ says that $\mathbf{v}$ is orthogonal to every row of $\mathbf{A}$, hence to their span. Apply this to $\mathbf{AT}$ for the second identity.
- (b) Put a basis of $\mathcal{S}$ (k vectors) into the columns of $\mathbf{B}$. Then $\mathcal{S}^\perp = \mathcal{N}(\mathbf{BT})$, which has dimension $n - k$ by rank-nullity. If $\mathbf{u} \in \mathcal{S} \cap \mathcal{S}^\perp$ then $\mathbf{u}^T\mathbf{u} = 0$, so the sum is direct and has dimension n , so it is $\mathbb{R}^n$.
- (c) $\mathcal{S} \subseteq (\mathcal{S}^\perp)^\perp$, and both have dimension $\dim \mathcal{S}$ by (b).

1.5.2 Gram–Schmidt and the QR factorization

Theorem 1.5.2 — Gram–Schmidt; QR factorization

Let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_p]$ be $n \times p$ with full column rank. Define recursively

$$\mathbf{w}_j = \mathbf{x}_j - \sum_{k < j} (\mathbf{u}_k^T \mathbf{x}_j) \mathbf{u}_k, \quad \mathbf{u}_j = \mathbf{w}_j / \|\mathbf{w}_j\|.$$

Then $\mathbf{u}_1, \dots, \mathbf{u}_p$ are orthonormal and, for each j , span the same subspace as $\mathbf{x}_1, \dots, \mathbf{x}_j$. Equivalently, $\mathbf{X} = \mathbf{Q}\mathbf{R}$ with $\mathbf{Q} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$ satisfying $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}_p$ and $\mathbf{R}$ upper triangular with positive diagonal entries.

Proof. By induction on j . Suppose $\mathbf{u}_1, \dots, \mathbf{u}_{j-1}$ are orthonormal and span $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_{j-1})$. Then $\mathbf{w}_j \neq \mathbf{0}$, since otherwise $\mathbf{x}_j$ would lie in that span, contradicting independence. For $i < j$, $\mathbf{u}_i^T \mathbf{w}_j = \mathbf{u}_i^T \mathbf{x}_j - \sum_{k < j} (\mathbf{u}_k^T \mathbf{x}_j) \mathbf{u}_k^T \mathbf{u}_i = 0$. So $\mathbf{u}_1, \dots, \mathbf{u}_j$ are orthonormal, and they span $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_j)$ because $\mathbf{x}_j = \|\mathbf{w}_j\| \mathbf{u}_j + \sum_{k < j} (\mathbf{u}_k^T \mathbf{x}_j) \mathbf{u}_k$. That last identity, read column by column, is $\mathbf{X} = \mathbf{Q}\mathbf{R}$ with $r_{jj} = \|\mathbf{w}_j\| > 0$ and $r_{kj} = \mathbf{u}_k^T \mathbf{x}_j$ for $k < j$. ■

The factorization is unique ([Exercise \(B2\)](#)). Numerical software computes it by a more stable route, but the object is the same, and [Chapter 6](#) and [Chapter 10](#) rely on it. Two consequences are used repeatedly. First, any orthonormal set $\mathbf{u}_1, \dots, \mathbf{u}_r$ in $\mathbb{R}^n$ extends to an orthonormal basis: apply Gram–Schmidt to $\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{e}_1, \dots, \mathbf{e}_n$, discarding each vector that depends on its predecessors. Second, every subspace has an orthonormal basis.

1.5.3 Orthogonal matrices

A square matrix $\mathbf{Q}$ is **orthogonal** if $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$. By [Proposition 1.3.1](#), $\mathbf{Q}^T$ is then the inverse of $\mathbf{Q}$, so also $\mathbf{Q}\mathbf{Q}^T = \mathbf{I}$: the rows of an orthogonal matrix are orthonormal as well as its columns.

Proposition 1.5.3

Let $\mathbf{Q}$ and $\mathbf{P}$ be $n \times n$ orthogonal matrices.

- (a) $(\mathbf{Q}\mathbf{x})^T (\mathbf{Q}\mathbf{v}) = \mathbf{x}^T \mathbf{v}$ and $\|\mathbf{Q}\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}, \mathbf{v}$.
- (b) $\mathbf{Q}^T$ and $\mathbf{Q}\mathbf{P}$ are orthogonal.
- (c) Every entry satisfies $|q_{ij}| \leq 1$, and $\det \mathbf{Q} = \pm 1$.
- (d) If $[\mathbf{Q}_1, \mathbf{Q}_2]$ is orthogonal with $\mathbf{Q}_1$ of size $n \times r$, then $\mathbf{Q}_1 \mathbf{Q}_1^T + \mathbf{Q}_2 \mathbf{Q}_2^T = \mathbf{I}_n$, and $\mathcal{C}(\mathbf{Q}_2) = \mathcal{C}(\mathbf{Q}_1)^\perp$.

Proof.

- (a) $(\mathbf{Q}\mathbf{x})^T\mathbf{Q}\mathbf{v} = \mathbf{x}^T\mathbf{Q}^T\mathbf{Q}\mathbf{v} = \mathbf{x}^T\mathbf{v}$. (b) $(\mathbf{Q}\mathbf{P})^T\mathbf{Q}\mathbf{P} = \mathbf{P}^T\mathbf{P} = \mathbf{I}$.
 (b) Each column has unit length. For the determinant, see [Proposition 1.6.2](#): $1 = \det(\mathbf{Q}^T\mathbf{Q}) = (\det \mathbf{Q})^2$. (d) The first statement is $\mathbf{Q}\mathbf{Q}^T = \mathbf{I}$ written in blocks. For the second, $\mathbf{Q}_1^T\mathbf{Q}_2 = \mathbf{0}$ gives $\mathbb{C}(\mathbf{Q}_2) \subseteq \mathbb{C}(\mathbf{Q}_1)^\perp$, and both have dimension $n - r$.

■

Part (a) says an orthogonal matrix is a rigid motion that fixes the origin: a rotation, a reflection, or a composition of the two. Changing coordinates by an orthogonal matrix changes neither lengths nor angles. This is why the distribution theory of [Chapter 4](#) can rotate a normal vector with independent components into a convenient coordinate system without changing its distribution.

Example. (Helmert matrices)

The $n \times n$ **Helmert matrix** $\mathbf{H}$ has first row $n^{-1/2}\mathbf{1}^T$ and, for $k = 2, \dots, n$, k th row

$$\mathbf{h}_k^T = \frac{1}{\sqrt{k(k-1)}} \underbrace{(1, \dots, 1)_{k-1}}_{k-1}, -(k-1), 0, \dots, 0).$$

Each row has unit length, since $(k-1) + (k-1)^2 = k(k-1)$. Row k is orthogonal to row 1 because its entries sum to zero. For $2 \leq j < k$, the nonzero entries of $\mathbf{h}_j$ sit in positions $1, \dots, j$, where $\mathbf{h}_k$ is constant, so $\mathbf{h}_j^T\mathbf{h}_k$ is proportional to the sum of the entries of $\mathbf{h}_j$, which is zero. So $\mathbf{H}$ is orthogonal. For $\mathbf{y} \in \mathbb{R}^n$, $(\mathbf{H}\mathbf{y})_1 = \sqrt{n}\bar{y}$ and, by [Proposition 1.5.3\(a\)](#),

$$\sum_{k=2}^n (\mathbf{h}_k^T\mathbf{y})^2 = \|\mathbf{H}\mathbf{y}\|^2 - n\bar{y}^2 = \sum_{i=1}^n (y_i - \bar{y})^2.$$

The sum of squared deviations is a sum of $n-1$ squares of orthonormal contrasts. With independent normal observations this gives a short proof that the sample mean and variance are independent ([Exercise \(B1\)](#)).

Two other families appear later. **Permutation matrices**, the identity with its rows reordered, are orthogonal. For a unit vector $\mathbf{u}$, the **Householder reflection** $\mathbf{I} - 2\mathbf{u}\mathbf{u}^T$ is symmetric and orthogonal and reflects $\mathbb{R}^n$ in the hyperplane $\mathbf{u}^\perp$. Householder reflections are how QR factorizations are computed in practice ([Exercise \(B1\)](#)).

1.5.4 Exercises

1.5.4.1 A. Check your understanding

Exercise A1.

Show that every 2×2 orthogonal matrix is either a rotation $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ (determinant 1) or a reflection $\begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$ (determinant -1). Find the line fixed by the reflection.

1.5.4.2 B. Practice

Exercise B1.

Let $\mathbf{u}$ be a unit vector and $\mathbf{H} = \mathbf{I} - 2\mathbf{u}\mathbf{u}^T$. Show that $\mathbf{H}$ is symmetric and orthogonal, that $\mathbf{H}\mathbf{u} = -\mathbf{u}$, and that $\mathbf{H}\mathbf{v} = \mathbf{v}$ for $\mathbf{v} \perp \mathbf{u}$. Given $\mathbf{x}$ not a nonnegative multiple of $\mathbf{e}_1$, find $\mathbf{u}$ with $\mathbf{H}\mathbf{x} = \|\mathbf{x}\|\mathbf{e}_1$. Explain how repeating this column by column produces a QR factorization.

Solution. $\mathbf{H}^T = \mathbf{H}$, and $\mathbf{H}\mathbf{H} = \mathbf{I} - 4\mathbf{u}\mathbf{u}^T + 4\mathbf{u}(\mathbf{u}^T\mathbf{u})\mathbf{u}^T = \mathbf{I}$. Also $\mathbf{H}\mathbf{u} = \mathbf{u} - 2\mathbf{u} = -\mathbf{u}$, and $\mathbf{H}\mathbf{v} = \mathbf{v}$ when $\mathbf{u}^T\mathbf{v} = 0$. Let $\mathbf{w} = \mathbf{x} - \|\mathbf{x}\|\mathbf{e}_1 \neq \mathbf{0}$ and $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|$. Then $\mathbf{w}^T\mathbf{w} = 2\|\mathbf{x}\|^2 - 2\|\mathbf{x}\|x_1 = 2\mathbf{w}^T\mathbf{x}$, so $\mathbf{H}\mathbf{x} = \mathbf{x} - 2\mathbf{w}(\mathbf{w}^T\mathbf{x})/(\mathbf{w}^T\mathbf{w}) = \mathbf{x} - \mathbf{w} = \|\mathbf{x}\|\mathbf{e}_1$. Applying such a reflection $\mathbf{H}_1$ to the first column of $\mathbf{X}$, then a reflection acting on coordinates $2, \dots, n$ to the second column of $\mathbf{H}_1\mathbf{X}$, and so on, gives $\mathbf{H}_p \cdots \mathbf{H}_1\mathbf{X} = \begin{pmatrix} \mathbb{R} \\ \mathbf{0} \end{pmatrix}$ with $\mathbb{R}$ upper triangular. Hence $\mathbf{X} = \mathbf{Q}\mathbb{R}$, where $\mathbf{Q}$ holds the first p columns of the orthogonal matrix $\mathbf{H}_1 \cdots \mathbf{H}_p$. In floating point one uses $\mathbf{w} = \mathbf{x} + \text{sgn}(x_1)\|\mathbf{x}\|\mathbf{e}_1$ instead, which avoids cancellation. It gives $\mathbf{H}\mathbf{x} = -\text{sgn}(x_1)\|\mathbf{x}\|\mathbf{e}_1$, so diagonal entries of $\mathbb{R}$ may come out negative; changing the sign of such a row of $\mathbb{R}$ and of the matching column of $\mathbf{Q}$ at the end restores the positive diagonal of [Theorem 1.5.2](#).

Exercise B2.

Show that the QR factorization of a matrix with full column rank, with $\mathbb{R}$ having positive diagonal, is unique, and that $\mathbb{R}\mathbf{T}$ is the Cholesky factor of $\mathbf{X}^T\mathbf{X}$.

Solution. If $\mathbf{X} = \mathbf{Q}_1\mathbb{R}_1 = \mathbf{Q}_2\mathbb{R}_2$, then $\mathbf{X}^T\mathbf{X} = \mathbb{R}_1^T\mathbf{Q}_1^T\mathbf{Q}_1\mathbb{R}_1 = \mathbb{R}_1^T\mathbb{R}_1 = \mathbb{R}_2^T\mathbb{R}_2$. Each $\mathbb{R}_k^T\mathbb{R}_k$ is lower triangular with positive diagonal, so both are the Cholesky factor of $\mathbf{X}^T\mathbf{X}$, which is unique by [Theorem 1.7.4\(d\)](#). So $\mathbb{R}_1 = \mathbb{R}_2$ and $\mathbf{Q}_1 = \mathbf{X}\mathbb{R}_1^{-1} = \mathbf{Q}_2$.

1.6 Trace, determinants and partitioned matrices

1.6.1 Trace

The **trace** of a square matrix is the sum of its diagonal entries, $\text{tr}(\mathbf{A}) = \sum_i a_{ii}$. It is linear, $\text{tr}(a\mathbf{A} + b\mathbf{B}) = a\text{tr}(\mathbf{A}) + b\text{tr}(\mathbf{B})$, and $\text{tr}(\mathbf{A}\mathbf{T}) = \text{tr}(\mathbf{A})$.

Theorem 1.6.1 — Cyclic property

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times m$, then $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$.

Proof. $\text{tr}(\mathbf{AB}) = \sum_{i=1}^m \sum_{k=1}^n a_{ik}b_{ki} = \sum_{k=1}^n \sum_{i=1}^m b_{ki}a_{ik} = \text{tr}(\mathbf{BA})$. ■

The two products need not have the same size. Applied repeatedly, the theorem allows cyclic

permutations such as $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{CAB}) = \text{tr}(\mathbf{BCA})$, but not arbitrary reorderings. Three consequences occur again and again:

$$\text{tr}(\mathbf{ATA}) = \sum_{i,j} a_{ij}^2 = \|\mathbf{A}\|_F^2, \quad \mathbf{xT Ax} = \text{tr}(\mathbf{AxxT}), \quad \text{tr}(\mathbf{P}^{-1}\mathbf{AP}) = \text{tr}(\mathbf{A}). \quad (1.6.1)$$

The middle one turns a scalar quadratic form into the trace of a matrix that is linear in $\mathbf{xxT}$. Taking expectations then gives the mean of a random quadratic form (Chapter 2).

1.6.2 Determinants

The **determinant** of an $n \times n$ matrix is

$$\det \mathbf{A} = \sum_{\pi} \text{sgn}(\pi) a_{1\pi(1)} a_{2\pi(2)} \cdots a_{n\pi(n)},$$

the sum over all permutations π of $\{1, \dots, n\}$, where $\text{sgn}(\pi) = \pm 1$ according to whether π is a product of an even or odd number of transpositions. We collect the standard properties. Parts (a)–(d) are proved in any linear algebra text (Harville 1997, chapter 13), and the others follow from them.

Proposition 1.6.2 — Determinants

Let $\mathbf{A}, \mathbf{B}$ be $n \times n$.

- (a) $\det \mathbf{AT} = \det \mathbf{A}$.
- (b) $\det \mathbf{A}$ is a linear function of each column separately. It changes sign when two columns are interchanged, and is unchanged when a multiple of one column is added to another. The same holds for rows.
- (c) A triangular matrix has determinant equal to the product of its diagonal entries.
- (d) $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$.
- (e) $\mathbf{A}$ is nonsingular iff $\det \mathbf{A} \neq 0$, and then $\det \mathbf{A}^{-1} = 1/\det \mathbf{A}$.
- (f) $\det(c\mathbf{A}) = c^n \det \mathbf{A}$, and $\det \mathbf{Q} = \pm 1$ for orthogonal $\mathbf{Q}$.

Proof (Proof of (e) and (f)). If $\mathbf{A}$ is nonsingular, $1 = \det \mathbf{I} = \det \mathbf{A} \det \mathbf{A}^{-1}$ by (d). If $\mathbf{A}$ is singular, some column is a combination of the others. Subtracting that combination leaves a zero column without changing the determinant, by (b), and the determinant of a matrix with a zero column is zero, since every term of the sum contains a factor from that column. For (f), apply (b) to each of the n columns, and use (a) and (d) on $\mathbf{QTQ} = \mathbf{I}$.

■

1.6.3 Schur complements and block determinants

Partitioned matrices arise whenever the regressors split into two groups, the model matrix $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ giving

$$\mathbf{X}\mathbf{T}\mathbf{X} = \begin{pmatrix} \mathbf{X}_1\mathbf{T}\mathbf{X}_1 & \mathbf{X}_1\mathbf{T}\mathbf{X}_2 \\ \mathbf{X}_2\mathbf{T}\mathbf{X}_1 & \mathbf{X}_2\mathbf{T}\mathbf{X}_2 \end{pmatrix},$$

and whenever a random vector splits into two blocks (Chapter 3). In what follows

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}, \quad (1.6.2)$$

with $\mathbf{A}_{11}$ of size $k \times k$ and $\mathbf{A}_{22}$ of size $(n - k) \times (n - k)$.

Definition 1.6.1 — Schur complement

If $\mathbf{A}_{11}$ is nonsingular, the **Schur complement** of $\mathbf{A}_{11}$ in $\mathbf{A}$ is $\mathbf{A}_{22 \cdot 1} = \mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}$.
If $\mathbf{A}_{22}$ is nonsingular, the Schur complement of $\mathbf{A}_{22}$ is $\mathbf{A}_{11 \cdot 2} = \mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}$.

Everything in this subsection comes from one identity, block Gaussian elimination. If $\mathbf{A}_{11}$ is nonsingular, multiplying out shows

$$\mathbf{A} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{A}_{21}\mathbf{A}_{11}^{-1} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22 \cdot 1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{A}_{11}^{-1}\mathbf{A}_{12} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}. \quad (1.6.3)$$

The outer factors are block triangular with identity blocks on the diagonal, and are nonsingular with inverses obtained by changing the sign of the off-diagonal block.

Theorem 1.6.3 — Block determinants

Let $\mathbf{A}$ be partitioned as in Eq. 1.6.2.

- (a) If $\mathbf{A}_{12} = \mathbf{0}$ or $\mathbf{A}_{21} = \mathbf{0}$, then $\det \mathbf{A} = \det \mathbf{A}_{11} \det \mathbf{A}_{22}$.
- (b) If $\mathbf{A}_{11}$ is nonsingular, $\det \mathbf{A} = \det \mathbf{A}_{11} \det \mathbf{A}_{22 \cdot 1}$. If $\mathbf{A}_{22}$ is nonsingular, $\det \mathbf{A} = \det \mathbf{A}_{22} \det \mathbf{A}_{11 \cdot 2}$.
- (c) For $m \times n$ matrices $\mathbf{U}$ and $n \times m$ matrices $\mathbf{V}$, $\det(\mathbf{I}_m + \mathbf{U}\mathbf{V}) = \det(\mathbf{I}_n + \mathbf{V}\mathbf{U})$. If $\mathbf{B}$ is $m \times m$ and nonsingular, $\det(\mathbf{B} + \mathbf{U}\mathbf{V}) = \det \mathbf{B} \det(\mathbf{I}_n + \mathbf{V}\mathbf{B}^{-1}\mathbf{U})$.

Proof.

- (a) Suppose $\mathbf{A}_{12} = \mathbf{0}$ (otherwise transpose). In the permutation expansion, a term that is nonzero cannot use an entry of the zero block $\mathbf{A}_{12}$. So each of the first k rows takes its entry from the first k columns, and π maps $\{1, \dots, k\}$ onto itself. Such a permutation splits into a permutation of the first k indices and one of the rest. Its sign is the product of their signs, so the sum factors as $\det \mathbf{A}_{11} \det \mathbf{A}_{22}$.
- (b) Take determinants in Eq. 1.6.3, using (a) and Proposition 1.6.2(d). The outer factors have determinant 1. The second statement follows in the same way from the elimination that starts with $\mathbf{A}_{22}$.

- (c) Apply (b) in both ways to $\begin{pmatrix} \mathbf{I}_m & -\mathbf{U} \\ \mathbf{V} & \mathbf{I}_n \end{pmatrix}$, whose Schur complements are $\mathbf{I}_n + \mathbf{V}\mathbf{U}$ and $\mathbf{I}_m + \mathbf{U}\mathbf{V}$. For the second statement, write $\mathbf{B} + \mathbf{U}\mathbf{V} = \mathbf{B}(\mathbf{I}_m + \mathbf{B}^{-1}\mathbf{U}\mathbf{V})$ and apply the first. ■

Part (c) with $n = 1$ is the **matrix determinant lemma** $\det(\mathbf{B} + \mathbf{u}\mathbf{v}^T) = \det \mathbf{B} (1 + \mathbf{v}^T \mathbf{B}^{-1} \mathbf{u})$.

1.6.4 Inverses of partitioned matrices

Theorem 1.6.4 — Partitioned inverse

Let $\mathbf{A}$ be partitioned as in Eq. 1.6.2 with $\mathbf{A}_{11}$ nonsingular. Then $\mathbf{A}$ is nonsingular iff $\mathbf{A}_{22 \cdot 1}$ is, and in that case

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22 \cdot 1}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & -\mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22 \cdot 1}^{-1} \\ -\mathbf{A}_{22 \cdot 1}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{A}_{22 \cdot 1}^{-1} \end{pmatrix}. \quad (1.6.4)$$

If $\mathbf{A}_{22}$ is also nonsingular, the leading block of $\mathbf{A}^{-1}$ equals $\mathbf{A}_{11 \cdot 2}^{-1}$:

$$(\mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21})^{-1} = \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1}. \quad (1.6.5)$$

Proof. The nonsingularity statement is Theorem 1.6.3(b). Inverting the three factors of Eq. 1.6.3 and multiplying them in reverse order gives

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{I} & -\mathbf{A}_{11}^{-1} \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A}_{11}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22 \cdot 1}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{I} \end{pmatrix},$$

which multiplies out to Eq. 1.6.4. Eliminating with $\mathbf{A}_{22}$ first gives a second expression for $\mathbf{A}^{-1}$, whose leading block is $\mathbf{A}_{11 \cdot 2}^{-1}$. The inverse is unique, so the two leading blocks agree. ■

For a symmetric positive definite $\mathbf{X}\mathbf{T}\mathbf{X}$ (Section 1.7), the theorem says that the block of $(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}$ belonging to $\mathbf{X}_1$ is $(\mathbf{X}_1 \mathbf{T} \mathbf{X}_1 - \mathbf{X}_1 \mathbf{T} \mathbf{X}_2 (\mathbf{X}_2 \mathbf{T} \mathbf{X}_2)^{-1} \mathbf{X}_2 \mathbf{T} \mathbf{X}_1)^{-1}$. Chapter 6 derives the same formula geometrically, from the Frisch–Waugh–Lovell theorem.

Identity Eq. 1.6.5 is a disguised form of a result that deserves its own name. Rename $\mathbf{A}_{11} = \mathbf{B}$, $\mathbf{A}_{12} = \mathbf{U}$, $\mathbf{A}_{21} = \mathbf{V}^T$ and $\mathbf{A}_{22} = -\mathbf{C}^{-1}$.

Theorem 1.6.5 — Sherman–Morrison–Woodbury

Let $\mathbf{B}$ ($n \times n$) and $\mathbf{C}$ ($k \times k$) be nonsingular, and let $\mathbf{U}, \mathbf{V}$ be $n \times k$. If $\mathbf{C}^{-1} + \mathbf{V}^T \mathbf{B}^{-1} \mathbf{U}$ is nonsingular, then so is $\mathbf{B} + \mathbf{U} \mathbf{C} \mathbf{V}^T$, and

$$(\mathbf{B} + \mathbf{UCVT})^{-1} = \mathbf{B}^{-1} - \mathbf{B}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{VTB}^{-1}\mathbf{U})^{-1}\mathbf{VTB}^{-1}. \quad (1.6.6)$$

In particular, for vectors $\mathbf{u}, \mathbf{v}$ with $1 + \mathbf{vTB}^{-1}\mathbf{u} \neq 0$,

$$(\mathbf{B} + \mathbf{uvT})^{-1} = \mathbf{B}^{-1} - \frac{\mathbf{B}^{-1}\mathbf{uvTB}^{-1}}{1 + \mathbf{vTB}^{-1}\mathbf{u}}. \quad (1.6.7)$$

Proof. Apply [Theorem 1.6.4](#) to $\mathbf{N} = \begin{pmatrix} \mathbf{B} & \mathbf{U} \\ \mathbf{vT} & \mathbf{C}^{-1} \end{pmatrix}$. Both diagonal blocks are nonsingular. The Schur complement of $\mathbf{B}$ is $-(\mathbf{C}^{-1} + \mathbf{VTB}^{-1}\mathbf{U})$, which is nonsingular by hypothesis, so $\mathbf{N}$ is nonsingular. Then the Schur complement of the lower block, $\mathbf{B} + \mathbf{UCVT}$, is nonsingular by the same theorem with the roles of the blocks exchanged, and [Eq. 1.6.5](#) becomes [Eq. 1.6.6](#). For [Eq. 1.6.7](#) take $k = 1$ and $\mathbf{C} = 1$. ■

Idea. Every result in this section comes from one block elimination, [Eq. 1.6.3](#): a partitioned matrix is a unit triangular matrix times $\text{diag}(\mathbf{A}_{11}, \mathbf{A}_{22.1})$ times another. Taking determinants gives [Theorem 1.6.3](#), inverting the factors gives [Theorem 1.6.4](#), and eliminating in the other order and comparing gives the Woodbury identity. When a formula about a block is forgotten, redo the elimination.

The identity replaces the inversion of an $n \times n$ matrix by that of a $k \times k$ matrix. That matters when $\mathbf{B}^{-1}$ is already known or cheap, as in the following two examples.

Example. (Deleting an observation)

Removing the i th row $\mathbf{x}_i\mathbf{T}$ from a model matrix changes $\mathbf{XTX}$ into $\mathbf{XTX} - \mathbf{x}_i\mathbf{x}_i\mathbf{T}$. With $\mathbf{u} = -\mathbf{x}_i$, $\mathbf{v} = \mathbf{x}_i$ and $h_i = \mathbf{x}_i\mathbf{T}(\mathbf{XTX})^{-1}\mathbf{x}_i$, [Eq. 1.6.7](#) gives

$$(\mathbf{XTX} - \mathbf{x}_i\mathbf{x}_i\mathbf{T})^{-1} = (\mathbf{XTX})^{-1} + \frac{(\mathbf{XTX})^{-1}\mathbf{x}_i\mathbf{x}_i\mathbf{T}(\mathbf{XTX})^{-1}}{1 - h_i},$$

valid whenever $h_i < 1$, and the determinant lemma gives $\det(\mathbf{XTX} - \mathbf{x}_i\mathbf{x}_i\mathbf{T}) = (1 - h_i)\det(\mathbf{XTX})$. The quantity h_i is the leverage of observation i ([Chapter 6](#)), and these formulas are what make the deletion diagnostics of Chapter 20 cheap. For Brownlee's stack-loss data ($n = 21$, $p = 4$), the listing updates the inverse for each of the 21 deletions. The largest relative discrepancy from direct inversion is 1×10^{-13} . Observation 17 has the largest leverage, 0.412, and removing it multiplies $\det(\mathbf{XTX})$ by 0.588.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.stackloss.load_pandas().data
X = np.column_stack([np.ones(len(data)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]]])
n, p = X.shape

A_inv = np.linalg.inv(X.T @ X)
h = np.einsum("ij,jk,ik->i", X, A_inv, X) # leverages x_i' (X'X)^{-1} x_i
```

```

worst = 0.0
for i in range(n):
    x = X[i]
    # Sherman-Morrison with u = -x, v = x: (X'X - x x')^{-1}
    B_update = A_inv + np.outer(A_inv @ x, x @ A_inv) / (1 - h[i])
    X_del = np.delete(X, i, axis=0)
    B_direct = np.linalg.inv(X_del.T @ X_del)
    worst = max(worst, np.abs(B_update - B_direct).max() / np.abs(B_direct).max())
print(f"largest relative discrepancy over all {n} deletions: {worst:.1e}")

```

Example. (Diagonal plus low rank)

In factor models and in the linear mixed models of Chapter 32, covariance matrices have the form $\Sigma = D + LLT$ with D diagonal and positive and L of size $N \times k$, $k \ll N$. By Eq. 1.6.6 and Theorem 1.6.3(c),

$$\Sigma^{-1} = D^{-1} - D^{-1}L(I_k + LTD^{-1}L)^{-1}LTD^{-1}, \quad \det \Sigma = \det D \det(I_k + LTD^{-1}L).$$

Each needs only a $k \times k$ factorization. The script checks this with $N = 1500$ and $k = 4$. The solution of $\Sigma w = v$ agrees with a direct solve to a relative error of 8×10^{-14} .

Example. (Equicorrelation)

The matrix $E = (1 - \rho)I_n + \rho J_n$ has unit diagonal and every off-diagonal entry equal to ρ . It is the correlation matrix of exchangeable observations. Taking $B = (1 - \rho)I$ and $u = \rho \mathbf{1}$, $v = \mathbf{1}$, for $\rho \neq 1$ and $1 + (n - 1)\rho \neq 0$,

$$E^{-1} = \frac{1}{1 - \rho} \left(I_n - \frac{\rho}{1 + (n - 1)\rho} J_n \right), \quad \det E = (1 - \rho)^{n-1} (1 + (n - 1)\rho).$$

1.6.5 Exercises

1.6.5.1 A. Check your understanding

Exercise A1.

Show that $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{ACB})$ when $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are symmetric, and find 2×2 matrices for which the two traces differ.

Solution. For symmetric matrices, $\text{tr}(\mathbf{ABC}) = \text{tr}((\mathbf{ABC})^T) = \text{tr}(\mathbf{CBA}) = \text{tr}(\mathbf{ACB})$, using transposition and then the cyclic property. For a counterexample take $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Then $\mathbf{ABC} = \mathbf{C}$ has trace 1, while $\mathbf{AC} = \mathbf{0}$ gives $\text{tr}(\mathbf{ACB}) = 0$.

Exercise A2.

Let $\mathbf{A}$ be nonsingular, $\mathbf{a}$ a vector and c a scalar with $s = c - \mathbf{a}^T \mathbf{A}^{-1} \mathbf{a} \neq 0$. Write down

the inverse and the determinant of $\begin{pmatrix} \mathbf{A} & \mathbf{a} \\ \mathbf{a}^T & c \end{pmatrix}$. Use it to update $(\mathbf{X}^T\mathbf{X})^{-1}$ when a column is added to $\mathbf{X}$.

1.6.5.2 B. Practice

Exercise B1.

Let $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$, $e_i = y_i - \mathbf{x}_i^T\hat{\boldsymbol{\beta}}$ and $h_i = \mathbf{x}_i^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_i < 1$. Show that the least squares coefficients computed without observation i are

$$\hat{\boldsymbol{\beta}}_{(i)} = \hat{\boldsymbol{\beta}} - \frac{(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_i e_i}{1 - h_i}.$$

Solution. Write $\mathbf{K} = (\mathbf{X}^T\mathbf{X})^{-1}$ and $\mathbf{x} = \mathbf{x}_i$. Deleting row i turns $\mathbf{X}^T\mathbf{y}$ into $\mathbf{X}^T\mathbf{y} - \mathbf{x}y_i$, so by [Example \(Deleting an observation\)](#)

$$\hat{\boldsymbol{\beta}}_{(i)} = \left(\mathbf{K} + \frac{\mathbf{K}\mathbf{x}\mathbf{x}^T\mathbf{K}}{1 - h_i} \right) (\mathbf{X}^T\mathbf{y} - \mathbf{x}y_i) = \hat{\boldsymbol{\beta}} - \mathbf{K}\mathbf{x}y_i + \frac{\mathbf{K}\mathbf{x}(\mathbf{x}^T\hat{\boldsymbol{\beta}} - h_i y_i)}{1 - h_i} = \hat{\boldsymbol{\beta}} - \mathbf{K}\mathbf{x} \frac{y_i - \mathbf{x}^T\hat{\boldsymbol{\beta}}}{1 - h_i},$$

since $(1 - h_i)y_i - \mathbf{x}^T\hat{\boldsymbol{\beta}} + h_i y_i = y_i - \mathbf{x}^T\hat{\boldsymbol{\beta}}$.

Exercise B2.

For an $n \times p$ matrix $\mathbf{X}$ and $\lambda > 0$, show that $(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I}_p)^{-1}\mathbf{X}^T = \mathbf{X}^T(\mathbf{X}\mathbf{X}^T + \lambda\mathbf{I}_n)^{-1}$. Explain why the right-hand side is preferable when p is much larger than n , and derive the identity a second way from [Theorem 1.6.5](#).

Solution. Both $\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I}$ and $\mathbf{X}\mathbf{X}^T + \lambda\mathbf{I}$ are positive definite. The identity $(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I}_p)\mathbf{X}^T = \mathbf{X}^T\mathbf{X}\mathbf{X}^T + \lambda\mathbf{X}^T = \mathbf{X}^T(\mathbf{X}\mathbf{X}^T + \lambda\mathbf{I}_n)$ gives the result after multiplying by the two inverses. When $p \gg n$, the right-hand side inverts an $n \times n$ matrix instead of a $p \times p$ one. For the second derivation, apply [Theorem 1.6.5](#) with $\mathbf{B} = \lambda\mathbf{I}_p$, $\mathbf{U} = \mathbf{X}^T$, $\mathbf{C} = \mathbf{I}_n$ and $\mathbf{V} = \mathbf{X}^T$: $(\lambda\mathbf{I}_p + \mathbf{X}^T\mathbf{X})^{-1} = \lambda^{-1}\mathbf{I}_p - \lambda^{-2}\mathbf{X}^T\mathbf{W}^{-1}\mathbf{X}$ with $\mathbf{W} = \mathbf{I}_n + \mathbf{X}\mathbf{X}^T/\lambda$. Multiplying on the right by $\mathbf{X}^T$ gives $\lambda^{-1}\mathbf{X}^T[\mathbf{I}_n - \mathbf{W}^{-1}(\mathbf{X}\mathbf{X}^T/\lambda)] = \lambda^{-1}\mathbf{X}^T\mathbf{W}^{-1}[\mathbf{W} - \mathbf{X}\mathbf{X}^T/\lambda] = \lambda^{-1}\mathbf{X}^T\mathbf{W}^{-1} = \mathbf{X}^T(\mathbf{X}\mathbf{X}^T + \lambda\mathbf{I}_n)^{-1}$.

1.7 Eigenvalues, the spectral theorem and positive definiteness

1.7.1 Eigenvalues

A scalar λ is an **eigenvalue** of the $n \times n$ matrix $\mathbf{A}$, with **eigenvector** $\mathbf{x} \neq \mathbf{0}$, if $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$. Equivalently, $\mathbf{A} - \lambda\mathbf{I}$ is singular, that is, λ is a root of the **characteristic polynomial** $p(\lambda) = \det(\lambda\mathbf{I} - \mathbf{A})$. This polynomial has degree n and leading coefficient one, so over the complex numbers it factors as $\prod_{i=1}^n (\lambda - \lambda_i)$. The eigenvalues $\lambda_1, \dots, \lambda_n$ are listed with multiplicity and may be complex when $\mathbf{A}$ is not symmetric.

Proposition 1.7.1

Let $\mathbf{A}$ be $n \times n$ with eigenvalues $\lambda_1, \dots, \lambda_n$.

- (a) $\det \mathbf{A} = \prod_i \lambda_i$ and $\operatorname{tr} \mathbf{A} = \sum_i \lambda_i$.
- (b) If $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$, then $f(\mathbf{A})\mathbf{x} = f(\lambda)\mathbf{x}$ for every polynomial f . If $\mathbf{A}$ is nonsingular, $\mathbf{A}^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$.
- (c) $\mathbf{A}$ and $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ have the same characteristic polynomial for nonsingular $\mathbf{P}$.
- (d) If $\mathbf{C}$ is $m \times n$ and $\mathbf{D}$ is $n \times m$, then $\lambda^n \det(\lambda \mathbf{I}_m - \mathbf{C}\mathbf{D}) = \lambda^m \det(\lambda \mathbf{I}_n - \mathbf{D}\mathbf{C})$. So $\mathbf{C}\mathbf{D}$ and $\mathbf{D}\mathbf{C}$ have the same nonzero eigenvalues, with the same multiplicities.

Proof.

- (a) Setting $\lambda = 0$ in $\det(\lambda \mathbf{I} - \mathbf{A}) = \prod(\lambda - \lambda_i)$ gives $(-1)^n \det \mathbf{A} = (-1)^n \prod \lambda_i$. For the trace, compare coefficients of λ^{n-1} . On the right it is $-\sum \lambda_i$. On the left, in the permutation expansion of $\det(\lambda \mathbf{I} - \mathbf{A})$, only the product of diagonal entries $\prod_i (\lambda - a_{ii})$ contains λ^{n-1} , because every other term omits at least two diagonal entries. Its coefficient is $-\sum a_{ii}$.
- (b) $\mathbf{A}^k \mathbf{x} = \lambda^k \mathbf{x}$ by induction, and $\mathbf{x} = \mathbf{A}^{-1} \mathbf{A} \mathbf{x} = \lambda \mathbf{A}^{-1} \mathbf{x}$ with $\lambda \neq 0$. (c) $\det(\lambda \mathbf{I} - \mathbf{P}^{-1} \mathbf{A} \mathbf{P}) = \det(\mathbf{P}^{-1}(\lambda \mathbf{I} - \mathbf{A})\mathbf{P}) = \det(\lambda \mathbf{I} - \mathbf{A})$.
- (c) For $\lambda \neq 0$, Theorem 1.6.3(c) gives $\det(\mathbf{I}_m - \lambda^{-1} \mathbf{C}\mathbf{D}) = \det(\mathbf{I}_n - \lambda^{-1} \mathbf{D}\mathbf{C})$. Multiplying by λ^{m+n} gives the identity for $\lambda \neq 0$. Both sides are polynomials, so it holds for all λ .

■

1.7.2 The spectral theorem

Symmetric matrices have the best possible eigenstructure, and almost every matrix whose eigenvalues matter in statistics is symmetric.

Theorem 1.7.2 — Spectral theorem

Let $\mathbf{A}$ be an $n \times n$ symmetric matrix. Then all eigenvalues of $\mathbf{A}$ are real, and there is an orthogonal matrix $\mathbf{Q} = [\mathbf{q}_1, \dots, \mathbf{q}_n]$ with

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T = \sum_{i=1}^n \lambda_i \mathbf{q}_i \mathbf{q}_i^T, \quad \mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \dots, \lambda_n). \quad (1.7.1)$$

The columns of $\mathbf{Q}$ are an orthonormal basis of eigenvectors, $\mathbf{A}\mathbf{q}_i = \lambda_i \mathbf{q}_i$. Eigenvectors belonging to distinct eigenvalues are orthogonal.

Proof. Real eigenvalues. Let $\mathbf{A}\mathbf{z} = \lambda\mathbf{z}$ with $\mathbf{z} \in \mathbb{C}^n$ nonzero, and let $\bar{\mathbf{z}}$ be the complex conjugate. The scalar $s = \bar{\mathbf{z}}^T \mathbf{A} \mathbf{z}$ is real, since its conjugate is $\mathbf{z}^T \mathbf{A} \bar{\mathbf{z}} = (\mathbf{z}^T \mathbf{A} \bar{\mathbf{z}})^T = \bar{\mathbf{z}}^T \mathbf{A} \mathbf{z} = s$ by symmetry. Also $s = \lambda \bar{\mathbf{z}}^T \mathbf{z} = \lambda \sum_i |z_i|^2$, with the sum real and positive. So λ is real. The real singular matrix $\mathbf{A} - \lambda \mathbf{I}$ then has a real nonzero null vector, so

there is a real eigenvector.

Diagonalization. By induction on n , the case $n = 1$ being trivial. Let $\mathbf{q}_1$ be a real unit eigenvector for λ_1 and extend it to an orthogonal matrix $[\mathbf{q}_1, \mathbb{Q}_2]$ (Section 1.5). Since $\mathbb{Q}_2 \mathbf{T} \mathbf{A} \mathbf{q}_1 = \lambda_1 \mathbb{Q}_2 \mathbf{T} \mathbf{q}_1 = \mathbf{0}$ and $\mathbf{A}$ is symmetric,

$$[\mathbf{q}_1, \mathbb{Q}_2] \mathbf{T} \mathbf{A} [\mathbf{q}_1, \mathbb{Q}_2] = \begin{pmatrix} \lambda_1 & \mathbf{0} \mathbf{T} \\ \mathbf{0} & \mathbb{Q}_2 \mathbf{T} \mathbf{A} \mathbb{Q}_2 \end{pmatrix}.$$

The $(n-1) \times (n-1)$ block is symmetric, so by induction $\mathbb{Q}_2 \mathbf{T} \mathbf{A} \mathbb{Q}_2 = \mathbf{P} \mathbf{\Lambda}_2 \mathbf{P} \mathbf{T}$ with $\mathbf{P}$ orthogonal. Then $\mathbb{Q} = [\mathbf{q}_1, \mathbb{Q}_2 \mathbf{P}]$ is orthogonal (Proposition 1.5.3(b)) and $\mathbb{Q} \mathbf{T} \mathbf{A} \mathbb{Q} = \text{diag}(\lambda_1, \mathbf{\Lambda}_2)$.

Orthogonality. If $\mathbf{A} \mathbf{u} = \lambda \mathbf{u}$ and $\mathbf{A} \mathbf{v} = \mu \mathbf{v}$, then $\lambda \mathbf{u} \mathbf{T} \mathbf{v} = \mathbf{u} \mathbf{T} \mathbf{A} \mathbf{v} = \mu \mathbf{u} \mathbf{T} \mathbf{v}$, so $\mathbf{u} \mathbf{T} \mathbf{v} = 0$ when $\lambda \neq \mu$. ■

Unless stated otherwise, the eigenvalues of symmetric $\mathbf{A}$ are ordered as $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$, and we write $\lambda_{\max} = \lambda_1$ and $\lambda_{\min} = \lambda_n$. The following consequences are used without comment later on.

Corollary 1.7.3

Let $\mathbf{A} = \mathbb{Q} \mathbf{\Lambda} \mathbb{Q} \mathbf{T}$ be symmetric.

- (a) $\text{rank}(\mathbf{A})$ counts the nonzero eigenvalues. $\mathbb{C}(\mathbf{A})$ is spanned by the $\mathbf{q}_i$ with $\lambda_i \neq 0$, and $\mathcal{N}(\mathbf{A})$ by those with $\lambda_i = 0$.
- (b) $\mathbf{A}^k = \mathbb{Q} \mathbf{\Lambda}^k \mathbb{Q} \mathbf{T}$ for $k = 1, 2, \dots$. If $\mathbf{A}$ is nonsingular, $\mathbf{A}^{-1} = \mathbb{Q} \mathbf{\Lambda}^{-1} \mathbb{Q} \mathbf{T}$, and $\mathbf{A}^+ = \mathbb{Q} \mathbf{\Lambda}^+ \mathbb{Q} \mathbf{T}$ in general, where $\mathbf{\Lambda}^+$ inverts the nonzero diagonal entries.
- (c) The eigenspace $\{\mathbf{x} : \mathbf{A} \mathbf{x} = \lambda \mathbf{x}\}$ has dimension equal to the multiplicity of λ as a root of the characteristic polynomial.

Proof. Multiplying by the nonsingular $\mathbb{Q}$ and $\mathbb{Q} \mathbf{T}$ changes neither rank (Proposition 1.2.3(b)) nor, after the change of variable $\mathbf{x} = \mathbb{Q} \mathbf{c}$, the structure of null spaces and eigenspaces. So each statement reduces to the diagonal matrix $\mathbf{\Lambda}$, where it is immediate. For (b), $\mathbb{Q} \mathbf{T} \mathbb{Q} = \mathbf{I}$ collapses the middle factors, and the four Penrose conditions for $\mathbb{Q} \mathbf{\Lambda}^+ \mathbb{Q} \mathbf{T}$ reduce to those for $\mathbf{\Lambda}^+$. ■

Example. (Eigenvalues of the equicorrelation matrix)

For $\mathbf{E} = (1 - \rho) \mathbf{I} + \rho \mathbf{J}_n$ of Example (Equicorrelation), $\mathbf{E} \mathbf{1} = (1 + (n-1)\rho) \mathbf{1}$, and $\mathbf{E} \mathbf{v} = (1 - \rho) \mathbf{v}$ for every $\mathbf{v} \perp \mathbf{1}$. So the eigenvalues are $1 + (n-1)\rho$ once and $1 - \rho$ with multiplicity $n-1$, with eigenvectors $\mathbf{1}/\sqrt{n}$ and the last $n-1$ rows of the Helmert matrix (Example (Helmert matrices)). The determinant and inverse found earlier follow at once from Proposition 1.7.1 and Corollary 1.7.3(b).

1.7.3 Positive definite and nonnegative definite matrices

Definition 1.7.1

A symmetric matrix $\mathbf{A}$ is **positive definite** if $\mathbf{x}^T \mathbf{A} \mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$, and **nonnegative definite** if $\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0$ for every $\mathbf{x}$. For symmetric $\mathbf{A}, \mathbf{B}$ of the same size we write $\mathbf{A} \succeq \mathbf{B}$ if $\mathbf{A} - \mathbf{B}$ is nonnegative definite, and $\mathbf{A} \succ \mathbf{B}$ if it is positive definite (the **Loewner order**).

In this book, “nonnegative definite” includes positive definite. Some texts, Rencher and Schaalje among them, reserve “positive semidefinite” for matrices that are nonnegative definite but singular. We use the terms only for symmetric matrices. Covariance matrices are nonnegative definite (Chapter 2), and *Gram matrices* $\mathbf{X}^T \mathbf{X}$ are too, since $\mathbf{x}^T \mathbf{X}^T \mathbf{X} \mathbf{x} = \|\mathbf{X} \mathbf{x}\|^2$.

Theorem 1.7.4 — Characterizations

For a symmetric $n \times n$ matrix $\mathbf{A}$ the following are equivalent:

- (a) $\mathbf{A}$ is positive definite;
- (b) every eigenvalue of $\mathbf{A}$ is positive;
- (c) $\mathbf{A} = \mathbf{B}^T \mathbf{B}$ for some matrix $\mathbf{B}$ with n columns and full column rank;
- (d) $\mathbf{A} = \mathbf{L} \mathbf{L}^T$ with $\mathbf{L}$ lower triangular with positive diagonal entries (the **Cholesky factorization**; $\mathbf{L}$ is unique);
- (e) every leading principal minor $\det \mathbf{A}_{[k]}$, $k = 1, \dots, n$, is positive, where $\mathbf{A}_{[k]}$ is the upper left $k \times k$ block.

Likewise, the following are equivalent: (a') $\mathbf{A}$ is nonnegative definite; (b') every eigenvalue is nonnegative; (c') $\mathbf{A} = \mathbf{B}^T \mathbf{B}$ for some $\mathbf{B}$, which can be taken of size $\text{rank}(\mathbf{A}) \times n$; (e') every principal minor (determinant of a submatrix with the same row and column indices) is nonnegative.

Proof. Write $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T$. (a) $\Rightarrow$ (b): $\lambda_i = \mathbf{q}_i^T \mathbf{A} \mathbf{q}_i > 0$. (b) $\Rightarrow$ (c): $\mathbf{B} = \mathbf{\Lambda}^{1/2} \mathbf{Q}^T$ is nonsingular. (c) $\Rightarrow$ (a): $\mathbf{x}^T \mathbf{B}^T \mathbf{B} \mathbf{x} = \|\mathbf{B} \mathbf{x}\|^2$, which is positive for $\mathbf{x} \neq \mathbf{0}$ by full column rank. (d) $\Rightarrow$ (c) is immediate, with $\mathbf{B} = \mathbf{L}^T$.

(a) $\Rightarrow$ (d), by induction on n . Write $\mathbf{A} = \begin{pmatrix} a & \mathbf{b}^T \\ \mathbf{b} & \mathbf{A}_2 \end{pmatrix}$. Then $a = \mathbf{e}_1^T \mathbf{A} \mathbf{e}_1 > 0$. The Schur complement $\mathbf{A}_2 - \mathbf{b} \mathbf{b}^T / a$ is positive definite: for $\mathbf{v} \neq \mathbf{0}$, put $\mathbf{x} = (-\mathbf{b}^T \mathbf{v} / a, \mathbf{v}^T)^T$ and expand to get $\mathbf{x}^T \mathbf{A} \mathbf{x} = \mathbf{v}^T (\mathbf{A}_2 - \mathbf{b} \mathbf{b}^T / a) \mathbf{v} > 0$. By induction $\mathbf{A}_2 - \mathbf{b} \mathbf{b}^T / a = \mathbf{L}_2 \mathbf{L}_2^T$, and $\mathbf{L} = \begin{pmatrix} \sqrt{a} & \mathbf{0}^T \\ \mathbf{b} / \sqrt{a} & \mathbf{L}_2 \end{pmatrix}$ satisfies $\mathbf{L} \mathbf{L}^T = \mathbf{A}$. For uniqueness, if $\mathbf{L} \mathbf{L}^T = \mathbf{K} \mathbf{K}^T$, then $\mathbf{K}^{-1} \mathbf{L} = \mathbf{K}^T (\mathbf{L}^T)^{-1}$ is both lower and upper triangular, hence diagonal, say $\mathbf{D}$. From $\mathbf{K}^{-1} \mathbf{L} = \mathbf{D}$ and $\mathbf{L} \mathbf{L}^T = \mathbf{K} \mathbf{K}^T$ we get $\mathbf{D} \mathbf{D}^T = \mathbf{I}$, so $\mathbf{D} = \mathbf{I}$ because its diagonal is positive.

(a) $\Rightarrow$ (e): $\mathbf{A}_{[k]} = \mathbf{E}^T \mathbf{A} \mathbf{E}$ with $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_k]$ is positive definite, since $\mathbf{x}^T \mathbf{A}_{[k]} \mathbf{x} = (\mathbf{E} \mathbf{x})^T \mathbf{A} (\mathbf{E} \mathbf{x})$. By (b) and Proposition 1.7.1(a) its determinant is positive. (e) $\Rightarrow$ (a), by induction on n . The block $\mathbf{A}_{[n-1]}$ satisfies (e) and so is positive definite, so by (a) $\Rightarrow$ (d),

which is already proved, $\mathbf{A}_{[n-1]} = \mathbf{L}\mathbf{L}\mathbf{T}$. Write $\mathbf{A} = \begin{pmatrix} \mathbf{A}_{[n-1]} & \mathbf{c} \\ \mathbf{c}^T & d \end{pmatrix}$ and $s = d - \mathbf{c}^T \mathbf{A}_{[n-1]}^{-1} \mathbf{c}$. By Theorem 1.6.3(b), $s = \det \mathbf{A} / \det \mathbf{A}_{[n-1]} > 0$. With $\mathbf{T} = \begin{pmatrix} \mathbf{I} & \mathbf{A}_{[n-1]}^{-1} \mathbf{c} \\ \mathbf{0}^T & 1 \end{pmatrix}$, a direct check gives $\mathbf{A} = \mathbf{T}^T \mathbf{T} \operatorname{diag}(\mathbf{A}_{[n-1]}, s) \mathbf{T} = \mathbf{B}^T \mathbf{B}$ with $\mathbf{B} = \operatorname{diag}(\mathbf{L}\mathbf{T}, \sqrt{s}) \mathbf{T}$ nonsingular, so $\mathbf{A}$ is positive definite by (c) $\Rightarrow$ (a).

The primed statements: (a') $\Leftrightarrow$ (b') $\Leftrightarrow$ (c') exactly as above, taking for $\mathbf{B}$ the rows of $\mathbf{A}^{1/2} \mathbf{Q} \mathbf{T}$ that belong to nonzero eigenvalues. (a') $\Rightarrow$ (e') as above, with an arbitrary index set in place of $\{1, \dots, k\}$. For (e') $\Rightarrow$ (a'), fix $t > 0$. Each column of $\mathbf{A}_{[k]} + t\mathbf{I}$ is a column of $\mathbf{A}_{[k]}$ plus t times a coordinate vector. Expanding the determinant by linearity in each column gives $\det(\mathbf{A}_{[k]} + t\mathbf{I}) = \sum_S t^{k-|S|} \det \mathbf{A}_{SS}$, the sum over all subsets $S \subseteq \{1, \dots, k\}$, with $\det \mathbf{A}_{\emptyset\emptyset} = 1$. Every term is nonnegative and the term for $S = \emptyset$ is $t^k > 0$. So $\mathbf{A} + t\mathbf{I}$ is positive definite by (e) $\Rightarrow$ (a), and $\mathbf{x}^T \mathbf{A} \mathbf{x} = \lim_{t \rightarrow 0} \mathbf{x}^T (\mathbf{A} + t\mathbf{I}) \mathbf{x} \geq 0$. ■

Two warnings. Leading principal minors do not suffice for nonnegative definiteness: $\operatorname{diag}(0, -1)$ has leading minors 0 and 0. And (c') holds with many matrices $\mathbf{B}$. The Cholesky factor and the symmetric square root below are two convenient choices.

Proposition 1.7.5

Let $\mathbf{A}$ be $n \times n$ nonnegative definite.

- (a) $a_{ii} \geq 0$, and if $a_{ii} = 0$ then row i and column i of $\mathbf{A}$ are zero.
- (b) $\mathbf{x}^T \mathbf{A} \mathbf{x} = 0$ iff $\mathbf{A} \mathbf{x} = \mathbf{0}$.
- (c) $\mathbf{B}^T \mathbf{A} \mathbf{B}$ is nonnegative definite for any $n \times q$ matrix $\mathbf{B}$. If $\mathbf{A}$ is positive definite, $\mathbf{B}^T \mathbf{A} \mathbf{B}$ is positive definite iff $\mathbf{B}$ has full column rank.
- (d) Every principal submatrix of $\mathbf{A}$ is nonnegative definite, and positive definite if $\mathbf{A}$ is.
- (e) If $\mathbf{A}$ is positive definite, it is nonsingular and $\mathbf{A}^{-1}$ is positive definite. If also $\mathbf{A}$ is partitioned as in Eq. 1.6.2, both Schur complements $\mathbf{A}_{11.2}$ and $\mathbf{A}_{22.1}$ are positive definite.
- (f) If $\mathbf{A} \succeq \mathbf{B} \succ \mathbf{0}$, then $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$.

Proof.

- (b) Write $\mathbf{A} = \mathbf{B}^T \mathbf{B}$ by Theorem 1.7.4(c'). Then $\mathbf{x}^T \mathbf{A} \mathbf{x} = \|\mathbf{B} \mathbf{x}\|^2$, which vanishes iff $\mathbf{B} \mathbf{x} = \mathbf{0}$, which implies $\mathbf{A} \mathbf{x} = \mathbf{0}$.
- (c) $a_{ii} = \mathbf{e}_i^T \mathbf{A} \mathbf{e}_i \geq 0$. If $a_{ii} = 0$ then $\mathbf{A} \mathbf{e}_i = \mathbf{0}$ by (b), so column i is zero, and row i by symmetry.
- (d) $\mathbf{x}^T \mathbf{B}^T \mathbf{A} \mathbf{B} \mathbf{x} = (\mathbf{B} \mathbf{x})^T \mathbf{A} (\mathbf{B} \mathbf{x})$ is nonnegative, and when $\mathbf{A}$ is positive definite it is zero only if $\mathbf{B} \mathbf{x} = \mathbf{0}$.
- (e) is (c) with $\mathbf{B}$ a set of columns of $\mathbf{I}$.
- (f) The eigenvalues of $\mathbf{A}^{-1}$ are $1/\lambda_i > 0$. The inverse in Eq. 1.6.4 has $\mathbf{A}_{22.1}^{-1}$ as a principal submatrix, which is positive definite by (d). The other Schur complement follows by symmetry of the argument.
- (g) By Theorem 1.7.6 below, $\mathbf{B} = \mathbf{B}^{1/2} \mathbf{B}^{1/2}$. Then $\mathbf{C} = \mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} \succeq \mathbf{I}$ by (c), so

every eigenvalue of $\mathbf{C}$ is at least 1 and every eigenvalue of $\mathbf{C}^{-1} = \mathbf{B}^{1/2}\mathbf{A}^{-1}\mathbf{B}^{1/2}$ is at most 1. So $\mathbf{I} \succeq \mathbf{B}^{1/2}\mathbf{A}^{-1}\mathbf{B}^{1/2}$, and multiplying on both sides by $\mathbf{B}^{-1/2}$ gives $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$ by (c). ■

Part (f) is used in Chapter 7: an estimator whose covariance matrix is smaller in the Loewner order has smaller variance for every linear combination of its components.

Theorem 1.7.6 — Square root

A nonnegative definite matrix $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$ has exactly one nonnegative definite matrix $\mathbf{B}$ with $\mathbf{B}^2 = \mathbf{A}$, namely $\mathbf{A}^{1/2} = \mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{Q}^T$. It satisfies $\mathbb{C}(\mathbf{A}^{1/2}) = \mathbb{C}(\mathbf{A})$. If $\mathbf{A}$ is positive definite, $\mathbf{A}^{1/2}$ is positive definite and its inverse is $\mathbf{A}^{-1/2} = \mathbf{Q}\mathbf{\Lambda}^{-1/2}\mathbf{Q}^T$.

Proof. $\mathbf{A}^{1/2}$ is symmetric with nonnegative eigenvalues and $(\mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{Q}^T)^2 = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$. Its column space is spanned by the $\mathbf{q}_i$ with $\lambda_i > 0$, as is $\mathbb{C}(\mathbf{A})$ (Corollary 1.7.3(a)). For uniqueness, let $\mathbf{B}$ be nonnegative definite with $\mathbf{B}^2 = \mathbf{A}$, and write $\mathbf{B} = \mathbf{P}\mathbf{\Delta}\mathbf{P}^T$ with $\mathbf{\Delta} = \text{diag}(\delta_i)$, $\delta_i \geq 0$. Then $\mathbf{A} = \mathbf{P}\mathbf{\Delta}^2\mathbf{P}^T$. Let $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ and expand $\mathbf{v} = \sum_i c_i \mathbf{p}_i$. Comparing $\mathbf{A}\mathbf{v} = \sum_i c_i \delta_i^2 \mathbf{p}_i$ with $\lambda\mathbf{v}$ gives $c_i(\delta_i^2 - \lambda) = 0$, so $c_i \neq 0$ only if $\delta_i = \sqrt{\lambda}$. Hence $\mathbf{B}\mathbf{v} = \sum_i c_i \delta_i \mathbf{p}_i = \sqrt{\lambda}\mathbf{v}$. Applying this to each $\mathbf{q}_j$ gives $\mathbf{B}\mathbf{Q} = \mathbf{Q}\mathbf{\Lambda}^{1/2}$, that is, $\mathbf{B} = \mathbf{A}^{1/2}$. The last statement is Corollary 1.7.3(b). ■

1.7.4 Extremal properties of eigenvalues

Theorem 1.7.7 — Rayleigh quotient

Let $\mathbf{A}$ be symmetric with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$ and orthonormal eigenvectors $\mathbf{q}_1, \dots, \mathbf{q}_n$. For every $\mathbf{x} \neq \mathbf{0}$,

$$\lambda_n \leq \frac{\mathbf{x}^T \mathbf{A} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} \leq \lambda_1, \quad (1.7.2)$$

with equality on the right at $\mathbf{x} = \mathbf{q}_1$ and on the left at $\mathbf{x} = \mathbf{q}_n$. More generally, the maximum of $\mathbf{x}^T \mathbf{A} \mathbf{x} / \mathbf{x}^T \mathbf{x}$ over nonzero $\mathbf{x}$ orthogonal to $\mathbf{q}_1, \dots, \mathbf{q}_{k-1}$ is λ_k , attained at $\mathbf{q}_k$.

Proof. Put $\mathbf{c} = \mathbf{Q}^T \mathbf{x}$, so that $\mathbf{x}^T \mathbf{x} = \mathbf{c}^T \mathbf{c}$ and

$$\frac{\mathbf{x}^T \mathbf{A} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} = \frac{\sum_i \lambda_i c_i^2}{\sum_i c_i^2},$$

a weighted average of the eigenvalues with weights $c_i^2 / \sum_j c_j^2$. It lies between the smallest and largest eigenvalue, and equals λ_1 when all weight is on c_1 , that is, at $\mathbf{x} = \mathbf{q}_1$. The condition $\mathbf{x} \perp \mathbf{q}_1, \dots, \mathbf{q}_{k-1}$ means $c_1 = \dots = c_{k-1} = 0$, so the average is then over $\lambda_k, \dots, \lambda_n$, whose largest member is λ_k .

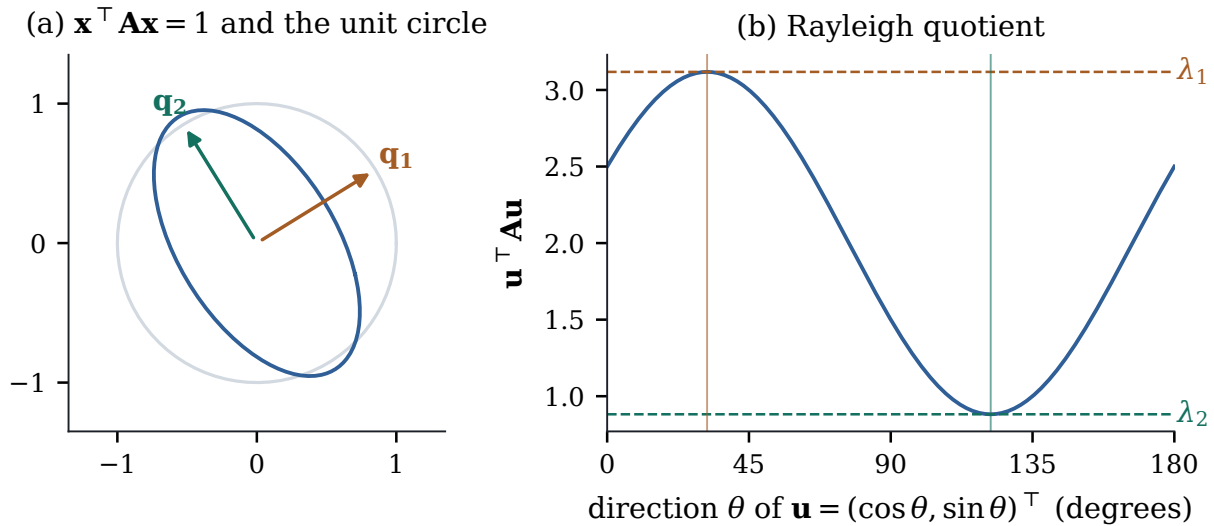


Figure 1.7.1: The Rayleigh quotient of a 2×2 symmetric matrix. (a) The level set $\mathbf{x}^T \mathbf{A} \mathbf{x} = 1$, an ellipse with axes along the orthonormal eigenvectors $\mathbf{q}_1, \mathbf{q}_2$, and the unit circle. (b) The quotient $\mathbf{u}^T \mathbf{A} \mathbf{u}$ along the unit circle. Its maximum λ_1 and minimum λ_2 are attained at the eigenvector directions (Theorem 1.7.7).

Figure 1.7.1 shows the theorem for $\mathbf{A} = \begin{pmatrix} 2.5 & 1 \\ 1 & 1.5 \end{pmatrix}$. The eigenvalues are 3.118 and 0.882, with eigenvectors at angles 31.7° and 121.7° . As $\mathbf{x}$ turns around the unit circle, the quotient swings between the two eigenvalues. The ellipse $\mathbf{x}^T \mathbf{A} \mathbf{x} = 1$ has its principal axes along the eigenvectors, with semi-axis lengths $\lambda_i^{-1/2}$. Consequently the Rayleigh quotient is largest in the direction in which the ellipse is narrowest.

Corollary 1.7.8 — Relative to a positive definite matrix

Let $\mathbf{A}$ be symmetric and $\mathbf{B}$ positive definite, both $n \times n$.

- (a) $\max_{\mathbf{x} \neq \mathbf{0}} \mathbf{x}^T \mathbf{A} \mathbf{x} / \mathbf{x}^T \mathbf{B} \mathbf{x}$ is the largest eigenvalue of $\mathbf{B}^{-1} \mathbf{A}$, which has the same eigenvalues as the symmetric $\mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2}$.
- (b) There is a nonsingular $\mathbf{P}$ with $\mathbf{P}^T \mathbf{B} \mathbf{P} = \mathbf{I}$ and $\mathbf{P}^T \mathbf{A} \mathbf{P}$ diagonal.
- (c) For $\mathbf{a} \in \mathbb{R}^n$, $\max_{\mathbf{x} \neq \mathbf{0}} (\mathbf{a}^T \mathbf{x})^2 / \mathbf{x}^T \mathbf{B} \mathbf{x} = \mathbf{a}^T \mathbf{B}^{-1} \mathbf{a}$, attained at $\mathbf{x} = \mathbf{B}^{-1} \mathbf{a}$.

Proof. Substitute $\mathbf{x} = \mathbf{B}^{-1/2} \mathbf{z}$. The quotient in (a) becomes $\mathbf{z}^T \mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} \mathbf{z} / \mathbf{z}^T \mathbf{z}$, to which Theorem 1.7.7 applies, and $\mathbf{B}^{-1} \mathbf{A} = \mathbf{B}^{-1/2} (\mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2}) \mathbf{B}^{1/2}$ is similar to it (Proposition 1.7.1(c)). For (b), if $\mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T$, take $\mathbf{P} = \mathbf{B}^{-1/2} \mathbf{Q}$. For (c), the matrix $\mathbf{A} = \mathbf{a} \mathbf{a}^T$ has $\mathbf{B}^{-1/2} \mathbf{a} \mathbf{a}^T \mathbf{B}^{-1/2}$ of rank at most one, whose only nonzero eigenvalue is its trace $\mathbf{a}^T \mathbf{B}^{-1} \mathbf{a}$. Substituting $\mathbf{x} = \mathbf{B}^{-1} \mathbf{a}$ attains it.

Part (c) is the Cauchy–Schwarz inequality in the inner product $\mathbf{x}^T \mathbf{B} \mathbf{v}$. It is the algebra behind simultaneous confidence intervals (Chapter 12), and part (b) is used for linear discriminants and for comparing two covariance matrices.

1.7.5 Exercises

1.7.5.1 A. Check your understanding

Exercise A1.

Show that $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ is positive definite iff $a > 0$ and $ac > b^2$, and nonnegative definite iff $a \geq 0$, $c \geq 0$ and $ac \geq b^2$. Why is $a \geq 0$ and $ac \geq b^2$ not enough?

1.7.5.2 B. Practice

Exercise B1.

Let $\mathbf{A} \succeq \mathbf{B} \succ \mathbf{0}$. Show that $\det \mathbf{A} \geq \det \mathbf{B}$ and $\operatorname{tr} \mathbf{A} \geq \operatorname{tr} \mathbf{B}$.

Exercise B2.

Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ have full column rank. Show that the leading block of $(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}$ satisfies $[(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}]_{11} \succeq (\mathbf{X}_1\mathbf{T}\mathbf{X}_1)^{-1}$, with equality iff $\mathbf{X}_1\mathbf{T}\mathbf{X}_2 = \mathbf{0}$. Interpret this for the variances of least squares coefficients when regressors are added to a model.

Solution. Write $\mathbf{A} = \mathbf{X}\mathbf{T}\mathbf{X}$ in blocks. By [Theorem 1.6.4](#), $[\mathbf{A}^{-1}]_{11} = \mathbf{A}_{11.2}^{-1}$. Now $\mathbf{A}_{11} - \mathbf{A}_{11.2} = \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}$ is nonnegative definite by [Proposition 1.7.5\(c,e\)](#), and $\mathbf{A}_{11.2}$ is positive definite. So $\mathbf{A}_{11} \succeq \mathbf{A}_{11.2} \succ \mathbf{0}$, and [Proposition 1.7.5\(f\)](#) gives $\mathbf{A}_{11.2}^{-1} \succeq \mathbf{A}_{11}^{-1}$. Equality forces $\mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21} = \mathbf{0}$, hence $\mathbf{A}_{21}\mathbf{x} = \mathbf{0}$ for all $\mathbf{x}$ by [Proposition 1.7.5\(b\)](#), that is, $\mathbf{X}_2\mathbf{T}\mathbf{X}_1 = \mathbf{0}$. When $\operatorname{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$, the covariance matrix of the coefficients of $\mathbf{X}_1$ is $\sigma^2[\mathbf{A}^{-1}]_{11}$ in the larger model and $\sigma^2\mathbf{A}_{11}^{-1}$ in the smaller one. Adding regressors never decreases the variance of any linear combination of the original coefficients, and leaves it unchanged only when the new regressors are orthogonal to the old ones.

1.7.5.3 C. Going deeper

Exercise C1.

Let $\mathbf{A}$ be symmetric with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$. Prove the Courant–Fischer theorem

$$\lambda_k = \max_{\dim S=k} \min_{\mathbf{0} \neq \mathbf{x} \in S} \frac{\mathbf{x}\mathbf{T}\mathbf{A}\mathbf{x}}{\mathbf{x}\mathbf{T}\mathbf{x}}.$$

Deduce that if $\mathbf{B}$ is obtained from $\mathbf{A}$ by deleting one row and the corresponding column, its eigenvalues $\mu_1 \geq \cdots \geq \mu_{n-1}$ interlace: $\lambda_k \geq \mu_k \geq \lambda_{k+1}$.

Exercise C2.

Let $\mathbf{S}$ be positive definite of size $p \times p$. Show that $g(\mathbf{\Sigma}) = -\log \det \mathbf{\Sigma} - \operatorname{tr}(\mathbf{\Sigma}^{-1}\mathbf{S})$, over positive definite $\mathbf{\Sigma}$, is maximized uniquely at $\mathbf{\Sigma} = \mathbf{S}$. *Hint:* write g in terms of the eigenvalues of $\mathbf{S}^{-1/2}\mathbf{\Sigma}\mathbf{S}^{-1/2}$.

Solution. Let $\mathbf{M} = \mathbf{S}^{-1/2}\boldsymbol{\Sigma}\mathbf{S}^{-1/2}$, which is positive definite with eigenvalues $\mu_1, \dots, \mu_p > 0$. Then $\det \boldsymbol{\Sigma} = \det \mathbf{S} \det \mathbf{M}$ and $\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\mathbf{S}) = \operatorname{tr}(\mathbf{S}^{1/2}\boldsymbol{\Sigma}^{-1}\mathbf{S}^{1/2}) = \operatorname{tr}(\mathbf{M}^{-1})$. So

$$g(\boldsymbol{\Sigma}) = -\log \det \mathbf{S} - \sum_{i=1}^p \left(\log \mu_i + \frac{1}{\mu_i} \right).$$

The function $\mu \mapsto \log \mu + 1/\mu$ has derivative $(\mu - 1)/\mu^2$ and a unique minimum at $\mu = 1$. So g is maximized iff every $\mu_i = 1$, that is, $\mathbf{M} = \mathbf{I}$ and $\boldsymbol{\Sigma} = \mathbf{S}$. The maximum is $-\log \det \mathbf{S} - p$.

1.8 The singular value decomposition and conditioning

The spectral theorem needs a symmetric matrix. A model matrix is rectangular, and the decomposition that fits it is the singular value decomposition.

Theorem 1.8.1 — Singular value decomposition

Let $\mathbf{A}$ be $m \times n$ of rank $r \geq 1$. There are an $m \times r$ matrix $\mathbf{U}$ and an $n \times r$ matrix $\mathbf{V}$, both with orthonormal columns, and numbers $\sigma_1 \geq \dots \geq \sigma_r > 0$, such that

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}\mathbf{T} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i \mathbf{T}, \quad \mathbf{D} = \operatorname{diag}(\sigma_1, \dots, \sigma_r). \quad (1.8.1)$$

The **singular values** σ_i are the positive square roots of the nonzero eigenvalues of $\mathbf{A}\mathbf{T}\mathbf{A}$ (equivalently of $\mathbf{A}\mathbf{A}\mathbf{T}$). The columns of $\mathbf{V}$ and $\mathbf{U}$ are eigenvectors of $\mathbf{A}\mathbf{T}\mathbf{A}$ and $\mathbf{A}\mathbf{A}\mathbf{T}$, with $\mathbb{C}(\mathbf{V}) = \mathbb{C}(\mathbf{A}\mathbf{T})$ and $\mathbb{C}(\mathbf{U}) = \mathbb{C}(\mathbf{A})$. Moreover $\mathbf{A}^+ = \mathbf{V}\mathbf{D}^{-1}\mathbf{U}\mathbf{T}$ is the Moore–Penrose inverse of $\mathbf{A}$.

Proof. $\mathbf{A}\mathbf{T}\mathbf{A}$ is nonnegative definite of rank r (Proposition 1.2.3(c)). By the spectral theorem its eigenvalues are $\sigma_1^2 \geq \dots \geq \sigma_r^2 > 0$ and $n-r$ zeros, with orthonormal eigenvectors $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_r]$ for the nonzero ones and $\mathbf{V}_0$ for the zeros. Define $\mathbf{U} = \mathbf{A}\mathbf{V}\mathbf{D}^{-1}$. Then $\mathbf{U}\mathbf{T}\mathbf{U} = \mathbf{D}^{-1}\mathbf{V}\mathbf{T}\mathbf{A}\mathbf{T}\mathbf{A}\mathbf{V}\mathbf{D}^{-1} = \mathbf{D}^{-1}\mathbf{D}^2\mathbf{D}^{-1} = \mathbf{I}_r$. For each column $\mathbf{v}$ of $\mathbf{V}_0$, $\|\mathbf{A}\mathbf{v}\|^2 = \mathbf{v}\mathbf{T}\mathbf{A}\mathbf{T}\mathbf{A}\mathbf{v} = 0$, so $\mathbf{A}\mathbf{v} = \mathbf{0}$. Since $\mathbf{V}\mathbf{V}\mathbf{T} + \mathbf{V}_0\mathbf{V}_0\mathbf{T} = \mathbf{I}$ (Proposition 1.5.3(d)),

$$\mathbf{A} = \mathbf{A}\mathbf{V}\mathbf{V}\mathbf{T} + \mathbf{A}\mathbf{V}_0\mathbf{V}_0\mathbf{T} = \mathbf{U}\mathbf{D}\mathbf{V}\mathbf{T}.$$

Then $\mathbf{A}\mathbf{A}\mathbf{T}\mathbf{U} = \mathbf{U}\mathbf{D}\mathbf{V}\mathbf{T}\mathbf{V}\mathbf{D}\mathbf{U}\mathbf{T}\mathbf{U} = \mathbf{U}\mathbf{D}^2$, so the columns of $\mathbf{U}$ are eigenvectors of $\mathbf{A}\mathbf{A}\mathbf{T}$. $\mathbb{C}(\mathbf{U}) \subseteq \mathbb{C}(\mathbf{A})$ by definition of $\mathbf{U}$, and both have dimension r . The statement for $\mathbf{V}$ is the same argument applied to $\mathbf{A}\mathbf{T} = \mathbf{V}\mathbf{D}\mathbf{U}\mathbf{T}$. The four Penrose conditions for $\mathbf{V}\mathbf{D}^{-1}\mathbf{U}\mathbf{T}$ follow from $\mathbf{U}\mathbf{T}\mathbf{U} = \mathbf{V}\mathbf{T}\mathbf{V} = \mathbf{I}_r$. For instance, $\mathbf{A}\mathbf{A}^+ = \mathbf{U}\mathbf{U}\mathbf{T}$ and $\mathbf{A}^+\mathbf{A} = \mathbf{V}\mathbf{V}\mathbf{T}$ are symmetric. ■

Completing $\mathbf{U}$ and $\mathbf{V}$ to orthogonal matrices gives the *full* SVD $\mathbf{A} = \tilde{\mathbf{U}}\tilde{\mathbf{D}}\tilde{\mathbf{V}}\mathbf{T}$, with $\tilde{\mathbf{D}}$ of size

$m \times n$ carrying $\sigma_1, \dots, \sigma_r$ on its diagonal and zeros elsewhere. By convention, $\sigma_{r+1} = \dots = \sigma_{\min(m,n)} = 0$. Geometrically, $\mathbf{A}$ maps the unit ball of $\mathbb{R}^n$ onto a solid ellipsoid in $\mathbb{C}(\mathbf{A})$ with semi-axes $\sigma_i \mathbf{u}_i$. The input direction $\mathbf{v}_i$ is stretched by the factor σ_i and turned into $\mathbf{u}_i$. The Moore–Penrose inverse undoes the stretching on $\mathbb{C}(\mathbf{A})$ and sends $\mathbb{C}(\mathbf{A})^\perp$ to zero. The matrix $\mathbf{U}\mathbf{U}^T$ is the orthogonal projection onto $\mathbb{C}(\mathbf{A})$, whose theory is in [Chapter 6](#).

Proposition 1.8.2 — Norms and extremal properties

Let $\mathbf{A}$ be $m \times n$ with singular values $\sigma_1 \geq \sigma_2 \geq \dots$, and let $\|\mathbf{A}\|_2 = \max_{\mathbf{x} \neq \mathbf{0}} \|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\|$ be the **spectral norm**.

- (a) $\|\mathbf{A}\|_2 = \sigma_1$ and $\|\mathbf{A}\|_F^2 = \sum_i \sigma_i^2$.
- (b) If $\mathbf{A}$ has full column rank n , then $\min_{\mathbf{x} \neq \mathbf{0}} \|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\| = \sigma_n$, attained at $\mathbf{v}_n$.
- (c) (Eckart–Young.) For $k < r$, let $\mathbf{A}_k = \sum_{i \leq k} \sigma_i \mathbf{u}_i \mathbf{v}_i^T$. Then $\|\mathbf{A} - \mathbf{B}\|_2 \geq \sigma_{k+1} = \|\mathbf{A} - \mathbf{A}_k\|_2$ for every $\mathbf{B}$ with $\text{rank}(\mathbf{B}) \leq k$.

Proof.

- (a) and (b): $\|\mathbf{A}\mathbf{x}\|^2/\|\mathbf{x}\|^2 = \mathbf{x}^T \mathbf{A}^T \mathbf{A} \mathbf{x} / \mathbf{x}^T \mathbf{x}$, and the eigenvalues of $\mathbf{A}^T \mathbf{A}$ are the σ_i^2 (padded with zeros), so [Theorem 1.7.7](#) applies. For the Frobenius norm, $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^T \mathbf{A}) = \sum \sigma_i^2$ by [Proposition 1.7.1\(a\)](#).
- (b) $\mathbf{A} - \mathbf{A}_k = \sum_{i > k} \sigma_i \mathbf{u}_i \mathbf{v}_i^T$ has largest singular value σ_{k+1} . If $\text{rank}(\mathbf{B}) \leq k$, then $\dim \mathcal{N}(\mathbf{B}) \geq n - k$, and since $(n - k) + (k + 1) > n$, a basis of $\mathcal{N}(\mathbf{B})$ together with $\mathbf{v}_1, \dots, \mathbf{v}_{k+1}$ is dependent. So the two subspaces share a unit vector $\mathbf{x} = \sum_{i \leq k+1} c_i \mathbf{v}_i$. Then $\|(\mathbf{A} - \mathbf{B})\mathbf{x}\|^2 = \|\mathbf{A}\mathbf{x}\|^2 = \sum_{i \leq k+1} \sigma_i^2 c_i^2 \geq \sigma_{k+1}^2$.

■

The Frobenius version of (c), $\|\mathbf{A} - \mathbf{B}\|_F^2 \geq \sum_{i > k} \sigma_i^2$, also holds (Eckart and Young 1936). It is the basis of principal components.

1.8.1 The condition number

Definition 1.8.1 — Condition number

The **condition number** of an $m \times n$ matrix $\mathbf{A}$ of full column rank is

$$\kappa(\mathbf{A}) = \frac{\sigma_{\max}(\mathbf{A})}{\sigma_{\min}(\mathbf{A})} = \frac{\sigma_1}{\sigma_n}.$$

For a nonsingular square matrix this equals $\|\mathbf{A}\|_2 \|\mathbf{A}^{-1}\|_2$. A matrix without full column rank has $\kappa(\mathbf{A}) = \infty$ by convention.

By [Proposition 1.8.2](#), $\kappa(\mathbf{A})$ is the ratio of the largest to the smallest stretching that $\mathbf{A}$ applies to a unit vector. It is at least 1, is unchanged by scaling $\mathbf{A}$ or multiplying it on the left by a matrix with orthonormal columns, and equals 1 iff $\mathbf{A}^T \mathbf{A} = c\mathbf{I}$, that is, iff the columns are orthogonal with equal lengths. Two further facts explain its role in regression. First, the singular values of $\mathbf{A}^T \mathbf{A}$ are the σ_i^2 , so

$$\kappa(\mathbf{ATA}) = \kappa(\mathbf{A})^2. \quad (1.8.2)$$

Second, κ bounds how much relative errors in the data can grow when a linear system is solved. If $\mathbf{A}$ is square and nonsingular, $\mathbf{Ax} = \mathbf{b}$ and $\mathbf{A}(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b} + \delta\mathbf{b}$, then $\delta\mathbf{x} = \mathbf{A}^{-1}\delta\mathbf{b}$, and $\|\delta\mathbf{b}\| \leq \sigma_1\|\mathbf{x}\|$ gives

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{\|\mathbf{A}^{-1}\|_2\|\delta\mathbf{b}\|}{\|\mathbf{b}\|/\sigma_1} = \kappa(\mathbf{A}) \frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|}. \quad (1.8.3)$$

In floating point, relative errors of about 10^{-16} in the data can therefore become errors of about $\kappa \times 10^{-16}$ in the solution. Solving the normal equations works with $\mathbf{XTX}$ and so, by [Eq. 1.8.2](#), squares the damage. [Chapter 6](#) and [Chapter 10](#) take this up.

A small $\sigma_{\min}$ has a direct statistical reading. By [Proposition 1.8.2\(b\)](#), $\|\mathbf{X}\mathbf{v}_n\| = \sigma_n$ with $\|\mathbf{v}_n\| = 1$. The coefficients $\mathbf{v}_n$ combine the columns of $\mathbf{X}$ into a vector that is nearly zero, which is a near-dependence among the regressors. This is the basis of the collinearity diagnostics of [Chapter 26](#). The size of κ depends on the units of the columns, so columns are usually rescaled before it is interpreted.

Example. (A macroeconomic design)

Take the 203 quarterly US observations of real GDP, real investment, real government spending, real disposable income, population and unemployment, with an intercept, as a model matrix with $p = 7$ columns. As recorded, $\kappa(\mathbf{X}) = 5.10 \times 10^5$. Most of that is units: GDP is measured in billions and unemployment in percent. Rescaling each column to unit length reduces it to 383, and centring the regressors before rescaling reduces it further to 124. By [Eq. 1.8.2](#) the corresponding cross-product matrix has condition number 15386.

What remains is not an artefact of units. [Figure 1.8.1\(b\)](#) shows the right singular vector for the smallest singular value, $\sigma_{\min} = 0.0175$, of the centred and scaled design. Its weight is concentrated on GDP (0.78) and disposable income (-0.61). The combination of the scaled columns with these coefficients is a vector of length 0.0175, although every column has length one. The two series move almost in lockstep: their correlation is 0.9990. No rescaling can separate their coefficients well. Centring is not a neutral choice, however: it removes near-dependences that involve the intercept, and [Belsley et al. \(1980\)](#) recommend scaling without centring for that reason, so their diagnostics would use the unit-length figure 383. [Chapter 26](#) returns to the choice. The listing computes the singular values, and the script also checks [Eq. 1.8.2](#), the Penrose conditions for $\mathbf{VD}^{-1}\mathbf{UT}$, and the Eckart–Young bound.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.macrodta.load_pandas().data
names = ["realgdp", "realinv", "realgovt", "realdpi", "pop", "unemp"]
labels = ["GDP", "investment", "government", "disp. income", "population", "unemployment"]
Z = data[names].to_numpy()
n = len(Z)

X = np.column_stack([np.ones(n), Z]) # as recorded
```

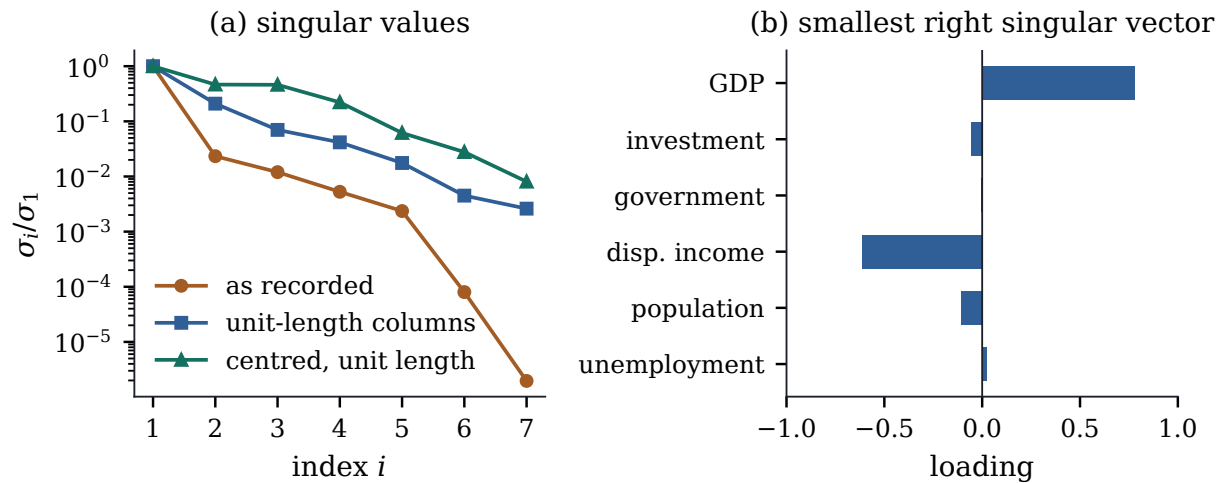


Figure 1.8.1: Singular value decomposition of a macroeconomic model matrix (Example (A macroeconomic design)). (a) Singular values relative to the largest, for the columns as recorded, rescaled to unit length, and centred then rescaled. The ratio of the first to the last is the condition number. (b) The right singular vector for the smallest singular value of the centred design. Its two large loadings reveal a near-dependence between GDP and disposable income.

```

X_unit = X / np.linalg.norm(X, axis=0)
Xc = np.column_stack([np.ones(n), Z - Z.mean(axis=0)])
X_std = Xc / np.linalg.norm(Xc, axis=0)

# unit-length columns
# regressors centred
# ... then length 1

for name, M in [("raw", X), ("unit length", X_unit), ("centred", X_std)]:
    U, s, Vt = np.linalg.svd(M, full_matrices=False)
    print(f"{name:12s} kappa = {s[0] / s[-1]:10.1f} singular values / s_1:",
          np.array2string(s / s[0], precision=4))
    # M = U diag(s) V'

```

1.8.2 Exercises

1.8.2.1 B. Practice

Exercise B1.

Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2]$ with $\|\mathbf{x}_1\| = \|\mathbf{x}_2\| = 1$ and $\mathbf{x}_1^T \mathbf{x}_2 = r$, $|r| < 1$. Show that $\kappa(\mathbf{X}) = \sqrt{(1 + |r|)/(1 - |r|)}$. How large must $|r|$ be for $\kappa(\mathbf{X}) > 30$?

Solution. $\mathbf{X}^T \mathbf{X} = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}$ has eigenvalues $1 \pm r$, with eigenvectors $(1, \pm 1)^T$. The singular values of $\mathbf{X}$ are their square roots, so $\kappa(\mathbf{X})^2 = (1 + |r|)/(1 - |r|)$. For $\kappa > 30$ we need $(1 + |r|)/(1 - |r|) > 900$, that is, $|r| > 899/901 \approx 0.998$. For two columns, even $\kappa(\mathbf{X}) = 100$ requires $|r| > 9999/10001 \approx 0.9998$.

Exercise B2.

Show that $\kappa(\mathbf{X}) \geq \max_j \|\mathbf{x}_j\| / \min_k \|\mathbf{x}_k\|$, so that columns on very different scales force a large condition number. Show by example that rescaling the columns can also increase κ .

1.9 Idempotent matrices

A square matrix $\mathbf{A}$ is **idempotent** if $\mathbf{A}^2 = \mathbf{A}$. Applying it twice has the same effect as applying it once, which is how a projection behaves. Examples are $\mathbf{I}$, $\mathbf{0}$, the centring matrix of [Example \(Sums of squares as quadratic forms\)](#), the products $\mathbf{A}\mathbf{G}$ and $\mathbf{G}\mathbf{A}$ for a generalized inverse $\mathbf{G}$ ([Proposition 1.3.3\(a\)](#)), and the non-symmetric $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$. In linear models, sums of squares are quadratic forms in *symmetric* idempotent matrices, and their degrees of freedom are ranks.

Proposition 1.9.1 — General idempotent matrices

Let $\mathbf{A}$ be $n \times n$ and idempotent of rank r .

- (a) Every eigenvalue of $\mathbf{A}$ is 0 or 1.
- (b) $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$.
- (c) $\mathbf{I} - \mathbf{A}$ is idempotent, $\mathbf{A}(\mathbf{I} - \mathbf{A}) = (\mathbf{I} - \mathbf{A})\mathbf{A} = \mathbf{0}$, $\mathbb{C}(\mathbf{I} - \mathbf{A}) = \mathcal{N}(\mathbf{A})$, and $\text{rank}(\mathbf{A}) + \text{rank}(\mathbf{I} - \mathbf{A}) = n$.
- (d) An idempotent matrix that is nonsingular equals $\mathbf{I}$.
- (e) $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ is idempotent for nonsingular $\mathbf{P}$, and $\mathbf{A}\mathbf{T}$ is idempotent.

Proof.

- (a) If $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, then $\lambda\mathbf{x} = \mathbf{A}\mathbf{x} = \mathbf{A}^2\mathbf{x} = \lambda^2\mathbf{x}$, so $\lambda^2 = \lambda$.
- (b) If $r = 0$ both sides vanish. Otherwise take a rank factorization $\mathbf{A} = \mathbf{B}\mathbf{F}$ ([Theorem 1.2.1](#)). Then $\mathbf{B}\mathbf{F}\mathbf{B}\mathbf{F} = \mathbf{B}\mathbf{F}$. Since $\mathbf{B}$ has full column rank and $\mathbf{F}$ has full row rank, $\mathbf{B}$ has a left inverse and $\mathbf{F}$ a right inverse ([Section 1.3](#)), and cancelling them gives $\mathbf{F}\mathbf{B} = \mathbf{I}_r$. So $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{B}\mathbf{F}) = \text{tr}(\mathbf{F}\mathbf{B}) = r$ by [Theorem 1.6.1](#).
- (c) The first two claims are one-line expansions. $\mathbb{C}(\mathbf{I} - \mathbf{A}) \subseteq \mathcal{N}(\mathbf{A})$ because $\mathbf{A}(\mathbf{I} - \mathbf{A}) = \mathbf{0}$, and $\mathbf{v} \in \mathcal{N}(\mathbf{A})$ gives $\mathbf{v} = (\mathbf{I} - \mathbf{A})\mathbf{v}$. The rank statement is then rank-nullity.
- (d) Multiply $\mathbf{A}^2 = \mathbf{A}$ by $\mathbf{A}^{-1}$.
- (e) $(\mathbf{P}^{-1}\mathbf{A}\mathbf{P})^2 = \mathbf{P}^{-1}\mathbf{A}^2\mathbf{P}$ and $(\mathbf{A}\mathbf{T})^2 = (\mathbf{A}^2)\mathbf{T}$.

■

By (c), every $\mathbf{x}$ splits as $\mathbf{A}\mathbf{x} + (\mathbf{I} - \mathbf{A})\mathbf{x}$ with the first part in $\mathbb{C}(\mathbf{A})$ and the second in $\mathcal{N}(\mathbf{A})$. So $\mathbb{R}^n = \mathbb{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A})$, and $\mathbf{A}$ projects onto $\mathbb{C}(\mathbf{A})$ along $\mathcal{N}(\mathbf{A})$. The projection is orthogonal exactly when the two subspaces are orthogonal, which happens iff $\mathbf{A}$ is symmetric ([Exercise \(B1\)](#)).

Theorem 1.9.2 — Symmetric idempotent matrices

Let $\mathbf{A}$ be an $n \times n$ symmetric matrix of rank r . Then $\mathbf{A}$ is idempotent iff all its eigenvalues are 0 or 1. In that case

- (a) $\mathbf{A} = \mathbf{Q}_1\mathbf{Q}_1^T$, where the $n \times r$ matrix $\mathbf{Q}_1$ has orthonormal columns spanning $\mathbb{C}(\mathbf{A})$;
- (b) $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}) = r$, the multiplicity of the eigenvalue 1;
- (c) $\mathbf{A}$ and $\mathbf{I} - \mathbf{A}$ are nonnegative definite, and $\mathbf{x}^T\mathbf{A}\mathbf{x} = \|\mathbf{A}\mathbf{x}\|^2 \leq \|\mathbf{x}\|^2$.

Proof. If $\mathbf{A}$ is idempotent its eigenvalues are 0 or 1 by [Proposition 1.9.1\(a\)](#). Conversely, if $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$ with each $\lambda_i \in \{0, 1\}$, then $\mathbf{A}^2 = \mathbf{Q}\mathbf{\Lambda}^2\mathbf{Q}^T = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T = \mathbf{A}$. In that case, order the eigenvalues so that the r ones come first and let $\mathbf{Q}_1$ hold the corresponding eigenvectors. Then $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T = \mathbf{Q}_1\mathbf{Q}_1^T$, and $\mathbb{C}(\mathbf{A}) = \mathbb{C}(\mathbf{Q}_1)$ by [Corollary 1.7.3\(a\)](#). This is (a). The trace is the sum of the eigenvalues, which is (b). For (c), the eigenvalues of $\mathbf{A}$ and of $\mathbf{I} - \mathbf{A}$ are nonnegative, and $\mathbf{x}^T\mathbf{A}\mathbf{x} = \mathbf{x}^T\mathbf{A}\mathbf{T}\mathbf{A}\mathbf{x}$. Also $\|\mathbf{x}\|^2 - \|\mathbf{A}\mathbf{x}\|^2 = \mathbf{x}^T(\mathbf{I} - \mathbf{A})\mathbf{x} \geq 0$.

■

[Chapter 6](#) shows that symmetric idempotent matrices are exactly the orthogonal projections, and $\mathbf{Q}_1\mathbf{Q}_1^T$ in (a) is the projection onto $\mathbb{C}(\mathbf{A})$. In [Chapter 4](#), part (b) becomes the degrees of freedom of a chi-squared distribution.

Theorem 1.9.3 — Sums of symmetric idempotent matrices

Let $\mathbf{A}_1, \dots, \mathbf{A}_k$ be symmetric $n \times n$ matrices and $\mathbf{A} = \sum_i \mathbf{A}_i$. Consider

- (i) $\mathbf{A}$ is idempotent;
- (ii) every $\mathbf{A}_i$ is idempotent;
- (iii) $\mathbf{A}_i\mathbf{A}_j = \mathbf{0}$ for all $i \neq j$.

Any two of these imply the third. Moreover, if $\mathbf{A} = \mathbf{I}_n$ and $\sum_i \text{rank}(\mathbf{A}_i) = n$, then (ii) and (iii) hold.

Proof.

(ii) and (iii) imply (i): $\mathbf{A}^2 = \sum_i \mathbf{A}_i^2 + \sum_{i \neq j} \mathbf{A}_i\mathbf{A}_j = \sum_i \mathbf{A}_i = \mathbf{A}$.

(iii) and (ii) imply (iii): By 1(c), $\|\mathbf{x}\|^2 \geq \mathbf{x}^T\mathbf{A}\mathbf{x}$ for all $\mathbf{x}$. Fix j and take $\mathbf{x} = \mathbf{A}_j\mathbf{y}$, so that $\mathbf{A}_j\mathbf{x} = \mathbf{x}$ and $\mathbf{x}^T\mathbf{A}_j\mathbf{x} = \|\mathbf{x}\|^2$. Then

$$\|\mathbf{x}\|^2 \geq \mathbf{x}^T\mathbf{A}\mathbf{x} = \|\mathbf{x}\|^2 + \sum_{i \neq j} \mathbf{x}^T\mathbf{A}_i\mathbf{x} = \|\mathbf{x}\|^2 + \sum_{i \neq j} \|\mathbf{A}_i\mathbf{x}\|^2.$$

So $\mathbf{A}_i\mathbf{A}_j\mathbf{y} = \mathbf{A}_i\mathbf{x} = \mathbf{0}$ for every $i \neq j$ and every $\mathbf{y}$.

(iv) and (iii) imply (ii): By (iii), $\mathbf{A}\mathbf{A}_i = \mathbf{A}_i^2$. Using $\mathbf{A}^2 = \mathbf{A}$, $\mathbf{A}_i^2 = \mathbf{A}\mathbf{A}_i = \mathbf{A}^2\mathbf{A}_i = \mathbf{A}\mathbf{A}_i^2 = \mathbf{A}_i^3$. The eigenvalues of the symmetric $\mathbf{A}_i$ therefore satisfy $\lambda^2 = \lambda^3$, so they lie in $\{0, 1\}$, and $\mathbf{A}_i$ is idempotent by

1.

For the last statement, let $r_i = \text{rank}(\mathbf{A}_i)$. For each i , $\mathbf{I} - \mathbf{A}_i = \sum_{j \neq i} \mathbf{A}_j$ has rank at most $\sum_{j \neq i} r_j = n - r_i$ ([Proposition 1.2.3\(d\)](#)). So the eigenvalue 1 of $\mathbf{A}_i$, whose eigenspace is $\mathcal{N}(\mathbf{I} - \mathbf{A}_i)$, has multiplicity at least r_i . But $\mathbf{A}_i$ has exactly r_i nonzero eigenvalues, so all of them equal 1 and $\mathbf{A}_i$ is idempotent. Now (i) holds trivially, and (iii) follows from the second part of the proof.

■

The last statement is the algebraic core of Cochran's theorem ([Chapter 4](#)): if $\mathbf{y}^T\mathbf{y} = \sum_i \mathbf{y}^T\mathbf{A}_i\mathbf{y}$

and the ranks add up to n , the quadratic forms are built from mutually orthogonal projections.

1.9.1 Exercises

1.9.1.1 A. Check your understanding

Exercise A1.

Let $\mathbf{A}, \mathbf{B}$ be idempotent of the same size. Show that $\mathbf{AB}$ is idempotent if $\mathbf{AB} = \mathbf{BA}$, and that $\mathbf{A} + \mathbf{B}$ is idempotent iff $\mathbf{AB} + \mathbf{BA} = \mathbf{0}$. Show that the latter forces $\mathbf{AB} = \mathbf{0}$ (symmetry is not needed).

1.9.1.2 B. Practice

Exercise B1.

Let $\mathbf{A}$ be idempotent. Show that $\mathbf{A}$ is symmetric iff $\mathbb{C}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A})$.

Solution. Suppose $\mathbf{A}$ is symmetric. For $\mathbf{u} = \mathbf{Ax} \in \mathbb{C}(\mathbf{A})$ and $\mathbf{v} \in \mathcal{N}(\mathbf{A})$, $\mathbf{u}^T \mathbf{v} = \mathbf{x}^T \mathbf{A} \mathbf{v} = 0$. Conversely, suppose $\mathbb{C}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A})$. For all $\mathbf{x}, \mathbf{y}$, $(\mathbf{I} - \mathbf{A})\mathbf{y} \in \mathcal{N}(\mathbf{A})$, so $\mathbf{x}^T \mathbf{A} \mathbf{T} (\mathbf{I} - \mathbf{A})\mathbf{y} = 0$. Hence $\mathbf{AT}(\mathbf{I} - \mathbf{A}) = \mathbf{0}$, that is, $\mathbf{AT} = \mathbf{ATA}$. The right side is symmetric, so $\mathbf{AT}$ is symmetric, and so is $\mathbf{A}$.

1.10 Kronecker products and the vec operator

Balanced designs have model matrices with a product structure, and multivariate observations have covariance matrices with one. The Kronecker product expresses both.

Definition 1.10.1

The **Kronecker product** of an $m \times n$ matrix $\mathbf{A}$ and a $p \times q$ matrix $\mathbf{B}$ is the $mp \times nq$ block matrix $\mathbf{A} \otimes \mathbf{B} = (a_{ij}\mathbf{B})$. The **vec operator** places the n columns of an $m \times n$ matrix one below another, giving a vector of length mn : $\text{vec}(\mathbf{A}) = (\mathbf{a}_1^T, \dots, \mathbf{a}_n^T)^T$.

Proposition 1.10.1

For matrices of compatible sizes:

- (a) $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$ (mixed-product rule);
- (b) $(\mathbf{A} \otimes \mathbf{B})\mathbf{T} = \mathbf{AT} \otimes \mathbf{BT}$, and $(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$ for nonsingular $\mathbf{A}, \mathbf{B}$. More generally $\mathbf{A}^- \otimes \mathbf{B}^-$ is a generalized inverse of $\mathbf{A} \otimes \mathbf{B}$;
- (c) $\text{tr}(\mathbf{A} \otimes \mathbf{B}) = \text{tr}(\mathbf{A}) \text{tr}(\mathbf{B})$ and $\text{rank}(\mathbf{A} \otimes \mathbf{B}) = \text{rank}(\mathbf{A}) \text{rank}(\mathbf{B})$;

- (d) if $\mathbf{A}$ ($m \times m$) and $\mathbf{B}$ ($p \times p$) are symmetric with eigenvalues λ_i and μ_j , the eigenvalues of $\mathbf{A} \otimes \mathbf{B}$ are the products $\lambda_i \mu_j$, and $\det(\mathbf{A} \otimes \mathbf{B}) = (\det \mathbf{A})^p (\det \mathbf{B})^m$;
- (e) Kronecker products of orthogonal, symmetric, idempotent, or nonnegative definite matrices are of the same kind;
- (f) $\text{vec}(\mathbf{ABC}) = (\mathbf{CT} \otimes \mathbf{A}) \text{vec}(\mathbf{B})$ and $\text{tr}(\mathbf{ATB}) = \text{vec}(\mathbf{A})^T \text{vec}(\mathbf{B})$.

Proof.

- (a) The (i, j) block of the left side is $\sum_k a_{ik} c_{kj} \mathbf{B} \mathbf{D}$, which is the (i, j) block of the right side. (b) Transposition is direct, and the rest follows from (a), for example $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-)(\mathbf{A} \otimes \mathbf{B}) = \mathbf{AA}^- \mathbf{A} \otimes \mathbf{BB}^- \mathbf{B}$.
- (b) With $\mathbf{A} = \mathbf{P} \mathbf{\Lambda} \mathbf{P}^T$ and $\mathbf{B} = \mathbf{Q} \mathbf{M} \mathbf{Q}^T$, (a) and (b) give $\mathbf{A} \otimes \mathbf{B} = (\mathbf{P} \otimes \mathbf{Q})(\mathbf{\Lambda} \otimes \mathbf{M})(\mathbf{P} \otimes \mathbf{Q})^T$, a spectral decomposition because $\mathbf{P} \otimes \mathbf{Q}$ is orthogonal and $\mathbf{\Lambda} \otimes \mathbf{M}$ is diagonal with entries $\lambda_i \mu_j$. The determinant is the product of these. (c) The trace statement is direct. For the rank, by (a) $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-) = \mathbf{AA}^- \otimes \mathbf{BB}^-$ is idempotent with the same rank as $\mathbf{A} \otimes \mathbf{B}$ (Proposition 1.3.3(a)). Its trace is $\text{tr}(\mathbf{AA}^-) \text{tr}(\mathbf{BB}^-) = \text{rank}(\mathbf{A}) \text{rank}(\mathbf{B})$ by Proposition 1.9.1(b).
- (c) follows from (a), (b) and (d). (f) The j th column of $\mathbf{ABC}$ is $\mathbf{ABc}_j = \sum_k c_{kj} \mathbf{Ab}_k$, which is the j th block of $(\mathbf{CT} \otimes \mathbf{A}) \text{vec}(\mathbf{B})$. The trace identity is $\sum_{ij} a_{ij} b_{ij}$ written two ways.

■

The determinant formula in (d) holds for all square $\mathbf{A}$ and $\mathbf{B}$, not only symmetric ones. We need only the symmetric case.

Example. (The balanced two-way layout)

Let observations y_{ij} , $i = 1, \dots, a$, $j = 1, \dots, b$, be stacked with j running inside i . The additive model $\mathbf{E}(y_{ij}) = \mu + \alpha_i + \beta_j$ has model matrix

$$\mathbf{X} = [\mathbf{1}_a \otimes \mathbf{1}_b, \mathbf{I}_a \otimes \mathbf{1}_b, \mathbf{1}_a \otimes \mathbf{I}_b],$$

because $\mathbf{I}_a \otimes \mathbf{1}_b$ has a 1 in column i exactly for the b observations in row i . With $\bar{\mathbf{J}}_k = k^{-1} \mathbf{J}_k$, a symmetric idempotent averaging matrix, the mixed-product rule shows that $\mathbf{I}_a \otimes \bar{\mathbf{J}}_b$, $\bar{\mathbf{J}}_a \otimes \mathbf{I}_b$ and $\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$ replace each observation by its row mean, its column mean and the grand mean. The matrix $\mathbf{M} = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes \mathbf{I}_b - \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$ is symmetric and idempotent, with trace $a + b - 1 = \text{rank}(\mathbf{X})$. It is the projection onto $\mathbb{C}(\mathbf{X})$ (Chapter 15). The script checks this for $a = 3$ and $b = 4$, together with each part of Proposition 1.10.1.

```
import numpy as np
rng = np.random.default_rng(11)
def vec(M):
    return M.reshape(-1, order="F") # stack the columns

A, B, C = rng.normal(size=(3, 4)), rng.normal(size=(4, 2)), rng.normal(size=(2, 5))
assert np.allclose(vec(A @ B @ C), np.kron(C.T, A) @ vec(B)) # vec(ABC)

# balanced two-way layout: a = 3 rows, b = 4 columns, one observation per cell
a, b = 3, 4
one = lambda k: np.ones((k, 1))
```

```
X = np.hstack([np.kron(one(a), one(b)),      # intercept
               np.kron(np.eye(a), one(b)),   # row indicators
               np.kron(one(a), np.eye(b))])   # column indicators
print("rank(X) =", np.linalg.matrix_rank(X), "= a + b - 1 =", a + b - 1)
```

1.11 Vector and matrix calculus

1.11.1 Gradients of linear and quadratic functions

For a differentiable scalar function f of $\mathbf{x} \in \mathbb{R}^n$, the **gradient** $\partial f / \partial \mathbf{x}$ is the column vector with entries $\partial f / \partial x_i$, and the **Hessian** $\partial^2 f / \partial \mathbf{x} \partial \mathbf{x}^T$ is the $n \times n$ matrix of second derivatives. For a vector-valued $\mathbf{g}(\mathbf{x}) \in \mathbb{R}^m$, the $m \times n$ *Jacobian* $\partial \mathbf{g} / \partial \mathbf{x}^T$ has (i, j) entry $\partial g_i / \partial x_j$.

Theorem 1.11.1 — Derivatives of linear and quadratic forms

Let $\mathbf{a} \in \mathbb{R}^n$, let $\mathbf{A}$ be $n \times n$ and $\mathbf{B}$ be $m \times n$. Then

$$\frac{\partial(\mathbf{a}^T \mathbf{x})}{\partial \mathbf{x}} = \mathbf{a}, \quad \frac{\partial(\mathbf{B} \mathbf{x})}{\partial \mathbf{x}^T} = \mathbf{B}, \quad \frac{\partial(\mathbf{x}^T \mathbf{A} \mathbf{x})}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^T) \mathbf{x}, \quad \frac{\partial^2(\mathbf{x}^T \mathbf{A} \mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}^T} = \mathbf{A} + \mathbf{A}^T.$$

In particular $\partial(\mathbf{x}^T \mathbf{A} \mathbf{x}) / \partial \mathbf{x} = 2\mathbf{A} \mathbf{x}$ when $\mathbf{A}$ is symmetric.

Proof. The first two are immediate from $\mathbf{a}^T \mathbf{x} = \sum_i a_i x_i$ and $(\mathbf{B} \mathbf{x})_i = \sum_j b_{ij} x_j$. For the quadratic form, expand

$$(\mathbf{x} + \mathbf{h})^T \mathbf{A} (\mathbf{x} + \mathbf{h}) = \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{h}^T (\mathbf{A} + \mathbf{A}^T) \mathbf{x} + \mathbf{h}^T \mathbf{A} \mathbf{h}.$$

The middle term is linear in $\mathbf{h}$ and the last is $O(\|\mathbf{h}\|^2)$, so $(\mathbf{A} + \mathbf{A}^T) \mathbf{x}$ is the gradient. The gradient is linear in $\mathbf{x}$ with matrix $\mathbf{A} + \mathbf{A}^T$, which is the Hessian by the second formula. ■

Example. (The least squares criterion)

Let $S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2 = \mathbf{y}^T \mathbf{y} - 2\mathbf{y}^T \mathbf{X} \mathbf{b} + \mathbf{b}^T \mathbf{X}^T \mathbf{X} \mathbf{b}$. By Theorem 1.11.1,

$$\frac{\partial S}{\partial \mathbf{b}} = -2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X} \mathbf{b} = -2\mathbf{X}^T (\mathbf{y} - \mathbf{X} \mathbf{b}), \quad \frac{\partial^2 S}{\partial \mathbf{b} \partial \mathbf{b}^T} = 2\mathbf{X}^T \mathbf{X}.$$

Setting the gradient to zero gives the normal equations $\mathbf{X}^T \mathbf{X} \mathbf{b} = \mathbf{X}^T \mathbf{y}$. The Hessian is nonnegative definite, so S is convex and every solution is a global minimizer. The next result proves this without calculus, and covers the rank-deficient case.

Proposition 1.11.2 — Minimizing a quadratic function

Let $\mathbf{A}$ be $n \times n$ nonnegative definite, $\mathbf{b} \in \mathbb{R}^n$ and $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} - 2\mathbf{b}^T \mathbf{x}$. If $\mathbf{b} \in \mathcal{C}(\mathbf{A})$,

the minimizers of f are exactly the solutions of $\mathbf{Ax} = \mathbf{b}$, and the minimum value is $-\mathbf{bTA}^{-}\mathbf{b}$ for any generalized inverse $\mathbf{A}^{-}$. If $\mathbf{b} \notin \mathbb{C}(\mathbf{A})$, then f is unbounded below.

Proof. Let $\mathbf{b} \in \mathbb{C}(\mathbf{A})$ and let $\mathbf{x}_0$ solve $\mathbf{Ax}_0 = \mathbf{b}$. Completing the square,

$$f(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_0)\mathbf{TA}(\mathbf{x} - \mathbf{x}_0) - \mathbf{x}_0\mathbf{TAx}_0.$$

The first term is nonnegative, and by Proposition 1.7.5(b) it is zero iff $\mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{0}$, that is, iff $\mathbf{Ax} = \mathbf{b}$. The minimum is $-\mathbf{x}_0\mathbf{TAx}_0 = -\mathbf{x}_0\mathbf{TA}\mathbf{A}^{-}\mathbf{Ax}_0 = -\mathbf{bTA}^{-}\mathbf{b}$. If $\mathbf{b} \notin \mathbb{C}(\mathbf{A})$, write $\mathbf{b} = \mathbf{b}_1 + \mathbf{b}_2$ with $\mathbf{b}_1 \in \mathbb{C}(\mathbf{A})$ and $\mathbf{0} \neq \mathbf{b}_2 \in \mathbb{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A})$ (Proposition 1.5.1). Then $f(t\mathbf{b}_2) = -2t\|\mathbf{b}_2\|^2 \rightarrow -\infty$ as $t \rightarrow \infty$. ■

With $\mathbf{A} = \mathbf{XTX}$ and $\mathbf{b} = \mathbf{XTy}$, the proposition gives least squares for every model matrix: the minimizers are the solutions of the normal equations, and the minimum of S is $\mathbf{yTy} - \mathbf{yTX}(\mathbf{XTX})^{-}\mathbf{XTy}$.

1.11.2 Constrained minimization

To minimize a differentiable $f(\mathbf{x})$ subject to linear constraints $\mathbf{Hx} = \mathbf{h}$, with $\mathbf{H}$ of size $q \times n$, the method of **Lagrange multipliers** looks for stationary points of $f(\mathbf{x}) + 2\theta\mathbf{T}(\mathbf{Hx} - \mathbf{h})$ in $(\mathbf{x}, \theta)$. The factor 2 is only for convenience. For quadratic f the stationary point can be written down and checked directly.

Proposition 1.11.3

Let $\mathbf{A}$ be positive definite, $\mathbf{H}$ of size $q \times n$ with full row rank, and $f(\mathbf{x}) = \mathbf{xTAx} - 2\mathbf{bTx}$. The unique minimizer of f subject to $\mathbf{Hx} = \mathbf{h}$ is

$$\mathbf{x}_* = \mathbf{x}_0 - \mathbf{A}^{-1}\mathbf{HT}(\mathbf{HA}^{-1}\mathbf{HT})^{-1}(\mathbf{Hx}_0 - \mathbf{h}), \quad \mathbf{x}_0 = \mathbf{A}^{-1}\mathbf{b}.$$

Proof. $\mathbf{HA}^{-1}\mathbf{HT}$ is positive definite by Proposition 1.7.5(c,e), so $\mathbf{x}_*$ is well defined, and $\mathbf{Hx}_* = \mathbf{h}$ by direct substitution. Stationarity of the Lagrangian gives $2\mathbf{Ax} - 2\mathbf{b} + 2\mathbf{HT}\theta = \mathbf{0}$. Indeed $\mathbf{Ax}_* - \mathbf{b} = -\mathbf{HT}\theta_*$ with $\theta_* = (\mathbf{HA}^{-1}\mathbf{HT})^{-1}(\mathbf{Hx}_0 - \mathbf{h})$. For any feasible $\mathbf{x}$, put $\mathbf{d} = \mathbf{x} - \mathbf{x}_*$, so that $\mathbf{Hd} = \mathbf{0}$. Then

$$f(\mathbf{x}) = f(\mathbf{x}_*) + 2\mathbf{dT}(\mathbf{Ax}_* - \mathbf{b}) + \mathbf{dTAd} = f(\mathbf{x}_*) - 2(\mathbf{Hd})\mathbf{T}\theta_* + \mathbf{dTAd} = f(\mathbf{x}_*) + \mathbf{dTAd},$$

which exceeds $f(\mathbf{x}_*)$ unless $\mathbf{d} = \mathbf{0}$. ■

With $\mathbf{A} = \mathbf{XTX}$ and $\mathbf{b} = \mathbf{XTy}$, this is least squares subject to a linear hypothesis $\mathbf{H}\beta = \mathbf{h}$, the starting point of Chapter 11.

1.11.3 Derivatives with respect to matrices and scalars

Likelihoods for normal models involve $\log \det \Sigma$ and $\text{tr}(\Sigma^{-1}S)$. For a scalar function f of an $m \times n$ matrix $\mathbf{W}$ with functionally independent entries, $\partial f / \partial \mathbf{W}$ is the $m \times n$ matrix with entries $\partial f / \partial w_{ij}$.

Proposition 1.11.4 (a) $\partial \text{tr}(\mathbf{A}\mathbf{W}) / \partial \mathbf{W} = \mathbf{A}^T$.

(b) If $\det \mathbf{W} > 0$, then $\partial \log \det \mathbf{W} / \partial \mathbf{W} = (\mathbf{W}^{-1})^T$.

(c) If $\mathbf{A}(t)$ is a nonsingular matrix whose entries are differentiable functions of a scalar t , with derivative $\dot{\mathbf{A}} = \partial \mathbf{A} / \partial t$, then

$$\frac{\partial \mathbf{A}^{-1}}{\partial t} = -\mathbf{A}^{-1} \dot{\mathbf{A}} \mathbf{A}^{-1}, \quad \frac{\partial \log |\det \mathbf{A}|}{\partial t} = \text{tr}(\mathbf{A}^{-1} \dot{\mathbf{A}}).$$

Proof.

(a) $\text{tr}(\mathbf{A}\mathbf{W}) = \sum_{i,j} a_{ji} w_{ij}$. (c) Differentiate $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ by the product rule to get $\dot{\mathbf{A}}\mathbf{A}^{-1} + \mathbf{A}\partial \mathbf{A}^{-1} / \partial t = \mathbf{0}$. For the determinant, $\mathbf{A}(t+h) = \mathbf{A}(t)(\mathbf{I} + h\mathbf{A}^{-1}\dot{\mathbf{A}} + o(h))$. In the permutation expansion of $\det(\mathbf{I} + h\mathbf{C})$ only the diagonal product contributes terms of first order in h , so $\det(\mathbf{I} + h\mathbf{C}) = 1 + h \text{tr} \mathbf{C} + O(h^2)$. Hence $\det \mathbf{A}(t+h) = \det \mathbf{A}(t)(1 + h \text{tr}(\mathbf{A}^{-1}\dot{\mathbf{A}}) + o(h))$, and taking logarithms gives the formula. (b) Apply (c) with $\mathbf{A}(t) = \mathbf{W} + te_i e_j^T$. Then $\text{tr}(\mathbf{W}^{-1}e_i e_j^T) = e_j^T \mathbf{W}^{-1} e_i$, the (j, i) entry of $\mathbf{W}^{-1}$.

■

When $\mathbf{W}$ is constrained to be symmetric, w_{ij} and w_{ji} are the same variable, so the derivatives with respect to off-diagonal entries double. Some books (Rencher and Schaalje 2008) report formulas of the type $2\mathbf{W}^{-1} - \text{diag}(\mathbf{W}^{-1})$ for this reason. The simplest way to avoid the issue is to differentiate along a symmetric direction $\mathbf{W} + t\mathbf{E}$, as in (c). This is the convenient route for the normal likelihoods of Chapter 32. The listing checks the gradient formulas against central finite differences, and the script also checks part (c) and Proposition 1.11.2. The largest discrepancy in the listing is 3×10^{-9} , which is the size one expects from the finite-difference approximation.

```
import numpy as np
rng = np.random.default_rng(5)
p = 4
def num_grad(f, x, h=1e-6):
    """Central-difference gradient of a scalar function of a vector or matrix."""
    g = np.zeros_like(x)
    for idx in np.ndindex(x.shape):
        e = np.zeros_like(x)
        e[idx] = h
        g[idx] = (f(x + e) - f(x - e)) / (2 * h)
    return g

a = rng.normal(size=p)
A = rng.normal(size=(p, p)) # not symmetric
x = rng.normal(size=p)
checks = {
    "a'x": (num_grad(lambda v: a @ v, x), a),
    "x'Ax": (num_grad(lambda v: v @ A @ v, x), (A + A.T) @ x),
```

```

}
X = rng.normal(size=(20, p))
y = rng.normal(size=20)
b = rng.normal(size=p)
grad_ls = -2 * X.T @ (y - X @ b)
checks["||y - Xb||^2"] = (num_grad(lambda v: np.sum((y - X @ v) ** 2), b), grad_ls)

S = rng.normal(size=(p, p)); S = S @ S.T + p * np.eye(p) # positive definite
checks["log det S"] = (num_grad(lambda W: np.linalg.slogdet(W)[1], S), np.linalg.inv(S).T)
checks["tr(AS)"] = (num_grad(lambda W: np.trace(A @ W), S), A.T)
for name, (numeric, formula) in checks.items():
    print(f"{name:14s} max error {np.abs(numeric - formula).max():.1e}")

```

1.11.4 Exercises

1.11.4.1 A. Check your understanding

Exercise A1.

Observations fall in a clusters of size b , and $\Sigma = \sigma^2 \mathbf{I}_a \otimes \mathbf{E}_b$, where $\mathbf{E}_b = (1 - \rho)\mathbf{I}_b + \rho\mathbf{J}_b$. Find the eigenvalues, determinant and inverse of Σ using [Proposition 1.10.1](#), [Example \(Equicorrelation\)](#) and [Example \(Eigenvalues of the equicorrelation matrix\)](#).

Exercise A2.

Use [Proposition 1.11.2](#) to show that $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda\|\mathbf{b}\|^2$ with $\lambda > 0$ has the unique minimizer $(\mathbf{X}\mathbf{T}\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}\mathbf{T}\mathbf{y}$, whatever the rank of $\mathbf{X}$.

1.11.4.2 B. Practice

Exercise B1.

Let $\mathbf{X}$ have full column rank, $\hat{\beta} = (\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{X}\mathbf{T}\mathbf{y}$, and let $\mathbf{H}$ be $q \times p$ with full row rank. Show that the minimizer of $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ subject to $\mathbf{H}\mathbf{b} = \mathbf{h}$ is

$$\hat{\beta}_H = \hat{\beta} - (\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{H}\mathbf{T}\mathbf{W}^{-1}(\mathbf{H}\hat{\beta} - \mathbf{h}), \quad \mathbf{W} = \mathbf{H}(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{H}\mathbf{T},$$

and that the increase in the residual sum of squares is $(\mathbf{H}\hat{\beta} - \mathbf{h})\mathbf{T}\mathbf{W}^{-1}(\mathbf{H}\hat{\beta} - \mathbf{h})$.

Solution. Apply [Proposition 1.11.3](#) with $\mathbf{A} = \mathbf{X}\mathbf{T}\mathbf{X}$ and $\mathbf{b} = \mathbf{X}\mathbf{T}\mathbf{y}$. Since $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \mathbf{y}\mathbf{T}\mathbf{y} + f(\mathbf{b})$ and $\mathbf{x}_0 = \hat{\beta}$, this gives $\hat{\beta}_H$. By completing the square as in [Proposition 1.11.2](#), $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X}\hat{\beta}\|^2 + (\mathbf{b} - \hat{\beta})\mathbf{T}\mathbf{X}\mathbf{T}\mathbf{X}(\mathbf{b} - \hat{\beta})$ for every $\mathbf{b}$. With $\mathbf{b} - \hat{\beta} = -(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{H}\mathbf{T}\mathbf{W}^{-1}(\mathbf{H}\hat{\beta} - \mathbf{h})$, the increase is

$$(\mathbf{H}\hat{\beta} - \mathbf{h})\mathbf{T}\mathbf{W}^{-1}\mathbf{H}(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{H}\mathbf{T}\mathbf{W}^{-1}(\mathbf{H}\hat{\beta} - \mathbf{h}) = (\mathbf{H}\hat{\beta} - \mathbf{h})\mathbf{T}\mathbf{W}^{-1}(\mathbf{H}\hat{\beta} - \mathbf{h}).$$

This quadratic form is the numerator of the F statistic in Chapter 11.

1.12 Summary and Notes

1.12.1 Summary

Idea.

1. Row rank equals column rank, and $\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A})$ is the number of columns. $\mathbb{C}(\mathbf{A}\mathbf{T}\mathbf{A}) = \mathbb{C}(\mathbf{A}\mathbf{T})$, so the normal equations are always consistent (Theorem 1.2.1, Theorem 1.2.2 and Proposition 1.2.3).
2. A generalized inverse satisfies $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$. Every matrix has one, built from any nonsingular $r \times r$ submatrix. The Moore–Penrose inverse is the unique one that is also reflexive and makes $\mathbf{A}\mathbf{A}^+$ and $\mathbf{A}^+\mathbf{A}$ symmetric (Definition 1.3.1, Theorem 1.3.2, Proposition 1.2.4 and Definition 1.3.2).
3. $\mathbf{A}\mathbf{x} = \mathbf{b}$ is consistent iff $\mathbf{A}\mathbf{A}^-\mathbf{b} = \mathbf{b}$, and then its solutions are $\mathbf{A}^-\mathbf{b} + (\mathbf{I} - \mathbf{A}^-\mathbf{A})\mathbf{z}$. A linear function $\mathbf{q}\mathbf{T}\mathbf{x}$ is the same for all solutions iff $\mathbf{q} \in \mathbb{C}(\mathbf{A}\mathbf{T})$ (Theorem 1.4.1 and Corollary 1.4.2).
4. Orthogonal matrices preserve lengths and angles. Gram–Schmidt gives the QR factorization, and $\mathbb{R}^n = \mathbb{C}(\mathbf{A}\mathbf{T}) \oplus \mathcal{N}(\mathbf{A})$ (Proposition 1.5.3, Theorem 1.5.2 and Proposition 1.5.1).
5. $\text{tr}(\mathbf{A}\mathbf{B}) = \text{tr}(\mathbf{B}\mathbf{A})$. Block elimination gives the determinant and inverse of a partitioned matrix through Schur complements, and the Sherman–Morrison–Woodbury identity as a special case (Theorem 1.6.1, Theorem 1.6.3, Theorem 1.6.4 and Theorem 1.6.5).
6. A symmetric matrix is $\mathbf{Q}\mathbf{A}\mathbf{Q}\mathbf{T}$ with $\mathbf{Q}$ orthogonal. Positive definiteness is equivalent to positive eigenvalues, to a Cholesky factorization, and to positive leading minors. Nonnegative definite matrices have a unique nonnegative definite square root (Theorem 1.7.2, Theorem 1.7.4 and Theorem 1.7.6).
7. The Rayleigh quotient $\mathbf{x}\mathbf{T}\mathbf{A}\mathbf{x}/\mathbf{x}\mathbf{T}\mathbf{x}$ ranges over $[\lambda_{\min}, \lambda_{\max}]$ (Theorem 1.7.7 and Corollary 1.7.8).
8. Every matrix is $\mathbf{U}\mathbf{D}\mathbf{V}\mathbf{T}$. The condition number $\kappa = \sigma_{\max}/\sigma_{\min}$ bounds error growth, is squared by forming $\mathbf{X}\mathbf{T}\mathbf{X}$, and its smallest singular vector exhibits near-dependences among the columns (Theorem 1.8.1, Definition 1.8.1 and Proposition 1.8.2).
9. Idempotent matrices have eigenvalues 0 and 1 and rank equal to trace. Symmetric ones are $\mathbf{Q}_1\mathbf{Q}_1\mathbf{T}$. If symmetric matrices summing to $\mathbf{I}$ have ranks summing to n , they are idempotent and mutually orthogonal (Proposition 1.9.1, 1 and Theorem 1.9.3).
10. Kronecker products factor balanced designs. $\text{vec}(\mathbf{A}\mathbf{B}\mathbf{C}) = (\mathbf{C}\mathbf{T} \otimes \mathbf{A})\text{vec}(\mathbf{B})$ (Proposition 1.10.1).
11. $\partial(\mathbf{x}\mathbf{T}\mathbf{A}\mathbf{x})/\partial\mathbf{x} = (\mathbf{A} + \mathbf{A}\mathbf{T})\mathbf{x}$. Quadratic functions with nonnegative definite matrices are minimized by solving a linear system, with or without linear constraints (Theorem 1.11.1, Proposition 1.11.2 and Proposition 1.11.3).

1.12.2 Notes and sources

Coverage and framing. The selection of results follows what the rest of the book uses. Rencher and Schaalje (2008, chapter 2) was the checklist for topics: its sections on positive definite matrices, generalized inverses, orthogonal matrices, idempotent matrices and matrix calculus correspond to [Section 1.7](#), [Section 1.3](#), [Section 1.5](#), [Section 1.9](#) and [Section 1.11](#). Several results that Rencher and Schaalje state without proof or with a reference, such as the general solution of a consistent system, the characterizations of positive definiteness and the sums of idempotent matrices, are proved here. The vector-space point of view, in which column spaces, null spaces and orthogonal complements come first and matrices second, is that of Christensen (2020, appendices A and B). Seber and Lee (2003, appendix A) is a compact list of further matrix facts. The Kronecker product results in [Section 1.10](#) are also in Christensen’s appendix B.

Linear algebra for statisticians. Harville (1997) is the most complete treatment written for statisticians, with full proofs of the determinant facts assumed in [Proposition 1.6.2](#) and a long development of generalized inverses. Searle (1982) is more elementary and has many worked examples from linear models. Horn and Johnson (2013) is the standard reference for eigenvalue inequalities, positive definite matrices and norms. Halmos (1958) gives the coordinate-free view of the vector-space facts used in [Section 1.2](#).

Generalized inverses. The four conditions of [Definition 1.3.2](#) are due to Penrose (1955), who proved existence and uniqueness. E.H. Moore had described an equivalent inverse decades earlier. Rao (1973) made the weaker inverse $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$ the basic tool of linear-model theory, and it is his approach that [Section 1.4](#) and Chapter 8 follow. The construction from a nonsingular submatrix in [Theorem 1.3.2](#) is a matrix-free way of stating an algorithm that is often given as a sequence of steps (Searle 1982).

Schur complements and updating formulas. The name “Schur complement” and the notation for it vary. Zhang (2005) collects the history and many applications. The rank-one formula [Eq. 1.6.7](#) is from Sherman and Morrison (1950), and the general identity [Eq. 1.6.6](#) from Woodbury (1950). Henderson and Searle (1981) trace earlier appearances and compare several derivations. Obtaining the identity by comparing two expressions for one block of a partitioned inverse, as in [Theorem 1.6.5](#), is a classical route.

Singular values and conditioning. Stewart (1993) tells the early history of the singular value decomposition, beginning with Beltrami and Jordan in the 1870s. The low-rank approximation theorem, [Proposition 1.8.2\(c\)](#) and its Frobenius form, is usually credited to Eckart and Young (1936), although Stewart points out that Schmidt had proved it earlier in the setting of integral equations. Golub and Van Loan (2013) is the reference for computing the decomposition and for perturbation bounds such as [Eq. 1.8.3](#), and Higham (2002) for their refinements. The use of scaled condition numbers and singular vectors to diagnose collinearity is due to Belsley et al. (1980). Chapter 26 returns to it.

Kronecker products and calculus. Henderson and Searle (1979) survey the vec operator and its uses in multivariate statistics. Magnus and Neudecker (2019) develop matrix calculus systematically through differentials, which avoid the bookkeeping of the entrywise definitions used in [Section 1.11](#) and handle symmetric matrix arguments cleanly.

1.12.3 References

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Chapter 2

Random Vectors: Expectation and Covariance Operators

A linear model is a statement about the first two moments of a random vector: its mean lies in a subspace and its covariance matrix has a given form. This chapter builds the calculus for such statements. Expectation and covariance become matrix operators that commute with linear maps, covariance matrices turn out to be exactly the nonnegative definite matrices, and a singular covariance means the vector is confined to a flat. The mean of a quadratic form needs only two moments; its variance needs four. Partitioning a vector gives cross-covariances, Schur complements and the best linear predictor, which foreshadow conditional distributions under normality.

What you need. [Chapter 1](#) (trace, spectral theorem, nonnegative definite matrices and their square roots, generalized inverses, partitioned matrices), and probability at the level of expectations, variances and independence of random variables.

2.0.1 Roadmap

- [Mean vectors and covariance matrices](#)
- [Linear and affine transformations](#)
- [Expectation of quadratic forms](#)
- [Correlation matrices](#)
- [Moments of partitioned random vectors](#)
- [Moment generating functions and independence](#)
- [Sample moments and their computation](#)
- [Summary and notes](#)

2.1 Mean vectors and covariance matrices

Random vectors enter regression in two different ways. The first is the response vector. In the model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

$\mathbf{Y} = (Y_1, \dots, Y_n)\mathbf{T}$ has one entry per observational unit, and so does the unobservable error ε . The assumptions that do the real work in Chapter 5, Chapter 6 and Chapter 7 are not about the shape of any distribution. They are statements about moments: $\mathbf{E}(\varepsilon) = \mathbf{0}$, and the errors have equal variances and are uncorrelated. The second way is the measurement vector. When several quantities are recorded on the same unit, such as income, age and years of schooling, the unit contributes a vector whose components are usually correlated and measured in different units. Random regressors, errors-in-variables models and multivariate responses are all built from such vectors.

Both uses need the same toolkit: expectation and covariance as operations on vectors and matrices, their behaviour under linear maps, and the expected values of quadratic forms. Throughout, a random vector *has finite second moments* if every component has finite variance.

2.1.1 Random matrices and their expectation

Definition 2.1.1 — Expectation of a random matrix

A **random matrix** $\mathbf{Z} = (Z_{ij})$ is an $m \times k$ array of random variables. If every Z_{ij} has a finite expectation, the **expectation** of $\mathbf{Z}$ is the $m \times k$ matrix of expectations,

$$\mathbf{E}(\mathbf{Z}) = (\mathbf{E}(Z_{ij})).$$

A random vector is the case $k = 1$.

The definition works entry by entry, so every rule that is linear in the entries carries over at once.

Proposition 2.1.1 — Linearity of expectation

Let $\mathbf{Z}$ and $\mathbf{Z}_2$ be $m \times k$ random matrices with integrable entries, and let $\mathbf{A}$ ($l \times m$), $\mathbf{B}$ ($k \times s$) and $\mathbf{C}$ ($l \times s$) be constant matrices. Then

- (a) $\mathbf{E}(\mathbf{AZB} + \mathbf{C}) = \mathbf{A} \mathbf{E}(\mathbf{Z}) \mathbf{B} + \mathbf{C}$;
- (b) $\mathbf{E}(\mathbf{Z} + \mathbf{Z}_2) = \mathbf{E}(\mathbf{Z}) + \mathbf{E}(\mathbf{Z}_2)$ and $\mathbf{E}(\mathbf{ZT}) = \mathbf{E}(\mathbf{Z})\mathbf{T}$;
- (c) if $m = k$, then $\mathbf{E} \operatorname{tr}(\mathbf{Z}) = \operatorname{tr} \mathbf{E}(\mathbf{Z})$.

Proof. The (i, j) entry of $\mathbf{AZB} + \mathbf{C}$ is $\sum_r \sum_t a_{ir} Z_{rt} b_{tj} + c_{ij}$, a finite linear combination of integrable random variables with constant coefficients. Its expectation is $\sum_r \sum_t a_{ir} \mathbf{E}(Z_{rt}) b_{tj} + c_{ij}$, which is the (i, j) entry of $\mathbf{A} \mathbf{E}(\mathbf{Z}) \mathbf{B} + \mathbf{C}$. Parts (b) and (c) are the same argument applied to $Z_{ij} + Z_{2,ij}$, to Z_{ji} , and to $\sum_i Z_{ii}$. ■

Two warnings go with this proposition. The multipliers must be constant: if $\mathbf{A}$ is random, $\mathbf{E}(\mathbf{AZ})$ is generally not $\mathbf{E}(\mathbf{A})\mathbf{E}(\mathbf{Z})$. And the expectation of a product of random matrices is generally not the product of expectations. It is when every entry of the first is independent of every entry of the second, because then $\mathbf{E}(Z_{ir}W_{rj}) = \mathbf{E}(Z_{ir})\mathbf{E}(W_{rj})$ for each term.

2.1.2 Mean vector and covariance matrix

Definition 2.1.2 — Mean vector and covariance matrices

Let $\mathbf{Y}$ be a $p \times 1$ random vector with finite second moments. Its **mean vector** is $\boldsymbol{\mu} = \mathbf{E}(\mathbf{Y})$, and its **covariance matrix** is the $p \times p$ matrix

$$\text{Cov}(\mathbf{Y}) = \mathbf{E}[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^T],$$

whose (i, j) entry is $\sigma_{ij} = \text{Cov}(Y_i, Y_j)$ and whose diagonal holds the variances $\sigma_{ii} = \text{Var}(Y_i)$. If $\mathbf{U}$ ($p \times 1$) and $\mathbf{V}$ ($q \times 1$) both have finite second moments, their **cross-covariance matrix** is the $p \times q$ matrix

$$\text{Cov}(\mathbf{U}, \mathbf{V}) = \mathbf{E}[(\mathbf{U} - \mathbf{E}\mathbf{U})(\mathbf{V} - \mathbf{E}\mathbf{V})^T],$$

with (i, j) entry $\text{Cov}(U_i, V_j)$.

The expectations exist: $|U_i V_j| \leq \frac{1}{2}(U_i^2 + V_j^2)$, so finite second moments make every product integrable. We often write $\boldsymbol{\Sigma}$ for $\text{Cov}(\mathbf{Y})$, and $\text{Cov}(\mathbf{Y}) = \text{Cov}(\mathbf{Y}, \mathbf{Y})$. Some books write $\text{Var}(\mathbf{Y})$ or $D(\mathbf{Y})$ for the covariance matrix. We keep Var for scalars. The symbol Cov then has three uses: $\text{Cov}(Y_i, Y_j)$ is a number, $\text{Cov}(\mathbf{Y})$ is a symmetric square matrix, and $\text{Cov}(\mathbf{U}, \mathbf{V})$ is a rectangular matrix that is usually not symmetric even when $p = q$.

Proposition 2.1.2 — Elementary properties

Let $\mathbf{Y}, \mathbf{U}, \mathbf{V}$ have finite second moments, with $\boldsymbol{\mu} = \mathbf{E}(\mathbf{Y})$ and $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$, and let $\mathbf{a}, \mathbf{b}$ be constant vectors of matching sizes. Then:

- (a) $\boldsymbol{\Sigma}$ is symmetric;
- (b) $\text{Cov}(\mathbf{U}, \mathbf{V}) = \mathbf{E}(\mathbf{U}\mathbf{V}^T) - \mathbf{E}(\mathbf{U})\mathbf{E}(\mathbf{V})^T$, and in particular $\boldsymbol{\Sigma} = \mathbf{E}(\mathbf{Y}\mathbf{Y}^T) - \boldsymbol{\mu}\boldsymbol{\mu}^T$;
- (c) $\text{Cov}(\mathbf{V}, \mathbf{U}) = \text{Cov}(\mathbf{U}, \mathbf{V})^T$;
- (d) $\text{Cov}(\mathbf{U} + \mathbf{a}, \mathbf{V} + \mathbf{b}) = \text{Cov}(\mathbf{U}, \mathbf{V})$;
- (e) $\mathbf{E}[(\mathbf{Y} - \mathbf{a})(\mathbf{Y} - \mathbf{a})^T] = \boldsymbol{\Sigma} + (\boldsymbol{\mu} - \mathbf{a})(\boldsymbol{\mu} - \mathbf{a})^T$, and hence $\mathbf{E}\|\mathbf{Y} - \mathbf{a}\|^2 = \text{tr}(\boldsymbol{\Sigma}) + \|\boldsymbol{\mu} - \mathbf{a}\|^2$.

Proof.

- (a) and (c) hold because $(\mathbf{u}\mathbf{v}^T)^T = \mathbf{v}\mathbf{u}^T$ and expectation commutes with transposition. For (b), expand $(\mathbf{U} - \boldsymbol{\mu}_U)(\mathbf{V} - \boldsymbol{\mu}_V)^T = \mathbf{U}\mathbf{V}^T - \mathbf{U}\boldsymbol{\mu}_V^T - \boldsymbol{\mu}_U\mathbf{V}^T + \boldsymbol{\mu}_U\boldsymbol{\mu}_V^T$ and take expectations with Proposition 2.1.1. The two middle terms each give $-\boldsymbol{\mu}_U\boldsymbol{\mu}_V^T$. (d) holds because adding a constant does not change the centred vectors. For (e), write $\mathbf{Y} - \mathbf{a} = (\mathbf{Y} - \boldsymbol{\mu}) + (\boldsymbol{\mu} - \mathbf{a})$ and expand. The cross terms contain the factor $\mathbf{E}(\mathbf{Y} - \boldsymbol{\mu}) = \mathbf{0}$ and vanish. The trace identity follows from $\|\mathbf{v}\|^2 = \text{tr}(\mathbf{v}\mathbf{v}^T)$ and Proposition 2.1.1(c). ■

Part (e) is the vector form of “mean squared error equals variance plus squared bias”. It shows that the mean is the constant vector closest to $\mathbf{Y}$ in expected squared distance, and that the

minimal value is $\text{tr}(\mathbf{\Sigma}) = \sum_i \text{Var}(Y_i)$. The matrix statement is stronger than the trace statement. It says that the second-moment matrix about any point exceeds $\mathbf{\Sigma}$ by a nonnegative definite matrix of rank at most one.

2.1.3 Uncorrelated is not independent

Random vectors $\mathbf{U}$ and $\mathbf{V}$ are **uncorrelated** if $\text{Cov}(\mathbf{U}, \mathbf{V}) = \mathbf{0}$. If $\mathbf{U}$ and $\mathbf{V}$ are independent, then $\mathbf{E}(U_i V_j) = \mathbf{E}(U_i) \mathbf{E}(V_j)$ for all i, j , so they are uncorrelated. The converse is false, and it fails in the most innocent-looking cases.

Example. (A point on a circle)

Let Θ be uniform on $[0, 2\pi)$ and $\mathbf{Y} = (\cos \Theta, \sin \Theta) \mathbf{T}$. Then $\mathbf{E}(\cos \Theta) = \mathbf{E}(\sin \Theta) = 0$, $\mathbf{E}(\cos^2 \Theta) = \mathbf{E}(\sin^2 \Theta) = \frac{1}{2}$, and $\mathbf{E}(\cos \Theta \sin \Theta) = \frac{1}{2} \mathbf{E}(\sin 2\Theta) = 0$. So $\boldsymbol{\mu} = \mathbf{0}$ and $\text{Cov}(\mathbf{Y}) = \frac{1}{2} \mathbf{I}_2$, exactly the moments of two independent coordinates. Yet $Y_1^2 + Y_2^2 = 1$: once Y_1 is known, Y_2 is known up to sign. Covariance measures *linear* association only, and the dependence here is entirely nonlinear.

For jointly normal vectors, zero covariance and independence coincide ([Theorem 3.4.1](#)). That equivalence rests on joint normality, not on normal marginals, and it is one of the main reasons normal theory is so tractable.

Scalar summaries. The **total variance** $\text{tr}(\mathbf{\Sigma})$ ignores correlation; the **generalized variance** $\det(\mathbf{\Sigma})$ does not. If $\lambda_1, \dots, \lambda_p$ are the eigenvalues of $\mathbf{\Sigma}$ ([Theorem 1.7.2](#)), these are $\sum_i \lambda_i$ and $\prod_i \lambda_i$, and both are unchanged by a rotation $\mathbf{Y} \mapsto \mathbf{QY}$. The determinant can be tiny even when every variance is large, if one eigenvalue is near zero. [Theorem 2.2.3](#) shows that a zero eigenvalue means some linear combination of $\mathbf{Y}$ does not vary at all.

2.1.4 Exercises

2.1.4.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Z}$ be an $n \times p$ random matrix whose entries are independent, with mean 0 and variance σ^2 . Show that $\mathbf{E}(\mathbf{ZTZ}) = n\sigma^2 \mathbf{I}_p$ and $\mathbf{E}(\mathbf{ZZT}) = p\sigma^2 \mathbf{I}_n$. For a constant $p \times p$ matrix $\mathbf{B}$, find $\mathbf{E}(\mathbf{ZBZT})$.

2.1.4.2 B. Practice

Exercise B1.

Let X have a distribution symmetric about 0 with $\text{Var}(X) = 1$ and $\mathbf{E}X^4 < \infty$, let S be independent of X with $\Pr(S = 1) = \Pr(S = -1) = \frac{1}{2}$, and put $\mathbf{Y} = (X, SX) \mathbf{T}$. Show that the two components have the same distribution and are uncorrelated, and that $\text{Cov}(Y_1^2, Y_2^2) = \text{Var}(X^2)$. Deduce that Y_1 and Y_2 are independent iff X is itself a random sign. ([Chapter 3](#) takes X standard normal.)

Solution. Since S is independent of X and X is symmetric, $\Pr(SX \leq y) = \frac{1}{2} \Pr(X \leq y) + \frac{1}{2} \Pr(-X \leq y) = \Pr(X \leq y)$. By independence, $\text{Cov}(X, SX) = \mathbf{E}(S)\mathbf{E}(X^2) = 0$. Both squares equal X^2 , so $\text{Cov}(Y_1^2, Y_2^2) = \text{Var}(X^2)$. Functions of independent variables are independent, hence uncorrelated, so $\text{Var}(X^2) > 0$ makes Y_1 and Y_2 dependent. If $\text{Var}(X^2) = 0$, then $X^2 = \mathbf{E}X^2 = 1$ with probability one, and by symmetry $X = \pm 1$ with probability $\frac{1}{2}$ each. Then $\Pr(Y_1 = a, Y_2 = b) = \Pr(X = a) \Pr(S = ab) = \frac{1}{4}$ for all signs a, b , which is the product of the marginal probabilities, as in [Example \(Pairwise but not mutually independent\)](#).

Exercise B2.

Show that $\mathbf{E}[(\mathbf{Y} - \mathbf{a})(\mathbf{Y} - \mathbf{a})^T] - \text{Cov}(\mathbf{Y})$ is nonnegative definite for every constant $\mathbf{a}$, with equality iff $\mathbf{a} = \boldsymbol{\mu}$. Deduce that $\mathbf{a} = \boldsymbol{\mu}$ minimizes $\mathbf{E}[(\mathbf{Y} - \mathbf{a})^T \mathbf{N} (\mathbf{Y} - \mathbf{a})]$ for every positive definite $\mathbf{N}$.

2.2 Linear and affine transformations

Almost every statistic in a linear model is a linear function of the response: least squares coefficients, fitted values, residuals, contrasts. Their means and covariance matrices follow from one theorem.

2.2.1 The transformation rules

Theorem 2.2.1 — Moments of affine transformations

Let $\mathbf{Y}$ be a $p \times 1$ random vector with finite second moments, $\mathbf{E}(\mathbf{Y}) = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$. Let $\mathbf{A}$ be $k \times p$, $\mathbf{B}$ be $m \times p$, and $\mathbf{b}$ be $k \times 1$, all constant. Then

- (a) $\mathbf{E}(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$;
- (b) $\text{Cov}(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T$;
- (c) $\text{Cov}(\mathbf{A}\mathbf{Y}, \mathbf{B}\mathbf{Y}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^T$.

More generally, if $\mathbf{U}$ and $\mathbf{V}$ have finite second moments and $\mathbf{A}, \mathbf{B}, \mathbf{b}, \mathbf{c}$ are constants of conformable sizes, then $\text{Cov}(\mathbf{A}\mathbf{U} + \mathbf{b}, \mathbf{B}\mathbf{V} + \mathbf{c}) = \mathbf{A} \text{Cov}(\mathbf{U}, \mathbf{V}) \mathbf{B}^T$.

Proof.

- (a) is [Proposition 2.1.1\(a\)](#). For the general statement, the centred vectors are $\mathbf{A}\mathbf{U} + \mathbf{b} - \mathbf{E}(\mathbf{A}\mathbf{U} + \mathbf{b}) = \mathbf{A}(\mathbf{U} - \mathbf{E}\mathbf{U})$ and similarly $\mathbf{B}(\mathbf{V} - \mathbf{E}\mathbf{V})$. Hence

$$\text{Cov}(\mathbf{A}\mathbf{U} + \mathbf{b}, \mathbf{B}\mathbf{V} + \mathbf{c}) = \mathbf{E}[\mathbf{A}(\mathbf{U} - \mathbf{E}\mathbf{U})(\mathbf{V} - \mathbf{E}\mathbf{V})^T \mathbf{B}^T] = \mathbf{A} \mathbf{E}[(\mathbf{U} - \mathbf{E}\mathbf{U})(\mathbf{V} - \mathbf{E}\mathbf{V})^T] \mathbf{B}^T,$$

again by [Proposition 2.1.1\(a\)](#). Parts (c) and (b) are the cases $\mathbf{U} = \mathbf{V} = \mathbf{Y}$, and $\mathbf{B} = \mathbf{A}$ in (b).

For a single linear combination, $\mathbf{A} = \mathbf{a}\mathbf{T}$ is a row vector, and (b) and (c) give

$$\text{Var}(\mathbf{a}\mathbf{T}\mathbf{Y}) = \mathbf{a}\mathbf{T}\mathbf{\Sigma}\mathbf{a} = \sum_{i=1}^p \sum_{j=1}^p a_i a_j \sigma_{ij}, \quad \text{Cov}(\mathbf{a}\mathbf{T}\mathbf{Y}, \mathbf{b}\mathbf{T}\mathbf{Y}) = \mathbf{a}\mathbf{T}\mathbf{\Sigma}\mathbf{b}. \quad (2.2.1)$$

Every variance and covariance enters $\text{Var}(\mathbf{a}\mathbf{T}\mathbf{Y})$, weighted by the products of the coefficients. Conversely, $\mathbf{\Sigma}$ is determined by the variances of linear combinations, because $\sigma_{ij} = \frac{1}{2}[\text{Var}(Y_i + Y_j) - \text{Var}(Y_i) - \text{Var}(Y_j)]$.

Proposition 2.2.2 — Covariance of sums

Let $\mathbf{U}, \mathbf{U}_2, \mathbf{V}$ have finite second moments, with $\mathbf{U}$ and $\mathbf{U}_2$ of the same size in (a), and $\mathbf{U}$ and $\mathbf{V}$ of the same size in (b). Then

- (a) $\text{Cov}(\mathbf{U} + \mathbf{U}_2, \mathbf{V}) = \text{Cov}(\mathbf{U}, \mathbf{V}) + \text{Cov}(\mathbf{U}_2, \mathbf{V})$;
- (b) $\text{Cov}(\mathbf{U} + \mathbf{V}) = \text{Cov}(\mathbf{U}) + \text{Cov}(\mathbf{U}, \mathbf{V}) + \text{Cov}(\mathbf{V}, \mathbf{U}) + \text{Cov}(\mathbf{V})$;
- (c) if $\mathbf{Y}_1, \dots, \mathbf{Y}_n$ are pairwise uncorrelated $p \times 1$ vectors and $c_1, \dots, c_n$ are constants, then $\text{Cov}(\sum_i c_i \mathbf{Y}_i) = \sum_i c_i^2 \text{Cov}(\mathbf{Y}_i)$. In particular, if every $\text{Cov}(\mathbf{Y}_i) = \mathbf{\Sigma}$, then $\text{Cov}(\bar{\mathbf{Y}}) = \mathbf{\Sigma}/n$.

Proof.

- (a) Centre each vector and use linearity of the outer product and of $\mathbf{E}$. (b) Apply (a) in each argument; note that the two cross terms are transposes of each other, not equal.
- (b) By (a) and (b), applied repeatedly, $\text{Cov}(\sum_i c_i \mathbf{Y}_i) = \sum_i \sum_j c_i c_j \text{Cov}(\mathbf{Y}_i, \mathbf{Y}_j)$, using $\text{Cov}(c\mathbf{U}, \mathbf{V}) = c \text{Cov}(\mathbf{U}, \mathbf{V})$ from [Theorem 2.2.1](#). The terms with $i \neq j$ vanish.

Example. (Centring a sample)

Let $\mathbf{Y}$ be $n \times 1$ with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$, and write $\mathbf{J} = \mathbf{1}\mathbf{1}\mathbf{T}$. The mean is $\bar{Y} = n^{-1} \mathbf{1}\mathbf{T}\mathbf{Y}$, and the vector of deviations is $\mathbf{d} = (\mathbf{I} - n^{-1} \mathbf{J})\mathbf{Y}$. The matrix $\mathbf{I} - n^{-1} \mathbf{J}$ is symmetric and idempotent, so [Theorem 2.2.1](#) gives

$$\text{Cov}(\mathbf{d}) = \sigma^2 (\mathbf{I} - n^{-1} \mathbf{J}), \quad \text{Cov}(\bar{Y}, \mathbf{d}) = \sigma^2 n^{-1} \mathbf{1}\mathbf{T}(\mathbf{I} - n^{-1} \mathbf{J}) = \mathbf{0}\mathbf{T}.$$

The deviations have variances $\sigma^2(1 - 1/n)$ and are negatively correlated, with correlation $-1/(n-1)$ between each pair, because they are forced to sum to zero. And each deviation is uncorrelated with the mean. Under normality, uncorrelated becomes independent ([Example \(Sample mean and deviations\)](#)), which is the first half of [Corollary 4.4.5](#).

2.2.2 Covariance matrices are nonnegative definite

A variance cannot be negative, and by 2 this simple fact constrains the whole matrix. Recall from [Definition 1.7.1](#) that a symmetric Σ is nonnegative definite if $\mathbf{a}^T \Sigma \mathbf{a} \geq 0$ for all $\mathbf{a}$, and positive definite if in addition $\mathbf{a}^T \Sigma \mathbf{a} > 0$ for $\mathbf{a} \neq \mathbf{0}$.

Theorem 2.2.3 — Covariance matrices

Let $\mathbf{Y}$ have finite second moments, mean $\boldsymbol{\mu}$ and covariance matrix Σ .

- (a) Σ is nonnegative definite.
- (b) Conversely, every $p \times p$ nonnegative definite matrix is the covariance matrix of some random vector.
- (c) For a constant $\mathbf{a}$, $\Sigma \mathbf{a} = \mathbf{0}$ iff $\mathbf{a}^T \mathbf{Y} = \mathbf{a}^T \boldsymbol{\mu}$ with probability one.
- (d) $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathbb{C}(\Sigma)) = 1$. If $\mathcal{S}$ is any subspace with $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{S}) = 1$, then $\mathbb{C}(\Sigma) \subseteq \mathcal{S}$.

In particular, Σ is singular iff some nontrivial linear combination of $\mathbf{Y}$ is constant with probability one. Then $\mathbf{Y}$ lies, with probability one, in the affine subspace $\boldsymbol{\mu} + \mathbb{C}(\Sigma)$ of dimension $\text{rank}(\Sigma) < p$, and this is the smallest such flat.

Proof.

- (a) Σ is symmetric, and $\mathbf{a}^T \Sigma \mathbf{a} = \text{Var}(\mathbf{a}^T \mathbf{Y}) \geq 0$ by 2.
- (b) Let Σ be nonnegative definite and let $\Sigma^{1/2}$ be its symmetric nonnegative definite square root ([Theorem 1.7.6](#)). Let $\mathbb{Z}$ have independent components with mean 0 and variance 1, for instance independent standard normal variables, so that $\text{Cov}(\mathbb{Z}) = \mathbf{I}$. Then $\text{Cov}(\Sigma^{1/2} \mathbb{Z}) = \Sigma^{1/2} \mathbf{I} \Sigma^{1/2} = \Sigma$.
- (c) If $\Sigma \mathbf{a} = \mathbf{0}$, the random variable $\mathbf{a}^T (\mathbf{Y} - \boldsymbol{\mu})$ has mean zero and variance $\mathbf{a}^T \Sigma \mathbf{a} = 0$, so it is zero with probability one. Conversely, if $\mathbf{a}^T (\mathbf{Y} - \boldsymbol{\mu}) = 0$ with probability one, then $\Sigma \mathbf{a} = \mathbf{E}[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^T \mathbf{a}] = \mathbf{0}$.
- (d) Let $\mathbf{a}_1, \dots, \mathbf{a}_s$ be a basis of $\mathcal{N}(\Sigma)$. By (c), each event $\{\mathbf{a}_j^T (\mathbf{Y} - \boldsymbol{\mu}) = 0\}$ has probability one, and so does their intersection. On that intersection $\mathbf{Y} - \boldsymbol{\mu}$ is orthogonal to $\mathcal{N}(\Sigma)$. Because Σ is symmetric, $\mathcal{N}(\Sigma)$ is the orthogonal complement of $\mathbb{C}(\Sigma)$, and so $\mathbf{Y} - \boldsymbol{\mu} \in \mathbb{C}(\Sigma)$. Now let $\mathcal{S}$ be a subspace with $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{S}) = 1$. For every $\mathbf{a} \in \mathcal{S}^\perp$ we have $\mathbf{a}^T (\mathbf{Y} - \boldsymbol{\mu}) = 0$ with probability one, hence $\mathbf{a} \in \mathcal{N}(\Sigma) = \mathbb{C}(\Sigma)^\perp$ by (c). So $\mathcal{S}^\perp \subseteq \mathbb{C}(\Sigma)^\perp$, and taking orthogonal complements gives $\mathbb{C}(\Sigma) \subseteq \mathcal{S}$. The final statements follow, with $\dim \mathbb{C}(\Sigma) = \text{rank}(\Sigma)$. For minimality among all flats, suppose $\Pr(\mathbf{Y} \in \mathbf{c} + \mathcal{S}) = 1$ for a vector $\mathbf{c}$ and a subspace $\mathcal{S}$. Every $\mathbf{a} \in \mathcal{S}^\perp$ has $\mathbf{a}^T (\mathbf{Y} - \mathbf{c}) = 0$ with probability one, and taking expectations gives $\mathbf{a}^T (\boldsymbol{\mu} - \mathbf{c}) = 0$. Hence $\boldsymbol{\mu} - \mathbf{c} \in \mathcal{S}$, so $\mathbf{c} + \mathcal{S} = \boldsymbol{\mu} + \mathcal{S}$, and the subspace case applies.

■

A covariance matrix is therefore positive definite exactly when no linear combination of the com-

ponents is degenerate. Singular covariance matrices are not pathological. They arise whenever the components satisfy an exact linear constraint, as with the deviations of [Example \(Centring a sample\)](#), whose covariance matrix has rank $n - 1$ because $\mathbf{1T}\mathbf{d} = 0$. The residual vector of a regression is another case: it is orthogonal to every column of $\mathbf{X}$, and when $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ its covariance matrix has rank $n - \text{rank}(\mathbf{X})$ ([Section 2.2, Chapter 6](#)).

Example. (Multinomial counts)

In m independent trials, each outcome falls into one of k categories with probabilities $\boldsymbol{\pi} = (\pi_1, \dots, \pi_k)\mathbf{T}$, where $\sum_j \pi_j = 1$. Let $\mathbf{Z}_t$ be the indicator vector of trial t , with a single 1 in the position of its category. Then $\mathbf{E}(\mathbf{Z}_t) = \boldsymbol{\pi}$, and $\mathbf{Z}_t\mathbf{Z}_t\mathbf{T} = \text{diag}(\mathbf{Z}_t)$ because the indicator has only one nonzero entry. So $\text{Cov}(\mathbf{Z}_t) = \text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}\mathbf{T}$. The count vector $\mathbf{Y} = \sum_t \mathbf{Z}_t$ is a sum of independent vectors, and [Proposition 2.2.2\(c\)](#) gives

$$\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y}) = m(\text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}\mathbf{T}).$$

Because $\boldsymbol{\Sigma}\mathbf{1} = m(\boldsymbol{\pi} - \boldsymbol{\pi}\boldsymbol{\pi}\mathbf{T}\mathbf{1}) = \mathbf{0}$, the matrix is singular, and [Theorem 2.2.3\(c\)](#) recovers the constraint $\mathbf{1T}\mathbf{Y} = m$. For $m = 20$ and $\boldsymbol{\pi} = (0.5, 0.3, 0.2)\mathbf{T}$ the eigenvalues are 7.762, 4.638 and 0. All 5000 simulated count vectors lie in the plane $\boldsymbol{\mu} + \mathbb{C}(\boldsymbol{\Sigma}) = \{\mathbf{y} : \mathbf{1T}\mathbf{y} = m\}$.

Linear maps cannot increase the rank of a covariance matrix, since $\text{rank}(\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\mathbf{T}) \leq \text{rank}(\boldsymbol{\Sigma})$ ([Exercise \(B2\)](#)). They can, however, discard degenerate directions. If $\mathbf{Y} = (Z, Z)\mathbf{T}$ with $\text{Var}(Z) = 1$, then $\boldsymbol{\Sigma}$ has all entries equal to 1 and is singular, while $\mathbf{A} = (1, 0)$ gives $\text{Var}(\mathbf{A}\mathbf{Y}) = 1 > 0$. By [Theorem 2.2.1](#), the covariance matrix of $\mathbf{A}\mathbf{Y}$ is $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\mathbf{T}$. If $\boldsymbol{\Sigma}$ is positive definite and $\mathbf{A}$ has full row rank $k \leq p$, then for $\mathbf{b} \neq \mathbf{0}$ the vector $\mathbf{A}\mathbf{T}\mathbf{b}$ is nonzero and $\mathbf{b}\mathbf{T}\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\mathbf{T}\mathbf{b} > 0$. So $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\mathbf{T}$ is positive definite. If $k > p$, then $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\mathbf{T}$ has rank at most p and is singular. A vector of more than p linear combinations of p variables always satisfies a linear constraint.

2.2.3 Standardization and Mahalanobis distance

A scalar with mean μ and variance $\sigma^2 > 0$ is standardized by $(Y - \mu)/\sigma$. The vector version must remove correlation as well as scale.

Proposition 2.2.4 — Whitening

Let $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$ be positive definite, with $\mathbf{E}(\mathbf{Y}) = \boldsymbol{\mu}$.

- (a) $\mathbb{Z} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{Y} - \boldsymbol{\mu})$ has $\mathbf{E}(\mathbb{Z}) = \mathbf{0}$ and $\text{Cov}(\mathbb{Z}) = \mathbf{I}_p$.
- (b) A $p \times p$ matrix $\mathbf{W}$ satisfies $\mathbf{W}\boldsymbol{\Sigma}\mathbf{W}\mathbf{T} = \mathbf{I}_p$ iff $\mathbf{W} = \mathbf{Q}\boldsymbol{\Sigma}^{-1/2}$ for some orthogonal $\mathbf{Q}$. In particular, if $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}\mathbf{T}$ is the Cholesky factorization, $\mathbf{W} = \mathbf{L}^{-1}$ is such a matrix.
- (c) For every $\mathbf{W}$ as in (b), $\|\mathbf{W}(\mathbf{Y} - \boldsymbol{\mu})\|^2 = (\mathbf{Y} - \boldsymbol{\mu})\mathbf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$.

Proof.

- (a) By [Theorem 2.2.1](#), $\text{Cov}(\mathbb{Z}) = \boldsymbol{\Sigma}^{-1/2}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-1/2} = \mathbf{I}$, since $\boldsymbol{\Sigma}^{-1/2}$ is symmetric and commutes with $\boldsymbol{\Sigma}$. (b) If $\mathbf{W} = \mathbf{Q}\boldsymbol{\Sigma}^{-1/2}$, then $\mathbf{W}\boldsymbol{\Sigma}\mathbf{W}\mathbf{T} = \mathbf{Q}\mathbf{Q}\mathbf{T} = \mathbf{I}$. Conversely, if $\mathbf{W}\boldsymbol{\Sigma}\mathbf{W}\mathbf{T} = \mathbf{I}$, put $\mathbf{Q} = \mathbf{W}\boldsymbol{\Sigma}^{1/2}$. Then $\mathbf{Q}\mathbf{Q}\mathbf{T} = \mathbf{W}\boldsymbol{\Sigma}\mathbf{W}\mathbf{T} = \mathbf{I}$, so the square matrix

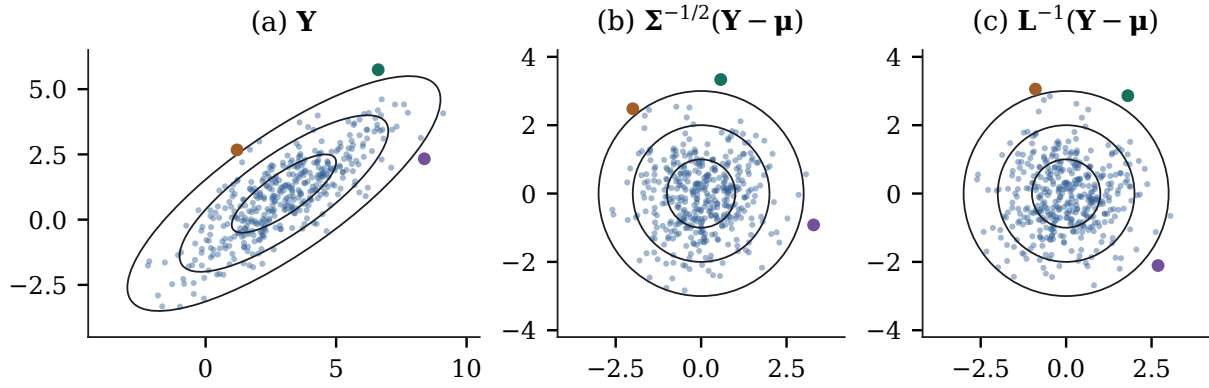


Figure 2.2.1: Whitening a correlated sample. (a) Draws with $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$ and the Mahalanobis contours $\Delta = 1, 2, 3$. (b) The symmetric whitening $\boldsymbol{\Sigma}^{-1/2}(\mathbf{Y} - \boldsymbol{\mu})$. (c) The Cholesky whitening $\mathbf{L}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$. The contours become circles in both, and the three coloured points keep their distance from the origin. Panels (b) and (c) differ by a rotation (Proposition 2.2.4).

$\mathbf{Q}$ is orthogonal, and $\mathbf{W} = \mathbf{Q}\boldsymbol{\Sigma}^{-1/2}$. For the Cholesky factor, $\mathbf{L}^{-1}\boldsymbol{\Sigma}(\mathbf{L}\mathbf{T})^{-1} = \mathbf{L}^{-1}\mathbf{L}\mathbf{L}\mathbf{T}(\mathbf{L}\mathbf{T})^{-1} = \mathbf{I}$. (c) $\mathbf{W}\mathbf{T}\mathbf{W} = \boldsymbol{\Sigma}^{-1/2}\mathbf{Q}\mathbf{T}\mathbf{Q}\boldsymbol{\Sigma}^{-1/2} = \boldsymbol{\Sigma}^{-1}$.

■

The quantity in (c),

$$\Delta^2(\mathbf{Y}) = (\mathbf{Y} - \boldsymbol{\mu})\mathbf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu}), \quad (2.2.2)$$

is the squared **Mahalanobis distance** of $\mathbf{Y}$ from $\boldsymbol{\mu}$. It measures distance in units of the covariance structure. Its level sets are ellipsoids with axes along the eigenvectors of $\boldsymbol{\Sigma}$, with half-lengths proportional to the square roots of the eigenvalues. Whitening maps these ellipsoids to spheres. Part (b) says that whitening is unique only up to a rotation or reflection. Different choices give different coordinates, but the same distances.

Example. (Two whitenings)

Take $\boldsymbol{\mu} = (3, 1)\mathbf{T}$ and $\boldsymbol{\Sigma} = \begin{pmatrix} 4 & 2.4 \\ 2.4 & 2.25 \end{pmatrix}$, so the standard deviations are 2 and 1.5 and the correlation is 0.8. The eigenvalues are 5.680 and 0.570. The symmetric and Cholesky whitening matrices are

$$\boldsymbol{\Sigma}^{-1/2} = \begin{pmatrix} 0.717 & -0.425 \\ -0.425 & 1.027 \end{pmatrix}, \quad \mathbf{L}^{-1} = \begin{pmatrix} 0.500 & 0 \\ -0.667 & 1.111 \end{pmatrix}.$$

The Cholesky version standardizes Y_1 alone and then removes from Y_2 its linear dependence on Y_1 , in the spirit of Gram–Schmidt. The symmetric version treats both coordinates alike and moves points as little as possible (Exercise (B3)). Figure 2.2.1 applies both to 400 simulated points. The two whitened clouds differ by a rotation through 22.5° , and every point keeps its Mahalanobis distance.

```
import numpy as np
rng = np.random.default_rng(2)

mu = np.array([3.0, 1.0])
```

```

Sigma = np.array([[4.0, 2.4],
                  [2.4, 2.25]])
                                # correlation 2.4 / (2 * 1.5) =
                                # 0.8

lam, U = np.linalg.eigh(Sigma)
W_sym = U @ np.diag(lam ** -0.5) @ U.T
L = np.linalg.cholesky(Sigma)
W_chol = np.linalg.inv(L)
                                # spectral decomposition
                                # Sigma^{-1/2}
                                # Sigma = L L', L lower triangular
                                # L^{-1}

n = 400
Y = mu + rng.standard_normal((n, 2)) @ L.T
do
Z_sym = (Y - mu) @ W_sym.T
Z_chol = (Y - mu) @ W_chol.T

Q = W_chol @ np.linalg.inv(W_sym)
print("Q'Q =", np.round(Q.T @ Q, 12))
                                # L^{-1} Sigma^{1/2}

```

Whitening runs in both directions. If $\text{Cov}(\mathbb{Z}) = \mathbf{I}$, then $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbb{Z}$, or $\boldsymbol{\mu} + \mathbf{L}\mathbb{Z}$, has mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. This is how correlated vectors are simulated, and with $\mathbb{Z}$ standard normal it is how [Chapter 3](#) constructs $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The Mahalanobis distance then appears in the exponent of the density ([Theorem 3.2.2](#)). When $\boldsymbol{\Sigma}$ is singular, no matrix whitens all of $\mathbf{Y}$, since a degenerate combination cannot be scaled to variance one. The best one can do is whiten within the support $\boldsymbol{\mu} + \mathbb{C}(\boldsymbol{\Sigma})$ using the Moore–Penrose inverse square root ([Exercise \(B4\)](#)).

2.2.4 Second moments in the linear model

Here is how the rules of this section enter regression. Suppose $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ an $n \times p$ constant matrix of rank p , $\mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$. Adding the constant $\mathbf{X}\boldsymbol{\beta}$ changes the mean but not the covariance, so $\mathbf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$. The least squares estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{X}\mathbf{T}\mathbf{Y}$ is $\mathbf{A}\mathbf{Y}$ with $\mathbf{A} = (\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{X}\mathbf{T}$, so

$$\mathbf{E}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{X}\mathbf{T}\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{X}\mathbf{T}\mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^{-1} = \sigma^2(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}.$$

No distributional assumption was used, and none is needed for the Gauss–Markov theorem of [Chapter 7](#). Fitted values and residuals are also linear in $\mathbf{Y}$, and their covariance matrices are computed the same way in [Chapter 5](#) and [Chapter 6](#). If instead $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ with $\mathbf{V}$ positive definite, then $\mathbf{V}^{-1/2}\mathbf{Y} = \mathbf{V}^{-1/2}\mathbf{X}\boldsymbol{\beta} + \mathbf{V}^{-1/2}\boldsymbol{\varepsilon}$ is a model with $\text{Cov}(\mathbf{V}^{-1/2}\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$. Whitening the data turns correlated errors back into the standard case. This is generalized least squares, which [Chapter 6](#) and [Chapter 31](#) develop.

2.2.5 Exercises

2.2.5.1 A. Check your understanding

Exercise A1.

Let $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_3$. Find the covariance and correlation matrices of the successive differences $(Y_2 - Y_1, Y_3 - Y_2)\mathbf{T}$, and of $(Y_2 - Y_1, Y_3 - Y_2, Y_3 - Y_1)\mathbf{T}$. Which is singular, and which linear combination is degenerate?

Exercise A2.

For $p = 2$ and $\Sigma = \mathbf{L}\mathbf{L}\mathbf{T}$, show that the second component of $\mathbf{L}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$ is a multiple of $Y_2 - \mu_2 - (\sigma_{12}/\sigma_{11})(Y_1 - \mu_1)$. Explain the connection with [Theorem 2.5.2](#).

2.2.5.2 B. Practice**Exercise B1.**

Let $\varepsilon_0, \dots, \varepsilon_n$ be uncorrelated with variance σ^2 , and let $Y_t = \varepsilon_t + \theta\varepsilon_{t-1}$ for $t = 1, \dots, n$. Find $\text{Cov}(\mathbf{Y})$, and show that it is positive definite for every real θ , including $\theta = \pm 1$.

Solution. $\mathbf{Y} = \mathbf{A}\boldsymbol{\varepsilon}$, where $\mathbf{A}$ is $n \times (n+1)$ with θ in column t and 1 in column $t+1$ of row t (columns indexed by $\varepsilon_0, \dots, \varepsilon_n$). So $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{A}\mathbf{A}\mathbf{T}$ is tridiagonal, with $\sigma^2(1 + \theta^2)$ on the diagonal and $\sigma^2\theta$ next to it. The last n columns of $\mathbf{A}$ form a lower bidiagonal matrix with unit diagonal, which is nonsingular. So $\mathbf{A}$ has full row rank n , and $\mathbf{A}\mathbf{A}\mathbf{T}$ is positive definite ([Section 2.2](#)) whatever the value of θ .

Exercise B2.

Show that $\mathbb{C}(\mathbf{A}\Sigma\mathbf{A}\mathbf{T}) = \mathbb{C}(\mathbf{A}\Sigma)$ for every nonnegative definite Σ and conformable $\mathbf{A}$. Deduce that $\text{rank Cov}(\mathbf{A}\mathbf{Y}) \leq \text{rank Cov}(\mathbf{Y})$.

Exercise B3.

Let Σ be positive definite. Among all $\mathbf{W}$ with $\mathbf{W}\Sigma\mathbf{W}\mathbf{T} = \mathbf{I}$, show that $\mathbf{E}\|\mathbf{W}(\mathbf{Y} - \boldsymbol{\mu}) - (\mathbf{Y} - \boldsymbol{\mu})\|^2$ is minimized uniquely by $\mathbf{W} = \Sigma^{-1/2}$. *Hint:* use [Proposition 2.2.4\(b\)](#) and show that $\text{tr}(\mathbf{Q}\Sigma^{1/2}) \leq \text{tr}(\Sigma^{1/2})$ for orthogonal $\mathbf{Q}$.

Solution. Write $\mathbf{W} = \mathbf{Q}\Sigma^{-1/2}$. By [Corollary 2.3.2\(a\)](#) with $\mathbf{A} = (\mathbf{W} - \mathbf{I})\mathbf{T}(\mathbf{W} - \mathbf{I})$,

$$\mathbf{E}\|(\mathbf{W} - \mathbf{I})(\mathbf{Y} - \boldsymbol{\mu})\|^2 = \text{tr}((\mathbf{W} - \mathbf{I})\Sigma(\mathbf{W} - \mathbf{I})\mathbf{T}) = \text{tr}(\mathbf{W}\Sigma\mathbf{W}\mathbf{T}) - 2\text{tr}(\mathbf{W}\Sigma) + \text{tr}(\Sigma) = p + \text{tr}(\Sigma) - 2\text{tr}(\mathbf{Q}\Sigma^{1/2}).$$

Let $\Sigma^{1/2} = \mathbf{\Gamma}\mathbf{\Lambda}^{1/2}\mathbf{\Gamma}\mathbf{T}$ and $\mathbf{G} = \mathbf{\Gamma}\mathbf{T}\mathbf{Q}\mathbf{\Gamma}$, which is orthogonal, so each $|g_{ii}| \leq 1$. Then $\text{tr}(\mathbf{Q}\Sigma^{1/2}) = \text{tr}(\mathbf{G}\mathbf{\Lambda}^{1/2}) = \sum_i g_{ii}\lambda_i^{1/2} \leq \sum_i \lambda_i^{1/2}$. Since every $\lambda_i > 0$, equality forces every $g_{ii} = 1$. An orthogonal matrix with unit diagonal is $\mathbf{I}$, because each column has norm one. So $\mathbf{G} = \mathbf{I}$, which means $\mathbf{Q} = \mathbf{I}$.

Exercise B4.

Let $\Sigma = \text{Cov}(\mathbf{Y})$ have rank r , with spectral decomposition $\Sigma = \mathbf{\Gamma}_r\mathbf{\Lambda}_r\mathbf{\Gamma}_r\mathbf{T}$, where $\mathbf{\Gamma}_r\mathbf{T}\mathbf{\Gamma}_r = \mathbf{I}_r$ and $\mathbf{\Lambda}_r$ holds the positive eigenvalues. Show that $\mathbb{Z} = \mathbf{\Lambda}_r^{-1/2}\mathbf{\Gamma}_r\mathbf{T}(\mathbf{Y} - \boldsymbol{\mu})$ has $\text{Cov}(\mathbb{Z}) = \mathbf{I}_r$ and that $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{\Gamma}_r\mathbf{\Lambda}_r^{1/2}\mathbb{Z}$ with probability one. Show that $(\mathbf{Y} - \boldsymbol{\mu})\mathbf{T}\Sigma^+(\mathbf{Y} - \boldsymbol{\mu}) = \|\mathbb{Z}\|^2$, where Σ^+ is the Moore–Penrose inverse, and that its expectation is r .

Solution. By [Theorem 2.2.1](#), $\text{Cov}(\mathbb{Z}) = \Lambda_r^{-1/2} \Gamma_r \mathbf{T} \Gamma_r \Lambda_r \Gamma_r \mathbf{T} \Gamma_r \Lambda_r^{-1/2} = \mathbf{I}_r$. By [Theorem 2.2.3\(d\)](#), $\mathbf{Y} - \boldsymbol{\mu} \in \mathbb{C}(\boldsymbol{\Sigma}) = \mathbb{C}(\Gamma_r)$ with probability one, and on that event $\mathbf{Y} - \boldsymbol{\mu} = \Gamma_r \Gamma_r \mathbf{T}(\mathbf{Y} - \boldsymbol{\mu}) = \Gamma_r \Lambda_r^{1/2} \mathbb{Z}$. The Moore–Penrose inverse is $\boldsymbol{\Sigma}^+ = \Gamma_r \Lambda_r^{-1} \Gamma_r \mathbf{T}$, so $(\mathbf{Y} - \boldsymbol{\mu}) \mathbf{T} \boldsymbol{\Sigma}^+ (\mathbf{Y} - \boldsymbol{\mu}) = \|\Lambda_r^{-1/2} \Gamma_r \mathbf{T}(\mathbf{Y} - \boldsymbol{\mu})\|^2 = \|\mathbb{Z}\|^2$, with expectation $\text{tr Cov}(\mathbb{Z}) = r$.

2.2.5.3 C. Going deeper

Exercise C1.

In $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ of full column rank, let $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ with $\mathbf{V}$ positive definite. Find $\text{Cov}(\hat{\boldsymbol{\beta}})$ for $\hat{\boldsymbol{\beta}} = (\mathbf{X} \mathbf{T} \mathbf{X})^{-1} \mathbf{X} \mathbf{T} \mathbf{Y}$ and $\text{Cov}(\tilde{\boldsymbol{\beta}})$ for $\tilde{\boldsymbol{\beta}} = (\mathbf{X} \mathbf{T} \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X} \mathbf{T} \mathbf{V}^{-1} \mathbf{Y}$. Show that $\text{Cov}(\hat{\boldsymbol{\beta}}) - \text{Cov}(\tilde{\boldsymbol{\beta}})$ is nonnegative definite. *Hint:* show that $\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}$ and $\tilde{\boldsymbol{\beta}}$ are uncorrelated.

Solution. By [Theorem 2.2.1](#),

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X} \mathbf{T} \mathbf{X})^{-1} \mathbf{X} \mathbf{T} \mathbf{V} \mathbf{X} (\mathbf{X} \mathbf{T} \mathbf{X})^{-1}, \quad \text{Cov}(\tilde{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X} \mathbf{T} \mathbf{V}^{-1} \mathbf{X})^{-1}.$$

Write $\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}} = \mathbf{D} \mathbf{Y}$ with $\mathbf{D} = (\mathbf{X} \mathbf{T} \mathbf{X})^{-1} \mathbf{X} \mathbf{T} - (\mathbf{X} \mathbf{T} \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X} \mathbf{T} \mathbf{V}^{-1}$, and note $\mathbf{D} \mathbf{X} = \mathbf{I} - \mathbf{I} = \mathbf{0}$. Then $\text{Cov}(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}, \tilde{\boldsymbol{\beta}}) = \sigma^2 \mathbf{D} \mathbf{V} \mathbf{V}^{-1} \mathbf{X} (\mathbf{X} \mathbf{T} \mathbf{V}^{-1} \mathbf{X})^{-1} = \mathbf{0}$, so by [Proposition 2.2.2\(b\)](#), $\text{Cov}(\hat{\boldsymbol{\beta}}) = \text{Cov}(\tilde{\boldsymbol{\beta}}) + \text{Cov}(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}})$, and the last term is nonnegative definite.

2.3 Expectation of quadratic forms

Sums of squares are quadratic forms in the response. The sample variance is $(n-1)^{-1} \mathbf{Y} \mathbf{T} (\mathbf{I} - n^{-1} \mathbf{J}) \mathbf{Y}$. The residual sum of squares of a regression is $\mathbf{Y} \mathbf{T} (\mathbf{I} - \mathbf{M}) \mathbf{Y}$, where $\mathbf{M}$ is the projection onto the column space of $\mathbf{X}$ ([Chapter 6](#)). Test statistics are ratios of such forms. Their expectations follow from a short trace argument that needs no distributional assumption.

2.3.1 The basic formula

Theorem 2.3.1 — Mean of a quadratic form

Let $\mathbf{Y}$ be a $p \times 1$ random vector with finite second moments, $\mathbf{E}(\mathbf{Y}) = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$, and let $\mathbf{A}$ be a constant $p \times p$ matrix. Then

$$\mathbf{E}(\mathbf{Y} \mathbf{T} \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu} \mathbf{T} \mathbf{A} \boldsymbol{\mu}.$$

Proof. A 1×1 matrix equals its trace, and $\text{tr}(\mathbf{a} \mathbf{b} \mathbf{T}) = \mathbf{b} \mathbf{T} \mathbf{a}$ ([Theorem 1.6.1](#)), so $\mathbf{Y} \mathbf{T} \mathbf{A} \mathbf{Y} = \text{tr}(\mathbf{A} \mathbf{Y} \mathbf{Y} \mathbf{T})$. By [Proposition 2.1.1](#) and [Proposition 2.1.2\(b\)](#),

$$\mathbf{E}(\mathbf{Y} \mathbf{T} \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \mathbf{E}(\mathbf{Y} \mathbf{Y} \mathbf{T})) = \text{tr}(\mathbf{A} (\boldsymbol{\Sigma} + \boldsymbol{\mu} \boldsymbol{\mu} \mathbf{T})) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu} \mathbf{T} \mathbf{A} \boldsymbol{\mu}.$$

■

The formula has two parts. The trace term is what the form picks up from random variation. The second term is the value of the form at the mean, which is what it would be without any randomness. The theorem does not require independence, normality or symmetry of $\mathbf{A}$. Since $\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y} = \mathbf{Y}\mathbf{T}\mathbf{A}_s\mathbf{Y}$ with $\mathbf{A}_s = \frac{1}{2}(\mathbf{A} + \mathbf{A}\mathbf{T})$, and $\text{tr}(\mathbf{A}\mathbf{\Sigma}) = \text{tr}(\mathbf{A}_s\mathbf{\Sigma})$ when $\mathbf{\Sigma}$ is symmetric, we may assume $\mathbf{A}$ symmetric whenever that is convenient.

Corollary 2.3.2

Under the conditions of [Theorem 2.3.1](#):

- (a) $\mathbf{E}[(\mathbf{Y} - \mathbf{b})\mathbf{T}\mathbf{A}(\mathbf{Y} - \mathbf{b})] = \text{tr}(\mathbf{A}\mathbf{\Sigma}) + (\boldsymbol{\mu} - \mathbf{b})\mathbf{T}\mathbf{A}(\boldsymbol{\mu} - \mathbf{b})$ for any constant $\mathbf{b}$;
- (b) if $\mathbf{\Sigma} = \sigma^2\mathbf{I}$, then $\mathbf{E}(\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}) = \sigma^2 \text{tr}(\mathbf{A}) + \boldsymbol{\mu}\mathbf{T}\mathbf{A}\boldsymbol{\mu}$;
- (c) if $\mathbf{\Sigma}$ is positive definite, the squared Mahalanobis distance [Eq. 2.2.2](#) has $\mathbf{E} \Delta^2(\mathbf{Y}) = p$;
- (d) for random vectors $\mathbf{U}$ ($p \times 1$) and $\mathbf{V}$ ($q \times 1$) with finite second moments and a constant $p \times q$ matrix $\mathbf{A}$, $\mathbf{E}(\mathbf{U}\mathbf{T}\mathbf{A}\mathbf{V}) = \text{tr}(\mathbf{A} \text{Cov}(\mathbf{V}, \mathbf{U})) + \mathbf{E}(\mathbf{U})\mathbf{T}\mathbf{A}\mathbf{E}(\mathbf{V})$.

Proof.

- (a) applies the theorem to $\mathbf{Y} - \mathbf{b}$, which has covariance matrix $\mathbf{\Sigma}$. (b) is immediate.
- (c) is (a) with $\mathbf{b} = \boldsymbol{\mu}$ and $\mathbf{A} = \mathbf{\Sigma}^{-1}$, so the trace is $\text{tr}(\mathbf{I}_p) = p$. For (d), $\mathbf{U}\mathbf{T}\mathbf{A}\mathbf{V} = \text{tr}(\mathbf{A}\mathbf{V}\mathbf{U}\mathbf{T})$ and $\mathbf{E}(\mathbf{V}\mathbf{U}\mathbf{T}) = \text{Cov}(\mathbf{V}, \mathbf{U}) + \mathbf{E}(\mathbf{V})\mathbf{E}(\mathbf{U})\mathbf{T}$.

■

Example. (Sample variance with unequal means and variances)

Let $Y_1, \dots, Y_n$ be uncorrelated with means μ_i and variances σ_i^2 , and let $S^2 = (n-1)^{-1}\mathbf{Y}\mathbf{T}\mathbf{C}\mathbf{Y}$ with $\mathbf{C} = \mathbf{I} - n^{-1}\mathbf{J}$. Here $\mathbf{\Sigma} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$, so $\text{tr}(\mathbf{C}\mathbf{\Sigma}) = (1 - 1/n) \sum_i \sigma_i^2$, and $\boldsymbol{\mu}\mathbf{T}\mathbf{C}\boldsymbol{\mu} = \sum_i (\mu_i - \bar{\mu})^2$. Therefore

$$\mathbf{E}(S^2) = \bar{\sigma}^2 + \frac{1}{n-1} \sum_{i=1}^n (\mu_i - \bar{\mu})^2, \quad \bar{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \sigma_i^2.$$

With a common mean, S^2 is unbiased for the *average* variance, even when the variances differ. Differences in the means inflate it. That is exactly why a residual mean square estimates σ^2 only when the mean structure is right.

Example. (Residual sum of squares under a wrong mean)

Let $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}_n$ and $\mathbf{E}(\mathbf{Y}) = \boldsymbol{\theta}$, where $\boldsymbol{\theta}$ need not lie in $\mathbb{C}(\mathbf{X})$. The residual sum of squares is $\mathbf{Y}\mathbf{T}(\mathbf{I} - \mathbf{M})\mathbf{Y}$, where $\mathbf{M}$ is symmetric and idempotent of rank $r = \text{rank}(\mathbf{X})$, so $\text{tr}(\mathbf{I} - \mathbf{M}) = n - r$ (1). By [Corollary 2.3.2\(b\)](#),

$$\mathbf{E}(\text{RSS}) = \sigma^2(n - r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2. \quad (2.3.1)$$

If the model is right, $\boldsymbol{\theta} \in \mathbb{C}(\mathbf{X})$, the second term vanishes and $\text{RSS}/(n - r)$ is unbiased for σ^2 . If the model is wrong, the estimate is biased upwards by the squared distance

from the true mean to the model space, divided by $n - r$. For instance, fit a straight line at $x = 1, \dots, 6$ when the true mean is the parabola $\theta_i = \frac{1}{4}(x_i - 3.5)^2$ and $\sigma^2 = 1$. Then $\sigma^2(n - r) = 4$, $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2 = 2.333$, and $\mathbf{E}(\text{RSS}) = 6.333$. Chapter 19 studies such departures systematically.

The theorem uses only two moments, so it holds equally for skewed and heavy-tailed data. The listing checks it for a correlated vector $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{B}\mathbf{U}$, where $\mathbf{U}$ has independent, centred exponential components, which are strongly skewed. With the matrices in the listing, $\text{tr}(\mathbf{A}\boldsymbol{\Sigma}) = 6.9$ and $\boldsymbol{\mu}\mathbf{T}\mathbf{A}\boldsymbol{\mu} = 17.0$, so $\mathbf{E}(\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}) = 23.9$. The average over 10^6 draws is 23.940, with Monte Carlo standard error 0.018. The script also verifies the formula exactly, by enumerating a discrete distribution with eight atoms.

```
import itertools
import numpy as np
rng = np.random.default_rng(20260917)

mu = np.array([1.0, -1.0, 2.0])
B = np.array([[1.0, 0.0, 0.0],
              [0.5, 1.0, 0.0],
              [-0.5, 0.8, 0.6]])
Sigma = B @ B.T                                     # Cov(mu + B U) when Cov(U) = I
A = np.array([[2.0, 1.0, 0.0],
              [1.0, 1.0, -1.0],
              [0.0, -1.0, 3.0]])

theory = np.trace(A @ Sigma) + mu @ A @ mu          # Theorem: no normality needed

N = 1_000_000
U = rng.exponential(size=(N, 3)) - 1.0             # skewed, mean 0, variance 1
Y = mu + U @ B.T
Q = np.einsum("ij,jk,ik->i", Y, A, Y)             # Y_i' A Y_i for every draw
print(f"theory {theory:.4f}    simulation {Q.mean():.4f} {Q.std() / np.sqrt(N):.4f}")
```

2.3.2 The variance needs fourth moments

The mean of $\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}$ involves only $\mathbf{E}(Y_i Y_j)$. Its second moment involves $\mathbf{E}(Y_i Y_j Y_k Y_l)$, so no formula for $\text{Var}(\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y})$ can be written in terms of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ alone. Two random vectors with the same mean and covariance matrix can give the same quadratic form very different variances. The next theorem handles independent components, the case that matters for residual sums of squares under the classical error assumptions. Chapter 4 gives the normal case with a general $\boldsymbol{\Sigma}$ (Theorem 4.2.3).

Theorem 2.3.3 — Variance of a quadratic form

Let $Y_1, \dots, Y_n$ be independent with means $\theta_1, \dots, \theta_n$ and common central moments $\sigma^2 = \mathbf{E}(Y_i - \theta_i)^2$, $\mu_3 = \mathbf{E}(Y_i - \theta_i)^3$ and $\mu_4 = \mathbf{E}(Y_i - \theta_i)^4 < \infty$. Let $\mathbf{A}$ be a constant symmetric $n \times n$ matrix, and let $\mathbf{a} = (a_{11}, \dots, a_{nn})\mathbf{T}$ be its diagonal. Then

$$\text{Var}(\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}) = (\mu_4 - 3\sigma^4) \mathbf{a}\mathbf{T}\mathbf{a} + 2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\sigma^2 \boldsymbol{\theta}\mathbf{T}\mathbf{A}^2\boldsymbol{\theta} + 4\mu_3 \boldsymbol{\theta}\mathbf{T}\mathbf{A}\mathbf{a}.$$

Proof. Write $\mathbf{Y} = \boldsymbol{\theta} + \boldsymbol{\varepsilon}$, where the ε_i are independent with mean zero. Then

$$\mathbf{Y}^T \mathbf{A} \mathbf{Y} = q + \ell + \boldsymbol{\theta}^T \mathbf{A} \boldsymbol{\theta}, \quad q = \boldsymbol{\varepsilon}^T \mathbf{A} \boldsymbol{\varepsilon}, \quad \ell = 2\mathbf{b}^T \boldsymbol{\varepsilon}, \quad \mathbf{b} = \mathbf{A} \boldsymbol{\theta},$$

so $\text{Var}(\mathbf{Y}^T \mathbf{A} \mathbf{Y}) = \text{Var}(q) + \text{Var}(\ell) + 2 \text{Cov}(q, \ell)$. We compute the three terms.

The linear term. $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, so $\text{Var}(\ell) = 4\sigma^2 \mathbf{b}^T \mathbf{b} = 4\sigma^2 \boldsymbol{\theta}^T \mathbf{A}^2 \boldsymbol{\theta}$.

The cross term. Since $\mathbf{E}(\ell) = 0$, $\text{Cov}(q, \ell) = \mathbf{E}(q\ell) = 2 \sum_{i,j,k} a_{ij} b_k \mathbf{E}(\varepsilon_i \varepsilon_j \varepsilon_k)$. By independence and zero means, $\mathbf{E}(\varepsilon_i \varepsilon_j \varepsilon_k)$ vanishes unless $i = j = k$, when it equals μ_3 . So $\text{Cov}(q, \ell) = 2\mu_3 \sum_i a_{ii} b_i = 2\mu_3 \boldsymbol{\theta}^T \mathbf{A} \mathbf{a}$.

The quadratic term. $\mathbf{E}(q^2) = \sum_{i,j,k,l} a_{ij} a_{kl} \mathbf{E}(\varepsilon_i \varepsilon_j \varepsilon_k \varepsilon_l)$. The fourth-order moment is μ_4 if all four indices agree. It is σ^4 if the indices form two distinct pairs, which happens in three patterns: $i = j \neq k = l$, $i = k \neq j = l$ and $i = l \neq j = k$. Otherwise some index occurs exactly once, and the moment is zero. Using the symmetry of $\mathbf{A}$,

$$\mathbf{E}(q^2) = \mu_4 \sum_i a_{ii}^2 + \sigma^4 \left[\sum_{i \neq k} a_{ii} a_{kk} + 2 \sum_{i \neq j} a_{ij}^2 \right] = \mu_4 \mathbf{a}^T \mathbf{A} \mathbf{a} + \sigma^4 [(\text{tr } \mathbf{A})^2 - \mathbf{a}^T \mathbf{A} \mathbf{a} + 2 \text{tr}(\mathbf{A}^2) - 2 \mathbf{a}^T \mathbf{A} \mathbf{a}],$$

since $\sum_{i,j} a_{ij}^2 = \text{tr}(\mathbf{A}^2)$. By [Theorem 2.3.1](#), $\mathbf{E}(q) = \sigma^2 \text{tr } \mathbf{A}$, and subtracting its square gives $\text{Var}(q) = (\mu_4 - 3\sigma^4) \mathbf{a}^T \mathbf{A} \mathbf{a} + 2\sigma^4 \text{tr}(\mathbf{A}^2)$. Adding the three terms proves the formula. ■

For normal errors, $\mu_3 = 0$ and $\mu_4 = 3\sigma^4$, and the formula collapses to $2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\sigma^2 \boldsymbol{\theta}^T \mathbf{A}^2 \boldsymbol{\theta}$. Both extra terms describe how non-normality leaks into second-order behaviour. The kurtosis term depends on $\mathbf{A}$ only through its diagonal. The skewness term needs both a skewed error distribution and a mean that the form does not annihilate. For the residual sum of squares of a correct model, $\mathbf{A} \boldsymbol{\theta} = (\mathbf{I} - \mathbf{M}) \boldsymbol{\theta} = \mathbf{0}$, so the skewness term disappears but the kurtosis term remains.

Example. (Skewed errors and a misspecified line)

Continue [Example \(Residual sum of squares under a wrong mean\)](#), now with independent centred exponential errors, for which $\sigma^2 = 1$, $\mu_3 = 2$ and $\mu_4 = 9$. [Theorem 2.3.3](#) gives $\text{Var}(\text{RSS}) = 29.80$. Of this, the kurtosis term contributes 16.73 and the skewness term -4.27 . Normal errors with the same mean and variance would give 17.33, so the exponential errors multiply the variance by 1.72 even though the mean of RSS is unchanged. Over 10^6 simulated data sets, the variance of RSS is 30.04. The script also confirms the formula exactly for a two-point error distribution.

```
se = Q.std() / np.sqrt(N)
vals, probs = np.array([-1.0, 2.0]), np.array([2 / 3, 1 / 3])
exact = 0.0
for idx in itertools.product(range(2), repeat=3):
    u = vals[list(idx)]
    y = mu + B @ u
    exact += np.prod(probs[list(idx)]) * (y @ A @ y)
x = np.arange(1.0, 7.0) # n = 6 design points
X = np.column_stack([np.ones_like(x), x])
M = X @ np.linalg.solve(X.T @ X, X.T)
A2 = np.eye(6) - M # RSS = Y'(I - M)Y for a straight
line
```

```

theta = 0.25 * (x - 3.5) ** 2          # true mean is curved: the line is
wrong
a = np.diag(A2)

def var_quadform(A, theta, m2, m3, m4):
    """Var(Y'AY) for independent Y_i with means theta_i and common central moments.
    """
    a = np.diag(A)
    return ((m4 - 3 * m2**2) * a @ a + 2 * m2**2 * np.trace(A @ A)
            + 4 * m2 * theta @ A @ A @ theta + 4 * m3 * theta @ A @ a)

# A2 = I - M for a straight line at x = 1..6; theta = true (curved) mean
mean_rss = np.trace(A2) + theta @ A2 @ theta          # sigma^2 = 1
var_expo = var_quadform(A2, theta, 1.0, 2.0, 9.0) # centred exponential errors
var_norm = var_quadform(A2, theta, 1.0, 0.0, 3.0) # what normal errors would give

E = rng.exponential(size=(N, 6)) - 1.0
R = np.einsum("ij,jk,ik->i", theta + E, A2, theta + E)
print(f"mean   {mean_rss:.3f} vs {R.mean():.3f}")
print(f"var    {var_expo:.3f} vs {R.var():.3f}   (normal-theory value {var_norm:.3f})")

```

2.3.3 Exercises

2.3.3.1 A. Check your understanding

Exercise A1.

For random vectors $\mathbf{U}$ and $\mathbf{V}$ of the same size, show that $\mathbf{E}\|\mathbf{U} - \mathbf{V}\|^2 = \text{tr Cov}(\mathbf{U}) + \text{tr Cov}(\mathbf{V}) - 2 \text{tr Cov}(\mathbf{U}, \mathbf{V}) + \|\mathbf{EU} - \mathbf{EV}\|^2$.

2.3.3.2 B. Practice

Exercise B1.

Let $\mathbf{E}(Y_t) = \mu$ and $\text{Cov}(Y_t, Y_s) = \sigma^2 \phi^{|t-s|}$ for $t, s = 1, \dots, n$, with $|\phi| < 1$. Show that the sample variance satisfies

$$\mathbf{E}(S^2) = \sigma^2 \left[1 - \frac{2}{n(n-1)} \sum_{k=1}^{n-1} (n-k) \phi^k \right].$$

Is S^2 biased up or down when $\phi > 0$? What happens as $n \rightarrow \infty$ with ϕ fixed?

Solution. With $\mathbf{C} = \mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^T$ and $\mathbf{E}(\mathbf{Y}) = \mu\mathbf{1}$, $\mathbf{CE}(\mathbf{Y}) = \mathbf{0}$, so $\mathbf{E}(\mathbf{YTCY}) = \text{tr}(\mathbf{C}\Sigma) = \text{tr}(\Sigma) - n^{-1}\mathbf{1}^T\Sigma\mathbf{1}$. Here $\text{tr}(\Sigma) = n\sigma^2$, and $\mathbf{1}^T\Sigma\mathbf{1}$ sums all entries. There are n diagonal entries and $2(n-k)$ entries at lag k , so $\mathbf{1}^T\Sigma\mathbf{1} = \sigma^2[n + 2\sum_{k=1}^{n-1}(n-k)\phi^k]$. Dividing $n\sigma^2 - n^{-1}\mathbf{1}^T\Sigma\mathbf{1}$ by $n-1$ gives the formula. For $\phi > 0$ the sum is positive, so S^2 is biased downwards: positively correlated observations cluster, and the sample looks less variable than the process is. As $n \rightarrow \infty$ the sum is $O(n)$, so the bias is $O(1/n)$ and vanishes.

Exercise B2.

Suppose $\mathbf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$. Show that $\mathbf{E}(\text{RSS}) = \sigma^2 \text{tr}((\mathbf{I} - \mathbf{M})\mathbf{V})$. For the equicorrelated $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^T$ and a model with $\mathbf{1} \in \mathbb{C}(\mathbf{X})$, show that $\mathbf{E}(\text{RSS})/(n - r) = \sigma^2(1 - \rho)$.

Exercise B3.

Let $Y_1, \dots, Y_n$ be independent and identically distributed with variance σ^2 and finite fourth central moment μ_4 . Use [Theorem 2.3.3](#) to show that $\text{Var}(S^2) = (\mu_4 - 3\sigma^4)/n + 2\sigma^4/(n - 1)$. Compare with the normal case.

Solution. Take $\mathbf{A} = \mathbf{C}/(n - 1)$ and $\boldsymbol{\theta} = \mu\mathbf{1}$. Then $\mathbf{A}\boldsymbol{\theta} = \mathbf{0}$, so the last two terms of [Theorem 2.3.3](#) vanish. Every diagonal entry of $\mathbf{A}$ is $(1 - 1/n)/(n - 1) = 1/n$, so $\mathbf{a}^T \mathbf{a} = 1/n$, and $\text{tr}(\mathbf{A}^2) = \text{tr}(\mathbf{C})/(n - 1)^2 = 1/(n - 1)$. This gives the formula. Normal data have $\mu_4 = 3\sigma^4$ and $\text{Var}(S^2) = 2\sigma^4/(n - 1)$. Heavy tails ($\mu_4 > 3\sigma^4$) add a term of the same order $1/n$, so the normal-theory variance is wrong even in large samples.

Exercise B4.

Under the conditions of [Theorem 2.3.3](#) with $\boldsymbol{\theta} = \mathbf{0}$, show that $\text{Cov}(\mathbf{Y}^T \mathbf{A} \mathbf{Y}, \mathbf{Y}^T \mathbf{B} \mathbf{Y}) = (\mu_4 - 3\sigma^4)\mathbf{a}^T \mathbf{b} + 2\sigma^4 \text{tr}(\mathbf{A}\mathbf{B})$, where $\mathbf{a}$ and $\mathbf{b}$ are the diagonals of the symmetric matrices $\mathbf{A}$ and $\mathbf{B}$. *Hint:* apply the theorem to $\mathbf{A} + \mathbf{B}$.

Exercise B5.

Let $Y_1, \dots, Y_n$ be independent $N(0, \sigma^2)$. Among symmetric $\mathbf{A}$ with $\mathbf{E}(\mathbf{Y}^T \mathbf{A} \mathbf{Y}) = \sigma^2$, show that $\text{Var}(\mathbf{Y}^T \mathbf{A} \mathbf{Y})$ is minimized by $\mathbf{A} = n^{-1}\mathbf{I}$. *Hint:* $\text{tr}(\mathbf{A}^2) \geq (\text{tr} \mathbf{A})^2/n$.

2.4 Correlation matrices

Covariances carry units: the covariance of income in dollars with age in years is in dollar-years. To compare the strength of linear association across pairs of variables, each covariance is divided by the two standard deviations.

Definition 2.4.1 — Correlation matrix

Let $\mathbf{Y}$ have covariance matrix $\boldsymbol{\Sigma} = (\sigma_{ij})$ with every $\sigma_{ii} > 0$, and let $\mathbf{D}_\sigma = \text{diag}(\sigma_{11}^{1/2}, \dots, \sigma_{pp}^{1/2})$ be the diagonal matrix of standard deviations. The **correlation matrix** of $\mathbf{Y}$ is

$$\mathbb{R} = \mathbf{D}_\sigma^{-1} \boldsymbol{\Sigma} \mathbf{D}_\sigma^{-1}, \quad \text{with entries} \quad \rho_{ij} = \frac{\sigma_{ij}}{\sqrt{\sigma_{ii}\sigma_{jj}}} = \text{corr}(Y_i, Y_j).$$

Equivalently $\boldsymbol{\Sigma} = \mathbf{D}_\sigma \mathbb{R} \mathbf{D}_\sigma$. For random vectors $\mathbf{U}$ and $\mathbf{V}$ with standard-deviation matrices $\mathbf{D}_U$ and $\mathbf{D}_V$, the cross-correlation matrix is $\mathbf{D}_U^{-1} \text{Cov}(\mathbf{U}, \mathbf{V}) \mathbf{D}_V^{-1}$.

Some books write $\mathbf{P}_\rho$ for the correlation matrix. We use $\mathbb{R}$ in this chapter, and $\hat{\mathbb{R}}$ for its sample counterpart in [Section 2.7](#).

Proposition 2.4.1 — Properties of correlation matrices

Let $\mathbb{R}$ be the correlation matrix of $\mathbf{Y}$.

- (a) $\mathbb{R}$ is the covariance matrix of the standardized vector $\mathbb{Z} = \mathbf{D}_\sigma^{-1}(\mathbf{Y} - \boldsymbol{\mu})$. Hence $\mathbb{R}$ is nonnegative definite with unit diagonal, and $\text{tr}(\mathbb{R}) = p$.
- (b) $|\rho_{ij}| \leq 1$, with $|\rho_{ij}| = 1$ iff $Z_j = \rho_{ij}Z_i$ with probability one.
- (c) $\text{rank}(\mathbb{R}) = \text{rank}(\boldsymbol{\Sigma})$. In particular, $\mathbb{R}$ is positive definite iff $\boldsymbol{\Sigma}$ is.
- (d) If $\mathbf{Y}^* = \mathbf{D}\mathbf{Y} + \mathbf{c}$ with $\mathbf{D} = \text{diag}(d_1, \dots, d_p)$ and every $d_i \neq 0$, then $\rho_{ij}^* = \text{sign}(d_i d_j) \rho_{ij}$. Positive rescaling and shifting leave $\mathbb{R}$ unchanged.
- (e) Conversely, every nonnegative definite matrix with unit diagonal is the correlation matrix of some random vector.

Proof.

- (a) [Theorem 2.2.1](#) gives $\text{Cov}(\mathbb{Z}) = \mathbf{D}_\sigma^{-1} \boldsymbol{\Sigma} \mathbf{D}_\sigma^{-1} = \mathbb{R}$, and [Theorem 2.2.3\(a\)](#) applies.
- (b) The 2×2 matrix of rows and columns i, j of $\mathbb{R}$ is the covariance matrix of (Z_i, Z_j) , so it is nonnegative definite and its determinant $1 - \rho_{ij}^2$ is nonnegative. If $\rho_{ij} = \pm 1$, the vector $(\rho_{ij}, -1)\mathbf{T}$ is in the null space of that 2×2 matrix, so $\rho_{ij}Z_i - Z_j = 0$ with probability one by [Theorem 2.2.3\(c\)](#). Conversely, $Z_j = cZ_i$ with $\text{Var}(Z_i) = \text{Var}(Z_j) = 1$ forces $c = \pm 1$ and $\rho_{ij} = c$.
- (c) $\mathbf{D}_\sigma$ is nonsingular.
- (d) $\text{Cov}(\mathbf{Y}^*) = \mathbf{D}\boldsymbol{\Sigma}\mathbf{D}$ has entries $d_i d_j \sigma_{ij}$ and standard deviations $|d_i| \sigma_{ii}^{1/2}$.
- (e) Such a matrix is a covariance matrix by [Theorem 2.2.3\(b\)](#), and its variances are one.

■

Part (d) is the reason correlation is the natural scale-free summary. Part (e) is less obvious than it looks. A symmetric matrix with unit diagonal and off-diagonal entries in $[-1, 1]$ need not be a correlation matrix, because the correlations must be jointly consistent.

Proposition 2.4.2 — Three correlations

Let $|\rho_{12}| < 1$ and $|\rho_{13}| < 1$. The matrix

$$\mathbb{R} = \begin{pmatrix} 1 & \rho_{12} & \rho_{13} \\ \rho_{12} & 1 & \rho_{23} \\ \rho_{13} & \rho_{23} & 1 \end{pmatrix}$$

is a correlation matrix iff

$$\rho_{12}\rho_{13} - \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)} \leq \rho_{23} \leq \rho_{12}\rho_{13} + \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}.$$

Proof. By [Proposition 2.4.1\(e\)](#) we must decide when $\mathbb{R}$ is nonnegative definite. Write $\mathbf{r} = (\rho_{12}, \rho_{13})\mathbf{T}$ and let $\mathbb{R}_2$ be the lower right 2×2 block. The nonsingular matrix

$\mathbf{T} = \begin{pmatrix} 1 & \mathbf{0}^T \\ -\mathbf{r} & \mathbf{I}_2 \end{pmatrix}$ gives

$$\mathbf{T}\mathbb{R}\mathbf{T}^T = \begin{pmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbb{R}_2 - \mathbf{r}\mathbf{r}^T \end{pmatrix},$$

and $\mathbf{a}^T\mathbb{R}\mathbf{a} = \mathbf{b}^T(\mathbf{T}\mathbb{R}\mathbf{T}^T)\mathbf{b}$ with $\mathbf{b} = (\mathbf{T}\mathbf{T})^{-1}\mathbf{a}$ ranges over all vectors as $\mathbf{a}$ does. So $\mathbb{R}$ is nonnegative definite iff the Schur complement

$$\mathbb{R}_2 - \mathbf{r}\mathbf{r}^T = \begin{pmatrix} 1 - \rho_{12}^2 & \rho_{23} - \rho_{12}\rho_{13} \\ \rho_{23} - \rho_{12}\rho_{13} & 1 - \rho_{13}^2 \end{pmatrix}$$

is. A symmetric 2×2 matrix with positive diagonal is nonnegative definite iff its determinant is nonnegative, because its eigenvalues have positive sum and their product is the determinant. The determinant is $(1 - \rho_{12}^2)(1 - \rho_{13}^2) - (\rho_{23} - \rho_{12}\rho_{13})^2$, and it is nonnegative exactly on the stated interval. ■

If Y_2 and Y_3 both correlate 0.8 with Y_1 , the interval is $[0.28, 1]$. Two variables that are strongly related to a third cannot be negatively correlated with each other, nor even uncorrelated. The quantity $(\rho_{23} - \rho_{12}\rho_{13}) / \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}$, which the proof forces into $[-1, 1]$, is the correlation of Y_2 and Y_3 after the linear effect of Y_1 has been removed from both. [Section 2.5](#) develops this residualization, and [Definition 3.6.1](#) names the result the partial correlation.

Correlation matrices do not transform simply under linear maps: the correlation matrix of $\mathbf{A}\mathbf{Y}$ must be recomputed from $\mathbf{A}\Sigma\mathbf{A}^T$. Theory is therefore done with covariance matrices, and correlation matrices are used for description.

2.4.1 Exercises

2.4.1.1 A. Check your understanding

Exercise A1.

Show that $(1 - \rho)\mathbf{I}_p + \rho\mathbf{1}\mathbf{1}^T$ is a correlation matrix iff $-1/(p - 1) \leq \rho \leq 1$. Explain the lower bound through the variance of $\sum_i Z_i$.

Solution. The matrix sends $\mathbf{1}$ to $(1 + (p - 1)\rho)\mathbf{1}$ and every $\mathbf{u} \perp \mathbf{1}$ to $(1 - \rho)\mathbf{u}$. Its eigenvalues are $1 + (p - 1)\rho$ (once) and $1 - \rho$ ($p - 1$ times). By [Proposition 2.4.1\(e\)](#) it is a correlation matrix iff both are nonnegative. If $\mathbb{Z}$ has this correlation matrix, $\text{Var}(\sum_i Z_i) = \mathbf{1}^T\mathbb{R}\mathbf{1} = p[1 + (p - 1)\rho]$, which must be nonnegative. Many variables cannot all be strongly negatively correlated with one another, because their sum would then have negative variance.

2.4.1.2 B. Practice

Exercise B1.

Show that $\det(\Sigma) \leq \prod_i \sigma_{ii}$ for every covariance matrix, with equality for positive definite Σ iff Σ is diagonal. *Hint:* factor out $\mathbf{D}_\sigma$ and use the eigenvalues of $\mathbb{R}$.

Solution. If some $\sigma_{ii} = 0$, the i th row of Σ vanishes (apply [Theorem 2.2.3\(c\)](#) with $\mathbf{a}$ the i th coordinate vector) and both sides are zero. Otherwise $\det \Sigma = \det(\mathbf{D}_\sigma)^2 \det \mathbb{R} = \prod_i \sigma_{ii} \cdot \det \mathbb{R}$. The eigenvalues $\lambda_i \geq 0$ of $\mathbb{R}$ sum to $\text{tr } \mathbb{R} = p$, so by the inequality of arithmetic and geometric means $\det \mathbb{R} = \prod_i \lambda_i \leq (p^{-1} \sum_i \lambda_i)^p = 1$. Equality holds iff every $\lambda_i = 1$, that is, iff $\mathbb{R} = \mathbf{I}$, that is, iff Σ is diagonal.

2.4.1.3 C. Going deeper

Exercise C1.

- (a) Show that a $p \times p$ matrix is a correlation matrix iff it is the Gram matrix $(\mathbf{u}_i \mathbf{T} \mathbf{u}_j)$ of p unit vectors in $\mathbb{R}^p$. (b) For $p = 3$, write $\rho_{ij} = \cos \vartheta_{ij}$ with $\vartheta_{ij} \in [0, \pi]$. Show that [Proposition 2.4.2](#) is equivalent to the spherical triangle inequalities $|\vartheta_{12} - \vartheta_{13}| \leq \vartheta_{23} \leq \min(\vartheta_{12} + \vartheta_{13}, 2\pi - \vartheta_{12} - \vartheta_{13})$.

2.5 Moments of partitioned random vectors

Regression with random regressors studies a response vector $\mathbf{Y}$ ($p \times 1$) together with a regressor vector $\mathbf{X}$ ($q \times 1$) observed on the same unit. In this section $\mathbf{X}$ is a random vector, not a model matrix. We stack the two into one vector and read off what the blocks of its moments mean.

2.5.1 Blocks of the mean and covariance

Theorem 2.5.1 — Partitioned moments

Let $\mathbf{W} = (\mathbf{Y} \mathbf{T}, \mathbf{X} \mathbf{T}) \mathbf{T}$ have finite second moments, and write

$$\mathbf{E}(\mathbf{W}) = \begin{pmatrix} \boldsymbol{\mu}_Y \\ \boldsymbol{\mu}_X \end{pmatrix}, \quad \text{Cov}(\mathbf{W}) = \begin{pmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix}.$$

- (a) $\boldsymbol{\mu}_Y = \mathbf{E}(\mathbf{Y})$, $\boldsymbol{\mu}_X = \mathbf{E}(\mathbf{X})$, $\Sigma_{YY} = \text{Cov}(\mathbf{Y})$, $\Sigma_{XX} = \text{Cov}(\mathbf{X})$, $\Sigma_{YX} = \text{Cov}(\mathbf{Y}, \mathbf{X})$ is $p \times q$, and $\Sigma_{XY} = \Sigma_{YX} \mathbf{T}$.
 (b) For constants $\mathbf{A}$, $\mathbf{B}$, $\mathbf{c}$ of conformable sizes, $\text{Cov}(\mathbf{A}\mathbf{Y}, \mathbf{B}\mathbf{X}) = \mathbf{A}\Sigma_{YX}\mathbf{B}\mathbf{T}$, and

$$\text{Cov}(\mathbf{A}\mathbf{Y} + \mathbf{B}\mathbf{X} + \mathbf{c}) = \mathbf{A}\Sigma_{YY}\mathbf{A}\mathbf{T} + \mathbf{A}\Sigma_{YX}\mathbf{B}\mathbf{T} + \mathbf{B}\Sigma_{XY}\mathbf{A}\mathbf{T} + \mathbf{B}\Sigma_{XX}\mathbf{B}\mathbf{T}.$$

- (c) $\mathbb{C}(\Sigma_{XY}) \subseteq \mathbb{C}(\Sigma_{XX})$ and $\mathbb{C}(\Sigma_{YX}) \subseteq \mathbb{C}(\Sigma_{YY})$. Equivalently, for every generalized inverse Σ_{XX}^- ,

$$\Sigma_{XX}\Sigma_{XX}^-\Sigma_{XY} = \Sigma_{XY}, \quad \Sigma_{YX}\Sigma_{XX}^-\Sigma_{XX} = \Sigma_{YX},$$

and similarly with the roles of $\mathbf{Y}$ and $\mathbf{X}$ exchanged.

Proof.

- (a) The outer product $(\mathbf{W} - \mathbf{E}\mathbf{W})(\mathbf{W} - \mathbf{E}\mathbf{W})\mathbf{T}$ has blocks $(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{Y} - \boldsymbol{\mu}_Y)\mathbf{T}$,

$(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{X} - \boldsymbol{\mu}_X)\mathbf{T}$, its transpose, and $(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)\mathbf{T}$. Take expectations blockwise. (b) $\mathbf{A}\mathbf{Y} + \mathbf{B}\mathbf{X} + \mathbf{c} = [\mathbf{A}, \mathbf{B}]\mathbf{W} + \mathbf{c}$, and $\mathbf{A}\mathbf{Y} = [\mathbf{A}, \mathbf{0}]\mathbf{W}$, $\mathbf{B}\mathbf{X} = [\mathbf{0}, \mathbf{B}]\mathbf{W}$. Apply [Theorem 2.2.1](#) and multiply out the blocks.

- (b) Let $\mathbf{a} \in \mathcal{N}(\boldsymbol{\Sigma}_{XX})$. By [Theorem 2.2.3\(c\)](#), $\mathbf{a}\mathbf{T}\mathbf{X}$ is constant with probability one, so its covariance with $\mathbf{Y}$ vanishes: $\boldsymbol{\Sigma}_{YX}\mathbf{a} = \text{Cov}(\mathbf{Y}, \mathbf{a}\mathbf{T}\mathbf{X}) = \mathbf{0}$. Thus $\mathcal{N}(\boldsymbol{\Sigma}_{XX}) \subseteq \mathcal{N}(\boldsymbol{\Sigma}_{YX})$. Taking orthogonal complements, and using $\mathcal{N}(\mathbf{B})^\perp = \mathbb{C}(\mathbf{B}\mathbf{T})$ and the symmetry of $\boldsymbol{\Sigma}_{XX}$, gives $\mathbb{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathbb{C}(\boldsymbol{\Sigma}_{XX})$. The same argument with $\mathbf{Y}$ and $\mathbf{X}$ exchanged gives the second inclusion. A column inclusion $\mathbb{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathbb{C}(\boldsymbol{\Sigma}_{XX})$ means $\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{XX}\mathbf{K}\mathbf{T}$ for some $\mathbf{K}$. Then $\boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XX}\mathbf{K}\mathbf{T} = \boldsymbol{\Sigma}_{XY}$. Conversely, that identity exhibits every column of $\boldsymbol{\Sigma}_{XY}$ as a combination of columns of $\boldsymbol{\Sigma}_{XX}$ ([Theorem 1.4.1](#)). Transposing gives $\boldsymbol{\Sigma}_{YX} = \mathbf{K}\boldsymbol{\Sigma}_{XX}$ and the second identity. ■

Part (c) looks technical, but it removes every difficulty with singular $\boldsymbol{\Sigma}_{XX}$ in what follows. A direction in which $\mathbf{X}$ does not vary cannot covary with anything.

2.5.2 Linear prediction and residual covariance

The most important use of the blocks is to split $\mathbf{Y}$ into a part explained linearly by $\mathbf{X}$ and a part uncorrelated with $\mathbf{X}$.

Theorem 2.5.2 — Best linear predictor

In the setting of [Theorem 2.5.1](#), fix a generalized inverse $\boldsymbol{\Sigma}_{XX}^-$ and put

$$\mathbf{B}_* = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^-, \quad \mathbf{E} = \mathbf{Y} - \boldsymbol{\mu}_Y - \mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X), \quad \boldsymbol{\Sigma}_{YY \cdot X} = \boldsymbol{\Sigma}_{YY} - \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XY}.$$

- (a) $\mathbf{E}(\mathbf{E}) = \mathbf{0}$ and $\text{Cov}(\mathbf{E}, \mathbf{X}) = \mathbf{0}$.
 (b) $\text{Cov}(\mathbf{E}) = \boldsymbol{\Sigma}_{YY \cdot X}$. This matrix is nonnegative definite and does not depend on the choice of generalized inverse. Neither does $\mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X)$, with probability one.
 (c) For every constant $p \times q$ matrix $\mathbf{B}$ and $p \times 1$ vector $\mathbf{c}$,

$$\mathbf{E}[(\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X})(\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X})\mathbf{T}] = \boldsymbol{\Sigma}_{YY \cdot X} + (\mathbf{B} - \mathbf{B}_*)\boldsymbol{\Sigma}_{XX}(\mathbf{B} - \mathbf{B}_*)\mathbf{T} + d\mathbf{d}\mathbf{T},$$

where $\mathbf{d} = \boldsymbol{\mu}_Y - \mathbf{c} - \mathbf{B}\boldsymbol{\mu}_X$. So the mean squared error matrix is never smaller, in the nonnegative definite ordering, than $\boldsymbol{\Sigma}_{YY \cdot X}$. Equality holds iff $\mathbf{c} = \boldsymbol{\mu}_Y - \mathbf{B}\boldsymbol{\mu}_X$ and $\mathbf{B}\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX}$.

Proof. By [Theorem 2.5.1\(c\)](#), write $\boldsymbol{\Sigma}_{YX} = \mathbf{K}\boldsymbol{\Sigma}_{XX}$. Then $\mathbf{B}_*\boldsymbol{\Sigma}_{XX} = \mathbf{K}\boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XX} = \mathbf{K}\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX}$.

- (a) The mean is clear. By [Theorem 2.5.1\(b\)](#), $\text{Cov}(\mathbf{E}, \mathbf{X}) = \boldsymbol{\Sigma}_{YX} - \mathbf{B}_*\boldsymbol{\Sigma}_{XX} = \mathbf{0}$.
 (b) Again by [Theorem 2.5.1\(b\)](#), $\text{Cov}(\mathbf{E}) = \boldsymbol{\Sigma}_{YY} - \mathbf{B}_*\boldsymbol{\Sigma}_{XY} - \boldsymbol{\Sigma}_{YX}\mathbf{B}_*\mathbf{T} + \mathbf{B}_*\boldsymbol{\Sigma}_{XX}\mathbf{B}_*\mathbf{T}$. Using $\mathbf{B}_*\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX}$, the last term is $\boldsymbol{\Sigma}_{YX}\mathbf{B}_*\mathbf{T}$, which cancels the third, so $\text{Cov}(\mathbf{E}) = \boldsymbol{\Sigma}_{YY} - \mathbf{B}_*\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{YY \cdot X}$. Next, $\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XY} = \mathbf{K}\boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XX}\mathbf{K}\mathbf{T} = \mathbf{K}\boldsymbol{\Sigma}_{XX}\mathbf{K}\mathbf{T}$, which involves no generalized inverse. A covariance matrix is nonnegative definite. Finally, by [Theorem 2.2.3\(d\)](#), $\mathbf{X} - \boldsymbol{\mu}_X = \boldsymbol{\Sigma}_{XX}\mathbf{v}$ for some random $\mathbf{v}$,

with probability one, and then $\mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X) = \boldsymbol{\Sigma}_{YX}\mathbf{v}$ for every choice of generalized inverse.

- (c) Decompose $\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X} = \mathbf{E} + (\mathbf{B}_* - \mathbf{B})(\mathbf{X} - \boldsymbol{\mu}_X) + \mathbf{d}$. The three pieces are, respectively, mean zero, mean zero and constant, and the first two are uncorrelated by (a). So the cross products have zero expectation, and the expected outer product is $\text{Cov}(\mathbf{E}) + (\mathbf{B}_* - \mathbf{B})\boldsymbol{\Sigma}_{XX}(\mathbf{B}_* - \mathbf{B})\mathbf{T} + \mathbf{d}\mathbf{d}\mathbf{T}$. Both added matrices are nonnegative definite. They vanish iff $\mathbf{d} = \mathbf{0}$ and $(\mathbf{B} - \mathbf{B}_*)\boldsymbol{\Sigma}_{XX}^{1/2} = \mathbf{0}$, and the latter is equivalent to $(\mathbf{B} - \mathbf{B}_*)\boldsymbol{\Sigma}_{XX} = \mathbf{0}$, that is, to $\mathbf{B}\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX}$.

■

The affine function $\boldsymbol{\mu}_Y + \mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X)$ is the **best linear predictor** of $\mathbf{Y}$ from $\mathbf{X}$. Because the ordering in (c) is the nonnegative definite ordering, it is simultaneously best for every linear combination $\mathbf{a}\mathbf{T}\mathbf{Y}$ and for the total squared error. The matrix $\boldsymbol{\Sigma}_{YY.X}$ is the **partial covariance matrix** of $\mathbf{Y}$ given $\mathbf{X}$. When $\boldsymbol{\Sigma}_{XX}$ is nonsingular, $\mathbf{B}_* = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}$ holds the population regression coefficients, and $\boldsymbol{\Sigma}_{YY.X}$ is a Schur complement. Two consequences follow from Chapter 1. If $\text{Cov}(\mathbf{W})$ is positive definite, the YY block of its inverse is $\boldsymbol{\Sigma}_{YY.X}^{-1}$ (Theorem 1.6.4). And $\det \text{Cov}(\mathbf{W}) = \det(\boldsymbol{\Sigma}_{XX})\det(\boldsymbol{\Sigma}_{YY.X})$ (Theorem 1.6.3), so the generalized variance factorizes into the part of $\mathbf{X}$ and the part of $\mathbf{Y}$ that $\mathbf{X}$ leaves unexplained.

Nothing here says that $\mathbf{E}(\mathbf{Y} | \mathbf{X})$ is linear. The best linear predictor is best among affine functions, whatever the joint distribution. For jointly normal vectors, the conditional mean happens to be this affine function, and $\boldsymbol{\Sigma}_{YY.X}$ is the conditional covariance matrix (Theorem 3.5.2). The partial correlation of Definition 3.6.1 is the correlation computed from $\boldsymbol{\Sigma}_{YY.X}$. Chapter 6 shows that least squares estimates $\mathbf{B}_*$ in large samples, whether or not the regression function is linear. It also recovers the predictor as an orthogonal projection in the space of square-integrable random variables. For scalar Y , the coefficient vector used there is $\mathbf{B}_*\mathbf{T}$.

Proposition 2.5.3 — Multiple correlation

Let Y be scalar with $\sigma_{YY} = \text{Var}(Y) > 0$, let $\boldsymbol{\Sigma}_{XX}$ be positive definite, and write $\boldsymbol{\sigma}_{XY} = \text{Cov}(\mathbf{X}, Y)$. Over all $\mathbf{a} \neq \mathbf{0}$,

$$\max_{\mathbf{a}} \text{corr}(Y, \mathbf{a}\mathbf{T}\mathbf{X})^2 = \rho_{Y.X}^2 = \frac{\boldsymbol{\sigma}_{XY}\mathbf{T}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\sigma}_{XY}}{\sigma_{YY}},$$

attained at $\mathbf{a} = \boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\sigma}_{XY}$, the coefficients of the best linear predictor (if $\boldsymbol{\sigma}_{XY} = \mathbf{0}$, every $\mathbf{a}$ attains the maximum 0). Moreover $1 - \rho_{Y.X}^2 = \sigma_{YY.X}/\sigma_{YY}$.

Proof. $\text{corr}(Y, \mathbf{a}\mathbf{T}\mathbf{X})^2 = (\mathbf{a}\mathbf{T}\boldsymbol{\sigma}_{XY})^2 / (\sigma_{YY} \mathbf{a}\mathbf{T}\boldsymbol{\Sigma}_{XX}\mathbf{a})$. Write $\mathbf{a}\mathbf{T}\boldsymbol{\sigma}_{XY} = (\boldsymbol{\Sigma}_{XX}^{1/2}\mathbf{a})\mathbf{T}(\boldsymbol{\Sigma}_{XX}^{-1/2}\boldsymbol{\sigma}_{XY})$. The Cauchy–Schwarz inequality gives $(\mathbf{a}\mathbf{T}\boldsymbol{\sigma}_{XY})^2 \leq (\mathbf{a}\mathbf{T}\boldsymbol{\Sigma}_{XX}\mathbf{a})(\boldsymbol{\sigma}_{XY}\mathbf{T}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\sigma}_{XY})$, with equality iff $\boldsymbol{\Sigma}_{XX}^{1/2}\mathbf{a}$ is proportional to $\boldsymbol{\Sigma}_{XX}^{-1/2}\boldsymbol{\sigma}_{XY}$. Dividing gives the bound and the maximizer. The last identity is the definition of $\sigma_{YY.X} = \sigma_{YY} - \boldsymbol{\sigma}_{XY}\mathbf{T}\boldsymbol{\Sigma}_{XX}^{-1}\boldsymbol{\sigma}_{XY}$, divided by σ_{YY} .

■

$\rho_{Y.X}^2$ is the population version of R^2 : the fraction of the variance of Y that the best linear predictor accounts for. Chapter 14 studies its sample version.

2.5.3 Conditioning: total expectation and total covariance

When the joint distribution is available, conditioning gives another decomposition of the covariance matrix. Conditional expectations and covariances are taken entry by entry, as in [Definition 2.1.1](#), and $\text{Cov}(\mathbf{Y} \mid \mathbf{X}) = \mathbf{E}[(\mathbf{Y} - \mathbf{E}(\mathbf{Y} \mid \mathbf{X}))(\mathbf{Y} - \mathbf{E}(\mathbf{Y} \mid \mathbf{X}))^T \mid \mathbf{X}]$.

Proposition 2.5.4 — Laws of total expectation and covariance

If $\mathbf{Y}$ has finite second moments, then $\mathbf{E}(\mathbf{Y}) = \mathbf{E}[\mathbf{E}(\mathbf{Y} \mid \mathbf{X})]$ and

$$\text{Cov}(\mathbf{Y}) = \mathbf{E}[\text{Cov}(\mathbf{Y} \mid \mathbf{X})] + \text{Cov}(\mathbf{E}(\mathbf{Y} \mid \mathbf{X})).$$

Proof. The first identity is the tower property of conditional expectation, applied to each component. For the second, let $\mathbf{m}(\mathbf{X}) = \mathbf{E}(\mathbf{Y} \mid \mathbf{X})$ and write $\mathbf{Y} - \boldsymbol{\mu} = (\mathbf{Y} - \mathbf{m}(\mathbf{X})) + (\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})$. The outer product has two cross terms. The conditional expectation of $(\mathbf{Y} - \mathbf{m}(\mathbf{X}))(\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})^T$ given $\mathbf{X}$ is $[\mathbf{E}(\mathbf{Y} \mid \mathbf{X}) - \mathbf{m}(\mathbf{X})](\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})^T = \mathbf{0}$, so its expectation is zero, and likewise for the other. The expectation of $(\mathbf{Y} - \mathbf{m})(\mathbf{Y} - \mathbf{m})^T$ is $\mathbf{E}[\text{Cov}(\mathbf{Y} \mid \mathbf{X})]$ by the tower property, and that of $(\mathbf{m} - \boldsymbol{\mu})(\mathbf{m} - \boldsymbol{\mu})^T$ is $\text{Cov}(\mathbf{m}(\mathbf{X}))$ because $\mathbf{E}\mathbf{m}(\mathbf{X}) = \boldsymbol{\mu}$. ■

Example. (A shared random effect)

Measurements $Y_1, \dots, Y_k$ on the same subject share a subject effect: $\mathbf{Y} = \alpha \mathbf{1} + \boldsymbol{\varepsilon}$, where α has variance τ^2 , $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_k$, and $\boldsymbol{\varepsilon}$ is independent of α with mean zero. Given α , the mean is $\alpha \mathbf{1}$ and the covariance is $\sigma^2 \mathbf{I}$. By [Proposition 2.5.4](#),

$$\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_k + \tau^2 \mathbf{1}\mathbf{1}^T,$$

so every pair of measurements has correlation $\tau^2/(\tau^2 + \sigma^2)$. Independent errors combined with a shared random component produce exactly this *equicorrelation* structure. It is the starting point of Chapter 32. Ordinary least squares ignores it, with consequences that Chapter 33 examines.

2.5.4 Exercises

2.5.4.1 A. Check your understanding

Exercise A1.

For scalar Y and X with $\text{Var}(X) > 0$, show that the best linear predictor has slope σ_{YX}/σ_{XX} , and that its mean squared error is $\sigma_{YY}(1 - \rho^2)$.

Exercise A2.

Let X be uniform on $(-1, 1)$ and $Y = X^2 + \varepsilon$, with ε independent of X , mean 0 and variance σ^2 . Find the best linear predictor of Y from X and its mean squared error, and compare with the mean squared error of $\mathbf{E}(Y | X)$.

2.5.4.2 B. Practice**Exercise B1.**

Let (Y_1, Y_2, Y_3) have correlation matrix $\mathbb{R}$. Compute the partial covariance matrix of (Y_2, Y_3) given Y_1 from [Theorem 2.5.2](#), and show that the corresponding correlation is

$$\frac{\rho_{23} - \rho_{12}\rho_{13}}{\sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}}.$$

Exercise B2.

Let $\mathbf{X} = (X_1, X_1)\mathbf{T}$ with $\text{Var}(X_1) = 1$, and let $\text{Cov}(Y, X_1) = c$. Find two different generalized inverses of Σ_{XX} that give different $\mathbf{B}_*$, and show that both give the same predictor $\mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X)$.

Exercise B3.

A random label K takes values $1, \dots, g$ with probabilities π_k . Given $K = k$, $\mathbf{Y}$ has mean $\mathbf{m}_k$ and covariance Σ_k . Show that $\text{Cov}(\mathbf{Y}) = \sum_k \pi_k \Sigma_k + \sum_k \pi_k (\mathbf{m}_k - \bar{\mathbf{m}})(\mathbf{m}_k - \bar{\mathbf{m}})\mathbf{T}$, where $\bar{\mathbf{m}} = \sum_k \pi_k \mathbf{m}_k$. Interpret the two terms as within-group and between-group covariance.

2.5.4.3 C. Going deeper**Exercise C1.**

Let $\mathbf{N}$ be the symmetric partitioned matrix

$$\mathbf{N} = \begin{pmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix}.$$

Show that $\mathbf{N}$ is nonnegative definite iff Σ_{XX} is nonnegative definite, $\mathcal{C}(\Sigma_{XY}) \subseteq \mathcal{C}(\Sigma_{XX})$, and $\Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^- \Sigma_{XY}$ is nonnegative definite for some generalized inverse Σ_{XX}^- (equivalently, given the column-space condition, for every one). Show by a 2×2 example that the column-space condition cannot be dropped.

Solution. If the matrix is nonnegative definite, it is the covariance matrix of some $\mathbf{W} = (\mathbf{Y}\mathbf{T}, \mathbf{X}\mathbf{T})\mathbf{T}$ ([Theorem 2.2.3\(b\)](#)). Then Σ_{XX} is a covariance matrix, the column condition is [Theorem 2.5.1\(c\)](#), and the Schur complement is $\text{Cov}(\mathbf{E})$ by [Theorem 2.5.2\(b\)](#). Conversely, the column condition gives $\Sigma_{YX} = \mathbf{K}\Sigma_{XX}$, and then

$\Sigma_{YX}\Sigma_{XX}^-\Sigma_{XY} = K\Sigma_{XX}KT$. With the nonsingular $T = \begin{pmatrix} I & -K \\ 0 & I \end{pmatrix}$,

$$T \begin{pmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix} T = \begin{pmatrix} \Sigma_{YY} - K\Sigma_{XX}KT & 0 \\ 0 & \Sigma_{XX} \end{pmatrix},$$

which is nonnegative definite. So is the original matrix, since $aTNa = bT(TNTT)b$ with $b = (TT)^{-1}a$. For the example, take $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. Here $\Sigma_{XX} = 0$ is nonnegative definite, and with the generalized inverse 0 the Schur complement is $1 \geq 0$. But the determinant is -1 , so the matrix is not nonnegative definite. The column condition fails, because $\mathbb{C}(1) \not\subseteq \mathbb{C}(0)$. Without that condition the Schur complement also depends on the choice: every scalar t is a generalized inverse of 0, and the Schur complement is then $1 - 1 \cdot t \cdot 1 = 1 - t$, which is negative for $t > 1$. Under the column condition, $\Sigma_{YX}\Sigma_{XX}^-\Sigma_{XY} = K\Sigma_{XX}KT$ for every choice, as shown above.

2.6 Moment generating functions and independence

Means and covariance matrices summarize a distribution. They do not determine it, and they cannot certify independence. For that we need a transform of the whole distribution. This section collects the facts that [Chapter 3](#) and [Chapter 4](#) use.

Definition 2.6.1 — Moment generating function

The **moment generating function** of a $p \times 1$ random vector $\mathbf{Y}$ is

$$M_{\mathbf{Y}}(\mathbf{t}) = \mathbf{E} \exp(\mathbf{tTY}), \quad \mathbf{t} \in \mathbb{R}^p,$$

where the expectation may be $+\infty$. We say that $M_{\mathbf{Y}}$ *exists* if it is finite for all $\mathbf{t}$ in some neighbourhood $\{\|\mathbf{t}\| < h\}$ of the origin, $h > 0$.

Theorem 2.6.1 — Properties of moment generating functions

Suppose $M_{\mathbf{Y}}$ exists.

- (a) *Uniqueness.* If $M_{\mathbf{V}}$ also exists and $M_{\mathbf{Y}} = M_{\mathbf{V}}$ on a neighbourhood of $\mathbf{0}$, then $\mathbf{Y}$ and $\mathbf{V}$ have the same distribution.
- (b) *Moments.* All moments of $\mathbf{Y}$ are finite. The gradient of $M_{\mathbf{Y}}$ at $\mathbf{0}$ is $\mathbf{E}(\mathbf{Y})$, and its Hessian at $\mathbf{0}$ is $\mathbf{E}(\mathbf{YYT})$. The Hessian of $\log M_{\mathbf{Y}}$ at $\mathbf{0}$ is $\text{Cov}(\mathbf{Y})$.
- (c) *Affine maps.* For a constant $k \times p$ matrix $\mathbf{A}$ and $k \times 1$ vector $\mathbf{b}$, $M_{\mathbf{AY}+\mathbf{b}}(\mathbf{s}) = \exp(\mathbf{sTb}) M_{\mathbf{Y}}(\mathbf{ATs})$.
- (d) *Independence.* Partition $\mathbf{Y} = (\mathbf{Y}_1\mathbf{T}, \mathbf{Y}_2\mathbf{T})\mathbf{T}$ and $\mathbf{t} = (\mathbf{t}_1\mathbf{T}, \mathbf{t}_2\mathbf{T})\mathbf{T}$ conformably. Then $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent iff $M_{\mathbf{Y}}(\mathbf{t}) = M_{\mathbf{Y}_1}(\mathbf{t}_1) M_{\mathbf{Y}_2}(\mathbf{t}_2)$ for all $\mathbf{t}$ in a neighbourhood of $\mathbf{0}$.

Proof. For (a), the scalar case is a theorem of analysis: a moment generating function that is finite near the origin extends analytically, which pins down the characteristic

function and hence the distribution (Billingsley 1995). For vectors, fix $\mathbf{a}$. For small $|s|$, $M_{\mathbf{a}\mathbf{T}\mathbf{Y}}(s) = M_{\mathbf{Y}}(s\mathbf{a}) = M_{\mathbf{V}}(s\mathbf{a}) = M_{\mathbf{a}\mathbf{T}\mathbf{V}}(s)$, so $\mathbf{a}\mathbf{T}\mathbf{Y}$ and $\mathbf{a}\mathbf{T}\mathbf{V}$ have the same distribution by the scalar case, and Proposition 2.6.2 below finishes the argument. For (b), finiteness near $\mathbf{0}$ implies $\mathbf{E} \exp(h'|Y_i|) < \infty$ for some $h' > 0$ and each i , so every polynomial in the components is integrable and differentiation under the expectation is justified. Then $\partial M / \partial t_i = \mathbf{E}(Y_i e^{t\mathbf{T}\mathbf{Y}})$ and $\partial^2 M / \partial t_i \partial t_j = \mathbf{E}(Y_i Y_j e^{t\mathbf{T}\mathbf{Y}})$. At $\mathbf{t} = \mathbf{0}$ these are $\mathbf{E}(Y_i)$ and $\mathbf{E}(Y_i Y_j)$. For $K = \log M$, the chain rule gives $\nabla^2 K = M^{-1} \nabla^2 M - M^{-2} (\nabla M)(\nabla M)\mathbf{T}$, which at $\mathbf{0}$, where $M = 1$, is $\mathbf{E}(\mathbf{Y}\mathbf{Y}\mathbf{T}) - \boldsymbol{\mu}\boldsymbol{\mu}\mathbf{T} = \text{Cov}(\mathbf{Y})$. For (c), $\mathbf{E} \exp(\mathbf{s}\mathbf{T}(\mathbf{A}\mathbf{Y} + \mathbf{b})) = e^{\mathbf{s}\mathbf{T}\mathbf{b}} \mathbf{E} \exp((\mathbf{A}\mathbf{T}\mathbf{s})\mathbf{T}\mathbf{Y})$. For (d), independence gives $\mathbf{E}(e^{t_1\mathbf{T}\mathbf{Y}_1} e^{t_2\mathbf{T}\mathbf{Y}_2}) = \mathbf{E}(e^{t_1\mathbf{T}\mathbf{Y}_1}) \mathbf{E}(e^{t_2\mathbf{T}\mathbf{Y}_2})$. Conversely, let $\mathbf{V}_1$ and $\mathbf{V}_2$ be independent with the distributions of $\mathbf{Y}_1$ and $\mathbf{Y}_2$. The vector $(\mathbf{V}_1\mathbf{T}, \mathbf{V}_2\mathbf{T})\mathbf{T}$ has moment generating function $M_{\mathbf{Y}_1}(\mathbf{t}_1)M_{\mathbf{Y}_2}(\mathbf{t}_2)$, which by hypothesis equals $M_{\mathbf{Y}}$ near $\mathbf{0}$. By (a) it has the same joint distribution as $\mathbf{Y}$, so $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent. ■

The next fact explains why Chapter 3 can define the multivariate normal through linear combinations.

Proposition 2.6.2 — Distributions are determined by linear combinations

If $\mathbf{a}\mathbf{T}\mathbf{Y}$ and $\mathbf{a}\mathbf{T}\mathbf{V}$ have the same distribution for every $\mathbf{a} \in \mathbb{R}^p$, then $\mathbf{Y}$ and $\mathbf{V}$ have the same distribution.

Proof. The characteristic function $\phi_{\mathbf{Y}}(\mathbf{t}) = \mathbf{E} \exp(i\mathbf{t}\mathbf{T}\mathbf{Y})$ exists for every random vector and determines its distribution (Billingsley 1995). Since $\phi_{\mathbf{Y}}(\mathbf{t}) = \phi_{\mathbf{t}\mathbf{T}\mathbf{Y}}(1) = \phi_{\mathbf{t}\mathbf{T}\mathbf{V}}(1) = \phi_{\mathbf{V}}(\mathbf{t})$ for every $\mathbf{t}$, the two distributions coincide. ■

If $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent, so are $g(\mathbf{Y}_1)$ and $h(\mathbf{Y}_2)$ for measurable g and h . Independence of statistics is usually proved this way. For normal vectors, Theorem 3.4.1 reduces the independence of two blocks to a zero cross-covariance, and Theorem 4.4.1 and Theorem 4.4.2 build on that.

Independence of *several* vectors means that the joint distribution factors into all the marginals, not merely that each pair is independent.

Example. (Pairwise but not mutually independent)

Let S_1 and S_2 be independent, each equal to ± 1 with probability $\frac{1}{2}$, and put $S_3 = S_1 S_2$. Each S_j is ± 1 with probability $\frac{1}{2}$. Each pair is independent: for instance, $\Pr(S_1 = s, S_3 = u) = \Pr(S_1 = s, S_2 = su) = \frac{1}{4}$ for all signs s, u . So $\text{Cov}(\mathbf{S}) = \mathbf{I}_3$. But the three are not mutually independent, since S_3 is a function of the other two. A third-order moment shows the failure directly: $\mathbf{E}(S_1 S_2 S_3) = 1$, whereas mutual independence would force it to be 0. No condition on the covariance matrix can detect this, because covariances involve only pairs.

2.6.1 Exercises

2.6.1.1 A. Check your understanding

Exercise A1.

Find the moment generating function of the multinomial count vector of [Example \(Multinomial counts\)](#), and recover its mean and covariance matrix from [Theorem 2.6.1\(b\)](#).

2.6.1.2 B. Practice

Exercise B1.

Let $S_1, \dots, S_k$ be independent random signs. For each nonempty $A \subseteq \{1, \dots, k\}$ put $T_A = \prod_{i \in A} S_i$. Show that the $2^k - 1$ variables T_A are pairwise independent with covariance matrix $\mathbf{I}$, and find a triple of them that is not mutually independent.

2.7 Sample moments and their computation

The population quantities of this chapter have sample analogues, and the operator rules give their expectations at once. Let $\mathbf{y}_1, \dots, \mathbf{y}_n$ be $p \times 1$ observations, and stack them as the rows of the $n \times p$ data matrix $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_n] \mathbf{T}$. The script letter keeps the data matrix apart from a single random vector $\mathbf{Y}$. The rows keep the lowercase $\mathbf{y}_i$ of observed data, but in [Proposition 2.7.1](#) they are treated as random. With $\mathbf{C} = \mathbf{I}_n - n^{-1} \mathbf{1} \mathbf{1}^T$, the sample mean vector and the **sample covariance matrix** are

$$\bar{\mathbf{y}} = \frac{1}{n} \sum_{i=1}^n \mathbf{y}_i = \frac{1}{n} \mathbf{Y} \mathbf{T} \mathbf{1}, \quad \mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})^T = \frac{1}{n-1} \mathbf{Y} \mathbf{T} \mathbf{C} \mathbf{Y}.$$

The second form of $\mathbf{S}$ holds because $\mathbf{C} \mathbf{Y}$ is the data matrix with its column means subtracted, and $\mathbf{C}$ is symmetric and idempotent. The sample correlation matrix is $\hat{\mathbf{R}} = \hat{\mathbf{D}}^{-1} \mathbf{S} \hat{\mathbf{D}}^{-1}$, where $\hat{\mathbf{D}}$ holds the sample standard deviations.

Proposition 2.7.1 — Moments of sample moments

Suppose the rows $\mathbf{y}_i$ are pairwise uncorrelated random vectors with common mean $\boldsymbol{\mu}$ and common covariance matrix $\boldsymbol{\Sigma}$. Then

- (a) $\mathbf{E}(\bar{\mathbf{y}}) = \boldsymbol{\mu}$ and $\text{Cov}(\bar{\mathbf{y}}) = \boldsymbol{\Sigma}/n$;
- (b) $\mathbf{E}(\mathbf{S}) = \boldsymbol{\Sigma}$;
- (c) $\mathbf{S}$ is nonnegative definite, and $\text{rank}(\mathbf{S}) \leq \min(p, n-1)$ for every realization.

Proof.

- (a) is [Proposition 2.2.2\(c\)](#) with $c_i = 1/n$. For (b), let $\mathbf{u}_j$ denote the j th column of $\mathbf{Y}$, the n observations of variable j . The (j, k) entry of $\mathbf{Y} \mathbf{T} \mathbf{C} \mathbf{Y}$ is the bilinear form $\mathbf{u}_j^T \mathbf{C} \mathbf{u}_k$. Because different rows are uncorrelated, $\text{Cov}(\mathbf{u}_k, \mathbf{u}_j) = \sigma_{kj} \mathbf{I}_n$, and

$\mathbf{E}(\mathbf{u}_j) = \mu_j \mathbf{1}$. By [Corollary 2.3.2\(d\)](#),

$$\mathbf{E}(\mathbf{u}_j \mathbf{T} \mathbf{C} \mathbf{u}_k) = \sigma_{kj} \operatorname{tr}(\mathbf{C}) + \mu_j \mu_k \mathbf{1} \mathbf{T} \mathbf{C} \mathbf{1} = (n-1) \sigma_{jk},$$

since $\operatorname{tr}(\mathbf{C}) = n-1$ and $\mathbf{C} \mathbf{1} = \mathbf{0}$. (c) For any $\mathbf{a}$, $\mathbf{a} \mathbf{T} \mathbf{S} \mathbf{a} = \|\mathbf{C} \mathbf{y} \mathbf{a}\|^2 / (n-1) \geq 0$. The rank of $\mathbf{y} \mathbf{T} \mathbf{C} \mathbf{y} = (\mathbf{C} \mathbf{y}) \mathbf{T} (\mathbf{C} \mathbf{y})$ equals $\operatorname{rank}(\mathbf{C} \mathbf{y})$, which is at most $\operatorname{rank}(\mathbf{C}) = n-1$ and at most p . ■

The divisor $n-1$ is the trace of $\mathbf{C}$, the dimension of the space the centred data live in, exactly as $n-r$ is for regression residuals in [Example \(Residual sum of squares under a wrong mean\)](#). Part (c) has a practical side. With $n \leq p$, as in genomics or finance, the sample covariance matrix is always singular, even when Σ is positive definite. It has a linear combination with zero sample variance whether or not one exists in the population. Chapter 28 returns to this.

Computation. In practice $\mathbf{C}$ is never formed: one subtracts column means. The eigenvalues of $\mathbf{S}$ are the squared singular values of $\mathbf{C} \mathbf{y} / \sqrt{n-1}$ ([Theorem 1.8.1](#)), which can be computed without forming a cross-product matrix (Chapter 10).

Example. (Macroeconomic series)

Longley's data give $n = 16$ annual observations (1947–1962) on $p = 7$ US series: total employment, the GNP deflator, GNP, unemployment, armed forces, population and the year. The eigenvalues of $\mathbf{S}$ run from 9.94×10^9 down to 1.07×10^{-2} , a ratio of 9.3×10^{11} . That spread says little about the structure of the data. It mostly reflects units: the standard deviation of GNP is 99395, that of the deflator 10.79.

The correlation matrix removes the units ([Proposition 2.4.1\(d\)](#)). Its eigenvalues sum to $p = 7$. The largest is 5.533, so one direction carries about 79% of the standardized variation, and the smallest is 2.59×10^{-4} . By [Theorem 1.7.7](#), the unit vector $\mathbf{v}$ belonging to the smallest eigenvalue minimizes the sample variance of $\mathbf{v} \mathbf{T} \mathbf{z}$ over standardized combinations. Its two largest weights are 0.681 on year and -0.674 on GNP, and the combination has standard deviation 0.016, against 1 for each standardized variable. In these sixteen years GNP is almost an exact linear function of time (their correlation is 0.995). This is the sample version of [Theorem 2.2.3\(c\)](#): a nearly zero eigenvalue marks a nearly constant linear combination. For regression it signals collinearity, which Chapter 26 studies. Finally, the first 5 rows alone give a sample covariance matrix of rank 4, as [Proposition 2.7.1\(c\)](#) requires.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data
names = list(data.columns)

Y = data.to_numpy(dtype=float)           # n x p data matrix, one row per
    year
n, p = Y.shape
C = np.eye(n) - np.ones((n, n)) / n     # centring matrix I - J/n
S = Y.T @ C @ Y / (n - 1)               # sample covariance matrix
d = np.sqrt(np.diag(S))
R = S / np.outer(d, d)                  # D^{-1} S D^{-1}, D = diag(d)
```

```

lam_S = np.linalg.eigvalsh(S)
lam_R, V = np.linalg.eigh(R)                                # ascending eigenvalues
print("eigenvalues of S:", lam_S)
print("eigenvalues of R:", lam_R)
v = V[:, 0]                                                  # direction of least variance
Z = (Y - Y.mean(0)) / d                                      # standardized variables
print("sample variance of Z v:", np.var(Z @ v, ddof=1))

```

2.7.1 Exercises

2.7.1.1 B. Practice

Exercise B1.

Suppose the rows of $\mathbf{Y}$ are observed in time order, with common mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, and that neighbouring rows are correlated: $\text{Cov}(\mathbf{y}_i, \mathbf{y}_{i+1}) = \rho\boldsymbol{\Sigma}$, while $\text{Cov}(\mathbf{y}_i, \mathbf{y}_j) = \mathbf{0}$ for $|i - j| \geq 2$. Show that $\mathbf{E}(\mathbf{S}) = (1 - 2\rho/n)\boldsymbol{\Sigma}$ and $\text{Cov}(\bar{\mathbf{y}}) = n^{-2}[n + 2(n - 1)\rho]\boldsymbol{\Sigma}$. Which of the two effects disappears as $n \rightarrow \infty$?

Solution. As in the proof of [Proposition 2.7.1](#), $\text{Cov}(\mathbf{u}_k, \mathbf{u}_j) = \sigma_{kj}\mathbf{T}$, where $\mathbf{T}$ is the $n \times n$ tridiagonal matrix with ones on the diagonal and ρ next to it. Since $\mathbf{C}\mathbf{1} = \mathbf{0}$, [Corollary 2.3.2\(d\)](#) gives $\mathbf{E}(\mathbf{u}_j\mathbf{T}\mathbf{C}\mathbf{u}_k) = \sigma_{kj} \text{tr}(\mathbf{C}\mathbf{T}) = \sigma_{kj}[\text{tr}(\mathbf{T}) - n^{-1}\mathbf{1}\mathbf{T}\mathbf{T}\mathbf{1}]$. Here $\text{tr}(\mathbf{T}) = n$ and $\mathbf{1}\mathbf{T}\mathbf{T}\mathbf{1} = n + 2(n - 1)\rho$, so the bracket is $(n - 1) - 2(n - 1)\rho/n$, and dividing by $n - 1$ gives $\mathbf{E}(\mathbf{S}) = (1 - 2\rho/n)\boldsymbol{\Sigma}$. For the mean, $\bar{\mathbf{y}} = n^{-1} \sum_i \mathbf{y}_i$ and [Proposition 2.2.2](#) gives n^{-2} times the sum of all $\text{Cov}(\mathbf{y}_i, \mathbf{y}_j)$: n diagonal terms $\boldsymbol{\Sigma}$ and $2(n - 1)$ neighbouring terms $\rho\boldsymbol{\Sigma}$. The bias of $\mathbf{S}$ is $O(1/n)$ and disappears, but $n \text{Cov}(\bar{\mathbf{y}}) \rightarrow (1 + 2\rho)\boldsymbol{\Sigma}$, not $\boldsymbol{\Sigma}$. For $\rho > 0$ the naive standard error $\sqrt{s_{jj}/n}$ stays too small by a fixed factor however large the sample.

2.8 Summary and Notes

2.8.1 Summary

Idea.

1. Expectation of a random matrix is taken entry by entry, and it commutes with constant linear maps, transposes and traces ([Proposition 2.1.1](#)).
2. $\text{Cov}(\mathbf{Y}) = \mathbf{E}[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})\mathbf{T}] = \mathbf{E}(\mathbf{Y}\mathbf{Y}\mathbf{T}) - \boldsymbol{\mu}\boldsymbol{\mu}\mathbf{T}$, and $\text{Cov}(\mathbf{U}, \mathbf{V})$ is the rectangular cross-covariance, with $\text{Cov}(\mathbf{V}, \mathbf{U}) = \text{Cov}(\mathbf{U}, \mathbf{V})\mathbf{T}$ ([Definition 2.1.2](#) and [Proposition 2.1.2](#)). Uncorrelated does not imply independent ([Example \(A point on a circle\)](#)).
3. Affine maps: $\mathbf{E}(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$, $\text{Cov}(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\mathbf{T}$, $\text{Cov}(\mathbf{A}\mathbf{U}, \mathbf{B}\mathbf{V}) = \mathbf{A} \text{Cov}(\mathbf{U}, \mathbf{V})\mathbf{B}\mathbf{T}$ ([Theorem 2.2.1](#)). In particular $\text{Var}(\mathbf{a}\mathbf{T}\mathbf{Y}) = \mathbf{a}\mathbf{T}\boldsymbol{\Sigma}\mathbf{a}$.
4. Covariance matrices are exactly the nonnegative definite matrices. $\boldsymbol{\Sigma}\mathbf{a} = \mathbf{0}$ iff $\mathbf{a}\mathbf{T}\mathbf{Y}$ is constant with probability one, and $\mathbf{Y}$ lives in the flat $\boldsymbol{\mu} + \mathbb{C}(\boldsymbol{\Sigma})$ ([Theorem 2.2.3](#)).
5. Whitening by $\mathbf{W}$ with $\mathbf{W}\boldsymbol{\Sigma}\mathbf{W}\mathbf{T} = \mathbf{I}$ is unique up to an orthogonal factor. Ev-

ery choice gives the same Mahalanobis distance $(\mathbf{Y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{Y} - \boldsymbol{\mu})$ (Proposition 2.2.4).

6. $\mathbf{E}(\mathbf{Y}^T \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}$ under second moments only (Theorem 2.3.1). Hence $\mathbf{E}(\text{RSS}) = \sigma^2(n - r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2$ (Example (Residual sum of squares under a wrong mean)). The variance of a quadratic form needs third and fourth moments (Theorem 2.3.3).
7. Correlation matrices are the nonnegative definite matrices with unit diagonal. They are invariant to rescaling, and their entries must be jointly consistent (Proposition 2.4.1 and Proposition 2.4.2).
8. For partitioned vectors, $\mathbb{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathbb{C}(\boldsymbol{\Sigma}_{XX})$. The best linear predictor $\boldsymbol{\mu}_Y + \boldsymbol{\Sigma}_{YX} \boldsymbol{\Sigma}_{XX}^{-1} (\mathbf{X} - \boldsymbol{\mu}_X)$ leaves an error uncorrelated with $\mathbf{X}$ with covariance matrix given by the Schur complement $\boldsymbol{\Sigma}_{YY.X}$ (Theorem 2.5.1 and Theorem 2.5.2).
9. Moment generating functions determine distributions and factor exactly under independence (Theorem 2.6.1). Pairwise independence is weaker than mutual independence (Example (Pairwise but not mutually independent)).
10. The sample covariance matrix is unbiased under uncorrelated rows and has rank at most $n - 1$. A tiny eigenvalue of the sample correlation matrix flags a nearly constant combination (Proposition 2.7.1 and Example (Macroeconomic series)).

2.8.2 Notes and sources

The operator calculus. Treating expectation and covariance as operators on random vectors and matrices is standard in linear-model texts. Rencher and Schaalje (2008, chapter 3) develops mean vectors, covariance and correlation matrices, partitioned vectors and linear functions in detail and at a gentle pace. Seber and Lee (2003, §§1.4–1.6) give a compact operator treatment, first and second moments of quadratic forms, and moment generating functions with their use in proving independence. This chapter covers the same ground. Rencher and Schaalje already introduce the standardized distance and the transformation by $\boldsymbol{\Sigma}^{-1/2}$. The chapter adds the characterization of singular covariance matrices through their column space (Theorem 2.2.3), the non-uniqueness of whitening and the comparison of symmetric and Cholesky whitening, the best linear predictor with generalized inverses, and the sample covariance matrix. Christensen (2020, chapter 1) states the covariance rules briefly in the notation of linear models. For the multivariate setting, Anderson (2003) remains the standard reference.

Mahalanobis distance and whitening. The distance Eq. 2.2.2 is due to Mahalanobis (1936), who introduced it to compare populations described by several correlated measurements. Kessy et al. (2018) compare whitening transformations. They show that the symmetric choice $\boldsymbol{\Sigma}^{-1/2}$ is the one closest to the original data in mean squared error (Exercise (B3)), and set it against Cholesky and principal-component whitening.

Quadratic forms. The trace argument for Theorem 2.3.1 is folklore. Theorem 2.3.3 is stated without proof by Atiqullah (1962), who used it to study how non-normal errors affect the residual variance estimate and the F -test. The kurtosis term depends on $\mathbf{A}$ only through its diagonal. For the residual sum of squares of a correct model it equals $(\mu_4 - 3\sigma^4) \sum_i (1 - h_{ii})^2$, which takes its simplest form, $(\mu_4 - 3\sigma^4)(n - r)^2/n$, when all leverages h_{ii} are equal. These are the quadratically balanced designs of Atiqullah's title. Seber and Lee (2003, theorem 1.6) give a proof by direct expansion. The proof here organizes the same expansion by splitting the form into quadratic, linear and constant parts. Mathai and Provost (1992) is a book-length treatment of moments and distributions of quadratic forms, including general covariance structures.

Correlation, partial covariance and prediction. The feasible range in [Proposition 2.4.2](#) is a special case of the fact that a symmetric matrix is nonnegative definite iff a Schur complement is, when the leading block is positive definite. Horn and Johnson (2013) give the general Schur-complement and Loewner-order results used in [Theorem 2.5.2](#). Stating the best linear predictor with a generalized inverse, so that nothing needs Σ_{XX} to be nonsingular, is in the spirit of Rao (1973), who uses generalized inverses throughout.

Probability background. Uniqueness of moment generating functions and characteristic functions, and the device of [Proposition 2.6.2](#), are proved in Billingsley (1995). The Longley data come from Longley (1967), who used them to test the numerical accuracy of regression programs. Their near-collinearity, visible in [Example \(Macroeconomic series\)](#), is the reason.

2.8.3 References

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Chapter 3

The Multivariate Normal Distribution

A random vector is normal when every linear combination of its entries is normal. From that definition come the moment generating function, the density, and closure under all linear maps, singular ones included. Two further properties make the normal distribution the working assumption of linear-model theory: zero covariance between jointly normal blocks means independence, and conditioning one block on another gives a linear regression with constant error variance. Partial correlation, the correlation that remains after linear adjustment, connects these facts to least squares.

What you need. [Chapter 1](#) (spectral theorem, square roots, generalized and partitioned inverses), [Chapter 2](#) (mean vectors and covariance matrices). Some examples refer forward to least squares ([Chapter 5](#) and [Chapter 6](#)).

3.0.1 Roadmap

- [Definition through linear combinations](#)
- [Moment generating function and density](#)
- [Linear transformations and marginals](#)
- [Independence and zero covariance](#)
- [Conditional distributions](#)
- [Partial correlation](#)
- [Summary and notes](#)

3.1 Definition through linear combinations

Nearly every exact distributional statement in the first half of this book has the same form. The response is $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with normal errors, and the object of interest is a linear function of $\mathbf{Y}$ (a coefficient estimate, a fitted value, a residual) or a quadratic function of $\mathbf{Y}$ (a sum of squares). A usable notion of normality for random vectors must therefore do two things. It must survive every linear map, including maps that collapse dimensions. And it must say when two linear functions of the same vector are independent.

The residual vector shows why the first requirement has to include singular cases. In a model with p linearly independent columns its covariance matrix has rank $n - p$ (Chapter 6), so the residuals have no density on $\mathbb{R}^n$. Defining multivariate normality through a density would exclude it. This chapter uses a definition that does not mention densities at all: a random vector is normal when every linear combination of its entries is normal. The density, the moment generating function and the representation $\boldsymbol{\mu} + \mathbf{A}\mathbb{Z}$ all follow as theorems.

3.1.1 The univariate normal, including the degenerate case

For $\mu \in \mathbb{R}$ and $\sigma^2 > 0$, a random variable X has the $N(\mu, \sigma^2)$ distribution when it has density $(2\pi\sigma^2)^{-1/2} \exp\{-(x - \mu)^2/(2\sigma^2)\}$. We also allow $\sigma^2 = 0$, and take $N(\mu, 0)$ to be the point mass at μ . Treating constants as normal variables with zero variance costs nothing and removes a great many special cases later.

Lemma 3.1.1

Let $X \sim N(\mu, \sigma^2)$ with $\sigma^2 \geq 0$.

- (a) $\mathbf{E}e^{sX} = \exp(\mu s + \frac{1}{2}\sigma^2 s^2)$ for every real s .
- (b) A random variable whose moment generating function is $\exp(\mu s + \frac{1}{2}\sigma^2 s^2)$ on an interval around 0 has the $N(\mu, \sigma^2)$ distribution.
- (c) $aX + b \sim N(a\mu + b, a^2\sigma^2)$, and if $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent, then $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Proof.

- (a) For $\sigma^2 = 0$ this is $e^{\mu s}$. Otherwise, completing the square, $sx - (x - \mu)^2/(2\sigma^2) = \mu s + \sigma^2 s^2/2 - (x - \mu - \sigma^2 s)^2/(2\sigma^2)$, and the remaining Gaussian integral equals $(2\pi\sigma^2)^{1/2}$.
- (b) is the uniqueness theorem for moment generating functions (Theorem 2.6.1(a)), in its scalar form.
- (c) The moment generating function of $aX + b$ is $e^{bs} \mathbf{E}e^{(as)X} = \exp\{(a\mu + b)s + \frac{1}{2}a^2\sigma^2 s^2\}$. For the sum, independence gives $\mathbf{E}e^{s(X_1 + X_2)} = \mathbf{E}e^{sX_1} \mathbf{E}e^{sX_2}$, which is the normal moment generating function with the added parameters. Apply (b). ■

3.1.2 The definition

Definition 3.1.1 — Multivariate normal distribution

A random vector $\mathbf{Y} = (Y_1, \dots, Y_n)\mathbf{T}$ has a **multivariate normal distribution** if $\mathbf{a}\mathbf{T}\mathbf{Y}$ has a univariate normal distribution (possibly degenerate) for every $\mathbf{a} \in \mathbb{R}^n$. Such a $\mathbf{Y}$ is also said to be **jointly normal**, and its entries are said to be jointly normal random

variables.

Taking $\mathbf{a}$ to be a coordinate vector shows that each Y_i is normal, so $\mathbf{E}Y_i^2 < \infty$. By the Cauchy–Schwarz inequality every product $Y_i Y_j$ is integrable, so the mean vector $\boldsymbol{\mu} = \mathbf{E}\mathbf{Y}$ and the covariance matrix $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$ of Definition 2.1.2 exist. By Theorem 2.2.1, $\mathbf{E}(\mathbf{aT}\mathbf{Y}) = \mathbf{aT}\boldsymbol{\mu}$ and $\text{Var}(\mathbf{aT}\mathbf{Y}) = \mathbf{aT}\boldsymbol{\Sigma}\mathbf{a}$, and a normal variable is determined by its mean and variance. Hence

$$\mathbf{aT}\mathbf{Y} \sim N(\mathbf{aT}\boldsymbol{\mu}, \mathbf{aT}\boldsymbol{\Sigma}\mathbf{a}) \quad \text{for every } \mathbf{a} \in \mathbb{R}^n. \quad (3.1.1)$$

We write $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The notation claims that the joint distribution of $\mathbf{Y}$ depends only on $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. That is true, but it is not obvious: the definition fixes the law of each one-dimensional projection, and it is not yet clear that these laws pin down the joint law. The claim is proved in Theorem 3.2.1. Until then, $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ should be read as “normal, with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$ ”.

3.1.3 Existence

Theorem 3.1.2 — Existence and the representation $\boldsymbol{\mu} + \mathbf{A}\mathbb{Z}$

Let $Z_1, \dots, Z_m$ be independent $N(0, 1)$ variables and $\mathbb{Z} = (Z_1, \dots, Z_m)\mathbf{T}$.

- (a) $\mathbb{Z} \sim N_m(\mathbf{0}, \mathbf{I}_m)$.
- (b) For any $n \times m$ matrix $\mathbf{A}$ and $\boldsymbol{\mu} \in \mathbb{R}^n$, $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{A}\mathbb{Z}$ is multivariate normal with mean $\boldsymbol{\mu}$ and covariance $\mathbf{AAT}$.
- (c) For every $\boldsymbol{\mu} \in \mathbb{R}^n$ and every nonnegative definite $n \times n$ matrix $\boldsymbol{\Sigma}$ there is a random vector with the $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution, for example $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbb{Z}$ with $m = n$.

No other matrices occur: the covariance matrix of a normal vector is nonnegative definite.

Proof.

- (a) $\mathbf{aT}\mathbb{Z} = \sum_i a_i Z_i$ is a sum of independent normal variables $a_i Z_i \sim N(0, a_i^2)$, which is normal by Lemma 3.1.1(c).
- (b) $\mathbf{aT}\mathbf{Y} = \mathbf{aT}\boldsymbol{\mu} + (\mathbf{ATa})\mathbf{T}\mathbb{Z}$ is a constant plus a linear combination of $\mathbb{Z}$, hence normal by (a). Its moments come from Theorem 2.2.1: $\mathbf{E}\mathbf{Y} = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \mathbf{AAT} = \boldsymbol{\Sigma}$.
- (c) By Theorem 1.7.6 a nonnegative definite $\boldsymbol{\Sigma}$ has a symmetric nonnegative definite square root with $\boldsymbol{\Sigma}^{1/2}\boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}$. Take $\mathbf{A} = \boldsymbol{\Sigma}^{1/2}$ in (b). The last sentence is Theorem 2.2.3.

■

Any factorization $\boldsymbol{\Sigma} = \mathbf{AAT}$ serves in (b). The symmetric square root is convenient in proofs. The Cholesky factor, a lower triangular $\mathbf{L}$ with $\mathbf{LLT} = \boldsymbol{\Sigma}$, is cheaper to compute and is what software uses. A factor with fewer columns than rows is natural when $\boldsymbol{\Sigma}$ is singular. Once Theorem 3.2.1 is proved, all of these give the same distribution. The first listing checks this by simulation. It draws 400,000 vectors from a three-dimensional normal distribution with each of

two factors. Both sample covariance matrices agree with Σ to within 0.016 in every entry, and the quantiles of a linear combination match the exact normal quantiles.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(301)
mu = np.array([1.0, -2.0, 0.5])
Sigma = np.array([[4.0, 1.2, -0.8],
                  [1.2, 2.0, 0.6],
                  [-0.8, 0.6, 1.5]])
N = 400_000

def draw_normal(mu, A, size, rng):
    """Rows are independent copies of mu + A Z with Z ~ N(0, I)."""
    Z = rng.standard_normal((size, A.shape[1]))
    return mu + Z @ A.T

L = np.linalg.cholesky(Sigma)                # lower triangular, L L^T = Sigma
lam, U = np.linalg.eigh(Sigma)
root = U @ np.diag(np.sqrt(lam)) @ U.T       # symmetric square root

Y_chol = draw_normal(mu, L, N, rng)
Y_root = draw_normal(mu, root, N, rng)
for Y in (Y_chol, Y_root):
    print(np.round(np.cov(Y, rowvar=False) - Sigma, 3))
```

3.1.4 Singular normal distributions

When Σ is singular, [Definition 3.1.1](#) still applies, and the distribution is called **singular** or degenerate. If $\Sigma \mathbf{v} = \mathbf{0}$, then $\text{Var}(\mathbf{v}^T \mathbf{Y}) = \mathbf{v}^T \Sigma \mathbf{v} = 0$, so $\mathbf{v}^T \mathbf{Y} = \mathbf{v}^T \boldsymbol{\mu}$ with probability one. Every vector in $\mathcal{N}(\Sigma)$ gives such a linear constraint, and together they say that $\mathbf{Y}$ lies in the affine subspace $\boldsymbol{\mu} + \mathbb{C}(\Sigma)$ with probability one ([Theorem 2.2.3](#)). A subspace of dimension $\text{rank}(\Sigma) < n$ has Lebesgue measure zero, so no density on $\mathbb{R}^n$ is possible.

Example. (Three overlapping sums)

Let $\mathbf{Z} \sim N_3(\mathbf{0}, \mathbf{I})$ and $\mathbf{U} = (Z_1 + Z_2, Z_2 - Z_3, Z_1 + Z_3)^T$. Then $\mathbf{U} = \mathbf{A}\mathbf{Z}$ with rows $(1, 1, 0)$, $(0, 1, -1)$ and $(1, 0, 1)$, so $\mathbf{U}$ is normal with mean zero and covariance

$$\mathbf{A}\mathbf{A}^T = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}.$$

The vector $(1, -1, -1)^T$ lies in $\mathcal{N}(\mathbf{A}\mathbf{A}^T)$, and the rank is 2. Correspondingly $U_1 - U_2 - U_3 = 0$ for every outcome, not merely with probability one. Each U_i is $N(0, 2)$, each pair (U_i, U_j) has a nonsingular covariance matrix (determinant 3) and a bivariate density, and yet the triple lives on a plane.

In regression the singular case is the rule. With $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$, the fitted values and the residuals are normal vectors of ranks $\text{rank}(\mathbf{X})$ and $n - \text{rank}(\mathbf{X})$ ([Example \(Least squares under normal errors\)](#)). The last part of the listing draws a normal vector in $\mathbb{R}^3$ from a 3×2 factor. Every draw lies on the plane $\boldsymbol{\mu} + \mathbb{C}(\mathbf{A})$ to within 2.7×10^{-15} , which is rounding error.

3.1.5 Why linear combinations suffice

Definition 3.1.1 describes a joint distribution entirely through its one-dimensional projections. That this loses no information is a general fact about random vectors, the Cramér–Wold device proved with characteristic functions in [Proposition 2.6.2](#): if $\mathbf{a}^T \mathbf{Y}$ and $\mathbf{a}^T \mathbf{W}$ have the same distribution for every $\mathbf{a} \in \mathbb{R}^n$, then $\mathbf{Y}$ and $\mathbf{W}$ have the same distribution. We do not need it in this generality. For normal vectors the moment generating function is finite everywhere, and the next section runs the same argument with it. The device explains why the definition is natural. It also warns that all directions $\mathbf{a}$ matter. Normality of the n coordinate variables, which are only n directions, is not enough ([Section 3.3](#) and [Exercise \(B3\)](#)).

3.1.6 Exercises

3.1.6.1 A. Check your understanding

Exercise A1.

Let $Y_1, \dots, Y_n$ be independent with $Y_i \sim N(\mu_i, \sigma_i^2)$. Show directly from [Definition 3.1.1](#) that $\mathbf{Y} = (Y_1, \dots, Y_n)^T$ is multivariate normal, and find its mean and covariance. Show also that a constant vector $\mathbf{c}$ is $N_n(\mathbf{c}, \mathbf{0})$.

Exercise A2.

For the bivariate normal of [Example \(The bivariate normal\)](#), find the Cholesky factor $\mathbf{L}$ of Σ and write Y_1 and Y_2 in terms of two independent standard normals. Do the same with the symmetric square root when $\sigma_1 = \sigma_2 = 1$. Why do the two representations define the same distribution?

3.1.6.2 B. Practice

Exercise B1.

Let $Z \sim N(0, 1)$ and $\mathbf{Y} = (Z, 2Z - 1, 3 - Z)^T$. Show that $\mathbf{Y}$ is normal, find $\boldsymbol{\mu}$ and Σ , and find the rank of Σ . Describe the set in which $\mathbf{Y}$ lies, and write it as the solution set of two linear equations.

Solution. $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{A}Z$ with $\boldsymbol{\mu} = (0, -1, 3)^T$ and $\mathbf{A} = (1, 2, -1)^T$, so $\mathbf{Y}$ is normal by [Theorem 3.1.2](#) with

$$\Sigma = \mathbf{A}\mathbf{A}^T = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{pmatrix}, \quad \text{rank}(\Sigma) = 1.$$

$\mathbf{Y}$ lies on the line $\{\boldsymbol{\mu} + s\mathbf{A} : s \in \mathbb{R}\}$. The null space of Σ is $\mathbf{A}^\perp$, spanned by $(2, -1, 0)^T$ and $(1, 0, 1)^T$. The corresponding constraints $\mathbf{v}^T \mathbf{Y} = \mathbf{v}^T \boldsymbol{\mu}$ are $2Y_1 - Y_2 = 1$ and $Y_1 + Y_3 = 3$, and both hold identically.

3.2 Moment generating function and density

3.2.1 Moment generating functions of random vectors

The moment generating function of a random vector $\mathbf{Y} \in \mathbb{R}^n$ is $M_{\mathbf{Y}}(\mathbf{t}) = \mathbf{E} \exp(\mathbf{t}^T \mathbf{Y})$, and [Theorem 2.6.1](#) proves the two properties we need when it is finite on a neighbourhood of $\mathbf{0}$. *Uniqueness*: two random vectors whose moment generating functions are finite and equal near $\mathbf{0}$ have the same distribution. *Factorization*: two blocks $\mathbf{Y}_1, \mathbf{Y}_2$ are independent iff $M_{\mathbf{Y}}(\mathbf{t}) = M_{\mathbf{Y}_1}(\mathbf{t}_1)M_{\mathbf{Y}_2}(\mathbf{t}_2)$ near $\mathbf{0}$. The second extends to k blocks by induction. If $M_{\mathbf{Y}}$ is the product of the k block functions, setting $\mathbf{t}_1 = \mathbf{0}$ shows that $(\mathbf{Y}_2, \dots, \mathbf{Y}_k)$ has the product of the last $k-1$, so $\mathbf{Y}_1$ is independent of $(\mathbf{Y}_2, \dots, \mathbf{Y}_k)$, whose blocks are mutually independent by the induction hypothesis. The converse is immediate.

Theorem 3.2.1 — Moment generating function

A random vector $\mathbf{Y}$ is $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ iff

$$M_{\mathbf{Y}}(\mathbf{t}) = \exp(\mathbf{t}^T \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^T \boldsymbol{\Sigma} \mathbf{t}) \quad \text{for every } \mathbf{t} \in \mathbb{R}^n. \quad (3.2.1)$$

Consequently the distribution of a normal vector is determined by its mean vector and covariance matrix.

Proof. Suppose $\mathbf{Y}$ is normal with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. By [Eq. 3.1.1](#), $\mathbf{t}^T \mathbf{Y} \sim N(\mathbf{t}^T \boldsymbol{\mu}, \mathbf{t}^T \boldsymbol{\Sigma} \mathbf{t})$, and $M_{\mathbf{Y}}(\mathbf{t})$ is the moment generating function of this scalar evaluated at $s = 1$. [Lemma 3.1.1\(a\)](#) gives [Eq. 3.2.1](#).

Conversely, suppose [Eq. 3.2.1](#) holds for a symmetric $\boldsymbol{\Sigma}$. Fix $\mathbf{a}$. The moment generating function of $\mathbf{a}^T \mathbf{Y}$ at s is $M_{\mathbf{Y}}(s\mathbf{a}) = \exp\{(\mathbf{a}^T \boldsymbol{\mu})s + \frac{1}{2}(\mathbf{a}^T \boldsymbol{\Sigma} \mathbf{a})s^2\}$. The coefficient $\mathbf{a}^T \boldsymbol{\Sigma} \mathbf{a}$ is nonnegative, since otherwise this function would tend to zero as $s \rightarrow \pm\infty$, whereas every random variable X has $\mathbf{E}e^{sX} + \mathbf{E}e^{-sX} \geq \mathbf{E}e^{s|X|} \geq 1$. By [Lemma 3.1.1\(b\)](#), $\mathbf{a}^T \mathbf{Y} \sim N(\mathbf{a}^T \boldsymbol{\mu}, \mathbf{a}^T \boldsymbol{\Sigma} \mathbf{a})$. So $\mathbf{Y}$ is normal, and matching means and variances of all linear combinations shows $\mathbf{E}\mathbf{Y} = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$.

Finally, two normal vectors with the same $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ have the same moment generating function [Eq. 3.2.1](#), which is finite everywhere, so they have the same distribution by the uniqueness property. ■

The last sentence is what makes $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ a well-defined distribution. Combined with [Theorem 3.1.2](#), it says that $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ holds iff $\mathbf{Y}$ has the same distribution as $\boldsymbol{\mu} + \mathbf{A}\mathbb{Z}$ for some, and then every, matrix $\mathbf{A}$ with $\mathbf{A}\mathbf{A}^T = \boldsymbol{\Sigma}$. The number of columns of $\mathbf{A}$, and so the length of $\mathbb{Z}$, does not matter. Some books take this representation as the definition ([Christensen 2020](#); [Seber and Lee 2003](#)). The proof above shows the two definitions agree.

The moment generating function also encodes moments. Differentiating [Eq. 3.2.1](#) at $\mathbf{t} = \mathbf{0}$ gives $\partial M_{\mathbf{Y}}/\partial \mathbf{t} = \boldsymbol{\mu}$ and $\partial^2 M_{\mathbf{Y}}/\partial \mathbf{t} \partial \mathbf{t}^T = \boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^T = \mathbf{E}(\mathbf{Y}\mathbf{Y}^T)$. Higher derivatives give the fourth moments needed for variances of quadratic forms in [Chapter 4](#) ([Exercise \(B1\)](#)).

3.2.2 The density

Theorem 3.2.2 — Density

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

- (a) If $\boldsymbol{\Sigma}$ is positive definite, $\mathbf{Y}$ has the density

$$f(\mathbf{y}) = (2\pi)^{-n/2} \det(\boldsymbol{\Sigma})^{-1/2} \exp\left\{-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})\right\}, \quad \mathbf{y} \in \mathbb{R}^n. \quad (3.2.2)$$

- (b) Conversely, a random vector with density [Eq. 3.2.2](#), for some positive definite $\boldsymbol{\Sigma}$, is $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.
- (c) If $\boldsymbol{\Sigma}$ is singular, $\mathbf{Y}$ has no density on $\mathbb{R}^n$.

Proof.

- (a) By [Theorem 3.2.1](#), $\mathbf{Y}$ has the same distribution as $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbb{Z}$ with $\mathbb{Z} \sim N_n(\mathbf{0}, \mathbf{I})$, so it suffices to find the density of the latter. The entries of $\mathbb{Z}$ are independent standard normals, so $\mathbb{Z}$ has density $\prod_i (2\pi)^{-1/2} e^{-z_i^2/2} = (2\pi)^{-n/2} \exp(-\frac{1}{2}\mathbf{z}^T \mathbf{z})$. Since $\boldsymbol{\Sigma}$ is positive definite, so is $\boldsymbol{\Sigma}^{1/2}$ (its eigenvalues are the positive square roots of those of $\boldsymbol{\Sigma}$), and the map $\mathbf{z} \mapsto \mathbf{y} = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{z}$ is an affine bijection of $\mathbb{R}^n$ with inverse $\mathbf{z} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{y} - \boldsymbol{\mu})$. The change-of-variables formula multiplies the density of $\mathbb{Z}$ at $\mathbf{z}(\mathbf{y})$ by $|\det \boldsymbol{\Sigma}^{-1/2}| = \det(\boldsymbol{\Sigma})^{-1/2}$. Finally $\mathbf{z}^T \mathbf{z} = (\mathbf{y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})$.
- (b) A density determines a distribution, and by (a) the $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution has this density.
- (c) $\mathbf{Y}$ lies with probability one in $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma})$, an affine subspace of dimension less than n . It has Lebesgue measure zero, so it would receive probability zero under any distribution with a density. ■

A singular normal vector does have a density with respect to the natural measure on the affine subspace where it lives ([Proposition 3.3.4](#) and [Exercise \(B2\)](#)), but that is rarely needed. In this book, arguments are made with the moment generating function or with linear transformations, which work in every case, and densities are used only for pictures and likelihoods.

Example. (The bivariate normal)

For $n = 2$, write $\sigma_1^2, \sigma_2^2 > 0$ for the variances and ρ for the correlation. Then

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}, \quad \det \boldsymbol{\Sigma} = \sigma_1^2\sigma_2^2(1 - \rho^2),$$

so Σ is positive definite iff $|\rho| < 1$. With standardized coordinates $u_i = (y_i - \mu_i)/\sigma_i$, the density Eq. 3.2.2 is

$$f(y_1, y_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{u_1^2 - 2\rho u_1 u_2 + u_2^2}{2(1-\rho^2)}\right\}.$$

When $\rho = 0$ it factors into the two univariate normal densities. When $\rho = \pm 1$ there is no density: $\text{Var}(u_2 \mp u_1) = 0$ in standardized units, so all the probability lies on a line.

3.2.3 Contours and the Mahalanobis distance

For positive definite Σ , the density Eq. 3.2.2 depends on $\mathbf{y}$ only through

$$\Delta^2(\mathbf{y}) = (\mathbf{y} - \boldsymbol{\mu})\mathbf{T}\Sigma^{-1}(\mathbf{y} - \boldsymbol{\mu}),$$

the squared Mahalanobis distance from $\mathbf{y}$ to $\boldsymbol{\mu}$ introduced in Section 2.2. The density is a decreasing function of Δ^2 , so its contours are the ellipsoids $\{\mathbf{y} : \Delta^2(\mathbf{y}) = c^2\}$. Their shape comes from the spectral decomposition $\Sigma = \sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^T$ (Theorem 1.7.2). In the coordinates $w_i = \mathbf{u}_i^T(\mathbf{y} - \boldsymbol{\mu})$ we have $\Delta^2 = \sum_i w_i^2/\lambda_i$, so the ellipsoid has principal axes along the eigenvectors $\mathbf{u}_i$, with half-lengths $c\sqrt{\lambda_i}$. Large-variance directions are long axes. The Mahalanobis distance measures distance from the centre in units of standard deviation along each principal direction. The volume of $\{\mathbf{y} : \Delta^2(\mathbf{y}) \leq c^2\}$ is proportional to $\prod_i \sqrt{\lambda_i} = \det(\Sigma)^{1/2}$, so the generalized variance $\det \Sigma$ (Section 2.1) measures the size of the region the distribution occupies. The factor $\det(\Sigma)^{-1/2}$ in Eq. 3.2.2 spreads the same total probability over that region: a larger generalized variance means a flatter density.

Proposition 3.2.3

If $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ with Σ positive definite, then $\Delta^2(\mathbf{Y})$ has the chi-squared distribution with n degrees of freedom, $\chi^2(n)$, which is $\chi^2(n, 0)$ in the notation of Definition 4.1.1. So the ellipsoid $\{\mathbf{y} : \Delta^2(\mathbf{y}) \leq q\}$, where q is the $1 - \alpha$ quantile of $\chi^2(n)$, has probability exactly $1 - \alpha$.

Proof. Put $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{Y} - \boldsymbol{\mu})$. For every $\mathbf{a}$, $\mathbf{a}^T \mathbf{Z} = (\Sigma^{-1/2} \mathbf{a})^T \mathbf{Y} - \mathbf{a}^T \Sigma^{-1/2} \boldsymbol{\mu}$ is a linear combination of $\mathbf{Y}$ minus a constant, hence normal by Definition 3.1.1, so $\mathbf{Z}$ is normal. Its mean is zero and its covariance is $\mathbf{I}$ (Proposition 2.2.4(a)). By Theorem 3.2.1 its moment generating function is $\exp(\frac{1}{2} \mathbf{t}^T \mathbf{t}) = \prod_i \exp(\frac{1}{2} t_i^2)$, which factorizes, so its entries are independent standard normals. $\Delta^2(\mathbf{Y}) = \mathbf{Z}^T \mathbf{Z} = \sum_i Z_i^2$ is a sum of n independent squared standard normals, which is the definition of $\chi^2(n)$. ■

Chapter 4 generalizes this considerably, to $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ with a nonzero mean and a general $\mathbf{A}$ (Theorem 4.3.2). Because the density is largest inside these ellipsoids, they are the smallest regions with their probability content, which is why they reappear as confidence regions for $\boldsymbol{\beta}$ in Chapter 12.

Computing with the density. Neither Σ^{-1} nor $\det \Sigma$ should be formed explicitly. With the Cholesky factorization $\Sigma = \mathbf{L} \mathbf{L}^T$, we have $\Delta^2(\mathbf{y}) = \|\mathbf{L}^{-1}(\mathbf{y} - \boldsymbol{\mu})\|^2$, computed by a triangular solve, and $\log \det \Sigma = 2 \sum_i \log l_{ii}$. The listing evaluates the log density this way, checks it

against `scipy`, and counts how many of the simulated vectors fall inside the 95% ellipsoid. The fraction is 0.9500. The average of Δ^2 over the draws is 3.004 and its variance is 5.994, against the $\chi^2(3)$ values 3 and 6.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(301)
mu = np.array([1.0, -2.0, 0.5])
Sigma = np.array([[4.0, 1.2, -0.8],
                  [1.2, 2.0, 0.6],
                  [-0.8, 0.6, 1.5]])

N = 400_000
a = np.array([0.3, -1.0, 2.0])
q = [0.05, 0.25, 0.5, 0.75, 0.95]
exact = stats.norm.ppf(q, loc=a @ mu, scale=np.sqrt(a @ Sigma @ a))
cov_err = max(np.abs(np.cov(Y, rowvar=False) - Sigma).max() for Y in (Y_chol, Y_root))

def draw_normal(mu, A, size, rng):
    """Rows are independent copies of mu + A Z with Z ~ N(0, I)."""
    Z = rng.standard_normal((size, A.shape[1]))
    return mu + Z @ A.T

L = np.linalg.cholesky(Sigma)                # lower triangular, L L^T = Sigma
lam, U = np.linalg.eigh(Sigma)
root = U @ np.diag(np.sqrt(lam)) @ U.T       # symmetric square root

Y_chol = draw_normal(mu, L, N, rng)
Y_root = draw_normal(mu, root, N, rng)
for Y in (Y_chol, Y_root):
    print(np.round(np.cov(Y, rowvar=False) - Sigma, 3))

def normal_logpdf(y, mu, Sigma):
    """log density of N(mu, Sigma) at the rows of y, Sigma positive definite."""
    L = np.linalg.cholesky(Sigma)
    w = np.linalg.solve(L, (y - mu).T)       # w = L^{-1}(y - mu), so w^T w =
    Delta^2
    D2 = np.sum(w**2, axis=0)
    logdet = 2.0 * np.sum(np.log(np.diag(L)))
    return -0.5 * (len(mu) * np.log(2 * np.pi) + logdet + D2), D2

logf, D2 = normal_logpdf(Y_chol, mu, Sigma)
inside = np.mean(D2 <= stats.chi2.ppf(0.95, df=3))
print(f"fraction inside the 95% ellipsoid: {inside:.4f}")
```

3.2.4 Exercises

3.2.4.1 A. Check your understanding

Exercise A1.

Show that $f(y_1, y_2) = k \exp\{-\frac{1}{2}(2y_1^2 + 2y_1y_2 + y_2^2 - 4y_1 - 2y_2)\}$ is a bivariate normal density for a suitable constant k . Find k , μ , Σ and the correlation.

Solution. The quadratic part is $\mathbf{y}^T \mathbf{P} \mathbf{y}$ with $\mathbf{P} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, which is positive definite with $\det \mathbf{P} = 1$. Matching $(\mathbf{y} - \boldsymbol{\mu})^T \mathbf{P} (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{y}^T \mathbf{P} \mathbf{y} - 2\boldsymbol{\mu}^T \mathbf{P} \mathbf{y} + \boldsymbol{\mu}^T \mathbf{P} \boldsymbol{\mu}$ with the linear terms requires $\mathbf{P}\boldsymbol{\mu} = (2, 1)^T$, so $\boldsymbol{\mu} = (1, 0)^T$ and $\boldsymbol{\mu}^T \mathbf{P} \boldsymbol{\mu} = 2$. The exponent is $-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^T \mathbf{P} (\mathbf{y} - \boldsymbol{\mu}) + 1$. By Theorem 3.2.2(b) this is the $N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ density with $\boldsymbol{\Sigma} = \mathbf{P}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ provided $ke = (2\pi)^{-1}(\det \boldsymbol{\Sigma})^{-1/2} = (2\pi)^{-1}$. So $k = e^{-1}/(2\pi)$, and the correlation is $-1/\sqrt{2}$.

Exercise A2.

Let $\boldsymbol{\Sigma}$ have unit variances and correlation ρ , with $|\rho| < 1$. Find the principal axes and half-lengths of the ellipse $\Delta^2 = c^2$, show that its area is $\pi c^2 \sqrt{1 - \rho^2}$, and show that it contains probability $1 - e^{-c^2/2}$.

3.2.4.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{X} = \mathbf{Y} - \boldsymbol{\mu}$. Expand the moment generating function of $\mathbf{X}$ in a power series to show that all odd-order moments of $\mathbf{X}$ vanish and

$$\mathbf{E}(X_i X_j X_k X_l) = \sigma_{ij} \sigma_{kl} + \sigma_{ik} \sigma_{jl} + \sigma_{il} \sigma_{jk}.$$

Deduce that $\text{Var}(\mathbf{X}^T \mathbf{A} \mathbf{X}) = 2 \text{tr}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma})$ for symmetric $\mathbf{A}$.

Solution. By Theorem 3.2.1, $M_{\mathbf{X}}(\mathbf{t}) = \exp(\frac{1}{2} \mathbf{t}^T \boldsymbol{\Sigma} \mathbf{t}) = \sum_{m \geq 0} (\mathbf{t}^T \boldsymbol{\Sigma} \mathbf{t})^m / (2^m m!)$, a series containing only terms of even degree in $\mathbf{t}$. Mixed moments of order d are the d th partial derivatives at $\mathbf{0}$, which come from the degree- d terms, so odd-order moments vanish. The degree-four term is $p(\mathbf{t}) = \frac{1}{8} (\mathbf{t}^T \boldsymbol{\Sigma} \mathbf{t})^2 = T(\mathbf{t}, \mathbf{t}, \mathbf{t}, \mathbf{t})$, where $T(\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{x}) = \frac{1}{8} (\mathbf{u}^T \boldsymbol{\Sigma} \mathbf{v})(\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{x})$ is 4-linear. Differentiating $T(\mathbf{t}, \mathbf{t}, \mathbf{t}, \mathbf{t})$ once in each of the directions $\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_k, \mathbf{e}_l$ gives the sum of T over the 24 ways of placing these four vectors in the four slots. This holds whether or not the indices are distinct, since it is the product rule applied to a multilinear form. A placement contributes $\frac{1}{8} \sigma_{ab} \sigma_{cd}$, where the first two slots receive the indices a, b and the last two receive c, d . Each of the three ways of splitting the four *directions* into two pairs, $\{ij, kl\}$, $\{ik, jl\}$, $\{il, jk\}$, arises from 8 placements (choose which pair goes first, and the order within each pair), which gives the formula. Then, for symmetric $\mathbf{A}$,

$$\mathbf{E}(\mathbf{X}^T \mathbf{A} \mathbf{X})^2 = \sum_{i,j,k,l} a_{ij} a_{kl} \mathbf{E}(X_i X_j X_k X_l) = (\text{tr } \mathbf{A} \boldsymbol{\Sigma})^2 + 2 \text{tr}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}),$$

and subtracting $(\mathbf{E} \mathbf{X}^T \mathbf{A} \mathbf{X})^2 = (\text{tr } \mathbf{A} \boldsymbol{\Sigma})^2$ gives the variance.

Exercise B2.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\text{rank}(\boldsymbol{\Sigma}) = r < n$, and use the notation of Proposition 3.3.4. Show that $\mathbf{V} = \mathbb{Q}^T (\mathbf{Y} - \boldsymbol{\mu})$ has a density on $\mathbb{R}^r$, and that this density can be written as $(2\pi)^{-r/2} (\lambda_1 \cdots \lambda_r)^{-1/2} \exp\{-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^+ (\mathbf{y} - \boldsymbol{\mu})\}$ with $\mathbf{y} = \boldsymbol{\mu} + \mathbb{Q} \mathbf{v}$, where $\boldsymbol{\Sigma}^+ = \mathbb{Q} \boldsymbol{\Lambda}^{-1} \mathbb{Q}^T$ is the Moore–Penrose inverse.

Exercise B3.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ positive definite and let $\boldsymbol{\nu}$ be a fixed vector. Show that $(\mathbf{Y} - \boldsymbol{\nu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{Y} - \boldsymbol{\nu})$ has the noncentral chi-squared distribution $\chi^2(n, \gamma)$ with $\gamma = (\boldsymbol{\mu} - \boldsymbol{\nu})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu} - \boldsymbol{\nu})$, in the convention of [Definition 4.1.1](#), and find its mean.

3.2.4.3 C. Going deeper**Exercise C1.**

Let $\mathbf{Y}$ have density [Eq. 3.2.2](#), and let $E = \{\mathbf{y} : \Delta^2(\mathbf{y}) \leq q\}$. Show that every Borel set A with $\Pr(\mathbf{Y} \in A) \geq \Pr(\mathbf{Y} \in E)$ has volume at least that of E . *Hint:* compare $\int_{A \setminus E} f$ with $\int_{E \setminus A} f$.

Solution. The density is a decreasing function of Δ^2 , so with $c = (2\pi)^{-n/2} \det(\boldsymbol{\Sigma})^{-1/2} e^{-q/2} > 0$ we have $f \geq c$ on E and $f < c$ off E . We may assume A has finite volume. Subtracting $\Pr(\mathbf{Y} \in A \cap E)$ from both sides of $\Pr(\mathbf{Y} \in A) \geq \Pr(\mathbf{Y} \in E)$ gives $\int_{A \setminus E} f \geq \int_{E \setminus A} f$. Hence

$$c \operatorname{vol}(A \setminus E) \geq \int_{A \setminus E} f \geq \int_{E \setminus A} f \geq c \operatorname{vol}(E \setminus A),$$

so $\operatorname{vol}(A \setminus E) \geq \operatorname{vol}(E \setminus A)$. Adding $\operatorname{vol}(A \cap E)$ to both sides gives $\operatorname{vol}(A) \geq \operatorname{vol}(E)$. The argument uses only that E is a set on which f is at least as large as anywhere outside it, so it applies to the level sets of any density.

3.3 Linear transformations and marginals

3.3.1 Affine maps

Theorem 3.3.1 — Affine transformations

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, let $\mathbf{A}$ be any $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$. Then

$$\mathbf{A}\mathbf{Y} + \mathbf{b} \sim N_m(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T).$$

Proof. For $\mathbf{c} \in \mathbb{R}^m$, $\mathbf{c}^T(\mathbf{A}\mathbf{Y} + \mathbf{b}) = (\mathbf{A}^T\mathbf{c})^T\mathbf{Y} + \mathbf{c}^T\mathbf{b}$ is a linear combination of $\mathbf{Y}$ plus a constant, which is normal by [Definition 3.1.1](#) and [Lemma 3.1.1\(c\)](#). So $\mathbf{A}\mathbf{Y} + \mathbf{b}$ is normal, and its mean and covariance are given by [Theorem 2.2.1](#). ■

The proof is one line because the definition was chosen to make it one line. There is no rank condition on $\mathbf{A}$. If $\mathbf{A}$ has more rows than $\operatorname{rank}(\boldsymbol{\Sigma})$, the result is a singular normal vector, and that is allowed. With a density-based definition, the same theorem needs $\mathbf{A}$ to have full row rank and $\boldsymbol{\Sigma}$ to be positive definite, and the most useful cases fall outside it. The moment generating function gives a second proof: $\mathbf{E}e^{t^T(\mathbf{A}\mathbf{Y} + \mathbf{b})} = e^{t^T\mathbf{b}} M_{\mathbf{Y}}(\mathbf{A}^T t) = \exp\{t^T(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}) + \frac{1}{2} t^T \mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^T t\}.$

Corollary 3.3.2 — Marginal distributions

Every subvector of a normal vector is normal. If $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{Y}_1$ consists of the entries with indices in a set I , then $\mathbf{Y}_1 \sim N(\boldsymbol{\mu}_I, \boldsymbol{\Sigma}_{II})$, where $\boldsymbol{\mu}_I$ and $\boldsymbol{\Sigma}_{II}$ keep the entries, rows and columns indexed by I . In particular $Y_i \sim N(\mu_i, \sigma_{ii})$.

Proof. $\mathbf{Y}_1 = \mathbf{A}\mathbf{Y}$, where the rows of $\mathbf{A}$ are the coordinate vectors $\mathbf{e}_i\mathbf{T}$, $i \in I$. Then $\mathbf{A}\boldsymbol{\mu} = \boldsymbol{\mu}_I$ and $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\mathbf{T} = \boldsymbol{\Sigma}_{II}$. ■

Corollary 3.3.3 — Standardization and rotation

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

- (a) If $\boldsymbol{\Sigma}$ is positive definite and $\boldsymbol{\Sigma} = \mathbf{A}\mathbf{A}\mathbf{T}$ with $\mathbf{A}$ square, then $\mathbf{A}^{-1}(\mathbf{Y} - \boldsymbol{\mu}) \sim N_n(\mathbf{0}, \mathbf{I})$. Its entries are independent standard normal variables.
- (b) If $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$ and $\mathbf{Q}$ is an $n \times n$ orthogonal matrix, then $\mathbf{Q}\mathbf{T}\mathbf{Y} \sim N_n(\mathbf{Q}\mathbf{T}\boldsymbol{\mu}, \sigma^2\mathbf{I})$.

Proof. Both follow from [Theorem 3.3.1](#): $\mathbf{A}^{-1}\mathbf{A}\mathbf{A}\mathbf{T}(\mathbf{A}^{-1})\mathbf{T} = \mathbf{I}$ and $\mathbf{Q}\mathbf{T}(\sigma^2\mathbf{I})\mathbf{Q} = \sigma^2\mathbf{I}$. Independence in (a) was shown in the proof of [Proposition 3.2.3](#). ■

Part (b) is the probabilistic basis of the canonical form of the linear model. Spherical normal errors look the same in every orthonormal coordinate system, so one may rotate to coordinates adapted to $\mathbb{C}(\mathbf{X})$, in which fitted values and residuals occupy separate independent coordinates. [Chapter 4](#) uses this rotation to find the distributions of sums of squares.

3.3.2 The rank of a normal vector

[Theorem 3.1.2](#) writes a normal vector as an affine function of independent standard normals *in distribution*. When $\boldsymbol{\Sigma}$ is singular the representation can be made exact, with as few standard normals as the rank.

Proposition 3.3.4

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\text{rank}(\boldsymbol{\Sigma}) = r \geq 1$, and let $\boldsymbol{\Sigma} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}\mathbf{T}$, where $\mathbf{Q}$ is $n \times r$ with orthonormal columns and $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_r)$ holds the positive eigenvalues. Put $\mathbf{W} = \boldsymbol{\Lambda}^{-1/2}\mathbf{Q}\mathbf{T}(\mathbf{Y} - \boldsymbol{\mu})$. Then $\mathbf{W} \sim N_r(\mathbf{0}, \mathbf{I}_r)$, and

$$\mathbf{Y} = \boldsymbol{\mu} + \mathbf{Q}\boldsymbol{\Lambda}^{1/2}\mathbf{W} \quad \text{with probability one.}$$

Proof. By [Theorem 3.3.1](#), $\mathbf{W}$ is normal with covariance $\boldsymbol{\Lambda}^{-1/2}\mathbf{Q}\mathbf{T}\mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}\mathbf{T}\mathbf{Q}\boldsymbol{\Lambda}^{-1/2} = \mathbf{I}_r$. Next, $\mathbf{Q}\boldsymbol{\Lambda}^{1/2}\mathbf{W} = \mathbf{Q}\mathbf{Q}\mathbf{T}(\mathbf{Y} - \boldsymbol{\mu})$, and $\mathbf{Q}\mathbf{Q}\mathbf{T}$ is the symmetric idempotent matrix with range $\mathbb{C}(\boldsymbol{\Sigma})$ (the orthogonal projection onto it, in the language of [Chapter 6](#)). Since $\mathbf{Y} - \boldsymbol{\mu} \in \mathbb{C}(\boldsymbol{\Sigma})$ with probability one ([Theorem 2.2.3](#)), $\mathbf{Q}\mathbf{Q}\mathbf{T}(\mathbf{Y} - \boldsymbol{\mu}) = \mathbf{Y} - \boldsymbol{\mu}$ with

probability one. ■

So a normal vector of rank r is exactly r independent standard normals placed in $\mathbb{R}^n$ by an injective affine map. [Exercise \(B4\)](#) gives the same construction for any random vector, with uncorrelated entries in place of independent normal ones. Singular normal vectors are lower-dimensional normal vectors in disguise.

Example. (Least squares under normal errors)

Let $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ with $\mathbf{X}$ of full column rank p , and let $\mathbf{M} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$, the symmetric idempotent matrix of rank p with $\mathbf{M}\mathbf{X} = \mathbf{X}$ (Chapter 5 and [Chapter 6](#)). By [Theorem 3.3.1](#):

- (a) $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y} \sim N_p(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1})$, a nonsingular normal vector;
- (b) $\boldsymbol{\lambda}^T\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\lambda}^T\boldsymbol{\beta}, \sigma^2\boldsymbol{\lambda}^T(\mathbf{X}^T\mathbf{X})^{-1}\boldsymbol{\lambda})$ for every $\boldsymbol{\lambda} \in \mathbb{R}^p$;
- (c) the fitted values $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{M})$, singular of rank p ;
- (d) the residuals $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{Y} \sim N_n(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{M}))$, singular of rank $n - p$.

No density for $\hat{\mathbf{Y}}$ or $\hat{\boldsymbol{\varepsilon}}$ was needed, and none exists.

3.3.3 Independent normal vectors

Lemma 3.3.5

If $\mathbf{Y}_1 \sim N_{n_1}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ and $\mathbf{Y}_2 \sim N_{n_2}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$ are independent, then $(\mathbf{Y}_1^T, \mathbf{Y}_2^T)^T$ is jointly normal with mean $(\boldsymbol{\mu}_1^T, \boldsymbol{\mu}_2^T)^T$ and block-diagonal covariance $\text{diag}(\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2)$. Consequently, if also $n_1 = n_2$, then $\mathbf{Y}_1 + \mathbf{Y}_2 \sim N(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)$.

Proof. $\mathbf{a}_1^T\mathbf{Y}_1 + \mathbf{a}_2^T\mathbf{Y}_2$ is a sum of independent normal variables, which is normal by [Lemma 3.1.1\(c\)](#). Independence makes the cross-covariance zero. The sum is $[\mathbf{I}, \mathbf{I}]$ applied to the stacked vector. ■

Independence is essential in this lemma. Two normal vectors defined on the same probability space need not be jointly normal, as we now see.

3.3.4 Normal marginals are not enough

[Corollary 3.3.2](#) cannot be reversed. A vector whose entries are each normal need not be jointly normal. Two examples make the point, one extreme and one realistic.

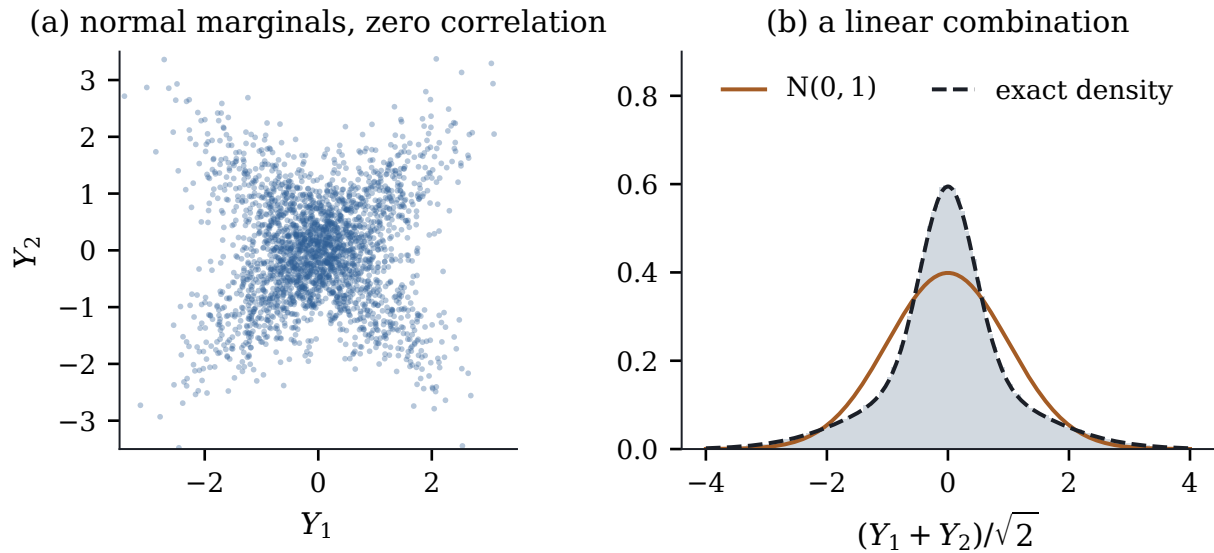


Figure 3.3.1: An equal mixture of bivariate normals with correlations ± 0.8 . (a) Both margins are exactly $N(0, 1)$ and the correlation is zero, but the scatter is an X, not an ellipse. (b) The linear combination $(Y_1 + Y_2)/\sqrt{2}$ has variance one but is sharply peaked and heavy-tailed compared with $N(0, 1)$. By [Definition 3.1.1](#), one non-normal direction is enough to rule out joint normality.

Example. (A random sign)

Let $Z \sim N(0, 1)$ and let S be independent of Z with $\Pr(S = 1) = \Pr(S = -1) = \frac{1}{2}$. Put $W = SZ$. By symmetry of Z , $W \sim N(0, 1)$. Also $\text{Cov}(Z, W) = \mathbf{E}(S)\mathbf{E}(Z^2) = 0$. But $Z + W = (1 + S)Z$ equals zero with probability $\frac{1}{2}$ and is otherwise $N(0, 4)$. A variable with an atom at zero and a continuous part is not normal, so (Z, W) is not jointly normal. And Z and W are far from independent: $|W| = |Z|$ always.

Example. (A mixture of two correlations)

Fix $0 < \rho < 1$. With probability $\frac{1}{2}$ draw (Y_1, Y_2) from the bivariate normal with unit variances and correlation ρ , and otherwise from the one with correlation $-\rho$. Both components have $N(0, 1)$ margins, so the mixture does too, exactly. The covariance is $\frac{1}{2}\rho + \frac{1}{2}(-\rho) = 0$. The pair is nevertheless dependent: since $\mathbf{E}(Y_1^2 Y_2^2) = 1 + 2\rho^2$ for a standard bivariate normal with correlation r , the squares have correlation ρ^2 , not zero. The direction $\mathbf{a} = (1, 1)\mathbf{T}/\sqrt{2}$ exposes the failure of joint normality. $S = (Y_1 + Y_2)/\sqrt{2}$ is an equal mixture of $N(0, 1 + \rho)$ and $N(0, 1 - \rho)$. It has variance 1 but kurtosis $\mathbf{E}S^4 = \frac{3}{2}\{(1 + \rho)^2 + (1 - \rho)^2\} = 3(1 + \rho^2)$, where a normal variable would have 3.

[Figure 3.3.1](#) shows a simulation with $\rho = 0.8$ and 200,000 draws. Kolmogorov–Smirnov tests of normality for the two margins give p -values 0.84 and 0.29. The sample correlation is -0.0033 , the correlation of the squares is 0.640 (theory 0.64), and the kurtosis of S is 4.88 (theory 4.92). The Kolmogorov–Smirnov distance of S from $N(0, 1)$ is 0.067, which with this many draws rejects normality at any conventional level.

```
import numpy as np
from scipy import integrate, stats
rng = np.random.default_rng(302)
```

```

rho, N = 0.8, 200_000

sign = rng.choice([-1.0, 1.0], size=N)          # which component each draw uses
Z1, Z2 = rng.standard_normal((2, N))
Y1 = Z1
Y2 = sign * rho * Z1 + np.sqrt(1 - rho**2) * Z2 # corr(Y1, Y2 | sign) = sign * rho

S = (Y1 + Y2) / np.sqrt(2)                     # a direction that is not normal
print("marginal KS p-values:", stats.kstest(Y1, "norm").pvalue, stats.kstest(Y2, "
    norm").pvalue)
print("correlation:", np.corrcoef(Y1, Y2)[0, 1])
print("corr of squares:", np.corrcoef(Y1**2, Y2**2)[0, 1])
print("kurtosis of S:", stats.kurtosis(S, fisher=False), " theory:", 3 * (1 + rho
    **2))
print("KS p-value of S:", stats.kstest(S, "norm").pvalue)

```

Idea. Normality of a vector is a property of *every direction*, not of the coordinate axes. Normal margins with zero correlation can hide strong dependence. The theorems of the next two sections, which turn zero covariance into independence and give linear conditional means, need *joint* normality. They are not about variables that merely happen to be normal one at a time.

3.3.5 Exercises

3.3.5.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Y} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu} = (0, 1, 2)\mathbf{T}$ and $\boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$. Find the joint distribution of $U = Y_1 + Y_2 + Y_3$, $V = Y_1 - Y_3$ and $W = Y_1 - 2Y_2 + Y_3$. Which of the pairs are independent? Is V independent of (U, W) ?

Exercise A2.

In [Example \(Least squares under normal errors\)](#), show that the i th residual is $N(0, \sigma^2(1 - m_{ii}))$, where m_{ii} is the i th diagonal entry of $\mathbf{M}$, and that it is zero with probability one iff $m_{ii} = 1$.

3.3.5.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$ and let $\mathbf{Q}$ be an orthogonal matrix whose first row is $n^{-1/2}\mathbf{1T}$. Show that the entries of $\mathbf{V} = \mathbf{QY}$ are independent, with $V_1 \sim N(\sqrt{n}\mu, \sigma^2)$ and $V_i \sim N(0, \sigma^2)$ for $i \geq 2$. Show that $\sum_i (Y_i - \bar{Y})^2 = \sum_{i \geq 2} V_i^2$, and conclude again that $\bar{Y}$ and S^2 are independent.

Exercise B2.

Let $Z \sim N(0, 1)$ and, for $c > 0$, let $W = Z$ if $|Z| \leq c$ and $W = -Z$ otherwise. Show that $W \sim N(0, 1)$, that there is a c with $\text{Cov}(Z, W) = 0$, and that (Z, W) is not jointly normal for any c .

Solution. Because Z and $-Z$ have the same distribution and the event $\{|Z| > c\}$ is unchanged by the sign change, $\Pr(W \leq w) = \Pr(Z \leq w, |Z| \leq c) + \Pr(-Z \leq w, |Z| > c) = \Pr(Z \leq w)$. Next, $\text{Cov}(Z, W) = \mathbf{E}(Z^2; |Z| \leq c) - \mathbf{E}(Z^2; |Z| > c) = 2\mathbf{E}(Z^2; |Z| \leq c) - 1$, which is continuous and increasing in c , tends to -1 as $c \rightarrow 0$ and to 1 as $c \rightarrow \infty$, and so vanishes at some c . Finally $Z + W = 2Z\mathbb{I}\{|Z| \leq c\}$ equals 0 with probability $\Pr(|Z| > c) \in (0, 1)$ but is not constant. A normal variable is either constant or has a density, so $Z + W$ is not normal.

Exercise B3.

For the pair (Z, W) of [Example \(A random sign\)](#), find every $\mathbf{a} \neq \mathbf{0}$ for which $a_1 Z + a_2 W$ is normal. *Hint:* an equal mixture of $N(0, v_1)$ and $N(0, v_2)$ has fourth moment $\frac{3}{2}(v_1^2 + v_2^2)$. Conclude that (Z, W) is normal along exactly two lines through the origin, the coordinate axes.

3.4 Independence and zero covariance

For any random vectors with finite second moments, independence implies zero cross-covariance, because the expectation of a product of functions of independent variables factors. The converse fails in general, as [Example \(A random sign\)](#) and [Example \(A mixture of two correlations\)](#) show. For jointly normal vectors it holds. This is the single most used property of the normal distribution in linear-model theory. Every independence statement behind a t or F statistic reduces to it.

Theorem 3.4.1 — Independence

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ be partitioned into blocks $\mathbf{Y} = (\mathbf{Y}_1 \mathbf{T}, \dots, \mathbf{Y}_k \mathbf{T}) \mathbf{T}$, with $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma} = (\boldsymbol{\Sigma}_{ij})$ partitioned to match.

- (a) The blocks $\mathbf{Y}_1, \dots, \mathbf{Y}_k$ are mutually independent iff $\boldsymbol{\Sigma}_{ij} = \text{Cov}(\mathbf{Y}_i, \mathbf{Y}_j) = \mathbf{0}$ for all $i \neq j$.
- (b) In particular, $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent iff $\text{Cov}(\mathbf{Y}_1, \mathbf{Y}_2) = \mathbf{0}$, and the scalar entries $Y_1, \dots, Y_n$ are mutually independent iff $\boldsymbol{\Sigma}$ is diagonal.

Proof. If the blocks are independent, every cross-covariance is zero. Conversely, suppose $\boldsymbol{\Sigma}_{ij} = \mathbf{0}$ for $i \neq j$, and partition $\mathbf{t}$ to match. Then

$$\mathbf{t} \mathbf{T} \boldsymbol{\mu} + \frac{1}{2} \mathbf{t} \mathbf{T} \boldsymbol{\Sigma} \mathbf{t} = \sum_{i=1}^k \left(\mathbf{t}_i \mathbf{T} \boldsymbol{\mu}_i + \frac{1}{2} \mathbf{t}_i \mathbf{T} \boldsymbol{\Sigma}_{ii} \mathbf{t}_i \right),$$

because all the cross terms $\mathbf{t}_i \mathbf{T} \boldsymbol{\Sigma}_{ij} \mathbf{t}_j$ vanish. Exponentiating, [Theorem 3.2.1](#) gives

$M_{\mathbf{Y}}(\mathbf{t}) = \prod_i M_i(\mathbf{t}_i)$, where M_i is the moment generating function of $N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_{ii})$. By [Corollary 3.3.2](#) that is the distribution of $\mathbf{Y}_i$, so $M_i = M_{\mathbf{Y}_i}$. The factorization property of [Section 3.2](#) gives mutual independence. Part (b) is the cases $k = 2$ and $k = n$. ■

A consequence worth stating explicitly: for jointly normal blocks, *pairwise* independence implies *mutual* independence, because condition (a) involves only pairs. For general random vectors this is false, as the example of three random signs in [Example \(Pairwise but not mutually independent\)](#) shows. Joint normality of the whole vector matters here: joint normality of each pair is not enough ([Exercise \(C1\)](#)).

The theorem is usually applied to two linear functions of the same normal vector.

Corollary 3.4.2

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and let $\mathbf{A}$ and $\mathbf{B}$ be matrices with n columns. Then $\mathbf{AY}$ and $\mathbf{BY}$ are independent iff $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^T = \mathbf{0}$. In that case $g(\mathbf{AY})$ and $h(\mathbf{BY})$ are independent for all measurable functions g and h .

Proof. The stacked vector $((\mathbf{AY})^T, (\mathbf{BY})^T)^T$ is a linear function of $\mathbf{Y}$, so it is normal by [Theorem 3.3.1](#). Its off-diagonal covariance block is $\text{Cov}(\mathbf{AY}, \mathbf{BY}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^T$ ([Theorem 2.2.1](#)). Apply [Theorem 3.4.1](#). Functions of independent random vectors are independent. ■

The condition $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^T = \mathbf{0}$ involves only second moments, but the conclusion is full independence. In particular, any quadratic form in $\mathbf{BY}$, such as $\|\mathbf{BY}\|^2$, is independent of $\mathbf{AY}$. [Chapter 4](#) extends this to quadratic forms that are not written as squared lengths ([Theorem 4.4.1](#)).

Example. (Sample mean and deviations)

Let $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$, so that $Y_1, \dots, Y_n$ is a random sample from $N(\mu, \sigma^2)$. The mean is $\bar{Y} = n^{-1}\mathbf{1}^T\mathbf{Y}$, and the vector of deviations is $\mathbf{d} = (\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^T)\mathbf{Y}$. As in [Example \(Centring a sample\)](#),

$$\text{Cov}(\bar{Y}, \mathbf{d}) = \sigma^2 n^{-1} \mathbf{1}^T (\mathbf{I} - n^{-1} \mathbf{1}\mathbf{1}^T) = \mathbf{0}^T,$$

By [Corollary 3.4.2](#), $\bar{Y}$ is independent of $\mathbf{d}$, and so of every function of the deviations: the sample variance $S^2 = \|\mathbf{d}\|^2/(n-1)$, the range, the sample skewness. The distribution of $(n-1)S^2/\sigma^2$ is found in [Corollary 4.4.5](#). The computation used $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ only through $\boldsymbol{\Sigma}\mathbf{1} \in \text{span}(\mathbf{1})$, which is the precise condition for the conclusion ([Exercise \(B1\)](#)).

Example. (Coefficients and residuals)

In the setting of [Example \(Least squares under normal errors\)](#),

$$\text{Cov}(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\varepsilon}}) = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T(\sigma^2\mathbf{I})(\mathbf{I} - \mathbf{M}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\{(\mathbf{I} - \mathbf{M})\mathbf{X}\}^T = \mathbf{0},$$

since $(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$. Hence $\hat{\boldsymbol{\beta}}$ is independent of the residual vector, and so of the residual sum of squares $\|\hat{\boldsymbol{\varepsilon}}\|^2$ and of the estimate $s^2 = \|\hat{\boldsymbol{\varepsilon}}\|^2/(n-p)$. The same argument shows that the fitted values $\hat{\mathbf{Y}}$ and the residuals are independent. This independence is what allows a t statistic $(\boldsymbol{\lambda}\mathbf{T}\hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}\mathbf{T}\boldsymbol{\beta})/\{s^2\boldsymbol{\lambda}\mathbf{T}(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\boldsymbol{\lambda}\}^{1/2}$ to have a t distribution (Chapter 11).

Remark (What joint normality buys). The two examples use only a covariance calculation. Without joint normality, zero covariance between $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\varepsilon}}$ still holds under the second-moment assumptions of Chapter 5, but independence does not follow, and neither do exact t and F distributions. Chapter 21 examines what survives approximately.

3.4.1 Exercises

3.4.1.1 A. Check your understanding

Exercise A1.

Let Y_1, Y_2 be independent $N(\mu, \sigma^2)$. Show that $Y_1 + Y_2$ and $Y_1 - Y_2$ are independent. What if the variances differ?

Exercise A2.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ and let $\mathbf{u}$ be a unit vector. Show that $\mathbf{u}\mathbf{T}\mathbf{Y}$ and $\|\mathbf{Y} - (\mathbf{u}\mathbf{T}\mathbf{Y})\mathbf{u}\|^2$ are independent.

3.4.1.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ arbitrary. Show that $\bar{Y}$ is independent of the deviations $Y_i - \bar{Y}$ iff $\boldsymbol{\Sigma}\mathbf{1} \in \text{span}(\mathbf{1})$, that is, iff all row sums of $\boldsymbol{\Sigma}$ are equal. Check the condition for the equicorrelation matrix $\sigma^2\{(1-\rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}\mathbf{T}\}$, and for a two-group matrix: the first m_1 and the last m_2 observations form groups, all variances are 1, the correlation is ρ within a group and τ between groups.

Solution. $\bar{Y} = n^{-1}\mathbf{1}\mathbf{T}\mathbf{Y}$ and the deviations are $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}\mathbf{T})\mathbf{Y}$, so by [Corollary 3.4.2](#) they are independent iff $n^{-1}\mathbf{1}\mathbf{T}\boldsymbol{\Sigma}(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}\mathbf{T}) = \mathbf{0}\mathbf{T}$, that is, $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}\mathbf{T})\boldsymbol{\Sigma}\mathbf{1} = \mathbf{0}$. The null space of $\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}\mathbf{T}$ is $\text{span}(\mathbf{1})$, and $\boldsymbol{\Sigma}\mathbf{1}$ is the vector of row sums. For the equicorrelation matrix, $\boldsymbol{\Sigma}\mathbf{1} = \sigma^2(1-\rho+n\rho)\mathbf{1}$, so independence holds for every admissible ρ . In the two-group matrix, a row in the first group sums to $1+(m_1-1)\rho+m_2\tau$ and a row in the second to $1+(m_2-1)\rho+m_1\tau$. The difference is $(m_1-m_2)(\rho-\tau)$, so independence holds iff the groups have equal size or $\rho = \tau$, and the latter is the equicorrelated case.

Exercise B2.

Let $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ with $\text{rank}(\mathbf{X}) = r < p$, let $\mathbf{G}$ be a generalized inverse of $\mathbf{X}\mathbf{T}\mathbf{X}$ and $\mathbf{M} = \mathbf{X}\mathbf{G}\mathbf{X}\mathbf{T}$. Show that the least squares estimate $\mathbf{G}\mathbf{X}\mathbf{T}\mathbf{Y}$ and the residual vector

$(\mathbf{I} - \mathbf{M})\mathbf{Y}$ are independent, and that every estimable function $\boldsymbol{\lambda}\mathbf{T}\boldsymbol{\beta}$ with $\boldsymbol{\lambda} \in \mathbb{C}(\mathbf{X}\mathbf{T})$ has an estimate independent of the residual sum of squares.

3.4.1.3 C. Going deeper

Exercise C1.

Let φ be the $N(0, 1)$ density, $0 < \varepsilon \leq e^{3/2}$, and

$$f(x, y, z) = \varphi(x)\varphi(y)\varphi(z)\{1 + \varepsilon xyz e^{-(x^2+y^2+z^2)/2}\}.$$

Show that f is a probability density, and that under f every pair of coordinates is $N_2(\mathbf{0}, \mathbf{I})$. So the three variables are pairwise jointly normal and pairwise independent. Show that they are not mutually independent and that (X, Y, Z) is not jointly normal. Which step of [Theorem 3.4.1](#) fails?

Solution. Since $|t|e^{-t^2/2} \leq e^{-1/2}$ for all t , the perturbation satisfies $|\varepsilon xyz e^{-(x^2+y^2+z^2)/2}| \leq \varepsilon e^{-3/2} \leq 1$, so $f \geq 0$. The function $z \mapsto z\varphi(z)e^{-z^2/2}$ is odd and integrable, so integrating f over z removes the perturbation and leaves $\varphi(x)\varphi(y)$. Integrating further shows that f integrates to 1, and by symmetry each pair has density $\varphi \otimes \varphi$, the $N_2(\mathbf{0}, \mathbf{I})$ density. Mutual independence would force $f = \varphi(x)\varphi(y)\varphi(z)$ almost everywhere, but the perturbation is nonzero off the coordinate planes. Directly, $\mathbf{E}(XYZ) = \varepsilon(\int t^2 \varphi(t) e^{-t^2/2} dt)^3 > 0$, whereas independence with mean-zero margins gives 0. If (X, Y, Z) were jointly normal, its covariance matrix would be $\mathbf{I}$ (unit variances from the margins, zero covariances from the pairs), and by [Theorem 3.4.1](#) the coordinates would be independent, which they are not. The theorem's hypothesis is joint normality of the whole vector. Here only the pairs are normal, and the moment generating function of the triple does not factor.

3.5 Conditional distributions

Throughout this section $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is split into $\mathbf{Y}_1 \in \mathbb{R}^{n_1}$ and $\mathbf{Y}_2 \in \mathbb{R}^{n_2}$, with

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}, \quad \boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{12}\mathbf{T},$$

as in [Theorem 2.5.1](#). The question is what $\mathbf{Y}_1$ looks like once $\mathbf{Y}_2$ has been observed. The answer is the probabilistic core of regression.

3.5.1 Partial covariance with a generalized inverse

The natural answer involves $\boldsymbol{\Sigma}_{22}^{-1}$, but $\boldsymbol{\Sigma}_{22}$ may be singular, for instance when $\mathbf{Y}_2$ contains a sum and its summands. A generalized inverse ([Definition 1.3.1](#)) removes the need for a separate treatment. The facts that make this work were proved for arbitrary random vectors in [Section 2.5](#). We collect them in the notation of this chapter.

Lemma 3.5.1

Let Σ be nonnegative definite and partitioned as above, and let $\mathbf{G}$ be any generalized inverse of Σ_{22} . Then:

- (a) $\mathbb{C}(\Sigma_{21}) \subseteq \mathbb{C}(\Sigma_{22})$, so $\Sigma_{21} = \Sigma_{22}\mathbf{H}$ for some matrix $\mathbf{H}$;
- (b) $\Sigma_{12}\mathbf{G}\Sigma_{22} = \Sigma_{12}$ and $\Sigma_{22}\mathbf{G}\Sigma_{21} = \Sigma_{21}$;
- (c) $\Sigma_{12}\mathbf{G}\Sigma_{21} = \mathbf{H}\mathbf{T}\Sigma_{22}\mathbf{H}$, which does not depend on the choice of $\mathbf{G}$ and is nonnegative definite.

The matrix $\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^-\Sigma_{21}$, the **Schur complement** of Σ_{22} in Σ , is therefore well defined. It is the partial covariance matrix of $\mathbf{Y}_1$ given $\mathbf{Y}_2$ in the sense of [Theorem 2.5.2](#).

Proof. By [Theorem 3.1.2](#), Σ is the covariance matrix of some random vector, so (a) and (b) are [Theorem 2.5.1\(c\)](#). For (c), $\Sigma_{12}\mathbf{G}\Sigma_{21} = \mathbf{H}\mathbf{T}\Sigma_{22}\mathbf{G}\Sigma_{22}\mathbf{H} = \mathbf{H}\mathbf{T}\Sigma_{22}\mathbf{H}$. ■

3.5.2 The conditional distribution**Theorem 3.5.2 — Conditional distribution**

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ be partitioned as above, let $\mathbf{G}$ be a generalized inverse of Σ_{22} , and put $\mathbf{B} = \Sigma_{12}\mathbf{G}$. Then:

- (a) $\mathbf{W} = \mathbf{Y}_1 - \boldsymbol{\mu}_1 - \mathbf{B}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$ is independent of $\mathbf{Y}_2$ and $\mathbf{W} \sim N_{n_1}(\mathbf{0}, \Sigma_{11.2})$;
- (b) the conditional distribution of $\mathbf{Y}_1$ given $\mathbf{Y}_2 = \mathbf{y}_2$ is

$$\mathbf{Y}_1 \mid \mathbf{Y}_2 = \mathbf{y}_2 \sim N_{n_1}\left(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^-(\mathbf{y}_2 - \boldsymbol{\mu}_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^-\Sigma_{21}\right), \quad (3.5.1)$$

and for $(\mathbf{y}_2)$ in $(\boldsymbol{\mu}_2 + \mathbb{C}(\Sigma_{22}))$, where $(\mathbf{Y}_2)$ lies with one, the conditional mean does not depend on the choice of $(\mathbf{G})$;

- (c) if Σ is positive definite, then $\Sigma_{22}^- = \Sigma_{22}^{-1}$, $\Sigma_{11.2}$ is positive definite, $\Sigma_{11.2}^{-1}$ is the leading $n_1 \times n_1$ block of Σ^{-1} , and $\det \Sigma = \det \Sigma_{22} \det \Sigma_{11.2}$.

Proof.

- (a) $\mathbf{W} = [\mathbf{I}, -\mathbf{B}](\mathbf{Y} - \boldsymbol{\mu})$, so $(\mathbf{W}\mathbf{T}, \mathbf{Y}_2\mathbf{T})\mathbf{T}$ is an affine function of $\mathbf{Y}$ and is jointly normal ([Theorem 3.3.1](#)). The vector $\mathbf{W}$ is the error of the best linear predictor of $\mathbf{Y}_1$ from $\mathbf{Y}_2$, so by [Theorem 2.5.2\(a,b\)](#) it has mean zero, $\text{Cov}(\mathbf{W}, \mathbf{Y}_2) = \mathbf{0}$ and $\text{Cov}(\mathbf{W}) = \Sigma_{11.2}$. These are moment calculations, valid for any random vector. What normality adds is that zero covariance becomes independence ([Theorem 3.4.1](#)), and that $\mathbf{W}$, a linear function of $\mathbf{Y}$, is normal.
- (b) Write $\mathbf{Y}_1 = \boldsymbol{\mu}_1 + \mathbf{B}(\mathbf{Y}_2 - \boldsymbol{\mu}_2) + \mathbf{W}$. The first two terms are a function of $\mathbf{Y}_2$, and $\mathbf{W}$ is independent of $\mathbf{Y}_2$. For independent $\mathbf{W}$ and $\mathbf{Y}_2$, the conditional distribution of

$h(\mathbf{W}, \mathbf{Y}_2)$ given $\mathbf{Y}_2 = \mathbf{y}_2$ is the distribution of $h(\mathbf{W}, \mathbf{y}_2)$ (Billingsley 1995). With $h(\mathbf{w}, \mathbf{y}_2) = \boldsymbol{\mu}_1 + \mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2) + \mathbf{w}$, this is the normal distribution

3. If $\mathbf{y}_2 - \boldsymbol{\mu}_2 = \boldsymbol{\Sigma}_{22}\mathbf{k}$, then $\mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2) = \boldsymbol{\Sigma}_{12}\mathbf{G}\boldsymbol{\Sigma}_{22}\mathbf{k} = \boldsymbol{\Sigma}_{12}\mathbf{k}$, which does not involve $\mathbf{G}$.
- (c) If $\boldsymbol{\Sigma}$ is positive definite, so is its principal submatrix $\boldsymbol{\Sigma}_{22}$, and its only generalized inverse is its inverse. By the block determinant formula (Theorem 1.6.3), which needs only $\boldsymbol{\Sigma}_{22}$ nonsingular, $\det \boldsymbol{\Sigma} = \det \boldsymbol{\Sigma}_{22} \det \boldsymbol{\Sigma}_{11.2}$. The left side and $\det \boldsymbol{\Sigma}_{22}$ are positive, so $\boldsymbol{\Sigma}_{11.2}$ is nonsingular. It is nonnegative definite, being $\text{Cov}(\mathbf{W})$ by (a), hence positive definite. With both $\boldsymbol{\Sigma}_{22}$ and its Schur complement nonsingular, the partitioned-inverse formula (Theorem 1.6.4) gives $\boldsymbol{\Sigma}_{11.2}^{-1}$ as the leading block of $\boldsymbol{\Sigma}^{-1}$.

When $\boldsymbol{\Sigma}$ is positive definite, (b) can also be read off the densities. By (c), $\boldsymbol{\Sigma}_{11.2}$ is positive definite, so $\mathbf{W}$ has a density. The map $(\mathbf{y}_1, \mathbf{y}_2) \mapsto (\mathbf{w}, \mathbf{y}_2)$ has unit Jacobian, so $f_{\mathbf{Y}}(\mathbf{y}_1, \mathbf{y}_2) = f_{\mathbf{W}}(\mathbf{y}_1 - \boldsymbol{\mu}_1 - \mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2)) f_{\mathbf{Y}_2}(\mathbf{y}_2)$ by independence, and dividing by $f_{\mathbf{Y}_2}(\mathbf{y}_2)$ gives the conditional density.

■

Idea. Every jointly normal vector splits as

$$\mathbf{Y}_1 = \underbrace{\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)}_{\text{function of } \mathbf{Y}_2} + \underbrace{\mathbf{W}}_{\text{independent of } \mathbf{Y}_2}.$$

The first term is a linear regression of $\mathbf{Y}_1$ on $\mathbf{Y}_2$. The second is a normal error whose distribution is the same whatever value $\mathbf{Y}_2$ takes.

3.5.3 What the theorem says

Three features of 3 are special to the normal distribution, and each is an assumption of the classical linear model.

- (i) *The regression is linear.* $\mathbf{E}(\mathbf{Y}_1 \mid \mathbf{Y}_2 = \mathbf{y}_2)$ is an affine function of $\mathbf{y}_2$, with coefficient matrix $\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}$.
- (ii) *The errors are homoscedastic.* The conditional covariance $\boldsymbol{\Sigma}_{11.2}$ is the same for every $\mathbf{y}_2$. Observing $\mathbf{Y}_2$ tells us where $\mathbf{Y}_1$ is centred, but not how spread out it is.
- (iii) *Conditioning cannot increase uncertainty.* $\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{11.2} = \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$ is nonnegative definite (Lemma 3.5.1(c)), so every linear combination $\mathbf{a}^T\mathbf{Y}_1$ has conditional variance at most its unconditional variance. The reduction is zero iff $\boldsymbol{\Sigma}_{12} = \mathbf{0}$.

None of these holds in general. In the mixture of Example (A mixture of two correlations), given $Y_1 = y_1$ the variable Y_2 is an equal mixture of $N(\rho y_1, 1 - \rho^2)$ and $N(-\rho y_1, 1 - \rho^2)$. Its conditional mean is 0 for every y_1 , but its conditional variance is $1 - \rho^2 + \rho^2 y_1^2$, which grows with $|y_1|$. Linearity and constant variance are not consequences of normal margins or of zero correlation. They are consequences of joint normality.

One response, several predictors. The case used most in this book has a scalar response Y and a vector $\mathbf{X}$ of k predictors, jointly normal, with $\text{Var}(Y) = \sigma_Y^2$, $\sigma_{XY} = \text{Cov}(\mathbf{X}, Y)$ and

$\Sigma_{XX} = \text{Cov}(\mathbf{X})$ positive definite. [Theorem 3.5.2](#) says

$$Y = \alpha + \beta \mathbf{T}\mathbf{X} + e, \quad \beta = \Sigma_{XX}^{-1} \sigma_{XY}, \quad \alpha = \mu_Y - \beta \mathbf{T} \mu_X, \quad e \sim N(0, \sigma^2) \text{ independent of } \mathbf{X}, \quad (3.5.2)$$

with $\sigma^2 = \sigma_Y^2 - \sigma_{XY} \mathbf{T} \Sigma_{XX}^{-1} \sigma_{XY} = \sigma_Y^2 (1 - \rho_{Y \cdot X}^2)$, where $\rho_{Y \cdot X}^2 = \sigma_{XY} \mathbf{T} \Sigma_{XX}^{-1} \sigma_{XY} / \sigma_Y^2$ is the population squared multiple correlation ([Proposition 2.5.3](#)). So if observations $(Y_i, \mathbf{X}_i)$ are drawn independently from a joint normal distribution, then conditionally on the predictors, the normal linear model with independent homoscedastic errors holds exactly. Inference that treats the predictors as fixed is valid for random predictors, as Chapter 14 develops.

The bivariate case. With $k = 1$, correlation ρ and standard deviations σ_X, σ_Y ,

$$\mathbf{E}(Y \mid X = x) = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X), \quad \text{Var}(Y \mid X = x) = \sigma_Y^2 (1 - \rho^2).$$

In standard units, $(\mathbf{E}(Y \mid X = x) - \mu_Y) / \sigma_Y = \rho (x - \mu_X) / \sigma_X$. The predicted response is fewer standard deviations from its mean than the predictor is from its mean, whenever $|\rho| < 1$. This is **regression toward the mean**, Galton's observation about the heights of parents and children (Galton 1886), and the origin of the word "regression". It is also a warning. Regressing X on Y is a different line, $\mathbf{E}(X \mid Y = y) = \mu_X + \rho(\sigma_X / \sigma_Y)(y - \mu_Y)$, and when $0 < |\rho| < 1$ neither line is the major axis of the density contours.

Example. (Two regression lines)

Take $\mu_X = 1$, $\mu_Y = 2$, $\sigma_X = 1$, $\sigma_Y = 1.5$ and $\rho = 0.7$. The regression of Y on X has slope 1.05 and residual standard deviation 1.071. Drawn in the same (x, y) plane, the regression of X on Y has slope 2.14, and the major axis of the ellipses lies between them, with slope 1.76. [Figure 3.5.1](#) shows the three lines. The line $\mathbf{E}(Y \mid X = x)$ passes through the points where each contour ellipse has a vertical tangent ([Exercise \(B2\)](#)): on a vertical line $X = x$, the density is highest at the conditional mean. A least squares line fitted to 60 draws has slope 1.12 (standard error 0.16). It estimates the regression of Y on X , not the major axis.

Example. (Holding a third variable fixed)

Let $\mathbf{Y} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\mu} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} 4 & 1 & 2 \\ 1 & 3 & 2 \\ 2 & 2 & 2 \end{pmatrix}.$$

Conditioning Y_1 on Y_2 alone gives slope $\sigma_{12} / \sigma_{22} = \frac{1}{3}$ and conditional variance $\frac{11}{3}$. Conditioning on (Y_2, Y_3) , the listing computes $\mathbf{B} = \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1}$ in exact arithmetic and finds

$$\mathbf{E}(Y_1 \mid Y_2 = y_2, Y_3 = y_3) = 3 - y_2 + 2y_3, \quad \text{Var}(Y_1 \mid Y_2, Y_3) = 1.$$

The coefficient of y_2 changes sign. Among outcomes with the same Y_3 , larger Y_2 goes with smaller Y_1 . Across all outcomes, larger Y_2 goes with larger Y_1 , because both tend to be large when Y_3 is large. The conditional variance 1 also equals $\det \boldsymbol{\Sigma} / \det \boldsymbol{\Sigma}_{22} = 2/2$, as [Theorem 3.5.2\(c\)](#) predicts.

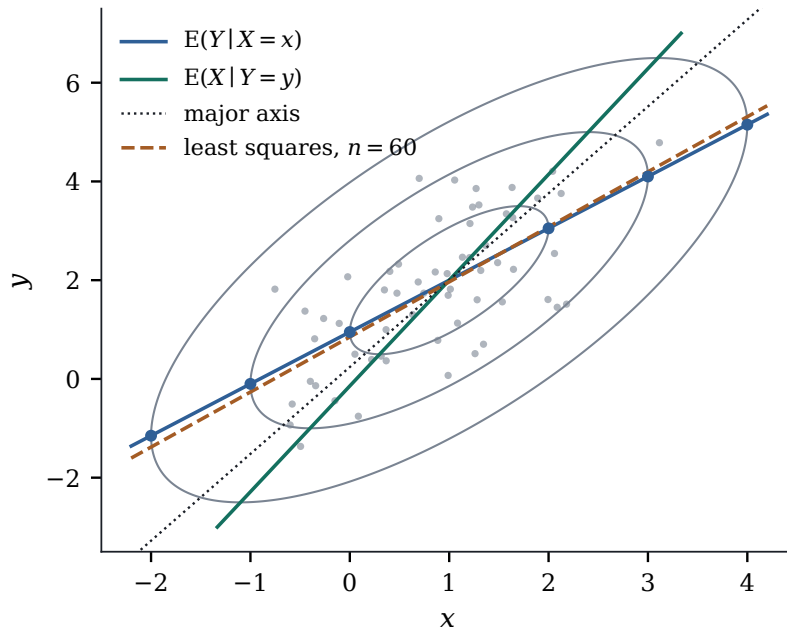


Figure 3.5.1: Contours $\Delta^2 = 1, 4, 9$ of a bivariate normal density with $\rho = 0.7$. The conditional mean line $\mathbf{E}(Y | X = x)$ passes through the points of vertical tangency (dots). The line $\mathbf{E}(X | Y = y)$ passes through the points of horizontal tangency and is steeper, and the major axis lies between the two. Least squares on a sample of 60 points estimates $\mathbf{E}(Y | X = x)$.

```

from fractions import Fraction as F
import numpy as np
rng = np.random.default_rng(305)
mx, my, sx, sy, rho = 1.0, 2.0, 1.0, 1.5, 0.7
Sigma = np.array([[sx**2, rho * sx * sy], [rho * sx * sy, sy**2]])
slope_y_on_x = rho * sy / sx                    # E(Y | X = x)
slope_x_on_y = rho * sx / sy                    # E(X | Y = y), as dx/dy
cond_sd = sy * np.sqrt(1 - rho**2)
n = 60
XY = rng.multivariate_normal([mx, my], Sigma, size=n)
b_ls, a_ls = np.polyfit(XY[:, 0], XY[:, 1], 1)
Linv = np.linalg.inv(np.linalg.cholesky(Sigma))
theta = np.linspace(0, 2 * np.pi, 200_001)
circle = np.vstack([np.cos(theta), np.sin(theta)])
L = np.linalg.cholesky(Sigma)
big = rng.multivariate_normal([mx, my], Sigma, size=4_000_000)
x0 = 2.0
sl = big[np.abs(big[:, 0] - x0) < 0.01, 1]
resid = XY[:, 1] - a_ls - b_ls * XY[:, 0]
se = np.sqrt(resid @ resid / (n - 2) / np.sum((XY[:, 0] - XY[:, 0].mean())**2))
lam, U = np.linalg.eigh(Sigma)
slope_major = U[1, 1] / U[0, 1]
levels = [1, 4, 9]
gx, gy = np.meshgrid(np.linspace(-3, 5, 300), np.linspace(-4, 8, 300))
D = np.stack([gx - mx, gy - my], axis=-1)
D2 = np.einsum("...i,ij,...j->...", D, np.linalg.inv(Sigma), D)
xs = np.linspace(-2.2, 4.2, 2)
ys = np.linspace(-3.0, 7.0, 2)

mu = [F(1), F(0), F(-1)]
S = [[F(4), F(1), F(2)],

```

```

[F(1), F(3), F(2)],
[F(2), F(2), F(2)]]

def inv2(M):
    det = M[0][0] * M[1][1] - M[0][1] * M[1][0]
    return [[M[1][1] / det, -M[0][1] / det], [-M[1][0] / det, M[0][0] / det]]

# Y1 given (Y2, Y3): B = sigma_12 Sigma_22^-1, variance sigma_11 - B sigma_21
S22inv = inv2([[S[1][1], S[1][2]], [S[2][1], S[2][2]]])
s12 = [S[0][1], S[0][2]]
B = [s12[0] * S22inv[0][j] + s12[1] * S22inv[1][j] for j in range(2)]
var_1_23 = S[0][0] - (B[0] * s12[0] + B[1] * s12[1])
const = mu[0] - B[0] * mu[1] - B[1] * mu[2]
print("E(Y1 | y2, y3) =", const, "+", B[0], "y2 +", B[1], "y3; variance", var_1_23)

# (Y1, Y2) given Y3, and the partial correlation of Y1 and Y2 given Y3
C = [[S[i][j] - S[i][2] * S[2][j] / S[2][2] for j in range(2)] for i in range(2)]
print("Cov((Y1, Y2) | Y3) =", C)

```

3.5.4 The conditional mean is the best predictor

Section 6.11 defines the best linear predictor $L(\mathbf{Y}_1 | \mathbf{Y}_2)$ as a projection, and shows that the conditional mean is the best predictor among *all* square-integrable functions of $\mathbf{Y}_2$. For a scalar response, the gap between the two is the approximation error of Proposition 6.11.2(c). Under joint normality the gap is zero.

Proposition 3.5.3

Let $\mathbf{Y}$ be jointly normal and partitioned as above, and let $\mathbf{m}(\mathbf{Y}_2) = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$. For every function $\mathbf{g}$ with $\mathbf{E}\|\mathbf{g}(\mathbf{Y}_2)\|^2 < \infty$,

$$\mathbf{E}\|\mathbf{Y}_1 - \mathbf{g}(\mathbf{Y}_2)\|^2 = \text{tr}(\boldsymbol{\Sigma}_{11.2}) + \mathbf{E}\|\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2)\|^2.$$

So $\mathbf{m}(\mathbf{Y}_2)$ minimizes the mean squared prediction error over all predictors, it equals the best linear predictor of Theorem 6.11.1 (entry by entry, when $\boldsymbol{\Sigma}_{22}$ is positive definite), and the minimum is $\text{tr}(\boldsymbol{\Sigma}_{11.2})$.

Proof. $\mathbf{Y}_1 - \mathbf{g}(\mathbf{Y}_2) = \mathbf{W} + (\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2))$ with $\mathbf{W}$ as in Theorem 3.5.2. The cross term $\mathbf{E}[\mathbf{W}\mathbf{T}(\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2))]$ vanishes, because $\mathbf{W}$ has mean zero and is independent of $\mathbf{Y}_2$. And $\mathbf{E}\|\mathbf{W}\|^2 = \text{tr Cov}(\mathbf{W}) = \text{tr}(\boldsymbol{\Sigma}_{11.2})$. The minimizer $\mathbf{m}$ is affine, so it is also the best affine predictor, which is the best linear predictor. ■

The proof is short because independence of $\mathbf{W}$ and $\mathbf{Y}_2$ is much stronger than the zero correlation that defines a projection. For non-normal vectors only zero correlation is available, and the best linear predictor is merely the best *linear* predictor.

3.5.5 Exercises

3.5.5.1 A. Check your understanding

Exercise A1.

For $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ of [Example \(Holding a third variable fixed\)](#), find the conditional distribution of (Y_1, Y_2) given $Y_3 = y_3$ and of Y_3 given $(Y_1, Y_2) = (y_1, y_2)$. Verify that the conditional variances are smaller than the unconditional ones.

Exercise A2.

In the setting of [Eq. 3.5.2](#), let $\boldsymbol{\Omega}$ be the inverse of $\text{Cov}((Y, \mathbf{X}\mathbf{T})\mathbf{T})$. Show that the row of $\boldsymbol{\Omega}$ belonging to Y is $(1, -\boldsymbol{\beta}\mathbf{T})/\sigma^2$, where $\sigma^2 = \sigma_Y^2(1 - \rho_{Y.X}^2)$. So the regression coefficients and the error variance can be read off one row of the precision matrix.

Solution. Put Y first and apply [Exercise \(B5\)](#) with the scalar block Y and $\mathbf{Y}_2 = \mathbf{X}$. The conditional variance is $\omega_{YY}^{-1} = \text{Var}(Y \mid \mathbf{X}) = \sigma^2$, and the coefficient vector of the conditional mean is $-\omega_{YY}^{-1}\boldsymbol{\Omega}_{YX} = \boldsymbol{\beta}\mathbf{T}$. Hence $\omega_{YX} = 1/\sigma^2$ and $\boldsymbol{\Omega}_{YX} = -\boldsymbol{\beta}\mathbf{T}/\sigma^2$. By [Eq. 3.5.2](#), $\sigma^2 = \sigma_Y^2(1 - \rho_{Y.X}^2)$.

3.5.5.2 B. Practice

Exercise B1.

Suppose $\mathbf{Y}_2 \sim N(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$ and that, given $\mathbf{Y}_2 = \mathbf{y}_2$, $\mathbf{Y}_1$ is $N(\mathbf{a} + \mathbf{B}\mathbf{y}_2, \mathbf{S})$ with $\mathbf{a}$, $\mathbf{B}$ and $\mathbf{S}$ not depending on $\mathbf{y}_2$. Show that $(\mathbf{Y}_1\mathbf{T}, \mathbf{Y}_2\mathbf{T})\mathbf{T}$ is jointly normal and find its mean and covariance. Explain why [Example \(A mixture of two correlations\)](#) does not contradict this.

Solution. Condition on $\mathbf{Y}_2$:

$$\mathbf{E}e^{t_1\mathbf{T}\mathbf{Y}_1 + t_2\mathbf{T}\mathbf{Y}_2} = \mathbf{E}\left[e^{t_2\mathbf{T}\mathbf{Y}_2} e^{t_1\mathbf{T}(\mathbf{a} + \mathbf{B}\mathbf{Y}_2) + \frac{1}{2}t_1\mathbf{T}\mathbf{S}t_1}\right] = e^{t_1\mathbf{T}\mathbf{a} + \frac{1}{2}t_1\mathbf{T}\mathbf{S}t_1} M_{\mathbf{Y}_2}(t_2 + \mathbf{B}\mathbf{T}t_1).$$

Inserting $M_{\mathbf{Y}_2}$ from [Theorem 3.2.1](#), the exponent is linear plus $\frac{1}{2}\mathbf{t}\mathbf{T}\mathbf{C}\mathbf{t}$ with $\mathbf{C} = \begin{pmatrix} \mathbf{S} + \mathbf{B}\boldsymbol{\Sigma}_{22}\mathbf{B}\mathbf{T} & \mathbf{B}\boldsymbol{\Sigma}_{22} \\ \boldsymbol{\Sigma}_{22}\mathbf{B}\mathbf{T} & \boldsymbol{\Sigma}_{22} \end{pmatrix}$, so the pair is normal with mean $((\mathbf{a} + \mathbf{B}\boldsymbol{\mu}_2)\mathbf{T}, \boldsymbol{\mu}_2\mathbf{T})\mathbf{T}$ and covariance $\mathbf{C}$. In [Example \(A mixture of two correlations\)](#), the conditional distribution of Y_2 given Y_1 is a mixture, not a normal distribution, and its variance depends on y_1 .

Exercise B2.

Show that for a bivariate normal density with $|\rho| < 1$, the points of each contour ellipse at which the tangent is vertical lie on the line $y = \mathbf{E}(Y \mid X = x)$. Explain the result in terms of the conditional density of Y given $X = x$.

Solution. Write the ellipse as $\{\boldsymbol{\mu} + \mathbf{u} : \mathbf{u}\mathbf{T}\boldsymbol{\Sigma}^{-1}\mathbf{u} = c^2\}$. At a point with vertical tangent, the normal vector $\boldsymbol{\Sigma}^{-1}\mathbf{u}$ is horizontal, so $\boldsymbol{\Sigma}^{-1}\mathbf{u} = \lambda(1, 0)\mathbf{T}$ and $\mathbf{u} = \lambda(\sigma_X^2, \rho\sigma_X\sigma_Y)\mathbf{T}$. The slope $u_2/u_1 = \rho\sigma_Y/\sigma_X$ is that of the conditional mean line, and both such points

($\lambda > 0$ and $\lambda < 0$) lie on it. Along the vertical line $X = x$, the joint density is proportional to the conditional density of Y , which is highest at $\mathbf{E}(Y \mid X = x)$. The contour that just touches the vertical line touches it at the point of highest density on that line.

Exercise B3.

Let $Z_1, \dots, Z_n$ be independent $N(0, 1)$ and $S_k = Z_1 + \dots + Z_k$. Show that $(S_1, \dots, S_n)$ is jointly normal with $\text{Cov}(S_j, S_k) = \min(j, k)$. Find the conditional distribution of S_j given $S_n = s$, and comment on its variance as a function of j .

Solution. $(S_1, \dots, S_n)\mathbf{T} = \mathbf{L}\mathbf{Z}$ with $\mathbf{L}$ lower triangular and all entries on and below the diagonal equal to one, so the vector is normal with covariance $\mathbf{L}\mathbf{L}\mathbf{T}$, whose (j, k) entry counts the common indices, $\min(j, k)$. By [Theorem 3.5.2](#) with $\mathbf{Y}_1 = S_j$, $\mathbf{Y}_2 = S_n$,

$$S_j \mid S_n = s \sim N\left(\frac{j}{n}s, j - \frac{j^2}{n}\right) = N\left(\frac{j}{n}s, \frac{j(n-j)}{n}\right).$$

The conditional mean interpolates linearly between 0 and s . The variance is zero at both ends, where the value is known, and largest in the middle.

Exercise B4.

Let (Y, X_1, X_2) be normal with positive definite covariance, and let $\mathbf{Y}_2 = (X_1, X_2, X_1 + X_2)\mathbf{T}$. Show that Σ_{22} is singular. Using a generalized inverse, show that the conditional distribution of Y given $\mathbf{Y}_2$ from [Theorem 3.5.2](#) coincides with its conditional distribution given (X_1, X_2) .

Exercise B5.

With $\Omega = \Sigma^{-1}$ partitioned like Σ , show that $\mathbf{E}(\mathbf{Y}_1 \mid \mathbf{Y}_2 = \mathbf{y}_2) = \boldsymbol{\mu}_1 - \Omega_{11}^{-1}\Omega_{12}(\mathbf{y}_2 - \boldsymbol{\mu}_2)$ and $\text{Cov}(\mathbf{Y}_1 \mid \mathbf{Y}_2) = \Omega_{11}^{-1}$.

Solution. By [Theorem 1.6.4](#), $\Omega_{11} = \Sigma_{11.2}^{-1}$ and $\Omega_{12} = -\Sigma_{11.2}^{-1}\Sigma_{12}\Sigma_{22}^{-1}$. Hence $-\Omega_{11}^{-1}\Omega_{12} = \Sigma_{12}\Sigma_{22}^{-1}$ and $\Omega_{11}^{-1} = \Sigma_{11.2}$. Substitute into 3.

3.5.5.3 C. Going deeper

Exercise C1.

Assume Σ is positive definite. Prove [Theorem 3.5.2\(b\)](#) by dividing the joint density by the marginal density of $\mathbf{Y}_2$ directly, using the partitioned inverse and the block determinant formula.

3.6 Partial correlation

Two variables can be strongly correlated only because both depend on a third. [Example \(Holding a third variable fixed\)](#) showed a positive association between Y_1 and Y_2 turning negative once Y_3 is held fixed. The conditional covariance matrix $\Sigma_{11 \cdot 2}$ measures association that remains after conditioning, and its correlation form is the partial correlation.

3.6.1 Definition and interpretations

Definition 3.6.1 — Partial correlation

Let $\mathbf{Y} = (\mathbf{Y}_1\mathbf{T}, \mathbf{Y}_2\mathbf{T})\mathbf{T}$ be a random vector with covariance matrix Σ , partitioned as in [Section 3.5](#), and write $\Sigma_{11 \cdot 2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = (\sigma_{ij \cdot 2})$. The **partial covariance** of Y_i and Y_j (two entries of $\mathbf{Y}_1$) given $\mathbf{Y}_2$ is $\sigma_{ij \cdot 2}$, and their **partial correlation** given $\mathbf{Y}_2$ is

$$\rho_{ij \cdot 2} = \frac{\sigma_{ij \cdot 2}}{\sqrt{\sigma_{ii \cdot 2} \sigma_{jj \cdot 2}}},$$

provided $\sigma_{ii \cdot 2}$ and $\sigma_{jj \cdot 2}$ are positive. When the conditioning variables are $Y_k, Y_l, \dots$ we also write $\rho_{ij \cdot kl \dots}$. The matrix of partial correlations is $\mathbf{D}^{-1/2}\Sigma_{11 \cdot 2}\mathbf{D}^{-1/2}$ with $\mathbf{D} = \text{diag}(\Sigma_{11 \cdot 2})$, the correlation matrix ([Definition 2.4.1](#)) built from $\Sigma_{11 \cdot 2}$.

The definition uses only Σ , so partial correlations exist for any random vector with finite second moments. The next result gives them two meanings, one that always applies and one that needs normality.

Proposition 3.6.1

In the setting of [Definition 3.6.1](#):

- (a) $\Sigma_{11 \cdot 2}$ is the covariance matrix of the prediction errors $\mathbf{W} = \mathbf{Y}_1 - \boldsymbol{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$, and $\mathbf{W}$ is uncorrelated with $\mathbf{Y}_2$. If Σ_{22} is positive definite, the entries of $\mathbf{W}$ are the errors of the best linear predictors $L(Y_i | \mathbf{Y}_2)$, so $\rho_{ij \cdot 2}$ is the correlation between $Y_i - L(Y_i | \mathbf{Y}_2)$ and $Y_j - L(Y_j | \mathbf{Y}_2)$.
- (b) If $\mathbf{Y}$ is jointly normal, $\rho_{ij \cdot 2}$ is the correlation of Y_i and Y_j in their conditional distribution given $\mathbf{Y}_2 = \mathbf{y}_2$, the same for every $\mathbf{y}_2$. Moreover $\rho_{ij \cdot 2} = 0$ iff Y_i and Y_j are conditionally independent given $\mathbf{Y}_2$.

Proof.

- (a) This is [Theorem 2.5.2](#), restated: $\text{Cov}(\mathbf{W}, \mathbf{Y}_2) = \mathbf{0}$ and $\text{Cov}(\mathbf{W}) = \Sigma_{11 \cdot 2}$ are the moment calculations in the proof of [Theorem 3.5.2\(a\)](#), which use no normality. When Σ_{22} is positive definite, the i th entry of $\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$ is $L(Y_i | \mathbf{Y}_2)$ by [Theorem 6.11.1](#).
- (b) By [Theorem 3.5.2](#), the conditional distribution of (Y_i, Y_j) given $\mathbf{Y}_2 = \mathbf{y}_2$ is bivariate normal with covariance matrix the corresponding 2×2 block of $\Sigma_{11 \cdot 2}$. Its correlation is $\rho_{ij \cdot 2}$, and by [Theorem 3.4.1](#) applied to this conditional normal distribution, the two are independent iff the off-diagonal entry is zero.

Part (a) is the definition to remember: *a partial correlation is the correlation of two residuals*. Remove from each variable the part that is linearly predictable from the conditioning variables, and correlate what is left. This is the population version of the Frisch–Waugh–Lovell construction ([Theorem 6.6.2](#)).

3.6.2 Computing partial correlations

Two routes avoid computing $\Sigma_{11 \cdot 2}$ in full.

With a single conditioning variable, $\Sigma_{11 \cdot 2}$ is a 2×2 matrix with entries $\sigma_{ij} - \sigma_{i3}\sigma_{j3}/\sigma_{33}$, and in correlation units this is the familiar three-variable formula of [Exercise \(B1\)](#),

$$\rho_{12 \cdot 3} = \frac{\rho_{12} - \rho_{13}\rho_{23}}{\sqrt{(1 - \rho_{13}^2)(1 - \rho_{23}^2)}}, \quad |\rho_{13}| < 1, \quad |\rho_{23}| < 1. \quad (3.6.1)$$

The residual interpretation shows that the same formula can be applied one conditioning variable at a time.

Proposition 3.6.2 — Recursion

Let S be a set of indices and $k \notin S \cup \{1, 2\}$ one more. Suppose $\sigma_{11 \cdot S}$, $\sigma_{22 \cdot S}$ and $\sigma_{kk \cdot S}$ are positive and $|\rho_{1k \cdot S}| < 1$, $|\rho_{2k \cdot S}| < 1$. Then

$$\sigma_{12 \cdot Sk} = \sigma_{12 \cdot S} - \frac{\sigma_{1k \cdot S} \sigma_{2k \cdot S}}{\sigma_{kk \cdot S}}, \quad \rho_{12 \cdot Sk} = \frac{\rho_{12 \cdot S} - \rho_{1k \cdot S} \rho_{2k \cdot S}}{\sqrt{(1 - \rho_{1k \cdot S}^2)(1 - \rho_{2k \cdot S}^2)}}.$$

With S empty this is [Eq. 3.6.1](#).

Proof. No normality is needed. For $i \in \{1, 2, k\}$ let e_i be the error of the best linear predictor of Y_i from $\mathbf{Y}_S$, as in [Proposition 3.6.1\(a\)](#). Then each e_i has mean zero, is uncorrelated with $\mathbf{Y}_S$, and $\text{Cov}(e_i, e_j) = \sigma_{ij \cdot S}$. Put $r_i = e_i - c_i e_k$ with $c_i = \sigma_{ik \cdot S}/\sigma_{kk \cdot S}$, for $i = 1, 2$.

We claim that r_i is the error of the best linear predictor of Y_i from $(\mathbf{Y}_S, Y_k)$. First, $Y_i - r_i$ is an affine function of $(\mathbf{Y}_S, Y_k)$, because e_i and e_k differ from Y_i and Y_k by affine functions of $\mathbf{Y}_S$. Second, r_i has mean zero and is uncorrelated with $\mathbf{Y}_S$ and with e_k , by the choice of c_i , hence with $Y_k = e_k + (\text{affine in } \mathbf{Y}_S)$. Now let r'_i be the error from [Proposition 3.6.1\(a\)](#) for predicting Y_i from $(\mathbf{Y}_S, Y_k)$, which has the same two properties. The difference $d = r_i - r'_i$ is an affine function of $(\mathbf{Y}_S, Y_k)$ with mean zero and is uncorrelated with $(\mathbf{Y}_S, Y_k)$, so $\text{Var}(d) = \text{Cov}(d, d) = 0$. Thus $r_i = r'_i$ with probability one.

Therefore $\sigma_{12 \cdot Sk} = \text{Cov}(r_1, r_2) = \sigma_{12 \cdot S} - c_2 \sigma_{1k \cdot S} - c_1 \sigma_{2k \cdot S} + c_1 c_2 \sigma_{kk \cdot S}$, which simplifies to the first formula, and similarly $\sigma_{ii \cdot Sk} = \sigma_{ii \cdot S}(1 - \rho_{ik \cdot S}^2)$, which is positive by hypothesis. Dividing $\sigma_{12 \cdot Sk}$ by $\sqrt{\sigma_{11 \cdot Sk} \sigma_{22 \cdot Sk}}$ and then numerator and denominator by $\sqrt{\sigma_{11 \cdot S} \sigma_{22 \cdot S}}$ gives the second.

Proposition 3.6.3 — Partial correlations from the precision matrix

Let Σ be positive definite and $\Omega = \Sigma^{-1} = (\omega_{ij})$. The partial correlation of Y_i and Y_j given all the other entries of $\mathbf{Y}$ is

$$\rho_{ij \cdot \text{rest}} = -\frac{\omega_{ij}}{\sqrt{\omega_{ii}\omega_{jj}}},$$

and the conditional variance of Y_i given all the others is $1/\omega_{ii}$.

Proof. Permute $\mathbf{Y}$ so that (Y_i, Y_j) comes first. By Theorem 3.5.2(c) (a statement about matrices, valid whether or not $\mathbf{Y}$ is normal), the leading 2×2 block of Ω is $\Sigma_{11 \cdot 2}^{-1}$. Inverting,

$$\Sigma_{11 \cdot 2} = \begin{pmatrix} \omega_{ii} & \omega_{ij} \\ \omega_{ij} & \omega_{jj} \end{pmatrix}^{-1} = \frac{1}{\omega_{ii}\omega_{jj} - \omega_{ij}^2} \begin{pmatrix} \omega_{jj} & -\omega_{ij} \\ -\omega_{ij} & \omega_{ii} \end{pmatrix},$$

whose correlation is $-\omega_{ij}/\sqrt{\omega_{ii}\omega_{jj}}$. The same argument with a 1×1 leading block gives the conditional variance. ■

Proposition 3.6.3 has a striking consequence for normal vectors: *a zero entry ω_{ij} of the inverse covariance matrix means that Y_i and Y_j are conditionally independent given all the other variables.* Zeros in Σ describe marginal independence, zeros in Σ^{-1} describe conditional independence. The pattern of zeros in Σ^{-1} defines a graph on the variables, which is the starting point of Gaussian graphical models (Dempster 1972; Lauritzen 1996).

Example. (The trivariate example, continued)

For Σ of Example (Holding a third variable fixed), the listing gives the covariance of (Y_1, Y_2) given Y_3 as $\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$. So $\rho_{12} = 0.289$ while $\rho_{12 \cdot 3} = -0.707$. The script confirms that Eq. 3.6.1, Proposition 3.6.3, and the correlation of simulated residuals all give this value.

3.6.3 Sample partial correlations

For data, replace Σ by the sample covariance matrix. Because the empirical distribution of the rows is a probability distribution in its own right (Section 6.11), all the identities above hold exactly for the sample versions. In particular, for two variables $\mathbf{y}, \mathbf{x}$ and conditioning variables collected with an intercept in $\mathbf{Z}$, the following numbers are *identical*:

- (1) the correlation of the residuals of $\mathbf{y}$ and of $\mathbf{x}$ after least squares regression on $\mathbf{Z}$;
- (2) the recursion Eq. 3.6.1 applied to sample correlations (one conditioning variable) or repeatedly (several);
- (3) $-w_{12}/\sqrt{w_{11}w_{22}}$ computed from the inverse of the sample covariance or correlation matrix of $(\mathbf{y}, \mathbf{x}, \text{conditioning variables})$;
- (4) $\text{sign}(t)\sqrt{t^2/(t^2 + n - p)}$, where t is the t statistic of $\mathbf{x}$ in the regression of $\mathbf{y}$ on $[\mathbf{Z}, \mathbf{x}]$ with p columns (Exercise (B3)).

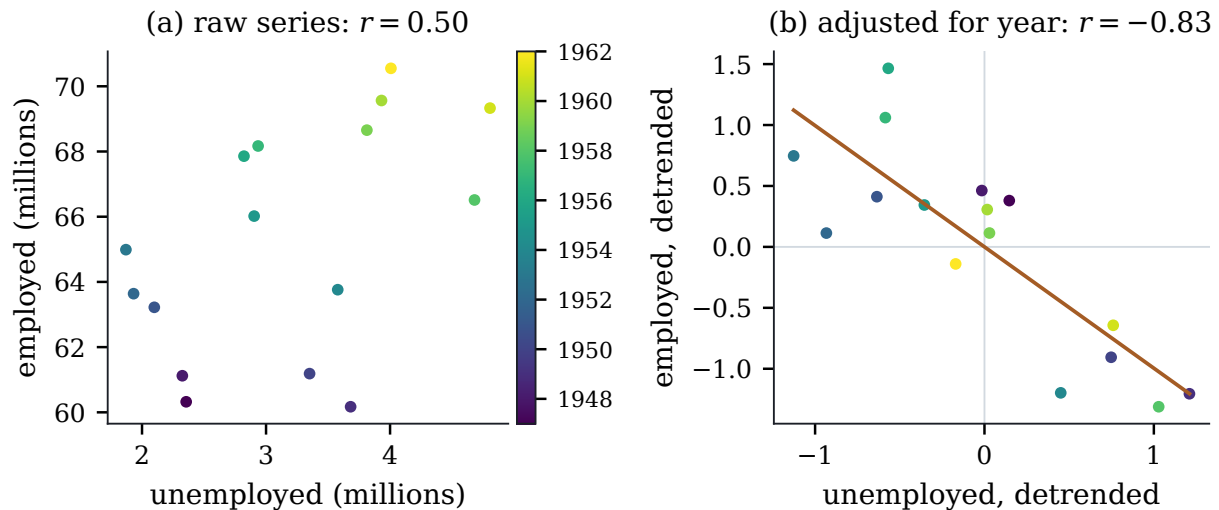


Figure 3.6.1: Employment and unemployment in the United States, 1947–1962, coloured by year. (a) The raw series are positively correlated because both trend upwards. (b) After removing the linear trend in year from each, the residuals are strongly negatively correlated. Their correlation is the partial correlation given year, and the fitted slope is the multiple-regression coefficient.

Example. (Employment and unemployment)

Longley’s macroeconomic series record, among other things, total employment and the number unemployed in the United States for the 16 years 1947–1962. Their correlation is 0.502: years with more people employed had more people unemployed. Both series grew with the population. Their correlations with the calendar year are 0.971 and 0.668. Adjusting for year, the partial correlation is -0.829 . Once the common trend is removed, a year with unemployment above its trend is a year with employment below its trend, as one would expect. Figure 3.6.1 shows both scatterplots. The listing computes the partial correlation in all four ways above. They agree to ten decimal places. The slope in panel (b), -0.996 , is the coefficient of unemployment in the regression of employment on unemployment and year, as Theorem 6.6.2 requires. Its t statistic is -5.34 .

Removing a linear time trend before correlating two series is the problem that Frisch and Waugh (1933) solved, and it is a special case of partial correlation.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data

emp = data["TOTEMP"].to_numpy() / 1000      # thousands -> millions of persons
unemp = data["UNEMP"].to_numpy() / 1000
year = data["YEAR"].to_numpy()
n = len(emp)
Z = np.column_stack([np.ones(n), year])     # the variables we adjust for

def resid(v, Z):
    coef, *_ = np.linalg.lstsq(Z, v, rcond=None)
    return v - Z @ coef

# 1. correlation of residuals (detrended series)
r_resid = np.corrcoef(resid(emp, Z), resid(unemp, Z))[0, 1]
```

```

# 2. the one-variable recursion from ordinary correlations
R = np.corrcoef(np.vstack([emp, unemp, year]))
r12, r13, r23 = R[0, 1], R[0, 2], R[1, 2]
r_formula = (r12 - r13 * r23) / np.sqrt((1 - r13**2) * (1 - r23**2))

# 3. from the inverse of the correlation (or covariance) matrix
W = np.linalg.inv(R)
r_precision = -W[0, 1] / np.sqrt(W[0, 0] * W[1, 1])

# 4. from the t statistic of unemployment in the regression of emp on (1, unemp,
    year)
fit = sm.OLS(emp, np.column_stack([Z, unemp])).fit()
t = fit.tvalues[2]
r_t = np.sign(t) * np.sqrt(t**2 / (t**2 + n - 3))

print(f"marginal r = {r12:.4f}")
print(f"partial r:  {r_resid:.10f} {r_formula:.10f} {r_precision:.10f} {r_t:.10f}")

```

Remark (Adjusting is not always right). A partial correlation answers a precise question: how are the parts of two variables not linearly explained by $\mathbf{Y}_2$ related? Whether that is the *right* question depends on what $\mathbf{Y}_2$ is. Adjusting for a common cause, as the year stands in for population growth here, removes a spurious association. Adjusting for a common effect can *create* an association between variables that are independent. Chapter 25 treats the difference. Under normality, the sampling distribution of a sample partial correlation given q variables is that of an ordinary sample correlation from $n - q$ observations, a result used for tests in Chapter 14.

3.6.4 Exercises

3.6.4.1 A. Check your understanding

Exercise A1.

Show that $\rho_{12.3} = 0$ iff $\rho_{12} = \rho_{13}\rho_{23}$. Find $\rho_{12.3}$ when $\rho_{12} = 0$ and $\rho_{13} = \rho_{23} = 0.6$, and interpret.

3.6.4.2 B. Practice

Exercise B1.

Let X , Z and e be independent, with $X, Z \sim N(0, 1)$ and $e \sim N(0, \tau^2)$, and let $C = X + Z + e$. Show that $\rho_{XZ} = 0$ but $\rho_{XZ.C} = -1/(1 + \tau^2)$. What happens as $\tau \rightarrow 0$ and as $\tau \rightarrow \infty$?

Solution. The covariance matrix of (X, Z, C) is $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2+\tau^2 \end{pmatrix}$. Conditioning on C ,

$$\sigma_{XZ.C} = 0 - \frac{1}{2 + \tau^2}, \quad \sigma_{XX.C} = \sigma_{ZZ.C} = 1 - \frac{1}{2 + \tau^2} = \frac{1 + \tau^2}{2 + \tau^2},$$

so $\rho_{XZ.C} = -1/(1 + \tau^2)$. As $\tau \rightarrow 0$ it tends to -1 : if $X + Z$ is known exactly, a larger

X forces a smaller Z . As $\tau \rightarrow \infty$ it tends to 0, since C carries little information about $X + Z$. Two independent causes become dependent once their common effect is held fixed.

Exercise B2.

Let Σ be positive definite. Show that the coefficient of Y_j in the best linear predictor of Y_1 from all the other entries is $-\omega_{1j}/\omega_{11}$. Deduce that $\rho_{1j \cdot \text{rest}}^2$ equals the product of the coefficient of Y_j for predicting Y_1 and the coefficient of Y_1 for predicting Y_j , each from all the other variables.

Solution. Put Y_1 first. By [Exercise \(B5\)](#) with a scalar first block, the coefficient vector of the best linear predictor of Y_1 from the rest is $-\Omega_{12}/\omega_{11}$, whose j th entry is $-\omega_{1j}/\omega_{11}$. Likewise the coefficient of Y_1 in the predictor of Y_j is $-\omega_{j1}/\omega_{jj}$. Their product is $\omega_{1j}^2/(\omega_{11}\omega_{jj})$, which is $\rho_{1j \cdot \text{rest}}^2$ by [Proposition 3.6.3](#).

Exercise B3.

Let D be an $n \times k$ data matrix whose columns are centred, and let $S = DTD$. Show, using [Theorem 6.6.2](#) and the partitioned inverse, that $-w_{12}/\sqrt{w_{11}w_{22}}$, computed from $S^{-1} = (w_{ij})$, equals the correlation of the residuals of columns 1 and 2 after regression on the remaining columns.

Exercise B4.

Using the Longley data of [Example \(Employment and unemployment\)](#), compute the partial correlation of employment and unemployment given both year and GNP, in at least two of the four ways. Draw the corresponding residual scatterplot. Does adjusting for GNP as well as year change the conclusion?

3.7 Summary and Notes

3.7.1 Summary

Idea.

1. $\mathbf{Y}$ is multivariate normal when every linear combination $\mathbf{aTY}$ is normal, constants included ([Definition 3.1.1](#)). The definition covers singular covariance matrices, which regression needs for fitted values and residuals.
2. $\mu + \mathbf{AZ}$, with Z standard normal, is $N(\mu, \mathbf{AA}^T)$. Every mean and every nonnegative definite covariance occurs ([Theorem 3.1.2](#)).
3. The moment generating function is $\exp(\mathbf{tT}\mu + \frac{1}{2}\mathbf{tT}\Sigma\mathbf{t})$, so a normal distribution is determined by μ and Σ ([Theorem 3.2.1](#)). When Σ is positive definite there is a density whose contours are the ellipsoids of constant Mahalanobis distance, and $\Delta^2(\mathbf{Y}) \sim \chi^2(n)$ ([Theorem 3.2.2](#) and [Proposition 3.2.3](#)).

4. Affine maps of normal vectors are normal, with no rank condition, and so are all subvectors ([Theorem 3.3.1](#) and [Corollary 3.3.2](#)). A normal vector of rank r is an exact affine image of r independent standard normals ([Proposition 3.3.4](#)).
5. Normal margins do not imply joint normality, and zero correlation between normal variables does not imply independence without it ([Example \(A random sign\)](#) and [Example \(A mixture of two correlations\)](#)).
6. For jointly normal blocks, zero cross-covariance is equivalent to independence. $\mathbf{AY}$ and $\mathbf{BY}$ are independent iff $\mathbf{A}\Sigma\mathbf{B}^T = \mathbf{0}$ ([Theorem 3.4.1](#) and [Corollary 3.4.2](#)).
7. $\mathbf{Y}_1 \mid \mathbf{Y}_2 = \mathbf{y}_2 \sim N(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{y}_2 - \boldsymbol{\mu}_2), \Sigma_{11.2})$: linear mean, constant covariance, reduced variance ([Theorem 3.5.2](#)). The conditional mean is the best predictor of all, and coincides with the best linear predictor ([Proposition 3.5.3](#)).
8. The partial correlation is the correlation of the residuals from linear prediction on the conditioning variables. Under normality it is the conditional correlation, and it is zero iff the variables are conditionally independent ([Definition 3.6.1](#) and [Proposition 3.6.1](#)). It can be computed by a recursion, from the inverse covariance matrix, or by FWL residuals ([Proposition 3.6.2](#) and [Proposition 3.6.3](#)).

3.7.2 Notes and sources

Three definitions. Linear-model texts reach the multivariate normal by different routes, and comparing them explains the choices made here. Rencher and Schaalje (2008, chapter 4) start from the positive definite case, obtain the density by transforming independent standard normals with $\Sigma^{1/2}$, and prove the properties of linear functions and independence with moment generating functions, for full-rank transformations. Seber and Lee (2003, chapter 2) also begin with the density, then extend the definition to singular covariance matrices through the representation $\mathbf{AZ} + \boldsymbol{\mu}$ and prove that normality of all linear combinations characterizes the family. Their moment-generating-function proof of the independence theorem is the model for [Section 3.4](#). Christensen (2020, §1.2) takes $\mathbf{AZ} + \boldsymbol{\mu}$ as the definition from the outset, shows with characteristic functions that the law depends only on $\boldsymbol{\mu}$ and Σ , and proves the independence theorem by building an explicit independent version. Defining the distribution through linear combinations, as this chapter does, goes back at least to Rao (1973, chapter 8). It has the advantage that the theorem on affine transformations needs no rank condition and a one-line proof.

Cramér–Wold. The device of reducing a vector to its linear combinations is from Cramér and Wold (1936), where it was used to reduce multivariate limit theorems to univariate ones. [Chapter 2](#) proves it, together with the uniqueness and factorization properties of moment generating functions used in [Section 3.2](#). Billingsley (1995) is the reference for these facts and for the conditioning property of independent variables used in the proof of [Theorem 3.5.2](#).

Conditional distributions and partial correlation. The proof of [Theorem 3.5.2](#) through the independent error $\mathbf{W}$ borrows one idea from Rencher and Schaalje (2008) and Seber and Lee (2003): both pass through the transformed vector $\mathbf{Y}_1 - \mathbf{BY}_2$, which is independent of $\mathbf{Y}_2$, although both still obtain the conditional density as a ratio of densities in the positive definite case. The proof here uses only independence and a generalized inverse, so it covers singular Σ_{22} , an extension that is standard in multivariate analysis (Rao 1973). Anderson (2003) is the classical reference for the multivariate normal in statistics, including the distribution theory of sample partial and multiple correlation coefficients that Chapter 14 uses. Rencher and Schaalje (2008, §4.5) define partial correlation as a conditional correlation, as in [Proposition 3.6.1\(b\)](#).

The residual interpretation (a) does not require normality and is the one that connects to least squares. Regression toward the mean was described by Galton (1886). The link between zeros of the inverse covariance matrix and conditional independence was made the basis of a modelling strategy by Dempster (1972), and Lauritzen (1996) develops the theory of graphical models that grew from it.

Data. [Example \(Employment and unemployment\)](#) uses the macroeconomic series published by Longley (1967) as a test of numerical accuracy, available in `statsmodels`. The same series appear in [Chapter 2](#), as an example of sample moments, and in [Chapter 6](#), to illustrate ill-conditioning.

3.7.3 References

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Chapter 4

Quadratic Forms and Their Distributions

Every sum of squares in a linear model is a quadratic form $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ in a normal vector, and every F and t statistic is a ratio built from such forms. This chapter works out their laws: the noncentral chi-squared, F and t distributions and the monotonicity of their tails; the mean, variance and a canonical representation of an arbitrary quadratic form; exactly when a quadratic form is chi-squared; when two forms, or a form and a linear function, are independent; and Cochran's theorem, which says when a sum of squares splits into independent chi-squared pieces.

What you need. [Chapter 1](#) (spectral theorem, symmetric idempotent matrices, trace), [Chapter 2](#) (means and covariances of random vectors) and [Chapter 3](#) (the multivariate normal distribution and its independence properties). Orthogonal projections appear here only as symmetric idempotent matrices. [Chapter 6](#) develops their geometry.

4.0.1 Roadmap

- [Noncentral chi-squared, t and F distributions](#)
- [Mean and variance of quadratic forms](#)
- [When a quadratic form is chi-squared](#)
- [Independence of quadratic forms and linear forms](#)
- [The Cochran theorem](#)
- [Summary and notes](#)

4.1 Noncentral chi-squared, t and F distributions

A regression analysis reports a handful of numbers whose sampling behaviour decides everything: the residual sum of squares, the reduction in it when a term is added, the ratio of the two. Each

is a function of the response vector of the form

$$\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} Y_i Y_j,$$

a **quadratic form** in $\mathbf{Y}$. The sample variance is one, with $\mathbf{A} = (n-1)^{-1}(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}\mathbf{T})$. So is the squared length $\|\mathbf{P}\mathbf{Y}\|^2 = \mathbf{Y}\mathbf{T}\mathbf{P}\mathbf{Y}$ of any symmetric idempotent image of $\mathbf{Y}$, and this is how the sums of squares of Chapter 6 will arise.

Since $\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}$ is a scalar, it equals its own transpose $\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{T}\mathbf{Y}$, and so also $\mathbf{Y}\mathbf{T}\frac{1}{2}(\mathbf{A} + \mathbf{A}\mathbf{T})\mathbf{Y}$. Replacing $\mathbf{A}$ by its symmetric part changes nothing, and *throughout this chapter the matrix of a quadratic form is symmetric*. Several formulas below, the variance formula among them, are false for a non-symmetric $\mathbf{A}$ (Exercise (A1)).

When $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ and $\mathbf{A}$ is symmetric idempotent, $\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}$ has a chi-squared distribution, which is the null distribution of a test. Under an alternative the mean is not zero, and a slightly larger family appears. We start with that family.

4.1.1 Noncentral chi-squared

Recall that $\chi^2(r)$, the chi-squared distribution with r degrees of freedom, is the law of $Z_1^2 + \cdots + Z_r^2$ for independent standard normal Z_i . It is the gamma distribution with shape $r/2$ and scale 2, with density

$$g_r(x) = \frac{x^{r/2-1} e^{-x/2}}{2^{r/2} \Gamma(r/2)}, \quad x > 0, \quad (4.1.1)$$

mean r , variance $2r$ and moment generating function $(1-2t)^{-r/2}$ for $t < 1/2$. By convention $\chi^2(0)$ is the point mass at zero. Everything about the noncentral version follows from one computation.

Lemma 4.1.1

Let $W \sim N(0, 1)$ and $\nu \in \mathbb{R}$. For $t < 1/2$,

$$\mathbf{E}[e^{t(W+\nu)^2}] = (1-2t)^{-1/2} \exp\left(\frac{\nu^2 t}{1-2t}\right).$$

Proof. With $u = w + \nu$ the expectation is $(2\pi)^{-1/2} \int \exp\{tu^2 - (u-\nu)^2/2\} du$. Completing the square in the exponent,

$$tu^2 - \frac{1}{2}(u-\nu)^2 = -\frac{1-2t}{2} \left(u - \frac{\nu}{1-2t}\right)^2 + \frac{\nu^2}{2} \left(\frac{1}{1-2t} - 1\right).$$

The last term equals $\nu^2 t / (1-2t)$. The remaining Gaussian integral is $(2\pi)^{1/2} (1-2t)^{-1/2}$ when $1-2t > 0$.

■

Definition 4.1.1 — Noncentral chi-squared

Let $\mathbb{Z} \sim N_r(\boldsymbol{\mu}, \mathbf{I}_r)$. The distribution of $\mathbb{Z}\mathbf{T}\mathbb{Z} = \sum_i Z_i^2$ is the **noncentral chi-squared distribution** with r degrees of freedom and **noncentrality** $\gamma = \boldsymbol{\mu}\mathbf{T}\boldsymbol{\mu} = \|\boldsymbol{\mu}\|^2$, written $\chi^2(r, \gamma)$.

The definition hides a claim: the law of $\|\mathbb{Z}\|^2$ depends on the mean vector $\boldsymbol{\mu}$ only through its length. [Theorem 4.1.2](#) proves it. Intuitively, a rotation $\mathbb{Q}\mathbb{Z}$ of a spherical normal is again spherical normal, with mean $\mathbb{Q}\boldsymbol{\mu}$, and it has the same length as $\mathbb{Z}$, so every mean vector of a given length gives the same answer.

Remark (Convention). In this book the noncentrality is $\gamma = \boldsymbol{\mu}\mathbf{T}\boldsymbol{\mu}$, with no factor $\frac{1}{2}$. Then $\mathbf{E}\chi^2(r, \gamma) = r + \gamma$, which is easy to remember, and γ agrees with the parameter `nc` of `scipy.stats.ncx2`. Several texts, including Rencher and Schaalje (2008) and Christensen (2020), use $\lambda = \boldsymbol{\mu}\mathbf{T}\boldsymbol{\mu}/2$ instead. When reading them, replace their λ by $\gamma/2$. The same convention applies to the noncentral F distribution below.

Theorem 4.1.2 — Properties of the noncentral chi-squared

Let $U \sim \chi^2(r, \gamma)$.

- (a) The moment generating function is

$$M_U(t) = (1 - 2t)^{-r/2} \exp\left(\frac{\gamma t}{1 - 2t}\right), \quad t < \frac{1}{2}, \quad (4.1.2)$$

so the law depends on $\boldsymbol{\mu}$ only through γ , and $\chi^2(r, 0) = \chi^2(r)$.

- (b) $\mathbf{E}(U) = r + \gamma$ and $\text{Var}(U) = 2r + 4\gamma$.
 (c) If $U_1, \dots, U_k$ are independent with $U_i \sim \chi^2(r_i, \gamma_i)$, then $\sum_i U_i \sim \chi^2(\sum_i r_i, \sum_i \gamma_i)$.
 (d) (*Poisson mixture.*) Let $K \sim \text{Poisson}(\gamma/2)$ and, given $K = k$, let $V \sim \chi^2(r + 2k)$. Then $V \sim \chi^2(r, \gamma)$. Consequently U has density

$$f_{r, \gamma}(x) = \sum_{k=0}^{\infty} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} g_{r+2k}(x), \quad x > 0. \quad (4.1.3)$$

Proof.

- (a) With $\mathbb{Z} \sim N_r(\boldsymbol{\mu}, \mathbf{I})$, the Z_i are independent, and $Z_i = W_i + \mu_i$ with $W_i \sim N(0, 1)$. By [Lemma 4.1.1](#),

$$\mathbf{E} e^{t\mathbb{Z}\mathbf{T}\mathbb{Z}} = \prod_{i=1}^r (1 - 2t)^{-1/2} \exp\left(\frac{\mu_i^2 t}{1 - 2t}\right) = (1 - 2t)^{-r/2} \exp\left(\frac{\gamma t}{1 - 2t}\right).$$

A moment generating function that is finite on an open interval around zero determines the distribution, so the law depends only on (r, γ) . Setting $\gamma = 0$ gives the central mgf.

- (b) The cumulant generating function is $K(t) = \log M_U(t) = -\frac{r}{2} \log(1 - 2t) + \gamma t / (1 - 2t)$. Differentiating, $K'(t) = r(1 - 2t)^{-1} + \gamma(1 - 2t)^{-2}$ and $K''(t) = 2r(1 - 2t)^{-2} + 4\gamma(1 - 2t)^{-3}$. The mean and variance are $K'(0)$ and $K''(0)$.

- (c) By independence the mgf of the sum is the product of the mgfs 4, which has the same form with $\sum r_i$ and $\sum \gamma_i$.
- (d) Conditioning on K ,

$$\mathbf{E} e^{tV} = \sum_{k=0}^{\infty} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} (1-2t)^{-(r+2k)/2} = (1-2t)^{-r/2} e^{-\gamma/2} \exp\left(\frac{\gamma/2}{1-2t}\right),$$

and $-\frac{\gamma}{2} + \frac{\gamma}{2(1-2t)} = \frac{\gamma t}{1-2t}$. This is

4. The density of a mixture is the mixture of the densities. ■

The mixture in (d) is more than a formula. It says that noncentrality acts like a random number of extra degrees of freedom, two at a time, with $\gamma/2$ of them on average. Since each extra pair of degrees of freedom pushes the distribution to the right, a larger γ should give a stochastically larger variable. That is the next result, and it is what makes tests based on these distributions sensible.

Example. (A noncentral chi-squared with four degrees of freedom)

Take $r = 4$ and $\gamma = 6$. The Poisson(3) weights of the central densities g_4, g_6, g_8, g_{10} in Eq. 4.1.3 are 0.050, 0.149, 0.224 and 0.224, and the mean and variance are 10 and 32. The listing checks the mixture against a library implementation of the density. The script also squares normal vectors with a mean of length $\sqrt{6}$ in an arbitrary direction, and their average squared length is 10.002. A test that rejects when a $\chi^2(4)$ statistic exceeds its upper 5% point 9.488 rejects with probability 0.470 when the statistic is in fact $\chi^2(4, 6)$. Figure 4.1.1 shows how the density moves right and spreads out as γ grows.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(20240401)

def ncx2_pdf_mixture(x, r, gamma, terms=200):
    """Density of chi^2(r, gamma) as a Poisson(gamma/2) mixture of central densities
    . """
    k = np.arange(terms)
    weights = stats.poisson.pmf(k, gamma / 2)          # P(K = k)
    dens = stats.chi2.pdf(np.asarray(x)[: , None], r + 2 * k)
    return dens @ weights

r, gamma = 4, 6.0
x = np.linspace(0.05, 40, 400)
print(np.max(np.abs(ncx2_pdf_mixture(x, r, gamma) - stats.ncx2.pdf(x, r, gamma))))
```

Proposition 4.1.3 — Monotonicity in the noncentrality

For fixed $r \geq 1$ and $x > 0$, the tail probability $\gamma \mapsto \Pr\{\chi^2(r, \gamma) > x\}$ is continuous and strictly increasing on $[0, \infty)$, and tends to 1 as $\gamma \rightarrow \infty$.

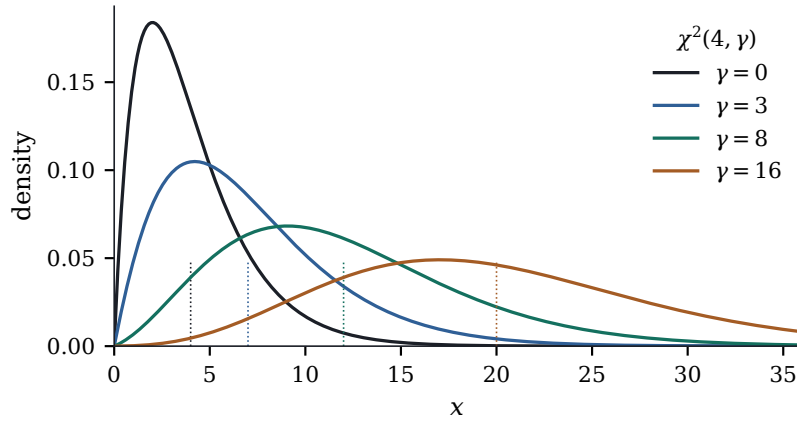


Figure 4.1.1: Densities of $\chi^2(4, \gamma)$. Dotted ticks mark the means $4 + \gamma$. The variance $2r + 4\gamma$ grows with the noncentrality, so the distributions flatten as they move right.

Proof. Let $h(k) = \Pr\{\chi^2(r + 2k) > x\}$ and $p_k(\lambda) = e^{-\lambda}\lambda^k/k!$. By Theorem 4.1.2(d), the tail probability is $\phi(\lambda) = \sum_k p_k(\lambda)h(k)$ with $\lambda = \gamma/2$. First, h is strictly increasing: if $V \sim \chi^2(r + 2k)$ and $E \sim \chi^2(2)$ are independent, then $V + E \sim \chi^2(r + 2k + 2)$ and $h(k + 1) - h(k) = \Pr(V \leq x < V + E) > 0$, since (V, E) has a positive joint density on $(0, \infty)^2$. Next, $p'_k(\lambda) = p_{k-1}(\lambda) - p_k(\lambda)$ (with $p_{-1} = 0$). Because $0 \leq h \leq 1$, the series of derivatives converges uniformly on bounded intervals, so term-by-term differentiation is valid and

$$\phi'(\lambda) = \sum_{k \geq 0} (p_{k-1}(\lambda) - p_k(\lambda))h(k) = \sum_{k \geq 0} p_k(\lambda)(h(k+1) - h(k)) > 0.$$

For the limit, fix $\varepsilon > 0$. Since $\chi^2(r + 2m)$ has mean $r + 2m$ and standard deviation $\sqrt{2r + 4m}$, Chebyshev's inequality gives $h(m) > 1 - \varepsilon$ for some m . Then $\phi(\lambda) \geq h(m) \Pr(K \geq m)$ for $K \sim \text{Poisson}(\lambda)$, and $\Pr(K \geq m) \rightarrow 1$ as $\lambda \rightarrow \infty$. ■

4.1.2 Noncentral F and t

Test statistics divide one quadratic form by another, to remove the unknown scale. The two ratio distributions that result are defined next.

Definition 4.1.2 — Noncentral F

Let $U \sim \chi^2(r, \gamma)$ and $V \sim \chi^2(s)$ be independent, $r, s \geq 1$. The law of

$$F = \frac{U/r}{V/s}$$

is the **noncentral F distribution** $F(r, s, \gamma)$, with numerator and denominator degrees of freedom r and s and noncentrality γ . The central $F(r, s) = F(r, s, 0)$ has upper α point $F_\alpha(r, s)$.

Definition 4.1.3 — Noncentral t

Let $X \sim N(\delta, 1)$ and $V \sim \chi^2(s)$ be independent. The law of $T = X/\sqrt{V/s}$ is the **noncentral t distribution** $t(s, \delta)$, and $t(s) = t(s, 0)$.

Only the numerator is allowed to be noncentral. A noncentral denominator gives a *doubly noncentral* F , which appears when the model used for the error sum of squares is itself wrong (Exercise (B3)). Two relations are immediate. First, $X^2 \sim \chi^2(1, \delta^2)$, so $T^2 \sim F(1, s, \delta^2)$: a two-sided t test is an F test with one numerator degree of freedom. Second, since $\mathbf{E}(V^{-1}) = 1/(s-2)$ for $s > 2$, independence gives

$$\mathbf{E}(F) = \frac{r + \gamma}{r} \cdot \frac{s}{s - 2}, \quad s > 2.$$

The central F is the null distribution of the tests of Chapter 11, and $F(r, s, \gamma)$ is their distribution under an alternative. The power of such a test is therefore $\Pr\{F(r, s, \gamma) > F_\alpha(r, s)\}$, and the following fact is what guarantees that it behaves as a test should.

Theorem 4.1.4 — Power of the F test increases with γ

For fixed $r, s \geq 1$ and $c > 0$, $\beta(\gamma) = \Pr\{F(r, s, \gamma) > c\}$ is continuous and strictly increasing in $\gamma \geq 0$, and $\beta(\gamma) \rightarrow 1$ as $\gamma \rightarrow \infty$. In particular, the level- α test that rejects when $F > F_\alpha(r, s)$ has power greater than α at every $\gamma > 0$.

Proof. With U, V as in Definition 4.1.2, $F > c$ iff $U > crV/s$. Condition on V , which is independent of U :

$$\beta(\gamma) = \mathbf{E}[\psi_\gamma(crV/s)], \quad \psi_\gamma(x) = \Pr\{\chi^2(r, \gamma) > x\}.$$

Since $V > 0$ with probability one, Proposition 4.1.3 gives, for $\gamma < \gamma'$, $\psi_\gamma(crV/s) < \psi_{\gamma'}(crV/s)$ almost surely, so $\beta(\gamma) < \beta(\gamma')$. Continuity and the limit follow from the same proposition by dominated convergence, since $0 \leq \psi_\gamma \leq 1$. ■

Two further monotonicity facts hold for fixed $\gamma > 0$ and fixed level. The power decreases as the numerator degrees of freedom r grow, and increases as the denominator degrees of freedom s grow (Ghosh 1973). The first says that spreading a fixed amount of signal over more dimensions makes it harder to detect. The second says that estimating σ^2 better helps. We use both only qualitatively, and Figure 4.4.1 in Section 4.4 shows the first. Because $T^2 \sim F(1, s, \delta^2)$, Theorem 4.1.4 also shows that the power of the two-sided t test increases with $|\delta|$ (Exercise (A2)).

4.1.3 Exercises**4.1.3.1 A. Check your understanding**

Exercise A1.

Show that $\chi^2(2)$ is the exponential distribution with mean 2. Use [Theorem 4.1.2\(d\)](#) to show that

$$\Pr\{\chi^2(2, \gamma) > x\} = \sum_{k \geq 0} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} e^{-x/2} \sum_{j=0}^k \frac{(x/2)^j}{j!}.$$

Exercise A2.

Show that $T \sim t(s, \delta)$ implies $T^2 \sim F(1, s, \delta^2)$. Deduce that the power of the two-sided level- α t test depends on δ only through $|\delta|$ and increases with it. Show directly, by conditioning on the denominator, that the power of the one-sided test that rejects for $T > c$ increases with δ .

Solution. $T^2 = X^2/(V/s)$ with $X^2 \sim \chi^2(1, \delta^2)$ independent of V , which is [Definition 4.1.2](#). The two-sided test rejects when $|T| > t_{\alpha/2}(s)$, that is, when $T^2 > F_{\alpha}(1, s)$. Its power is therefore $\Pr\{F(1, s, \delta^2) > F_{\alpha}(1, s)\}$, a strictly increasing function of δ^2 by [Theorem 4.1.4](#). For the one-sided test, $\Pr(T > c) = \mathbf{E} \Pr\{X > c\sqrt{V/s} \mid V\} = \mathbf{E} \Phi(\delta - c\sqrt{V/s})$, and Φ is strictly increasing.

4.1.3.2 B. Practice**Exercise B1.**

Show that the k th cumulant of $\chi^2(r, \gamma)$ is $2^{k-1}(k-1)!(r+k\gamma)$. Deduce that its skewness tends to zero like $3/\sqrt{\gamma}$ as $\gamma \rightarrow \infty$ with r fixed.

Solution. Using $-\frac{1}{2} \log(1-2t) = \sum_{k \geq 1} (2t)^k / (2k)$ and $t/(1-2t) = \sum_{k \geq 1} 2^{k-1} t^k$, the cumulant generating function is

$$K(t) = \sum_{k \geq 1} t^k 2^{k-1} \left(\frac{r}{k} + \gamma \right) = \sum_{k \geq 1} \frac{t^k}{k!} 2^{k-1} (k-1)! (r+k\gamma).$$

The k th cumulant is $k!$ times the coefficient of t^k . In particular $\kappa_2 = 2(r+2\gamma)$ and $\kappa_3 = 8(r+3\gamma)$, so the skewness is $\kappa_3/\kappa_2^{3/2} = 8(r+3\gamma)/\{2(r+2\gamma)\}^{3/2}$. As $\gamma \rightarrow \infty$ this behaves like $24\gamma/(4\gamma)^{3/2} = 3/\sqrt{\gamma}$.

Exercise B2.

Show that $\chi^2(1, \gamma)$ has density $(2\pi x)^{-1/2} e^{-(x+\gamma)/2} \cosh \sqrt{\gamma x}$ for $x > 0$. Verify that expanding $\cosh$ in a power series reproduces [Eq. 4.1.3](#). *Hint:* $\Gamma(k+\frac{1}{2}) = \sqrt{\pi} (2k)!/(4^k k!)$.

Solution. Let $X = (W + \sqrt{\gamma})^2$ with $W \sim N(0, 1)$. For $x > 0$, $X \leq x$ iff $-\sqrt{x} - \sqrt{\gamma} \leq W \leq \sqrt{x} - \sqrt{\gamma}$. Differentiating with ϕ the standard normal density,

$$f(x) = \frac{\phi(\sqrt{x} - \sqrt{\gamma}) + \phi(\sqrt{x} + \sqrt{\gamma})}{2\sqrt{x}} = \frac{e^{-(x+\gamma)/2}}{\sqrt{2\pi x}} \cdot \frac{e^{\sqrt{\gamma x}} + e^{-\sqrt{\gamma x}}}{2}.$$

Expanding, $f(x) = (2\pi x)^{-1/2} e^{-(x+\gamma)/2} \sum_k (\gamma x)^k / (2k)!$. The k th term of Eq. 4.1.3 is

$$\frac{e^{-\gamma/2} (\gamma/2)^k x^{k-1/2} e^{-x/2}}{k! 2^{k+1/2} \Gamma(k + \frac{1}{2})}.$$

By the hint the denominator is $k! 2^{k+1/2} \sqrt{\pi} (2k)! / (4^k k!) = \sqrt{2\pi} 2^{-k} (2k)!$, and the k th term becomes $(2\pi x)^{-1/2} e^{-(x+\gamma)/2} (\gamma x)^k / (2k)!$, as required.

Exercise B3.

Let $U_\gamma \sim \chi^2(r, \gamma)$ with r fixed. Show that $(U_\gamma - r - \gamma) / \sqrt{2r + 4\gamma}$ converges in distribution to $N(0, 1)$ as $\gamma \rightarrow \infty$. *Hint:* U_γ has the law of $(W_1 + \sqrt{\gamma})^2 + W_2^2 + \cdots + W_r^2$ with $W_1, \dots, W_r$ independent standard normal.

Exercise B4.

Let $U \sim \chi^2(a)$ and $V \sim \chi^2(b)$ be independent. Show that $B = U/(U + V)$ has the $\text{Beta}(a/2, b/2)$ distribution and is independent of $U + V$, and that $F = (U/a)/(V/b) = (b/a) B/(1 - B)$. Deduce the null distribution of R^2 asked for in Chapter 6.

Solution. The joint density of (U, V) is proportional to $u^{a/2-1} v^{b/2-1} e^{-(u+v)/2}$. Change variables to $w = u/(u + v) \in (0, 1)$ and $s = u + v > 0$, so that $u = ws$, $v = (1 - w)s$ and the Jacobian is s . The density of (W, S) is proportional to

$$w^{a/2-1} (1 - w)^{b/2-1} \cdot s^{(a+b)/2-1} e^{-s/2},$$

a product of a function of w and a function of s . So B and $U + V$ are independent, $B \sim \text{Beta}(a/2, b/2)$ and $U + V \sim \chi^2(a + b)$. Also $U/V = B/(1 - B)$. For R^2 in a model with intercept, $R^2 = \text{SSR}/(\text{SSR} + \text{SSE})$, where $\text{SSR} = \|(\mathbf{M} - \mathbf{P}_1)\mathbf{Y}\|^2$ and $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ come from orthogonal projections of ranks $r - 1$ and $n - r$. If $\boldsymbol{\mu} = \mu \mathbf{1}$, both noncentralities vanish, and Theorem 4.4.4 with $a = r - 1$, $b = n - r$ gives $R^2 \sim \text{Beta}((r - 1)/2, (n - r)/2)$.

Exercise B5.

Show that $f_{r,\gamma}(x)/g_r(x)$ is strictly increasing in $x > 0$ for every $\gamma > 0$. Conclude that among tests based on one observation $U \sim \chi^2(r, \gamma)$ of $\gamma = 0$ against a fixed $\gamma > 0$, rejecting for large U is most powerful.

4.2 Mean and variance of quadratic forms

4.2.1 The mean needs no normality

Theorem 2.3.1 computed $\mathbf{E}(\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y})$ for any random vector with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$:

$$\mathbf{E}(\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}) = \text{tr}(\mathbf{A}\mathbf{\Sigma}) + \boldsymbol{\mu}\mathbf{T}\mathbf{A}\boldsymbol{\mu}. \quad (4.2.1)$$

The formula has two parts with distinct roles. The trace term is what the noise contributes. The term $\boldsymbol{\mu}\mathbf{T}\mathbf{A}\boldsymbol{\mu}$ is what the signal contributes. For nonnegative definite $\mathbf{A}$ it vanishes exactly when $\mathbf{A}\boldsymbol{\mu} = \mathbf{0}$, which describes the mean vectors a well-designed estimator is meant to ignore.

Example. (Two estimators of σ^2 under a drifting mean)

Let $Y_1, \dots, Y_n$ be uncorrelated with common variance σ^2 , and consider

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2, \quad Q = \frac{1}{2(n-1)} \sum_{i=1}^{n-1} (Y_{i+1} - Y_i)^2,$$

the sample variance and the **mean square successive difference**. In matrix form $S^2 = \mathbf{Y}\mathbf{T}\mathbf{C}\mathbf{Y}/(n-1)$ with $\mathbf{C} = \mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^T$, and $Q = \mathbf{Y}\mathbf{T}\mathbf{D}\mathbf{T}\mathbf{D}\mathbf{Y}/\{2(n-1)\}$, where $\mathbf{D}$ is the $(n-1) \times n$ first-difference matrix with rows $(\dots, -1, 1, \dots)$. Both matrices annihilate $\mathbf{1}$, so both estimators ignore a constant mean. Since $\text{tr}(\mathbf{C}) = n-1$ and $\text{tr}(\mathbf{D}\mathbf{T}\mathbf{D}) = \text{tr}(\mathbf{D}\mathbf{D}\mathbf{T}) = 2(n-1)$, Eq. 4.2.1 with $\mathbf{\Sigma} = \sigma^2\mathbf{I}$ shows that both are unbiased when the mean is constant.

Now let the mean drift linearly, $\mu_i = \mu_0 + \theta i$. Then $\mathbf{C}\boldsymbol{\mu}$ is the centred trend and $\mathbf{D}\boldsymbol{\mu} = \theta\mathbf{1}_{n-1}$, so

$$\mathbf{E}(S^2) = \sigma^2 + \theta^2 \frac{n(n+1)}{12}, \quad \mathbf{E}(Q) = \sigma^2 + \frac{\theta^2}{2}.$$

The bias of S^2 grows quadratically with the length of the series, while that of Q does not grow at all. With $n = 20$, $\sigma = 1$ and $\theta = 0.1$, the means are 1.350 and 1.005. Differencing removes smooth trends, and this is why Q was proposed for series whose level may wander (von Neumann 1941). The price of robustness to drift is paid in variance, which we compute below.

4.2.2 A canonical representation

Higher moments and distributions need normality, and one representation answers all such questions at once.

Theorem 4.2.1 — Canonical form of a normal quadratic form

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{\Sigma})$ and let $\mathbf{A}$ be symmetric. Write $\mathbf{\Sigma} = \mathbf{L}\mathbf{L}^T$ with $\mathbf{L}$ of size $n \times k$, and take a spectral decomposition $\mathbf{L}^T\mathbf{A}\mathbf{L} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$ with $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_k)$. Put $\mathbf{b} = \mathbf{Q}^T\mathbf{L}^T\mathbf{A}\boldsymbol{\mu}$. Then

$$\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y} \stackrel{d}{=} \boldsymbol{\mu}\mathbf{T}\mathbf{A}\boldsymbol{\mu} + \sum_{j=1}^k (\lambda_j W_j^2 + 2b_j W_j), \quad \mathbf{W} \sim N_k(\mathbf{0}, \mathbf{I}_k). \quad (4.2.2)$$

For $\lambda_j \neq 0$ the j th term equals $\lambda_j(W_j + b_j/\lambda_j)^2 - b_j^2/\lambda_j$. The nonzero λ_j are the nonzero

eigenvalues of $\mathbf{A}\Sigma$.

Proof. Such an $\mathbf{L}$ exists. By [Theorem 1.7.2](#), $\Sigma = \mathbf{U}\mathbf{D}\mathbf{U}^T$ with $\mathbf{U}$ orthogonal and $\mathbf{D}$ diagonal with entries $d_i \geq 0$, and $\mathbf{L} = \mathbf{U}\mathbf{D}^{1/2}$ works with $k = n$. Let $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{I})$. The vector $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$ is normal with mean $\boldsymbol{\mu}$ and covariance $\mathbf{L}\mathbf{L}^T = \Sigma$ ([Theorem 3.3.1](#)). A normal distribution is determined by its mean and covariance ([Theorem 3.2.1](#)), so $\mathbf{Y}$ and $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$ have the same law, and so do $\mathbf{Y}^T\mathbf{A}\mathbf{Y}$ and

$$(\boldsymbol{\mu} + \mathbf{L}\mathbf{Z})^T\mathbf{A}(\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}) = \boldsymbol{\mu}^T\mathbf{A}\boldsymbol{\mu} + 2\boldsymbol{\mu}^T\mathbf{A}\mathbf{L}\mathbf{Z} + \mathbf{Z}^T\mathbf{L}^T\mathbf{A}\mathbf{L}\mathbf{Z}.$$

Put $\mathbf{W} = \mathbf{Q}^T\mathbf{Z}$, again $N_k(\mathbf{0}, \mathbf{I})$. Then $\mathbf{Z}^T\mathbf{L}^T\mathbf{A}\mathbf{L}\mathbf{Z} = \mathbf{W}^T\mathbf{A}\mathbf{W} = \sum \lambda_j W_j^2$ and $\boldsymbol{\mu}^T\mathbf{A}\mathbf{L}\mathbf{Z} = \mathbf{b}^T\mathbf{W}$, which is [Eq. 4.2.2](#). The completed square is algebra. Finally, $\mathbf{L}^T\mathbf{A}\mathbf{L} = \mathbf{H}\mathbf{F}$ with $\mathbf{F} = \mathbf{A}\mathbf{L}$ and $\mathbf{H} = \mathbf{L}^T$, and $\mathbf{F}\mathbf{H} = \mathbf{A}\Sigma$. By [Proposition 1.7.1\(d\)](#) these two products have the same nonzero eigenvalues, with the same multiplicities. ■

So a normal quadratic form is a constant plus a sum of independent pieces. Each piece is either a scaled noncentral $\chi^2(1)$ variable, with scale λ_j , or a normal variable with mean zero, when $\lambda_j = 0$ but $b_j \neq 0$. Because the λ_j may be negative, $\mathbf{Y}^T\mathbf{A}\mathbf{Y}$ need not be positive. The linear pieces are special to singular covariances. If Σ is positive definite, $\mathbf{L}$ can be taken square and nonsingular, and then $\lambda_j = 0$ gives $\mathbf{A}\mathbf{L}\mathbf{q}_j = \mathbf{0}$ for the j th column $\mathbf{q}_j$ of $\mathbf{Q}$, so $b_j = (\mathbf{A}\mathbf{L}\mathbf{q}_j)^T\boldsymbol{\mu} = 0$. A linear piece needs a direction in which $\mathbf{A}$ responds to the mean while the noise is absent.

Corollary 4.2.2 — Moment generating function

In the setting of [Theorem 4.2.1](#), for $|t| < 1/(2\max_j |\lambda_j|)$ (any t if all $\lambda_j = 0$),

$$\mathbf{E} e^{t\mathbf{Y}^T\mathbf{A}\mathbf{Y}} = e^{t\boldsymbol{\mu}^T\mathbf{A}\boldsymbol{\mu}} \prod_{j=1}^k (1 - 2\lambda_j t)^{-1/2} \exp\left(\frac{2b_j^2 t^2}{1 - 2\lambda_j t}\right).$$

Proof. The terms of [Eq. 4.2.2](#) are independent. If $\lambda_j = 0$, the term $2b_j W_j$ has mgf $e^{2b_j^2 t^2}$. If $\lambda_j \neq 0$, apply [Lemma 4.1.1](#) to $W_j + \nu_j$ with $\nu_j = b_j/\lambda_j$ at the argument $\lambda_j t$, and multiply by $e^{-tb_j^2/\lambda_j}$. The exponent is $b_j^2 t / \{\lambda_j(1 - 2\lambda_j t)\} - b_j^2 t / \lambda_j = 2b_j^2 t^2 / (1 - 2\lambda_j t)$. ■

4.2.3 Variance and covariances

Theorem 4.2.3 — Moments of normal quadratic forms

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$, let $\mathbf{A}$ and $\mathbf{B}$ be symmetric $n \times n$ matrices, and let $\mathbf{K}$ be an $m \times n$ matrix. Then

- (a) $\mathbf{E}(\mathbf{Y}^T\mathbf{A}\mathbf{Y}) = \text{tr}(\mathbf{A}\Sigma) + \boldsymbol{\mu}^T\mathbf{A}\boldsymbol{\mu}$;
- (b) $\text{Var}(\mathbf{Y}^T\mathbf{A}\mathbf{Y}) = 2\text{tr}\{(\mathbf{A}\Sigma)^2\} + 4\boldsymbol{\mu}^T\mathbf{A}\Sigma\mathbf{A}\boldsymbol{\mu}$;


```

def qf_moments(A, mu, Sigma):
    """Mean and variance of  $Y^T A Y$  for  $Y \sim N(\mu, \Sigma)$ ,  $A$  symmetric."""
    AS = A @ Sigma
    mean = np.trace(AS) + mu @ A @ mu
    var = 2 * np.trace(AS @ AS) + 4 * mu @ AS @ A @ mu
    return mean, var

C = np.eye(n) - np.ones((n, n)) / n # centring matrix
D = np.diff(np.eye(n), axis=0)      # (n-1) x n first differences
A_s2 = C / (n - 1)
A_q = D.T @ D / (2 * (n - 1))
for name, A in [("S2", A_s2), ("Q", A_q)]:
    print(name, qf_moments(A, mu, Sigma), qf_moments(A, 0 * mu, Sigma))

```

Normality enters (b)–(d) through the third and fourth moments of $\mathbf{Y}$. For independent coordinates with excess kurtosis the variance acquires an extra term that involves only the diagonal of $\mathbf{A}$ (Theorem 2.3.3). Variance formulas for sums of squares derived under normality can be badly wrong for heavy-tailed errors, a point taken up in Chapter 21.

4.2.4 Exercises

4.2.4.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$ and $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Compute $\text{Var}(\mathbf{Y}^T \mathbf{A} \mathbf{Y})$ directly, and compare with $2 \text{tr}(\mathbf{A}^2)$ and with $2 \text{tr}(\mathbf{A}_s^2)$, where $\mathbf{A}_s = \frac{1}{2}(\mathbf{A} + \mathbf{A}^T)$.

4.2.4.2 B. Practice

Exercise B1.

Let $(U_i, W_i), i = 1, \dots, n$, be independent copies of a bivariate normal pair with variances σ_U^2, σ_W^2 and covariance σ_{UW} , and let $s_{UW} = (n-1)^{-1} \sum_i (U_i - \bar{U})(W_i - \bar{W})$. Write $s_{UW} = \mathbf{Y}^T \mathbf{A} \mathbf{Y}$ for the stacked vector $\mathbf{Y} = (\mathbf{U}^T, \mathbf{W}^T)^T$ and a symmetric $2n \times 2n$ matrix $\mathbf{A}$, and use Theorem 4.2.3 to show that $\text{Var}(s_{UW}) = (\sigma_U^2 \sigma_W^2 + \sigma_{UW}^2) / (n-1)$. Check the answer against $\text{Var}(S^2)$ when $\mathbf{U} = \mathbf{W}$.

Exercise B2.

Let $Y_1, \dots, Y_n$ be independent with mean 0, variance σ^2 and $\mathbf{E}Y_i^4 = \mu_4$. By Exercise (B4), for symmetric $\mathbf{A}$ and $\mathbf{B}$,

$$\text{Cov}(\mathbf{Y}^T \mathbf{A} \mathbf{Y}, \mathbf{Y}^T \mathbf{B} \mathbf{Y}) = (\mu_4 - 3\sigma^4) \sum_i a_{ii} b_{ii} + 2\sigma^4 \text{tr}(\mathbf{A} \mathbf{B}).$$

Show that in a one-way layout with $g \geq 2$ groups, not all of size one, the between-group and within-group sums of squares are correlated unless $\mu_4 = 3\sigma^4$. Which sign does the correlation have for heavy-tailed errors?

Solution. In the one-way layout the between-group matrix is $\mathbf{A} = \mathbf{M} - \mathbf{P}_1$ and the within-group matrix is $\mathbf{B} = \mathbf{I} - \mathbf{M}$. They satisfy $\mathbf{AB} = \mathbf{0}$, so $\text{tr}(\mathbf{AB}) = 0$ and the covariance reduces to $(\mu_4 - 3\sigma^4) \sum_i a_{ii}b_{ii}$. An observation in group ℓ has $a_{ii} = 1/n_\ell - 1/n$ and $b_{ii} = 1 - 1/n_\ell$, so

$$\sum_i a_{ii}b_{ii} = \sum_\ell n_\ell \left(\frac{1}{n_\ell} - \frac{1}{n} \right) \left(1 - \frac{1}{n_\ell} \right) = \sum_\ell \left(1 - \frac{n_\ell}{n} \right) \left(1 - \frac{1}{n_\ell} \right).$$

With $g \geq 2$ every first factor is positive, and the second is positive for any group with two or more observations. So the sum is positive, and the covariance vanishes iff $\mu_4 = 3\sigma^4$. For heavy-tailed errors, $\mu_4 > 3\sigma^4$, the two sums of squares are positively correlated: a large error inflates both.

Exercise B3.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and let $\mathbf{M}$ be symmetric idempotent of rank r with $\mathbf{M}\boldsymbol{\mu} \neq \boldsymbol{\mu}$, as when a regression model omits a needed regressor. Show that $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/\sigma^2 \sim \chi^2(n - r, \|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2/\sigma^2)$, and find the bias of $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/(n - r)$ as an estimator of σ^2 . In the setting of [Theorem 4.4.6](#) without the assumption $\mathbf{P}\boldsymbol{\mu} = \boldsymbol{\mu}$, which parts of the conclusion survive?

Exercise B4.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \boldsymbol{\Sigma})$. Use [Corollary 4.2.2](#) and [Proposition 1.7.1\(d\)](#) to show that $\mathbf{E} e^{t\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}} = \det(\mathbf{I}_n - 2t\mathbf{A}\boldsymbol{\Sigma})^{-1/2}$ for $|t|$ small.

4.2.4.3 C. Going deeper

Exercise C1.

Let $\mathbf{Y} \sim N_n(\mu \mathbf{1}, \sigma^2 \mathbf{I})$ with μ and σ^2 unknown. Among symmetric $\mathbf{A}$ with $\mathbf{E}(\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}) = \sigma^2$ for all μ and σ^2 , show that $\mathbf{A} = \mathbf{C}/(n - 1)$, which gives S^2 , uniquely minimizes $\text{Var}(\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y})$ at every parameter value. (Compare [Exercise \(B5\)](#), where the mean is known to be zero.)

Solution. Since $\mathbf{E}(\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}) = \sigma^2 \text{tr} \mathbf{A} + \mu^2 \mathbf{1}\mathbf{T}\mathbf{A}\mathbf{1}$, unbiasedness means $\text{tr} \mathbf{A} = 1$ and $\mathbf{1}\mathbf{T}\mathbf{A}\mathbf{1} = 0$. By [Theorem 4.2.3](#), $\text{Var}(\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}) = 2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\mu^2 \sigma^2 \|\mathbf{A}\mathbf{1}\|^2$. The map $\mathbf{A} \mapsto \mathbf{C}\mathbf{A}\mathbf{C}$ is an orthogonal projection for the inner product $\text{tr}(\mathbf{AB})$ on symmetric matrices, so $\text{tr}(\mathbf{A}^2) \geq \text{tr}\{(\mathbf{C}\mathbf{A}\mathbf{C})^2\}$, with equality iff $\mathbf{A} = \mathbf{C}\mathbf{A}\mathbf{C}$. Also $\text{tr}(\mathbf{C}\mathbf{A}\mathbf{C}) = \text{tr} \mathbf{A} - \mathbf{1}\mathbf{T}\mathbf{A}\mathbf{1}/n = 1$. The symmetric matrix $\mathbf{C}\mathbf{A}\mathbf{C}$ has rank at most $n - 1$. If $\lambda_1, \dots, \lambda_{n-1}$ are its eigenvalues on $\mathbf{1}^\perp$, the Cauchy-Schwarz inequality gives $1 = (\sum \lambda_i)^2 \leq (n - 1) \sum \lambda_i^2$. Hence $\text{tr}(\mathbf{A}^2) \geq 1/(n - 1)$, with equality iff $\mathbf{A} = \mathbf{C}\mathbf{A}\mathbf{C}$ and all $\lambda_i = 1/(n - 1)$, that is, iff $\mathbf{A} = \mathbf{C}/(n - 1)$. That matrix also has $\mathbf{A}\mathbf{1} = \mathbf{0}$, so it minimizes both variance terms at once.

4.3 When a quadratic form is chi-squared

A quadratic form is a weighted sum of independent noncentral $\chi^2(1)$ pieces, and the weights are eigenvalues. It is itself chi-squared when every nonzero weight equals one. This section turns that observation into checkable conditions on $\mathbf{A}$ and Σ , and shows that for the spherical case the condition is necessary as well as sufficient.

4.3.1 Spherical normal vectors

Theorem 4.3.1 — Chi-squared forms in a spherical normal vector

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{I}_n)$, let $\mathbf{A}$ be symmetric, and let $r \geq 1$ be an integer and $\gamma \geq 0$. Then $\mathbf{YTA}\mathbf{Y} \sim \chi^2(r, \gamma)$ if and only if $\mathbf{A}$ is idempotent with $\text{rank}(\mathbf{A}) = r$ and $\gamma = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}$.

Proof. Sufficiency. By 1, $\mathbf{A} = \mathbf{Q}_1 \mathbf{Q}_1^T$, where the $n \times r$ matrix $\mathbf{Q}_1$ has orthonormal columns and $r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$. Then $\mathbf{YTA}\mathbf{Y} = \|\mathbf{Q}_1^T \mathbf{Y}\|^2$ and $\mathbf{Q}_1^T \mathbf{Y} \sim N_r(\mathbf{Q}_1^T \boldsymbol{\mu}, \mathbf{Q}_1^T \mathbf{I}_n \mathbf{Q}_1) = N_r(\mathbf{Q}_1^T \boldsymbol{\mu}, \mathbf{I}_r)$, so by definition $\mathbf{YTA}\mathbf{Y} \sim \chi^2(r, \|\mathbf{Q}_1^T \boldsymbol{\mu}\|^2)$, and $\|\mathbf{Q}_1^T \boldsymbol{\mu}\|^2 = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}$.

Necessity. Apply Theorem 4.2.1 with $\mathbf{L} = \mathbf{I}$, so that $\mathbf{A} = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^T$. Let $\boldsymbol{\nu} = \mathbf{Q}^T \boldsymbol{\mu}$. Then $\mathbf{b} = \mathbf{Q}^T \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\Lambda} \boldsymbol{\nu}$, so $b_j = \lambda_j \nu_j$. The constant left after completing squares is $\boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu} - \sum_{\lambda_j \neq 0} \lambda_j \nu_j^2 = \boldsymbol{\nu}^T \boldsymbol{\Lambda} \boldsymbol{\nu} - \sum_j \lambda_j \nu_j^2 = 0$. Hence $\mathbf{YTA}\mathbf{Y}$ has the law of $\sum_{j \in J} \lambda_j (W_j + \nu_j)^2$, where J indexes the nonzero eigenvalues. J is not empty, because a $\chi^2(r, \gamma)$ variable with $r \geq 1$ is not identically zero.

By Lemma 4.1.1, near $t = 0$ the logarithm of the mgf of this sum is $\sum_{j \in J} \{-\frac{1}{2} \log(1 - 2\lambda_j t) + \lambda_j \nu_j^2 t / (1 - 2\lambda_j t)\}$. Expanding $-\frac{1}{2} \log(1 - x) = \sum_{k \geq 1} x^k / (2k)$ and $x / (1 - x) = \sum_{k \geq 1} x^k$, the coefficient of t^k is $(2^{k-1}/k) \sum_{j \in J} \lambda_j^k (1 + k\nu_j^2)$. The same expansion of 4 gives $(2^{k-1}/k)(r + k\gamma)$. Equal distributions have equal mgfs, and two power series that agree on an interval have equal coefficients, so

$$\sum_{j \in J} \lambda_j^k (1 + k\nu_j^2) = r + k\gamma \quad \text{for every } k \geq 1. \quad (4.3.1)$$

Let $\rho = \max_{j \in J} |\lambda_j|$. For even k every term on the left is nonnegative and one of them is at least ρ^k . The right side grows only linearly in k , so $\rho \leq 1$. Split J into $J_+ = \{\lambda_j = 1\}$, $J_- = \{\lambda_j = -1\}$ and $J_0 = \{0 < |\lambda_j| < 1\}$, and write $N_\pm = |J_\pm|$ and $S_\pm = \sum_{j \in J_\pm} \nu_j^2$. The terms over J_0 tend to zero as $k \rightarrow \infty$, so

$$N_+ + (-1)^k N_- + k(S_+ + (-1)^k S_-) - r - k\gamma \rightarrow 0.$$

Dividing by k and letting $k \rightarrow \infty$ through even and through odd values gives $S_+ + S_- = \gamma = S_+ - S_-$. Then the same limit without the factor k gives $N_+ + N_- = r = N_+ - N_-$. So $N_- = S_- = 0$, $N_+ = r$ and $S_+ = \gamma$. The terms over J_+ in Eq. 4.3.1 now account for all of $r + k\gamma$, and with $k = 2$ the remaining sum $\sum_{j \in J_0} \lambda_j^2 (1 + 2\nu_j^2)$ is zero. Every

term of that sum is positive, so J_0 is empty. Thus every eigenvalue of $\mathbf{A}$ is 0 or 1, $\mathbf{A}$ is idempotent, $\text{rank}(\mathbf{A}) = N_+ = r$, and $\boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu} = \sum_{j \in J_+} \nu_j^2 = \gamma$. ■

The necessity half is the one that makes the theorem a tool for *recognizing* distributions. If a quadratic form in a spherical normal vector is chi-squared at all, then it is the squared length of a projection. There is no other way.

4.3.2 General covariance matrices

Theorem 4.3.2 — When a quadratic form is chi-squared

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and let $\mathbf{A}$ be symmetric.

- (a) (*Projections.*) If $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ with $\sigma > 0$ and $\mathbf{P}$ is symmetric and idempotent, then

$$\frac{\|\mathbf{P}\mathbf{Y}\|^2}{\sigma^2} = \frac{\mathbf{Y}^T \mathbf{P} \mathbf{Y}}{\sigma^2} \sim \chi^2(\text{rank}(\mathbf{P}), \frac{\|\mathbf{P}\boldsymbol{\mu}\|^2}{\sigma^2}).$$

- (b) (*Positive definite $\boldsymbol{\Sigma}$.*) If $\boldsymbol{\Sigma}$ is positive definite, then $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ has a $\chi^2(r, \gamma)$ distribution for some $r \geq 1$ and $\gamma \geq 0$ iff $\mathbf{A}\boldsymbol{\Sigma}$ is idempotent and $\mathbf{A} \neq \mathbf{0}$. In that case $r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$ and $\gamma = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}$.

- (c) (*Any $\boldsymbol{\Sigma}$.*) If

$$(i) \ \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}, \quad (ii) \ \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}, \quad (iii) \ \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu},$$

then $\mathbf{Y}^T \mathbf{A} \mathbf{Y} \sim \chi^2(\text{tr}(\mathbf{A}\boldsymbol{\Sigma}), \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu})$. The three conditions hold in particular when $\mathbf{A}\boldsymbol{\Sigma} \mathbf{A} = \mathbf{A}$, and when $\mathbf{A}\boldsymbol{\Sigma}$ is idempotent and $\boldsymbol{\mu} \in \mathbb{C}(\boldsymbol{\Sigma})$.

Proof.

- (a) $\sigma^{-1} \mathbf{Y} \sim N_n(\sigma^{-1} \boldsymbol{\mu}, \mathbf{I})$. Apply Theorem 4.3.1 with $\mathbf{A} = \mathbf{P}$, using $\mathbf{P}^T \mathbf{P} = \mathbf{P}$ to write $\mathbf{Y}^T \mathbf{P} \mathbf{Y} = \|\mathbf{P}\mathbf{Y}\|^2$ and $\boldsymbol{\mu}^T \mathbf{P} \boldsymbol{\mu} = \|\mathbf{P}\boldsymbol{\mu}\|^2$.
- (b) Let $\boldsymbol{\Sigma}^{1/2}$ be the positive definite square root (Theorem 1.7.6). Then $\mathbf{Z} = \boldsymbol{\Sigma}^{-1/2} \mathbf{Y} \sim N_n(\boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu}, \mathbf{I})$ and $\mathbf{Y}^T \mathbf{A} \mathbf{Y} = \mathbf{Z}^T \mathbf{B} \mathbf{Z}$ with $\mathbf{B} = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$ symmetric. By Theorem 4.3.1, $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ is noncentral chi-squared iff $\mathbf{B}$ is idempotent and nonzero. Now $\mathbf{B}^2 = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$, so $\mathbf{B}^2 = \mathbf{B}$ iff $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} = \mathbf{A}$ (multiply by $\boldsymbol{\Sigma}^{-1/2}$ on both sides) iff $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \mathbf{A} \boldsymbol{\Sigma}$ (multiply by $\boldsymbol{\Sigma}$ or $\boldsymbol{\Sigma}^{-1}$ on the right). Also $\mathbf{B} \neq \mathbf{0}$ iff $\mathbf{A} \neq \mathbf{0}$. The degrees of freedom are $\text{rank}(\mathbf{B}) = \text{rank}(\mathbf{A})$, which equals $\text{tr}(\mathbf{B}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$, and the noncentrality is $\boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1/2} \mathbf{B} \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu} = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}$.
- (c) Take $\mathbf{L}$ of size $n \times k$ with full column rank $k = \text{rank}(\boldsymbol{\Sigma})$ and $\mathbf{L} \mathbf{L}^T = \boldsymbol{\Sigma}$, for instance $\mathbf{L} = \mathbf{U}_1 \mathbf{D}_1^{1/2}$ from the eigenvectors $\mathbf{U}_1$ for the positive eigenvalues $\mathbf{D}_1$ of $\boldsymbol{\Sigma}$. (If $\boldsymbol{\Sigma} = \mathbf{0}$ the claim is trivial.) As in Theorem 4.2.1, $\mathbf{Y} \stackrel{d}{=} \boldsymbol{\mu} + \mathbf{L} \mathbf{Z}$ with $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{I})$. Since $\mathbf{L}$ has full column rank, $\mathbf{L}^T$ has full row rank, so $\mathbf{L} \mathbf{G} \mathbf{L}^T = \mathbf{L} \mathbf{H} \mathbf{L}^T$ implies $\mathbf{G} = \mathbf{H}$, and $\mathbf{L} \mathbf{g} = \mathbf{L} \mathbf{h}$ implies $\mathbf{g} = \mathbf{h}$. Put $\mathbf{B} = \mathbf{L}^T \mathbf{A} \mathbf{L}$ and $\mathbf{c} = \mathbf{L}^T \mathbf{A} \boldsymbol{\mu}$. Condition (i) reads $\mathbf{L} \mathbf{B}^2 \mathbf{L}^T = \mathbf{L} \mathbf{B} \mathbf{L}^T$, so $\mathbf{B}^2 = \mathbf{B}$. Condition (ii) reads $\mathbf{L} \mathbf{B} \mathbf{c} = \mathbf{L} \mathbf{c}$, so $\mathbf{B} \mathbf{c} = \mathbf{c}$. Condition (iii) reads $\mathbf{c}^T \mathbf{c} = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}$. Hence

$$(\boldsymbol{\mu} + \mathbf{L} \mathbf{Z})^T \mathbf{A} (\boldsymbol{\mu} + \mathbf{L} \mathbf{Z}) = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu} + 2 \mathbf{c}^T \mathbf{Z} + \mathbf{Z}^T \mathbf{B} \mathbf{Z} = (\mathbf{Z} + \mathbf{c})^T \mathbf{B} (\mathbf{Z} + \mathbf{c}),$$

because $(\mathbb{Z} + \mathbf{c})\mathbf{T}\mathbf{B}(\mathbb{Z} + \mathbf{c}) = \mathbb{Z}\mathbf{T}\mathbf{B}\mathbb{Z} + 2\mathbf{c}\mathbf{T}\mathbf{B}\mathbb{Z} + \mathbf{c}\mathbf{T}\mathbf{B}\mathbf{c}$ and $\mathbf{B}\mathbf{c} = \mathbf{c}$. Now $\mathbb{Z} + \mathbf{c} \sim N_k(\mathbf{c}, \mathbf{I})$, and [Theorem 4.3.1](#) gives the law $\chi^2(\text{rank } \mathbf{B}, \mathbf{c}\mathbf{T}\mathbf{B}\mathbf{c})$, where $\text{rank } \mathbf{B} = \text{tr } \mathbf{B} = \text{tr}(\mathbf{A}\mathbf{\Sigma})$ and $\mathbf{c}\mathbf{T}\mathbf{B}\mathbf{c} = \mathbf{c}\mathbf{T}\mathbf{c} = \boldsymbol{\mu}\mathbf{T}\mathbf{A}\boldsymbol{\mu}$. If $\mathbf{B} = \mathbf{0}$, then $\mathbf{c} = \mathbf{B}\mathbf{c} = \mathbf{0}$ and the form is identically 0, which is $\chi^2(0)$. If $\mathbf{A}\mathbf{\Sigma}\mathbf{A} = \mathbf{A}$, all three conditions follow by substitution. If $\mathbf{A}\mathbf{\Sigma}$ is idempotent and $\boldsymbol{\mu} = \mathbf{\Sigma}\mathbf{v}$, then (i) follows from multiplying $\mathbf{A}\mathbf{\Sigma}\mathbf{A}\mathbf{\Sigma} = \mathbf{A}\mathbf{\Sigma}$ by $\mathbf{\Sigma}$ on the left, and (ii) and (iii) from the same identity applied to $\mathbf{v}$.

■

Part (a) is the case used most often in this book. For the linear model $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ and the projection $\mathbf{M}$ onto $\mathbb{C}(\mathbf{X})$, [Chapter 6](#) shows that the residual sum of squares is $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$. Since $(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$, part (a) gives

$$\text{SSE}/\sigma^2 \sim \chi^2(n - \text{rank}(\mathbf{X})),$$

a central distribution whatever the value of $\boldsymbol{\beta}$. The projections that define test statistics, by contrast, do not annihilate the mean under the alternative, and their noncentrality $\|\mathbf{P}\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2$ measures how far the truth is from the hypothesis in the metric that the test can see.

In part (c) idempotence of $\mathbf{A}\mathbf{\Sigma}$ alone is *not* enough when $\mathbf{\Sigma}$ is singular. A mean with a component outside $\mathbb{C}(\mathbf{\Sigma})$ is a deterministic shift that the form may pick up as a constant or as a linear term ([Exercise \(B2\)](#)). Part (c) also covers the natural generalization of the Mahalanobis distance.

Corollary 4.3.3 — Mahalanobis forms

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{\Sigma})$.

- (a) If $\mathbf{\Sigma}$ is positive definite, $(\mathbf{Y} - \boldsymbol{\mu})\mathbf{T}\mathbf{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu}) \sim \chi^2(n)$ and $\mathbf{Y}\mathbf{T}\mathbf{\Sigma}^{-1}\mathbf{Y} \sim \chi^2(n, \boldsymbol{\mu}\mathbf{T}\mathbf{\Sigma}^{-1}\boldsymbol{\mu})$.
- (b) If $\mathbf{G}$ is a symmetric generalized inverse of $\mathbf{\Sigma}$ and $\boldsymbol{\mu} \in \mathbb{C}(\mathbf{\Sigma})$, then $\mathbf{Y}\mathbf{T}\mathbf{G}\mathbf{Y} \sim \chi^2(\text{rank}(\mathbf{\Sigma}), \boldsymbol{\mu}\mathbf{T}\mathbf{G}\boldsymbol{\mu})$.

Proof.

- (a) $\mathbf{A} = \mathbf{\Sigma}^{-1}$ satisfies $\mathbf{A}\mathbf{\Sigma}\mathbf{A} = \mathbf{A}$, and $\mathbf{Y} - \boldsymbol{\mu} \sim N_n(\mathbf{0}, \mathbf{\Sigma})$. The central statement was already obtained by whitening in [Proposition 3.2.3](#).
- (b) $\mathbf{G}\mathbf{\Sigma}$ is idempotent because $\mathbf{\Sigma}\mathbf{G}\mathbf{\Sigma} = \mathbf{\Sigma}$, so part (c) of the theorem applies and the degrees of freedom are $\text{tr}(\mathbf{G}\mathbf{\Sigma}) = \text{tr}(\mathbf{L}\mathbf{T}\mathbf{G}\mathbf{L})$. With $\mathbf{L}$ as in that proof, $\mathbf{L}\mathbf{L}\mathbf{T}\mathbf{G}\mathbf{L}\mathbf{L}\mathbf{T} = \mathbf{L}\mathbf{L}\mathbf{T}$ gives $\mathbf{L}\mathbf{T}\mathbf{G}\mathbf{L} = \mathbf{I}_k$, whose trace is $k = \text{rank}(\mathbf{\Sigma})$.

■

A symmetric generalized inverse always exists ([Proposition 1.3.3\(d\)](#)). The value of $\mathbf{Y}\mathbf{T}\mathbf{G}\mathbf{Y}$ does not depend on which one is chosen, with probability one ([Exercise \(B3\)](#)).

Example. (Correlated observations about a common level)

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{\Sigma})$ with $\mathbf{\Sigma}$ positive definite, and suppose we want to measure how far $\mathbf{Y}$ is from a common level $m\mathbf{1}$. The generalized least squares level is $\hat{m} = \mathbf{1}\mathbf{T}\mathbf{\Sigma}^{-1}\mathbf{Y}/\mathbf{1}\mathbf{T}\mathbf{\Sigma}^{-1}\mathbf{1}$,

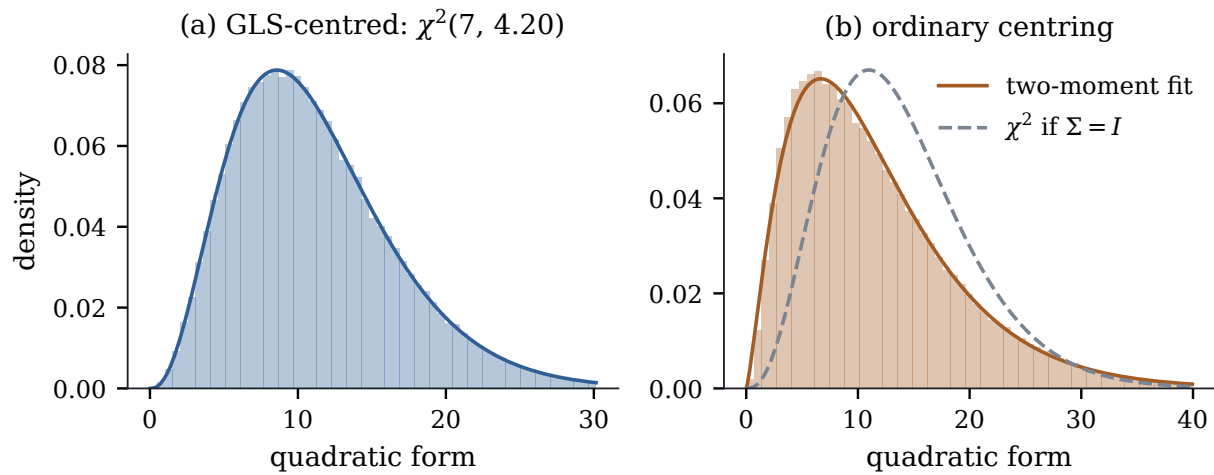


Figure 4.3.1: Two quadratic forms measuring spread about a common level, for $n = 8$ autoregressive observations with a trending mean. (a) With generalized least squares centring, $\mathbf{A}\Sigma$ is idempotent and the form is exactly noncentral chi-squared. (b) With ordinary centring it is a weighted sum of noncentral $\chi^2(1)$ terms. A two-moment scaled chi-squared fits closely. The chi-squared law that would hold for independent observations does not.

and the natural discrepancy is $(\mathbf{Y} - \hat{m}\mathbf{1})\mathbf{T}\Sigma^{-1}(\mathbf{Y} - \hat{m}\mathbf{1}) = \mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}$ with

$$\mathbf{A} = \Sigma^{-1} - \frac{\Sigma^{-1}\mathbf{1}\mathbf{T}\Sigma^{-1}}{\mathbf{1}\mathbf{T}\Sigma^{-1}\mathbf{1}}.$$

A direct multiplication gives $\mathbf{A}\Sigma\mathbf{A} = \mathbf{A}$, and $\text{tr}(\mathbf{A}\Sigma) = n - 1$. By Theorem 4.3.2(b) the form is noncentral chi-squared with $n - 1$ degrees of freedom and noncentrality $\boldsymbol{\mu}\mathbf{T}\mathbf{A}\boldsymbol{\mu}$, which is zero exactly when $\boldsymbol{\mu}$ is constant. The listing takes $n = 8$, an autoregressive correlation $\Sigma = (\rho^{|i-j|})$ with $\rho = 0.6$, and a linear trend in the mean, for which $\gamma = 4.200$. Figure 4.3.1(a) compares 200,000 simulated values with the $\chi^2(7, 4.200)$ density.

```
import numpy as np
from scipy import integrate, stats
rng = np.random.default_rng(31415)
n, rho = 8, 0.6
idx = np.arange(n)
Sigma = rho ** np.abs(idx[:, None] - idx[None, :]) # AR(1) correlation matrix
mu = 0.4 * (idx - idx.mean()) # linear trend, mean zero
one = np.ones(n)

W = np.linalg.inv(Sigma)
A = W - np.outer(W @ one, W @ one) / (one @ W @ one) # GLS-centred form
AS = A @ Sigma
print("A Sigma idempotent:", np.allclose(AS @ AS, AS))
df, gamma = np.trace(AS), mu @ A @ mu # chi^2(df, gamma)

L = np.linalg.cholesky(Sigma) # Y = mu + L Z
Z = rng.standard_normal((200_000, n))
Y = mu + Z @ L.T
q_gls = np.einsum("ij,jk,ik->i", Y, A, Y)
print(df, gamma, stats.kstest(q_gls, stats.ncx2(df, gamma).cdf).pvalue)
```

4.3.3 Forms that are not chi-squared

Most quadratic forms fail the conditions of [Theorem 4.3.2](#). The ordinary sum of squares about the mean, $\mathbf{YTCY}$ with $\mathbf{C} = \mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^T$, is an example as soon as the observations are correlated. Then $\mathbf{C}\Sigma$ is not idempotent, and by [Theorem 4.2.1](#) the law is that of $\sum_j \lambda_j (W_j + \nu_j)^2$, with weights λ_j equal to the nonzero eigenvalues of $\mathbf{C}\Sigma$. In [Example \(Correlated observations about a common level\)](#) these range from 0.259 to 1.832. Two practical ways to handle such a law are the following.

Exact computation. The characteristic function of the weighted sum is known in closed form (put $t = iu$ in [Corollary 4.2.2](#)), and a distribution function can be recovered from a characteristic function φ by the inversion formula of Gil-Pelaez (1951),

$$\Pr(X > x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\operatorname{Im}\{e^{-iux} \varphi(u)\}}{u} du.$$

For a weighted sum of k chi-squared terms the integrand decays like $u^{-1-k/2}$, so ordinary quadrature works well. Imhof (1961) gave the real-variable form of this integral that most software uses. The listing implements it directly and checks it against simulation.

Two-moment approximation. Match $a\chi^2(\nu)$ to the mean m and variance v from [Theorem 4.2.3](#). Since $a\chi^2(\nu)$ has mean $a\nu$ and variance $2a^2\nu$, this gives $a = v/(2m)$ and $\nu = 2m^2/v$, with ν usually not an integer (Satterthwaite 1946; Patnaik 1949).

In [Example \(Correlated observations about a common level\)](#), the simulated 95th percentile of $\mathbf{YTCY}$ is 26.51. The exact tail probability beyond it is 0.0490, and the approximation with $a = 2.488$ and $\nu = 4.68$ gives 0.0484. Treating the observations as independent, and so using $\chi^2(7, 6.720)$, gives 0.040. [Figure 4.3.1\(b\)](#) shows the three laws. Here the error from ignoring correlation is modest at the 5% point but visible in the shape. It is not a monotone function of ρ , but strong correlation makes it severe: with $\rho = 0.9$ and the same mean, the naive tail probability at the true 95th percentile is 0.197.

```
def canonical(A, mu, L):
    """Y^T A Y = sum_j lam_j (W_j + nu_j)^2 + c (all lam_j nonzero here), W ~ N(0, I)"""
    lam, Qm = np.linalg.eigh(L.T @ A @ L)
    b = Qm.T @ L.T @ A @ mu
    keep = np.abs(lam) > 1e-10
    assert np.allclose(b[~keep], 0) # no linear terms in this example
    c = mu @ A @ mu - np.sum(b[keep] ** 2 / lam[keep])
    return lam[keep], b[keep] / lam[keep], c

def tail_prob(x, lam, nu, c=0.0):
    """P(sum lam_j (W_j + nu_j)^2 + c > x) by Gil-Pelaez inversion."""
    def integrand(u):
        z = 1 - 2j * lam * u
        phi = np.prod(z ** -0.5 * np.exp(1j * u * lam * nu**2 / z))
        return (np.exp(-1j * u * (x - c)) * phi).imag / u
    edges = np.linspace(0, 400, 401) # integrand decays like u
    # ^(-1-k/2)
    val = sum(integrate.quad(integrand, a, b)[0] for a, b in zip(edges[:-1], edges[1:]))
    return 0.5 + val / np.pi
```

4.3.4 Exercises

4.3.4.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ with $n \geq 2$, and $\mathbf{A} = \alpha \mathbf{I} + \beta \mathbf{1}\mathbf{1}^T$. Find all (α, β) for which $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ has a chi-squared distribution, and identify each form.

Solution. $\mathbf{A}$ has eigenvalue $\alpha + n\beta$ on $\mathbf{1}$ and α on $\mathbf{1}^\perp$. By [Theorem 4.3.1](#), the form is chi-squared iff $\mathbf{A} \neq \mathbf{0}$ and both eigenvalues lie in $\{0, 1\}$. The three solutions are $(1, 0)$, giving $\|\mathbf{Y}\|^2 \sim \chi^2(n)$; $(0, 1/n)$, giving $n\bar{Y}^2 \sim \chi^2(1)$; and $(1, -1/n)$, giving $\sum (Y_i - \bar{Y})^2 \sim \chi^2(n-1)$.

Exercise A2.

Two independent samples of sizes m and k come from $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$. Write the pooled within-sample sum of squares as $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ for a suitable projection, and show that the two-sample t statistic has the $t(m+k-2, (\mu_1 - \mu_2)/(\sigma\sqrt{1/m + 1/k}))$ distribution.

4.3.4.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_n(\mu \mathbf{1}, \sigma^2\{(1-\rho)\mathbf{I} + \rho \mathbf{1}\mathbf{1}^T\})$ with $-1/(n-1) < \rho < 1$, and let $\bar{Y}$ and S^2 be the sample mean and variance. Show that $(n-1)S^2/\{\sigma^2(1-\rho)\} \sim \chi^2(n-1)$ and that S^2 is independent of $\bar{Y}$. Deduce that $\sqrt{n}(\bar{Y} - \mu)/S$ has the law of $\{(1 + (n-1)\rho)/(1-\rho)\}^{1/2}$ times a $t(n-1)$ variable, and describe what a positive ρ does to the coverage of the usual 95% t interval for μ .

Exercise B2.

Let $\Sigma = \text{diag}(1, 0)$, $\mu = (0, 1)^T$, $\mathbf{A} = \mathbf{I}_2$ and $\mathbf{Y} \sim N_2(\mu, \Sigma)$. Show that $\mathbf{A}\Sigma$ is idempotent but $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ is not chi-squared. Which of the conditions of [Theorem 4.3.2\(c\)](#) fail? What changes if $\mu = (m, 0)^T$? Repeat the first part with $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, where the shift enters through a linear term.

Solution. $\mathbf{A}\Sigma = \Sigma$ is idempotent. But $Y_2 = 1$ with probability one, so $\mathbf{Y}^T \mathbf{A} \mathbf{Y} = Y_1^2 + 1 \geq 1$, while every $\chi^2(r, \gamma)$ with $r \geq 1$ gives positive probability to $(0, 1)$. Conditions (i) and (ii) hold, since $\Sigma^3 = \Sigma$ and $\Sigma^2 \mu = \mathbf{0} = \Sigma \mu$. Condition (iii) fails, since $\mu^T \Sigma \mu = 0 \neq 1 = \mu^T \mu$. The part of the mean outside $\mathcal{C}(\Sigma)$ adds a constant. If $\mu = (m, 0)^T \in \mathcal{C}(\Sigma)$, then $\mathbf{Y}^T \mathbf{A} \mathbf{Y} = Y_1^2 \sim \chi^2(1, m^2)$, as [Theorem 4.3.2\(c\)](#) predicts. For $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ and $\mu = (0, 1)^T$, $\mathbf{A}\Sigma = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ is again idempotent, and $\mathbf{Y}^T \mathbf{A} \mathbf{Y} = Y_1^2 + 2Y_1Y_2 = Y_1^2 + 2Y_1 = (Y_1 + 1)^2 - 1$, which is negative with positive probability. Conditions (i) and (ii) hold, since $\Sigma \mathbf{A} \Sigma = \Sigma$ and $\Sigma \mathbf{A} \Sigma \mu = \Sigma \mu = (1, 0)^T$, but (iii) fails because $\mu^T \mathbf{A} \Sigma \mu = 1 \neq 0 = \mu^T \mu$. The shift now enters through the linear term $2Y_1$.

Exercise B3.

In [Corollary 4.3.3\(b\)](#), show that $\mathbf{Y}\mathbf{T}\mathbf{G}\mathbf{Y}$ takes the same value for every generalized inverse $\mathbf{G}$ of $\mathbf{\Sigma}$, with probability one. *Hint:* [Theorem 2.2.3](#).

Solution. By [Theorem 2.2.3](#), $\mathbf{Y} - \boldsymbol{\mu} \in \mathbb{C}(\mathbf{\Sigma})$ with probability one. Since $\boldsymbol{\mu} \in \mathbb{C}(\mathbf{\Sigma})$, also $\mathbf{Y} \in \mathbb{C}(\mathbf{\Sigma})$, so $\mathbf{Y} = \mathbf{\Sigma}\mathbf{v}$ for some (random) $\mathbf{v}$. For any generalized inverse $\mathbf{G}$, $\mathbf{Y}\mathbf{T}\mathbf{G}\mathbf{Y} = \mathbf{v}\mathbf{T}\mathbf{\Sigma}\mathbf{G}\mathbf{\Sigma}\mathbf{v} = \mathbf{v}\mathbf{T}\mathbf{\Sigma}\mathbf{v}$, which does not involve $\mathbf{G}$.

Exercise B4.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{\Sigma})$ with $\mathbf{\Sigma}$ positive definite, and let $\mathbf{A} \neq \mathbf{0}$ be nonnegative definite. Show that the two-moment approximation $a\chi^2(\nu)$ to $\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}$ has $\nu = \{\text{tr}(\mathbf{A}\mathbf{\Sigma})\}^2 / \text{tr}\{(\mathbf{A}\mathbf{\Sigma})^2\}$, and that $\nu \leq \text{rank}(\mathbf{A})$, with equality iff all nonzero eigenvalues of $\mathbf{A}\mathbf{\Sigma}$ are equal.

4.3.4.3 C. Going deeper

Exercise C1.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{\Sigma})$ with $\mathbf{\Sigma}$ possibly singular, and suppose $\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y} \sim \chi^2(r, \gamma)$ with $r \geq 1$. Show that $\mathbf{\Sigma}\mathbf{A}\mathbf{\Sigma}\mathbf{A}\mathbf{\Sigma} = \mathbf{\Sigma}\mathbf{A}\mathbf{\Sigma}$. *Hint:* in [Corollary 4.2.2](#), compare the coefficients of t^k for $k \geq 3$ with those of 4, and adapt the argument of [Theorem 4.3.1](#).

4.4 Independence of quadratic forms and linear forms

An F or t statistic is a ratio, and its distribution is known only if numerator and denominator are independent. For normal vectors, uncorrelated linear functions are independent ([Theorem 3.4.1](#)). Quadratic forms are not linear, but the following observation brings them within reach of that result.

Idea. A quadratic form is a function of a linear form: for any generalized inverse $\mathbf{A}^-$ of $\mathbf{A}$,

$$\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y} = \mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{A}^-\mathbf{A}\mathbf{Y} = (\mathbf{A}\mathbf{Y})\mathbf{T}\mathbf{A}^-(\mathbf{A}\mathbf{Y}).$$

So $\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}$ is independent of anything that is independent of the vector $\mathbf{A}\mathbf{Y}$.

4.4.1 The independence theorems

Theorem 4.4.1 — Linear and quadratic forms

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{\Sigma})$, let $\mathbf{B}$ be $m \times n$ and $\mathbf{A}$ symmetric $n \times n$. If $\mathbf{B}\mathbf{\Sigma}\mathbf{A} = \mathbf{0}$, then $\mathbf{B}\mathbf{Y}$ and $\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}$ are independent. Conversely, if $\mathbf{B}\mathbf{Y}$ and $\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}$ are independent for every $\boldsymbol{\mu} \in \mathbb{R}^n$, with $\mathbf{\Sigma}$ fixed, then $\mathbf{B}\mathbf{\Sigma}\mathbf{A} = \mathbf{0}$.

Proof. The vector $(\mathbf{BY}, \mathbf{AY})$ is a linear function of $\mathbf{Y}$, hence jointly normal (Theorem 3.3.1), and its cross-covariance is $\text{Cov}(\mathbf{BY}, \mathbf{AY}) = \mathbf{B}\Sigma\mathbf{A}^T = \mathbf{B}\Sigma\mathbf{A}$ (Theorem 2.2.1). If this is zero, $\mathbf{BY}$ and $\mathbf{AY}$ are independent by Theorem 3.4.1, and so are $\mathbf{BY}$ and the function $(\mathbf{AY})^T\mathbf{A}^-(\mathbf{AY})$ of $\mathbf{AY}$, which is $\mathbf{YTAY}$ by the key idea above. Conversely, independent variables with finite variances are uncorrelated, so Theorem 4.2.3(c) gives $2\mathbf{B}\Sigma\mathbf{A}\boldsymbol{\mu} = \mathbf{0}$ for every $\boldsymbol{\mu}$, that is, $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$. ■

Theorem 4.4.2 — Two quadratic forms

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ and let $\mathbf{A}, \mathbf{B}$ be symmetric. If $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$, equivalently $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$, then $\mathbf{YTAY}$ and $\mathbf{YTB Y}$ are independent.

Proof. The two conditions are transposes of each other. As before, $\mathbf{AY}$ and $\mathbf{BY}$ are jointly normal with $\text{Cov}(\mathbf{AY}, \mathbf{BY}) = \mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$, hence independent, and each form is a function of one of them. ■

The condition is sufficient for every $\boldsymbol{\mu}$ and every Σ . When Σ is singular one might hope that the weaker requirement $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$ suffices, since only directions in $\mathcal{C}(\Sigma)$ carry noise. For nonnegative definite $\mathbf{A}$ and $\mathbf{B}$ the two requirements turn out to be equivalent (Exercise (B2)). In the other direction, the condition is necessary under mild assumptions, a result usually credited to Craig (1943). Its general proof is surprisingly delicate, and the history of incorrect published proofs is recounted by Driscoll and Gundberg (1986). The case that matters most for sums of squares has a short proof.

Proposition 4.4.3 — Necessity for nonnegative definite forms

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ and let $\mathbf{A}, \mathbf{B}$ be nonnegative definite. If $\mathbf{YTAY}$ and $\mathbf{YTB Y}$ are independent, then $\mathbf{AB} = \mathbf{0}$.

Proof. Independence makes the covariance zero, and by Theorem 4.2.3(d) the covariance is $2\text{tr}(\mathbf{AB})$. With the square roots of Theorem 1.7.6,

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{A}^{1/2}\mathbf{A}^{1/2}\mathbf{B}^{1/2}\mathbf{B}^{1/2}) = \text{tr}\{(\mathbf{A}^{1/2}\mathbf{B}^{1/2})^T(\mathbf{A}^{1/2}\mathbf{B}^{1/2})\},$$

the sum of squares of the entries of $\mathbf{A}^{1/2}\mathbf{B}^{1/2}$. So $\mathbf{A}^{1/2}\mathbf{B}^{1/2} = \mathbf{0}$, and $\mathbf{AB} = \mathbf{A}^{1/2}(\mathbf{A}^{1/2}\mathbf{B}^{1/2})\mathbf{B}^{1/2} = \mathbf{0}$. ■

For nonnegative definite forms in a standard normal vector, then, zero correlation and independence coincide, just as for jointly normal linear forms. This fails for general symmetric matrices (Exercise (B1)).

4.4.2 Orthogonal projections of a spherical normal vector

The situation in regression is the tidiest one: a spherical normal vector and several projections onto mutually orthogonal subspaces.

Theorem 4.4.4 — Independent projections

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and let $\mathbf{P}_1, \dots, \mathbf{P}_k$ be symmetric idempotent matrices with $\mathbf{P}_i \mathbf{P}_j = \mathbf{0}$ for $i \neq j$, and ranks $r_1, \dots, r_k$. Then $\mathbf{P}_1 \mathbf{Y}, \dots, \mathbf{P}_k \mathbf{Y}$ are mutually independent normal vectors, $\mathbf{P}_i \mathbf{Y} \sim N_n(\mathbf{P}_i \boldsymbol{\mu}, \sigma^2 \mathbf{P}_i)$, and the sums of squares

$$\frac{\|\mathbf{P}_i \mathbf{Y}\|^2}{\sigma^2} \sim \chi^2\left(r_i, \frac{\|\mathbf{P}_i \boldsymbol{\mu}\|^2}{\sigma^2}\right), \quad i = 1, \dots, k,$$

are mutually independent.

Proof. Stack the vectors $\mathbf{P}_i \mathbf{Y}$ into one nk -vector. It is a linear function of $\mathbf{Y}$, so it is normal, and the covariance between blocks i and j is $\mathbf{P}_i(\sigma^2 \mathbf{I})\mathbf{P}_j \mathbf{T} = \sigma^2 \mathbf{P}_i \mathbf{P}_j$. This is $\mathbf{0}$ for $i \neq j$ and $\sigma^2 \mathbf{P}_i$ for $i = j$. By the block form of [Theorem 3.4.1](#) the blocks are mutually independent, and functions of them, the squared lengths, are then mutually independent. Their laws are [Theorem 4.3.2\(a\)](#). ■

In geometric language, which [Chapter 6](#) develops: the coordinates of a spherical normal vector in mutually orthogonal subspaces are independent. It is the reason the rows of an analysis of variance table can be treated separately.

Corollary 4.4.5 — Sample mean and sample variance

Let $Y_1, \dots, Y_n$ ($n \geq 2$) be independent $N(\mu, \sigma^2)$, with sample mean $\bar{Y}$ and sample variance $S^2 = (n-1)^{-1} \sum_i (Y_i - \bar{Y})^2$. Then

- (a) $\bar{Y}$ and S^2 are independent, and $\bar{Y} \sim N(\mu, \sigma^2/n)$;
- (b) $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$;
- (c) for any μ_0 , $T = \sqrt{n}(\bar{Y} - \mu_0)/S \sim t(n-1, \delta)$ with $\delta = \sqrt{n}(\mu - \mu_0)/\sigma$. In particular $T \sim t(n-1)$ when $\mu = \mu_0$.

Proof. Here $\mathbf{Y} \sim N_n(\mu \mathbf{1}, \sigma^2 \mathbf{I})$. Let $\mathbf{P}_1 = n^{-1} \mathbf{1} \mathbf{1}^T$ and $\mathbf{C} = \mathbf{I} - \mathbf{P}_1$. Both are symmetric and idempotent, $\mathbf{P}_1 \mathbf{C} = \mathbf{0}$, and $\text{rank}(\mathbf{C}) = \text{tr}(\mathbf{C}) = n-1$. Since $\mathbf{P}_1 \mathbf{Y} = \bar{Y} \mathbf{1}$ and $\|\mathbf{C} \mathbf{Y}\|^2 = (n-1)S^2$, [Theorem 4.4.4](#) gives (a), which [Example \(Sample mean and deviations\)](#) obtained directly, and, because $\mathbf{C} \mu \mathbf{1} = \mathbf{0}$, also (b). For (c), $X = \sqrt{n}(\bar{Y} - \mu_0)/\sigma \sim N(\delta, 1)$ and $V = (n-1)S^2/\sigma^2 \sim \chi^2(n-1)$ are independent, and $T = X/\sqrt{V/(n-1)}$. Apply [Definition 4.1.3](#). ■

The independence in (a) characterizes the normal distribution: for $n \geq 2$ independent identically distributed observations with finite variance, $\bar{Y}$ and S^2 are independent only under normality (Lukacs 1942). We will not need this, but it shows that normality is used essentially here. It is not a technicality that more careful arguments could remove.

4.4.3 Preview: the F statistic for nested models

The main use of these results comes in Chapter 11. Its core fits in a few lines.

Theorem 4.4.6 — F statistic for nested subspaces

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and let $\mathbf{P}_0$ and $\mathbf{P}$ be symmetric idempotent matrices of ranks $r_0 < r < n$ with $\mathbf{P}\mathbf{P}_0 = \mathbf{P}_0$. Suppose that $\mathbf{P}\boldsymbol{\mu} = \boldsymbol{\mu}$. Then

$$F = \frac{\mathbf{Y}\mathbf{T}(\mathbf{P} - \mathbf{P}_0)\mathbf{Y}/(r - r_0)}{\mathbf{Y}\mathbf{T}(\mathbf{I} - \mathbf{P})\mathbf{Y}/(n - r)} \sim F\left(r - r_0, n - r, \frac{\|(\mathbf{P} - \mathbf{P}_0)\boldsymbol{\mu}\|^2}{\sigma^2}\right),$$

and the noncentrality is zero iff $\mathbf{P}_0\boldsymbol{\mu} = \boldsymbol{\mu}$.

Proof. Transposing $\mathbf{P}\mathbf{P}_0 = \mathbf{P}_0$ gives $\mathbf{P}_0\mathbf{P} = \mathbf{P}_0$. Hence $(\mathbf{P} - \mathbf{P}_0)^2 = \mathbf{P} - 2\mathbf{P}_0 + \mathbf{P}_0 = \mathbf{P} - \mathbf{P}_0$, so $\mathbf{P} - \mathbf{P}_0$ is symmetric idempotent with rank $\text{tr}(\mathbf{P}) - \text{tr}(\mathbf{P}_0) = r - r_0$. Also $(\mathbf{P} - \mathbf{P}_0)(\mathbf{I} - \mathbf{P}) = \mathbf{P} - \mathbf{P} - \mathbf{P}_0 + \mathbf{P}_0 = \mathbf{0}$, and $\mathbf{I} - \mathbf{P}$ is symmetric idempotent of rank $n - r$. By Theorem 4.4.4 the numerator and denominator sums of squares are independent, with σ^{-2} times them distributed as $\chi^2(r - r_0, \|(\mathbf{P} - \mathbf{P}_0)\boldsymbol{\mu}\|^2/\sigma^2)$ and $\chi^2(n - r, \|(\mathbf{I} - \mathbf{P})\boldsymbol{\mu}\|^2/\sigma^2)$. The second noncentrality is 0 since $\mathbf{P}\boldsymbol{\mu} = \boldsymbol{\mu}$. The unknown σ^2 cancels in the ratio, and Definition 4.1.2 applies. Finally, $(\mathbf{P} - \mathbf{P}_0)\boldsymbol{\mu} = \boldsymbol{\mu} - \mathbf{P}_0\boldsymbol{\mu}$. ■

In a linear model, $\mathbf{P} = \mathbf{M}$ projects onto the column space of the full model matrix $\mathbf{X}$ and $\mathbf{P}_0 = \mathbf{M}_0$ onto that of a submodel $\mathbf{X}_0$. The numerator is the drop in residual sum of squares from the submodel to the full model (Chapter 6). The theorem says three things. The test that rejects for large F has an exactly known null distribution. Its power depends on the unknown parameters only through $\gamma = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2$. And, by Theorem 4.1.4, that power increases with γ .

Example. (Detecting curvature)

Let $x_1, \dots, x_n$, $n = 15$, be equally spaced on $[0, 1]$, and suppose $Y_i = 1 + 0.5x_i + \beta_2 x_i^2 + \varepsilon_i$ with independent $N(0, 1)$ errors. To test for curvature, compare the straight line ($\mathbf{X}_0 = [\mathbf{1}, \mathbf{x}]$) with the quadratic ($\mathbf{X} = [\mathbf{1}, \mathbf{x}, \mathbf{x}^2]$). The F statistic has 1 and 12 degrees of freedom, and the 5% critical value is 4.747. For $\beta_2 = 8.0$, the listing finds $\gamma = 6.873$ and power 0.673 from the noncentral F , and the rejection rate in 100,000 simulated data sets is 0.674.

The noncentrality has a clear meaning. It is β_2^2 times the squared length of the part of $\mathbf{x}^2$ orthogonal to $\mathbf{1}$ and $\mathbf{x}$, divided by σ^2 . It grows roughly in proportion to n for a fixed design pattern. Keeping the same design on $[0, 1]$, the smallest n with power at least 0.8 is 21, where $\gamma = 8.97$. Figure 4.4.1 places the example on the power curve and shows the loss of power when the same noncentrality is spread over more numerator degrees

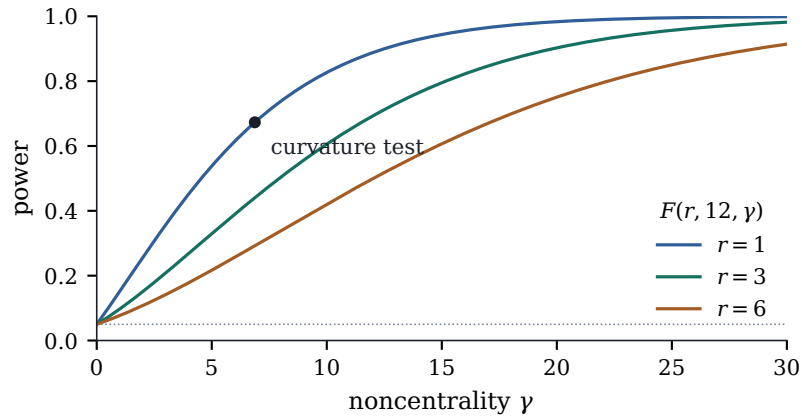


Figure 4.4.1: Power $\Pr\{F(r, 12, \gamma) > F_{0.05}(r, 12)\}$ of the level-0.05 F test as a function of the noncentrality. Every curve starts at 0.05 and increases to 1 (Theorem 4.1.4). For a fixed γ , power falls as the number r of numerator degrees of freedom grows. The dot is the curvature test of Example (Detecting curvature).

of freedom r . At $\gamma = 10$ the power is 0.827, 0.606 and 0.421 for $r = 1, 3, 6$.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(1104)
alpha, s = 0.05, 12
def power(gamma, q, s, alpha=0.05):
    crit = stats.f.ppf(1 - alpha, q, s)
    return stats.ncf.sf(crit, q, s, np.maximum(gamma, 1e-12))
gammas = np.linspace(0, 30, 301)
gamma_ref, rs = 10.0, (1, 3, 6) # r = numerator degrees of
                                # freedom
curves = {q: power(gammas, q, s) for q in rs}

n, sigma, beta0, beta1, beta2, reps = 15, 1.0, 1.0, 0.5, 8.0, 100_000
x = np.linspace(0, 1, n)
X0 = np.column_stack([np.ones(n), x]) # straight line
X = np.column_stack([X0, x**2]) # adds curvature

def proj(Z):
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T

M, M0 = proj(X), proj(X0)
mean = X @ np.array([beta0, beta1, beta2])
gamma = mean @ (M - M0) @ mean / sigma**2 # noncentrality
q, df_resid = 1, n - 3
crit = stats.f.ppf(0.95, q, df_resid)
exact_power = stats.ncf.sf(crit, q, df_resid, gamma)

Y = mean + sigma * rng.standard_normal((reps, n))
num = np.einsum("ij,jk,ik->i", Y, M - M0, Y) / q
den = np.einsum("ij,jk,ik->i", Y, np.eye(n) - M, Y) / df_resid
print(gamma, exact_power, np.mean(num / den > crit))
```

4.4.4 Exercises

4.4.4.1 A. Check your understanding

Exercise A1.

In [Example \(Two estimators of \$\sigma^2\$ under a drifting mean\)](#) let $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$ with $n \geq 3$. Use [Theorem 4.2.3\(d\)](#) to show that $\text{Cov}(S^2, Q) = \text{Var}(S^2)$, and deduce that S^2 and Q are dependent with correlation $\{2(n-1)/(3n-4)\}^{1/2}$. Which linear forms $\mathbf{aTY}$ are independent of Q for every mean vector $\mu \in \mathbb{R}^n$?

Solution. With $\Sigma = \sigma^2\mathbf{I}$, part (d) of [Theorem 4.2.3](#) gives $\text{Cov}(S^2, Q) = 2\sigma^4 \text{tr}(\mathbf{C}\mathbf{D}\mathbf{T}\mathbf{D})/\{2(n-1)^2\}$. Since $\mathbf{D}\mathbf{1} = \mathbf{0}$, $\mathbf{C}\mathbf{D}\mathbf{T}\mathbf{D} = \mathbf{D}\mathbf{T}\mathbf{D}$, whose trace is $2(n-1)$, so $\text{Cov}(S^2, Q) = 2\sigma^4/(n-1) = \text{Var}(S^2)$. The mean terms vanish because $\mathbf{C}\mu = \mathbf{D}\mu = \mathbf{0}$. Using $\text{Var}(Q)$ from [Example](#), the correlation is $\{\text{Var}(S^2)/\text{Var}(Q)\}^{1/2} = \{2(n-1)/(3n-4)\}^{1/2}$, which is nonzero, so the forms are dependent. By [Theorem 4.4.1](#), $\mathbf{aTY}$ is independent of Q for every μ iff $\mathbf{aT}\mathbf{D}\mathbf{T}\mathbf{D} = \mathbf{0T}$. Multiplying by $\mathbf{a}$ gives $\|\mathbf{D}\mathbf{a}\|^2 = 0$, so $\mathbf{D}\mathbf{a} = \mathbf{0}$ and $\mathbf{a}$ is a multiple of $\mathbf{1}$; conversely such $\mathbf{a}$ work. The only such linear forms are the multiples of $\bar{Y}$.

4.4.4.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$. Show that $Y_1^2 - Y_2^2$ and $2Y_1Y_2$ are uncorrelated but not independent. Why does [Proposition 4.4.3](#) not apply?

Solution. With $\mathbf{A} = \text{diag}(1, -1)$ and $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\text{tr}(\mathbf{AB}) = 0$, so the covariance $2\text{tr}(\mathbf{AB})$ is zero. In polar coordinates $Y_1 = R \cos \Theta$, $Y_2 = R \sin \Theta$, with $R^2 \sim \chi^2(2)$ independent of Θ uniform on $[0, 2\pi)$, the two forms are $R^2 \cos 2\Theta$ and $R^2 \sin 2\Theta$. Now $\mathbf{E}(R^4) = 8$ and $\mathbf{E}(R^8) = 384$, so $\mathbf{E}[(R^2 \cos 2\Theta)^2 (R^2 \sin 2\Theta)^2] = 384/8 = 48$, but the product of the expected squares is $(8/2)^2 = 16$. The forms are dependent. [Proposition 4.4.3](#) needs nonnegative definite matrices, and $\mathbf{A}$ and $\mathbf{B}$ are indefinite.

Exercise B2.

Let $\mathbf{Y} \sim N_n(\mu, \Sigma)$ and let $\mathbf{A}, \mathbf{B}$ be nonnegative definite with $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$. Show that $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$, so that $\mathbf{YTAY}$ and $\mathbf{YTB Y}$ are independent. Show by an example that the implication can fail when $\mathbf{A}$ is not nonnegative definite.

Solution. Write $\mathbf{A} = \mathbf{R}\mathbf{R}\mathbf{T}$ and $\mathbf{B} = \mathbf{S}\mathbf{S}\mathbf{T}$, for instance with symmetric square roots, and $\Sigma = \mathbf{L}\mathbf{L}\mathbf{T}$ with $\mathbf{L}$ of full column rank. Cancelling $\mathbf{L}$ on both sides of $\mathbf{L}(\mathbf{L}\mathbf{T}\mathbf{A}\mathbf{L})(\mathbf{L}\mathbf{T}\mathbf{B}\mathbf{L})\mathbf{L}\mathbf{T} = \mathbf{0}$ gives $\mathbf{L}\mathbf{T}\mathbf{A}\mathbf{L}\mathbf{L}\mathbf{T}\mathbf{B}\mathbf{L} = \mathbf{0}$. The trace of this matrix is

$$\text{tr}(\mathbf{L}\mathbf{T}\mathbf{R}\mathbf{R}\mathbf{T}\mathbf{L}\mathbf{L}\mathbf{T}\mathbf{S}\mathbf{S}\mathbf{T}\mathbf{L}) = \text{tr}\{(\mathbf{R}\mathbf{T}\Sigma\mathbf{S})(\mathbf{R}\mathbf{T}\Sigma\mathbf{S})\mathbf{T}\} = \|\mathbf{R}\mathbf{T}\Sigma\mathbf{S}\|_F^2,$$

so $\mathbf{R}\mathbf{T}\Sigma\mathbf{S} = \mathbf{0}$. Hence $\mathbf{A}\Sigma\mathbf{B} = \mathbf{R}(\mathbf{R}\mathbf{T}\Sigma\mathbf{S})\mathbf{S}\mathbf{T} = \mathbf{0}$, and [Theorem 4.4.2](#) applies. For nonnegative definite matrices the two conditions are therefore equivalent, since $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$ trivially implies $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$. Without nonnegative definiteness the argument breaks at the trace step. Take $\Sigma = \text{diag}(1, 0)$, $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{B} = \mathbf{I}_2$. Then $\Sigma\mathbf{A}\Sigma = \mathbf{0}$, so

$\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$, but $\mathbf{A}\Sigma\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \neq \mathbf{0}$. With $\boldsymbol{\mu} = (0, m)\mathbf{T}$ and $m \neq 0$ we have $Y_2 = m$ with probability one, so the forms are $2mY_1$ and $Y_1^2 + m^2$. The second is a nonconstant function of the first, so they are dependent.

Exercise B3.

Let $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ with $\mathbf{X}$ of full column rank $p < n$, let $\hat{\boldsymbol{\beta}} = (\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{X}\mathbf{T}\mathbf{Y}$, $\mathbf{M} = \mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{X}\mathbf{T}$ and $s^2 = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/(n - p)$. With v_{jj} the j th diagonal entry of $(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}$, show that $(\hat{\beta}_j - b)/(s\sqrt{v_{jj}}) \sim t(n - p, (\beta_j - b)/(\sigma\sqrt{v_{jj}}))$.

Solution. Here $\mathbf{M}$ is symmetric and idempotent with $\text{tr } \mathbf{M} = \text{tr}\{(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{X}\mathbf{T}\mathbf{X}\} = p$, and $\mathbf{M}\mathbf{X} = \mathbf{X}$. So $(n - p)s^2/\sigma^2 = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/\sigma^2 \sim \chi^2(n - p)$ by Theorem 4.3.2(a), because $(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$. With $\mathbf{B} = (\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{X}\mathbf{T}$, $\mathbf{B}(\sigma^2\mathbf{I})(\mathbf{I} - \mathbf{M}) = \sigma^2(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}(\mathbf{X}\mathbf{T} - \mathbf{X}\mathbf{T}\mathbf{M}) = \mathbf{0}$, so $\hat{\boldsymbol{\beta}} = \mathbf{B}\mathbf{Y}$ is independent of s^2 (Theorem 4.4.1). Since $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2(\mathbf{X}\mathbf{T}\mathbf{X})^{-1})$, the variable $X = (\hat{\beta}_j - b)/(\sigma\sqrt{v_{jj}})$ is $N((\beta_j - b)/(\sigma\sqrt{v_{jj}}), 1)$. The statistic is $X/\sqrt{V/(n - p)}$ with $V = (n - p)s^2/\sigma^2$, and Definition 4.1.3 applies.

Exercise B4.

Show that $F_\alpha(r, s) \rightarrow \chi_\alpha^2(r)/r$ as $s \rightarrow \infty$, where $\chi_\alpha^2(r)$ is the upper α point of $\chi^2(r)$, and that the power $\Pr\{F(r, s, \gamma) > F_\alpha(r, s)\}$ tends to $\Pr\{\chi^2(r, \gamma) > \chi_\alpha^2(r)\}$, the power of the test that knows σ^2 .

4.5 The Cochran theorem

An analysis of variance starts from an identity

$$\mathbf{Y}\mathbf{T}\mathbf{Y} = \mathbf{Y}\mathbf{T}\mathbf{A}_1\mathbf{Y} + \cdots + \mathbf{Y}\mathbf{T}\mathbf{A}_k\mathbf{Y}, \quad \mathbf{A}_1 + \cdots + \mathbf{A}_k = \mathbf{I},$$

that splits the total sum of squares into pieces attributed to sources of variation. Theorem 4.4.4 says the pieces are independent chi-squared variables when the $\mathbf{A}_i$ are mutually orthogonal projections. Checking that directly means checking k idempotence conditions and $\binom{k}{2}$ orthogonality conditions. Cochran's theorem shows that one count of ranks does all of it.

4.5.1 The algebra

The matrix algebra behind this was done in Chapter 1. For symmetric matrices, any two of the three properties “each $\mathbf{A}_i$ is idempotent”, “ $\mathbf{A}_i\mathbf{A}_j = \mathbf{0}$ for $i \neq j$ ” and “the sum is idempotent” imply the third, and symmetric matrices that add to $\mathbf{I}$ with ranks adding to n are idempotent and mutually orthogonal (Theorem 1.9.3). The next theorem adds the rank count for a sum that is idempotent but not necessarily $\mathbf{I}$, which is what nested models need.

Theorem 4.5.1 — Cochran, algebraic form

Let $\mathbf{A}_1, \dots, \mathbf{A}_k$ be symmetric $n \times n$ matrices and $\mathbf{A} = \sum_i \mathbf{A}_i$. Consider

- (a) each $\mathbf{A}_i$ is idempotent;
- (b) $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$ for all $i \neq j$;
- (c) $\mathbf{A}$ is idempotent;
- (d) $\text{rank}(\mathbf{A}) = \sum_i \text{rank}(\mathbf{A}_i)$.

Then any two of (a), (b), (c) imply all four, and (c) together with (d) implies all four.

Proof. Any two of (a), (b), (c). By [Theorem 1.9.3](#), any two of them imply the third. When all three hold, rank equals trace for symmetric idempotent matrices (1), so $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}) = \sum_i \text{tr}(\mathbf{A}_i) = \sum_i \text{rank}(\mathbf{A}_i)$, which is (d).

(c) and (d) imply (a) and (b). Let $r = \text{rank}(\mathbf{A})$ and $r_i = \text{rank}(\mathbf{A}_i)$. Every $\mathbf{A}\mathbf{v} = \sum_i \mathbf{A}_i \mathbf{v}$ lies in $\mathbb{C}(\mathbf{A}_1) + \cdots + \mathbb{C}(\mathbf{A}_k)$, whose dimension is at most $\sum_i r_i = r = \dim \mathbb{C}(\mathbf{A})$. So the two spaces coincide, and in particular $\mathbb{C}(\mathbf{A}_i) \subseteq \mathbb{C}(\mathbf{A})$. By 1(a), $\mathbf{A} = \mathbf{Q}\mathbf{Q}^T$ with $\mathbf{Q}$ of size $n \times r$ and $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}_r$. Since $\mathbf{A}$ is symmetric and idempotent it fixes every vector of its column space, so $\mathbf{A}\mathbf{A}_i = \mathbf{A}_i$, and transposing, $\mathbf{A}_i \mathbf{A} = \mathbf{A}_i$. Hence $\mathbf{A}_i = \mathbf{A}\mathbf{A}_i \mathbf{A} = \mathbf{Q}\mathbf{B}_i \mathbf{Q}^T$ with $\mathbf{B}_i = \mathbf{Q}^T \mathbf{A}_i \mathbf{Q}$. The $r \times r$ matrices $\mathbf{B}_i$ are symmetric, satisfy $\sum_i \mathbf{B}_i = \mathbf{Q}^T \mathbf{A} \mathbf{Q} = \mathbf{I}_r$, and have $\text{rank}(\mathbf{B}_i) = r_i$ because $\mathbf{Q}$ has orthonormal columns. By the last statement of [Theorem 1.9.3](#), $\mathbf{B}_i^2 = \mathbf{B}_i$ and $\mathbf{B}_i \mathbf{B}_j = \mathbf{0}$ for $i \neq j$, and multiplying by $\mathbf{Q}$ on the left and $\mathbf{Q}^T$ on the right transfers both identities to the $\mathbf{A}_i$.

■

4.5.2 The distributional statement

Theorem 4.5.2 — Cochran

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and let $\mathbf{A}_1, \dots, \mathbf{A}_k$ be symmetric with ranks $r_1, \dots, r_k$, whose sum $\mathbf{A} = \sum_i \mathbf{A}_i$ is idempotent of rank r (for example $\mathbf{A} = \mathbf{I}$, $r = n$). The following are equivalent:

- (i) the forms $\mathbf{Y}^T \mathbf{A}_i \mathbf{Y} / \sigma^2$ are mutually independent, and $\mathbf{Y}^T \mathbf{A}_i \mathbf{Y} / \sigma^2 \sim \chi^2(r_i, \boldsymbol{\mu}^T \mathbf{A}_i \boldsymbol{\mu} / \sigma^2)$ for each i ;
- (ii) $r_1 + \cdots + r_k = r$;
- (iii) each $\mathbf{A}_i$ is idempotent;
- (iv) $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$ for all $i \neq j$.

Proof. Condition (c) of [Theorem 4.5.1](#) holds by assumption, so that theorem shows that (ii), (iii) and (iv) are equivalent, and each implies the other two. If (iii) and (iv) hold, [Theorem 4.4.4](#) gives (i). If (i) holds, apply [Theorem 4.3.1](#) to $\sigma^{-1} \mathbf{Y} \sim N_n(\sigma^{-1} \boldsymbol{\mu}, \mathbf{I})$. Each $\mathbf{A}_i$ with $r_i \geq 1$ is idempotent, and an $\mathbf{A}_i$ of rank zero is $\mathbf{0}$. This is (iii).

■

In words: *if the degrees of freedom add up, the sums of squares are independent chi-squared variables.* Independence in (i) is not needed to reach (iii). The marginal chi-squared laws alone force the whole structure.

Example. (The one-way layout)

Let observations fall into g groups of sizes $n_1, \dots, n_g$, with $Y_{\ell j}$ independent $N(\mu_\ell, \sigma^2)$. Let $\mathbf{M}$ replace each observation by its group mean, and $\mathbf{P}_1 = n^{-1}\mathbf{1}\mathbf{1}^T$ by the grand mean. Then

$$\mathbf{I} = \mathbf{P}_1 + (\mathbf{M} - \mathbf{P}_1) + (\mathbf{I} - \mathbf{M})$$

splits $\|\mathbf{Y}\|^2$ into $n\bar{Y}^2$, the between-group sum of squares $\sum_\ell n_\ell(\bar{Y}_\ell - \bar{Y})^2$ and the within-group sum of squares $\sum_{\ell,j} (Y_{\ell j} - \bar{Y}_\ell)^2$. The three matrices are symmetric with ranks 1, $g - 1$ and $n - g$, which add to n . Cochran's theorem needs nothing more. The three sums of squares, divided by σ^2 , are independent noncentral chi-squared variables. Their noncentralities are $n\bar{\mu}^2/\sigma^2$, $\sum_\ell n_\ell(\mu_\ell - \bar{\mu})^2/\sigma^2$ with $\bar{\mu} = \sum_\ell n_\ell\mu_\ell/n$, and 0. The last is zero because $\mathbf{M}\boldsymbol{\mu} = \boldsymbol{\mu}$. The ratio of the between and within mean squares is $F(g - 1, n - g, \gamma)$ by [Theorem 4.4.6](#), and it is central iff all group means are equal.

The script uses $g = 3$ groups of sizes 3, 5, 4, means 2, 3, 1.5, and $\sigma = 1$, and draws 200,000 data sets. [Table 4.5.1](#) compares the theoretical means $r_i + \gamma_i$ and variances $2r_i + 4\gamma_i$ with the simulated ones. The largest absolute correlation between the three sums of squares is 0.0013. As a check that goes beyond correlation, the proportion of data sets in which both the between and the within sums of squares exceed their medians is 0.2497, against 0.25 for independent variables.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(5150)

sizes, means, sigma = [3, 5, 4], [2.0, 3.0, 1.5], 1.0
groups = np.repeat(np.arange(3), sizes)
n = len(groups)
Z = (groups[:, None] == np.arange(3)).astype(float) # group indicators
M = Z @ np.diag(1 / np.array(sizes)) @ Z.T # group-mean projection
J = np.ones((n, n)) / n
pieces = [J, M - J, np.eye(n) - M]
ranks = [np.linalg.matrix_rank(P) for P in pieces]
print("ranks", ranks, "sum", sum(ranks), "n", n)

mu = np.array(means)[groups]
gammas = [mu @ P @ mu / sigma**2 for P in pieces]
Y = mu + sigma * rng.standard_normal((200_000, n))
Q = np.stack([np.einsum("ij,jk,ik->i", Y, P, Y) for P in pieces], axis=1) / sigma**2
print(np.corrcoef(Q, rowvar=False).round(3))
```

Table 4.5.1. Cochran decomposition for a one-way layout with $n = 12$: theory against 200,000 simulated data sets.

Sum of squares	r_i	γ_i	mean	simulated	variance	simulated
grand mean	1	60.750	61.750	61.763	245.00	244.95
between groups	2	5.250	7.250	7.230	25.00	24.85
within groups	9	0.000	9.000	8.994	18.00	17.96

Remark (What the theorem does not cover). Two assumptions carry the weight. The first is normality. Without it, [Theorem 4.2.3\(d\)](#) fails, and with independent non-normal errors the between and within sums of squares are in general correlated, with a correlation driven by the fourth cumulant ([Exercise \(B2\)](#)). The second is the spherical covariance $\sigma^2 \mathbf{I}$. With $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$, $\mathbf{V}$ positive definite, apply the theorem to $\mathbf{V}^{-1/2} \mathbf{Y}$, whose forms have matrices $\mathbf{V}^{1/2} \mathbf{A}_i \mathbf{V}^{1/2}$. Their ranks are those of the $\mathbf{A}_i$, so the rank count is unchanged. What changes is the condition on the sum: $\mathbf{V}^{1/2} \mathbf{A} \mathbf{V}^{1/2}$ is idempotent iff $\mathbf{A} \mathbf{V} \mathbf{A} = \mathbf{A}$ (for example $\sum_i \mathbf{A}_i = \mathbf{V}^{-1}$), and conditions (iii) and (iv) become $\mathbf{A}_i \mathbf{V} \mathbf{A}_i = \mathbf{A}_i$ and $\mathbf{A}_i \mathbf{V} \mathbf{A}_j = \mathbf{0}$. These are the generalized least squares decompositions of Chapter 31.

Most analysis of variance tables in this book come from a chain of nested model spaces. The successive differences of their projections are mutually orthogonal projections whose ranks add up, so Cochran's theorem applies without further work ([Exercise \(B2\)](#), Chapter 9).

4.5.3 Exercises

4.5.3.1 A. Check your understanding

Exercise A1.

Let $\mathbf{A}_1 = \text{diag}(2, 0)$ and $\mathbf{A}_2 = \text{diag}(-1, 1)$, so that $\mathbf{A}_1 + \mathbf{A}_2 = \mathbf{I}_2$. Which hypotheses of [Theorem 4.5.2](#) fail, and what are the laws of the two forms when $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$?

Solution. The sum $\mathbf{I}_2$ is idempotent, but every one of the conditions (ii)–(iv) fails. The ranks are 1 and 2, which add to $3 \neq 2$. Neither matrix is idempotent, since $\mathbf{A}_1^2 = \text{diag}(4, 0)$ and $\mathbf{A}_2^2 = \mathbf{I}_2$. And $\mathbf{A}_1 \mathbf{A}_2 = \text{diag}(-2, 0) \neq \mathbf{0}$. The forms are $2Y_1^2$, which is twice a $\chi^2(1)$ variable, and $Y_2^2 - Y_1^2$, the difference of two independent $\chi^2(1)$ variables, which takes negative values. Neither is chi-squared. They are dependent, because

$$\text{Cov}(2Y_1^2, Y_2^2 - Y_1^2) = 2 \text{Cov}(Y_1^2, Y_2^2) - 2 \text{Var}(Y_1^2) = 0 - 2 \cdot 2 = -4 \neq 0.$$

4.5.3.2 B. Practice

Exercise B1.

A two-way layout has one observation Y_{ij} for each of a rows and b columns. Write $\|\mathbf{Y}\|^2$ as the sum of the grand-mean, row, column and residual sums of squares, identify the four matrices, and use [Theorem 4.5.2](#) to show that under $Y_{ij} \sim N(\mu + \alpha_i + \beta_j, \sigma^2)$ independently the four sums of squares are independent, with degrees of freedom 1, $a - 1$, $b - 1$ and $(a - 1)(b - 1)$.

Exercise B2.

Let $\mathbf{P}_1, \dots, \mathbf{P}_m$ be symmetric idempotent with $\mathbb{C}(\mathbf{P}_1) \subset \mathbb{C}(\mathbf{P}_2) \subset \dots \subset \mathbb{C}(\mathbf{P}_m)$, and put $\mathbf{P}_0 = \mathbf{0}$, $\mathbf{P}_{m+1} = \mathbf{I}$. Show that the matrices $\mathbf{P}_i - \mathbf{P}_{i-1}$, $i = 1, \dots, m + 1$, satisfy the conditions of [Theorem 4.5.2](#), so that the sequential sums of squares $\|(\mathbf{P}_i - \mathbf{P}_{i-1})\mathbf{Y}\|^2$ are independent noncentral chi-squared variables when $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$.

4.5.3.3 C. Going deeper

Exercise C1.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$, and $Q = Q_1 + Q_2$ with $Q = \mathbf{Y}^T \mathbf{A} \mathbf{Y} \sim \chi^2(r)$, $Q_1 = \mathbf{Y}^T \mathbf{A}_1 \mathbf{Y} \sim \chi^2(r_1)$ and $Q_2 = \mathbf{Y}^T \mathbf{A}_2 \mathbf{Y}$ for a nonnegative definite $\mathbf{A}_2$. Show that Q_1 and Q_2 are independent and $Q_2 \sim \chi^2(r - r_1)$.

Solution. By [Theorem 4.3.1](#), $\mathbf{A}$ and $\mathbf{A}_1$ are idempotent, with ranks r and r_1 . If $\mathbf{A}\mathbf{v} = \mathbf{0}$, then $0 = \mathbf{v}^T \mathbf{A}_1 \mathbf{v} + \mathbf{v}^T \mathbf{A}_2 \mathbf{v}$ with both terms nonnegative, so $\|\mathbf{A}_1 \mathbf{v}\|^2 = \mathbf{v}^T \mathbf{A}_1 \mathbf{v} = 0$. Thus $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A}_1)$, hence $\mathbb{C}(\mathbf{A}_1) \subseteq \mathbb{C}(\mathbf{A})$ and $\mathbf{A}\mathbf{A}_1 = \mathbf{A}_1 = \mathbf{A}_1 \mathbf{A}$. Then $\mathbf{A}_2^2 = (\mathbf{A} - \mathbf{A}_1)^2 = \mathbf{A} - 2\mathbf{A}_1 + \mathbf{A}_1 = \mathbf{A}_2$, so $\mathbf{A}_2$ is idempotent, and $\mathbf{A}_1 \mathbf{A}_2 = \mathbf{A}_1 \mathbf{A} - \mathbf{A}_1 = \mathbf{0}$. By [Theorem 4.4.4](#), Q_1 and Q_2 are independent and $Q_2 \sim \chi^2(\text{tr } \mathbf{A}_2) = \chi^2(r - r_1)$.

4.6 Summary and Notes

4.6.1 Summary

Idea.

1. If $\mathbf{Z} \sim N_r(\boldsymbol{\mu}, \mathbf{I})$ then $\|\mathbf{Z}\|^2 \sim \chi^2(r, \gamma)$ with $\gamma = \|\boldsymbol{\mu}\|^2$, mean $r + \gamma$ and variance $2r + 4\gamma$. The law is a $\text{Poisson}(\gamma/2)$ mixture of $\chi^2(r + 2k)$ laws, and its tails increase strictly with γ ([Theorem 4.1.2](#) and [Proposition 4.1.3](#)).
2. $F(r, s, \gamma)$ and $t(s, \delta)$ put a noncentral numerator over an independent central denominator. The power of an F test increases strictly with γ , and $T^2 \sim F(1, s, \delta^2)$ ([Theorem 4.1.4](#)).
3. $\mathbf{E}(\mathbf{Y}^T \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}$ for any distribution. Under normality

$$\text{Var}(\mathbf{Y}^T \mathbf{A} \mathbf{Y}) = 2 \text{tr}\{(\mathbf{A}\boldsymbol{\Sigma})^2\} + 4\boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}, \quad \text{Cov}(\mathbf{K}^T \mathbf{Y}, \mathbf{Y}^T \mathbf{A} \mathbf{Y}) = 2\mathbf{K}^T \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}$$

([Theorem 4.2.3](#)).

4. A normal quadratic form is a constant plus independent terms $\lambda_j(W_j + \nu_j)^2$, and for singular $\boldsymbol{\Sigma}$ possibly normal terms. The weights λ_j are the nonzero eigenvalues of $\mathbf{A}\boldsymbol{\Sigma}$ ([Theorem 4.2.1](#)).
5. For $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{I})$, $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ is chi-squared iff $\mathbf{A}$ is idempotent. Then the degrees of freedom are $\text{rank } \mathbf{A} = \text{tr } \mathbf{A}$ and the noncentrality is $\boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu}$ ([Theorem 4.3.1](#)). For positive definite $\boldsymbol{\Sigma}$ the condition is that $\mathbf{A}\boldsymbol{\Sigma}$ be idempotent. For projections, $\|\mathbf{P}\mathbf{Y}\|^2/\sigma^2 \sim \chi^2(\text{rank } \mathbf{P}, \|\mathbf{P}\boldsymbol{\mu}\|^2/\sigma^2)$ ([Theorem 4.3.2](#)).
6. $\mathbf{B}\mathbf{Y}$ and $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ are independent if $\mathbf{B}\boldsymbol{\Sigma}\mathbf{A} = \mathbf{0}$, and $\mathbf{Y}^T \mathbf{A} \mathbf{Y}$ and $\mathbf{Y}^T \mathbf{B} \mathbf{Y}$ are independent if $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B} = \mathbf{0}$, because a quadratic form is a function of $\mathbf{A}\mathbf{Y}$ ([Theorem 4.4.1](#) and [Theorem 4.4.2](#)).
7. Projections of a spherical normal vector onto mutually orthogonal subspaces are independent. This gives the independence of $\bar{Y}$ and S^2 and the exact $F(r - r_0, n - r, \gamma)$ law of the nested-model F statistic ([Theorem 4.4.4](#), [Corollary 4.4.5](#) and [Theorem 4.4.6](#)).
8. Cochran: if symmetric matrices add to an idempotent matrix and their ranks add up, the corresponding sums of squares are independent chi-squared variables

(Theorem 4.5.1 and Theorem 4.5.2).

4.6.2 Notes and sources

Source texts. Rencher and Schaalje (2008, chapter 5) cover the material of Section 4.2, Section 4.3, Section 4.4 and Section 4.5 for positive definite covariance matrices. They derive the mgf of a quadratic form by integrating against the normal density and obtain its variance by differentiating the logarithm of the mgf; the converse of the chi-squared theorem is referred to the literature. They use the noncentrality $\boldsymbol{\mu}^T \boldsymbol{\mu}/2$. Seber and Lee (2003, section 2.4) reduce the general case to a spherical normal vector by an orthogonal diagonalization. They prove necessity of idempotence for central forms by factorizing the moment generating function, and give the eigenvalue criterion for singular covariance matrices. Christensen (2020, section 1.3) states the three conditions of Theorem 4.3.2(c) for arbitrary nonnegative definite $\boldsymbol{\Sigma}$, and obtains independence by viewing quadratic forms as functions of linear ones, the approach followed in Section 4.4. The canonical representation (Theorem 4.2.1), the proof of necessity through the coefficients Eq. 4.3.1, the monotonicity proof through the Poisson mixture, and the treatment of the Cochran algebra are this book's own arrangements of standard material.

Noncentral distributions. The noncentral chi-squared distribution first appeared in Fisher (1928). Patnaik (1949) studied its approximation by a scaled central chi-squared variable, and Satterthwaite (1946) used the same two-moment idea for linear combinations of mean squares, which reappears in Chapter 32. Johnson et al. (1995, chapters 29–31) collect the properties of the noncentral chi-squared, F and t families. The monotonicity of power in the noncentrality and in the degrees of freedom is treated systematically by Ghosh (1973).

General quadratic forms. The exact distribution of a quadratic form that is not chi-squared can be computed by numerical inversion of its characteristic function (Gil-Pelaez 1951; Imhof 1961). Mathai and Provost (1992) give a book-length treatment of quadratic forms in normal and non-normal variables. The mean square successive difference of Example (Two estimators of σ^2 under a drifting mean) goes back to von Neumann (1941), who derived the exact distribution of its ratio to S^2 .

Independence and Cochran's theorem. Cochran (1934) proved the decomposition theorem that bears his name, in the context of the analysis of covariance. The necessity of $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B} = \mathbf{0}$ for independence of two forms is attributed to Craig (1943). Its full proof for arbitrary symmetric matrices and arbitrary means turned out to be subtle, and Driscoll and Gundberg (1986) trace the flawed and correct arguments. Searle (1971) gives general proofs of the independence and Cochran theorems in the style most linear-models texts follow. For nonnegative definite $\mathbf{A}$ and $\mathbf{B}$, the condition $\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}\boldsymbol{\Sigma} = \mathbf{0}$ of Christensen (2020, Theorem 1.3.8) is in fact equivalent to $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B} = \mathbf{0}$ (Exercise (B2)), so that theorem adds nothing to Theorem 4.4.2 even for singular $\boldsymbol{\Sigma}$, whereas his Theorem 1.3.9, for general symmetric $\mathbf{A}$ and $\mathbf{B}$, does. That the independence of $\bar{Y}$ and S^2 characterizes the normal distribution was shown by Lukacs (1942).

4.6.3 References

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Chapter 6

Projections: The Geometry of Least Squares

Least squares chooses the point of a subspace that is closest to the data. Everything else in this chapter follows from that one statement: the normal equations, why fitted values are unique when coefficients are not, the orthogonal decompositions behind every sum of squares, the Frisch–Waugh–Lovell theorem, R^2 as a squared cosine, leverage, generalized least squares, and how the projection should actually be computed.

What you need. [Chapter 1](#) (rank, inverses, eigenvalues), [Chapter 5](#) (the linear model and least squares). Random vectors ([Chapter 2](#)) appear only in a few remarks.

6.0.1 Roadmap

- [Least squares as a nearest-point problem](#)
- [The subspaces we need](#)
- [Orthogonal projection](#)
- [Least squares via projection](#)
- [Nested models and sequential projection](#)
- [The Frisch–Waugh–Lovell theorem](#)
- [Reparameterization, centring and \$R^2\$](#)
- [Leverage as geometry](#)
- [Other inner products](#)
- [Computation](#)
- [Projection in the population: best linear prediction](#)
- [Summary and notes](#)

6.1 Least squares as a nearest-point problem

Chapter 5 introduced least squares as a calculus problem: write down the sum of squared errors as a function of the coefficients, differentiate, set the derivative to zero. That route gets the right answer, but it hides the structure of the answer. It treats the coefficient vector β as the basic unknown. It needs $\mathbf{X}\mathbf{T}\mathbf{X}$ to be invertible before it can say anything. And it gives no hint

of why some quantities, such as fitted values, residuals and sums of squares, behave well in every model, while others, such as individual coefficients, can be fragile or not defined at all.

This chapter starts from a different place. Instead of the p -dimensional space where β lives, it works in the n -dimensional space where the data live.

6.1.1 Two pictures of the same data

A regression data set with n observations and a single regressor is usually drawn as a scatterplot: n points in the plane, one for each observation, with the regressor on one axis and the response on the other. Call this the **variable space** picture. It is the right picture for seeing the shape of a relationship, but it scales badly. With k regressors it needs $k + 1$ dimensions, and it has no natural place for the model's coefficients.

The other picture turns this around. Stack the n responses into a single vector $\mathbf{y} = (y_1, \dots, y_n)\mathbf{T}$, and regard it as *one* point in $\mathbb{R}^n$. Each column of the model matrix $\mathbf{X}$ is also a vector in $\mathbb{R}^n$: the intercept column is $\mathbf{1}$, and the column for a regressor holds its n observed values. This is the **observation space** picture. Its dimension is the sample size, which is usually far more than three. But it contains only $p + 1$ vectors of interest, and those always span a subspace of dimension at most $p + 1$. Most of the geometry happens in a subspace small enough to draw.

In observation space, the linear model

$$\mathbf{E}(\mathbf{Y}) = \mathbf{X}\beta, \quad \beta \in \mathbb{R}^p,$$

says one thing: the mean vector $\mathbf{E}(\mathbf{Y})$ lies in the set

$$\mathbb{C}(\mathbf{X}) = \{\mathbf{X}\mathbf{b} : \mathbf{b} \in \mathbb{R}^p\},$$

the **column space** of $\mathbf{X}$. That set is a subspace of $\mathbb{R}^n$. The coefficients β are the coordinates of $\mathbf{E}(\mathbf{Y})$ with respect to the columns of $\mathbf{X}$, but the claim itself is only about *where* $\mathbf{E}(\mathbf{Y})$ is. It does not depend on which spanning vectors are used to describe the subspace.

6.1.2 The least squares principle, restated

The observed $\mathbf{y}$ is almost never in $\mathbb{C}(\mathbf{X})$, because noise moves it off the subspace. A natural estimate of $\mathbf{E}(\mathbf{Y})$ is the point of $\mathbb{C}(\mathbf{X})$ that is closest to $\mathbf{y}$. With Euclidean distance, this is exactly least squares:

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \sum_{i=1}^n (y_i - \mathbf{x}_{(i)}\mathbf{T}\mathbf{b})^2,$$

where $\mathbf{x}_{(i)}\mathbf{T}$ is the i th row of $\mathbf{X}$. Minimizing over $\mathbf{b}$ is the same as searching $\mathbb{C}(\mathbf{X})$ for the point nearest to $\mathbf{y}$.

Definition 6.1.1 — Least squares

A vector $\hat{\beta} \in \mathbb{R}^p$ is a **least squares estimate** of β if

$$\|\mathbf{y} - \mathbf{X}\hat{\beta}\| \leq \|\mathbf{y} - \mathbf{X}\mathbf{b}\| \quad \text{for every } \mathbf{b} \in \mathbb{R}^p.$$

The vector $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ is the **fitted value** vector, and $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{Y}}$ is the **residual** vector.

The definition does not assume that the minimizer is unique, and deliberately so. In [Section 6.4](#) we will see that the *nearest point* $\hat{\mathbf{Y}}$ is always unique. The coefficient vector $\hat{\boldsymbol{\beta}}$ that produces it is unique only when the columns of $\mathbf{X}$ are linearly independent.

6.1.3 A picture with three observations

The smallest case that still shows all the geometry has $n = 3$ and $p = 2$. Then $\mathbb{C}(\mathbf{X})$ is a plane through the origin in $\mathbb{R}^3$, and $\mathbf{y}$ is a point somewhere off the plane.

Example. (Three observations)

Take responses $\mathbf{y} = (1, 4, 2)\mathbf{T}$ at regressor values $\mathbf{x} = (0, 1, 3)\mathbf{T}$, with an intercept, so that $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$. The least squares fit (computed by the listing below) is

$$\hat{\mathbf{Y}} = (2.143, 2.286, 2.571)\mathbf{T}, \quad \hat{\boldsymbol{\varepsilon}} = (-1.143, 1.714, -0.571)\mathbf{T},$$

with intercept 2.143 and slope 0.143. Two facts stand out. First, the residual is orthogonal to both columns of $\mathbf{X}$: the entries of $\hat{\boldsymbol{\varepsilon}}$ sum to zero, and $\mathbf{x}\mathbf{T}\hat{\boldsymbol{\varepsilon}} = 0$. Second, the squared lengths add:

$$\|\mathbf{y}\|^2 = 21.000 = 16.429 + 4.571 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2.$$

[Figure 6.1.1](#) shows the configuration. The fitted vector is the foot of the perpendicular dropped from $\mathbf{y}$ to the plane.

```
import numpy as np

y = np.array([1.0, 4.0, 2.0])
X = np.column_stack([np.ones(3), [0.0, 1.0, 3.0]]) # intercept and one regressor

beta_hat, *_ = np.linalg.lstsq(X, y, rcond=None)
y_hat = X @ beta_hat # the point of C(X) nearest to y
e_hat = y - y_hat # the residual vector

print("fitted ", y_hat)
print("residual", e_hat)
print("X^T e ", X.T @ e_hat) # zero: the residual is orthogonal to C(X)
```

The two facts in [Example \(Three observations\)](#) are not coincidences of this data set. They are the whole theory in miniature. The residual is orthogonal to the subspace *because* $\hat{\mathbf{Y}}$ is the nearest point, and the squared lengths add *because* the two pieces are orthogonal. The rest of the chapter makes these two statements precise and follows their consequences.

Idea. Fitted values are determined by the *subspace* $\mathbb{C}(\mathbf{X})$. Coefficients are determined by the *basis* used to describe it. Anything that depends only on $\mathbb{C}(\mathbf{X})$ is unchanged by reparameterization, by rescaling or recoding regressors, and by rank deficiency. This covers fitted values, residuals, sums of squares, R^2 and leverages.

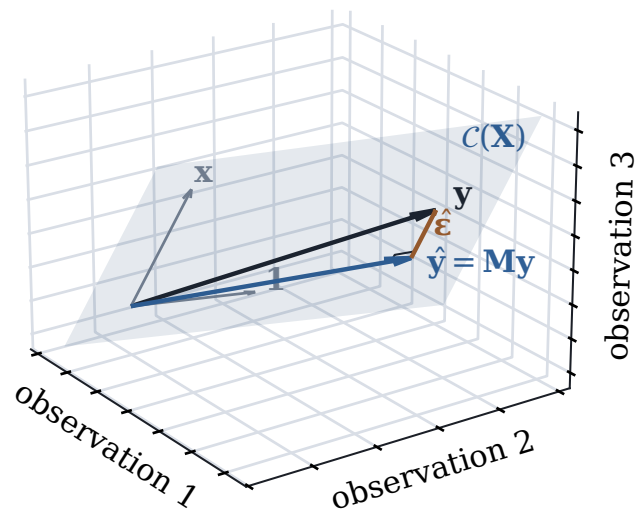


Figure 6.1.1: Least squares with $n = 3$ observations. The model's column space $\mathbb{C}(\mathbf{X})$ is the plane spanned by $\mathbf{1}$ and $\mathbf{x}$. The fitted vector $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ is the point of the plane nearest to $\mathbf{y}$, and the residual $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{Y}}$ is perpendicular to the plane.

6.1.4 Plan of the chapter

Section 6.2 collects the facts about subspaces that the geometry needs. Section 6.3 proves that nearest points exist and are unique, and identifies the matrices that compute them: the symmetric idempotent matrices, including $\mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^{-}\mathbf{X}\mathbf{T}$ for every generalized inverse. Section 6.4 applies this to least squares. Section 6.5 studies nested subspaces, which is where sums of squares come from. Section 6.6 proves the Frisch–Waugh–Lovell theorem, which says what a single coefficient in a multiple regression measures. Section 6.7 covers reparameterization and R^2 , and Section 6.8 covers leverage. Section 6.9 replaces the Euclidean inner product with a general one, the step that leads to generalized least squares. Section 6.10 shows how the projection should be computed in floating point.

6.2 The subspaces we need

This section gathers the vector-space facts used in the rest of the chapter. Readers who have worked through Chapter 1 can skim the definitions. The results on orthogonal complements, especially Lemma 6.2.1 and Theorem 6.2.3, are used constantly afterwards and are proved in full.

6.2.1 Subspaces, spans and dimension

A nonempty set $\mathcal{S} \subseteq \mathbb{R}^n$ is a **subspace** if it is closed under addition and scalar multiplication: $\mathbf{u}, \mathbf{v} \in \mathcal{S}$ and $a, b \in \mathbb{R}$ imply $a\mathbf{u} + b\mathbf{v} \in \mathcal{S}$. Every subspace contains $\mathbf{0}$. The **span** of vectors

$\mathbf{v}_1, \dots, \mathbf{v}_k$ is the set of all their linear combinations. It is the smallest subspace containing them. A set of vectors is *linearly independent* if the only linear combination equal to $\mathbf{0}$ is the one with all coefficients zero. A **basis** for $\mathcal{S}$ is a linearly independent set that spans $\mathcal{S}$. All bases of $\mathcal{S}$ have the same number of elements, the **dimension** $\dim \mathcal{S}$.

For an $n \times p$ matrix $\mathbf{A}$ there are two subspaces we use constantly:

$$\begin{aligned}\mathbb{C}(\mathbf{A}) &= \{\mathbf{A}\mathbf{b} : \mathbf{b} \in \mathbb{R}^p\} \subseteq \mathbb{R}^n && \text{(column space),} \\ \mathcal{N}(\mathbf{A}) &= \{\mathbf{b} \in \mathbb{R}^p : \mathbf{A}\mathbf{b} = \mathbf{0}\} \subseteq \mathbb{R}^p && \text{(null space).}\end{aligned}$$

The **rank** of $\mathbf{A}$ is $\text{rank}(\mathbf{A}) = \dim \mathbb{C}(\mathbf{A})$. Two standard facts from [Chapter 1](#): row rank equals column rank, so $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}\mathbf{T})$; and the rank–nullity theorem

$$\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = p. \quad (6.2.1)$$

In regression, $\mathbb{C}(\mathbf{X})$ is the **model space**. $\mathcal{N}(\mathbf{X})$ is the set of directions in which β can move without changing $\mathbf{X}\beta$. It is $\{\mathbf{0}\}$ exactly when $\mathbf{X}$ has full column rank.

Example. (A rank-deficient layout)

Suppose n observations fall into g groups and $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g]$, where $\mathbf{z}_k$ is the indicator of group k . Since $\mathbf{z}_1 + \dots + \mathbf{z}_g = \mathbf{1}$, the vector $(1, -1, \dots, -1)\mathbf{T}$ lies in $\mathcal{N}(\mathbf{X})$. The g indicators are linearly independent (their supports are disjoint), so $\text{rank}(\mathbf{X}) = g$ while $p = g + 1$. By [Eq. 6.2.1](#), $\mathcal{N}(\mathbf{X})$ is exactly the line spanned by $(1, -1, \dots, -1)\mathbf{T}$. The model space $\mathbb{C}(\mathbf{X}) = \mathbb{C}([\mathbf{z}_1, \dots, \mathbf{z}_g])$ is the set of vectors that are constant within groups.

6.2.2 Inner products, orthogonality and complements

Unless stated otherwise $\mathbb{R}^n$ carries the Euclidean inner product $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}\mathbf{T}\mathbf{v}$ and norm $\|\mathbf{u}\| = (\mathbf{u}\mathbf{T}\mathbf{u})^{1/2}$. Vectors are **orthogonal**, written $\mathbf{u} \perp \mathbf{v}$, if $\mathbf{u}\mathbf{T}\mathbf{v} = 0$. The Cauchy–Schwarz inequality $|\mathbf{u}\mathbf{T}\mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$, with equality iff one vector is a multiple of the other, lets us define the angle $\theta \in [0, \pi]$ between nonzero vectors by

$$\cos \theta = \frac{\mathbf{u}\mathbf{T}\mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}. \quad (6.2.2)$$

The angle comes back in [Section 6.7](#), where R^2 turns out to be a squared cosine. The Pythagorean identity

$$\mathbf{u} \perp \mathbf{v} \implies \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \quad (6.2.3)$$

follows by expanding $(\mathbf{u} + \mathbf{v})\mathbf{T}(\mathbf{u} + \mathbf{v})$. The cross terms are the ones that vanish.

Definition 6.2.1 — Orthogonal complement

The **orthogonal complement** of a subspace $\mathcal{S} \subseteq \mathbb{R}^n$ is

$$\mathcal{S}^\perp = \{\mathbf{v} \in \mathbb{R}^n : \mathbf{v}^T \mathbf{u} = 0 \text{ for all } \mathbf{u} \in \mathcal{S}\}.$$

It is immediate that $\mathcal{S}^\perp$ is a subspace and that $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$: a vector orthogonal to itself has zero length. To test whether $\mathbf{v} \in \mathcal{S}^\perp$ it is enough to check orthogonality to a spanning set of $\mathcal{S}$. In particular, $\mathbf{v} \in \mathbb{C}(\mathbf{A})^\perp$ iff $\mathbf{A}^T \mathbf{v} = \mathbf{0}$.

Lemma 6.2.1

For any $n \times p$ matrix $\mathbf{A}$,

$$\mathcal{N}(\mathbf{A}^T) = \mathbb{C}(\mathbf{A})^\perp \quad \text{and} \quad \mathcal{N}(\mathbf{A}) = \mathbb{C}(\mathbf{A}^T)^\perp.$$

Proof. $\mathbf{v} \in \mathcal{N}(\mathbf{A}^T)$ means $\mathbf{A}^T \mathbf{v} = \mathbf{0}$, that is, $\mathbf{v}$ is orthogonal to every column of $\mathbf{A}$, which is the same as $\mathbf{v} \perp \mathbb{C}(\mathbf{A})$. The second identity is the first applied to $\mathbf{A}^T$. ■

6.2.3 Orthonormal bases and Gram–Schmidt

Vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$ are **orthonormal** if $\mathbf{u}_i^T \mathbf{u}_j$ is 1 for $i = j$ and 0 otherwise. Orthonormal vectors are linearly independent: taking the inner product of $\sum_j a_j \mathbf{u}_j = \mathbf{0}$ with $\mathbf{u}_i$ gives $a_i = 0$. The coordinates of a vector $\mathbf{v} \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ are simply the inner products $a_i = \mathbf{u}_i^T \mathbf{v}$, with no linear system to solve. This is why orthonormal bases are so convenient.

Proposition 6.2.2 — Gram–Schmidt

Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be linearly independent. Define recursively

$$\mathbf{w}_j = \mathbf{v}_j - \sum_{i < j} (\mathbf{u}_i^T \mathbf{v}_j) \mathbf{u}_i, \quad \mathbf{u}_j = \mathbf{w}_j / \|\mathbf{w}_j\|, \quad j = 1, \dots, k.$$

Then each $\mathbf{w}_j \neq \mathbf{0}$. The vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$ are orthonormal, and $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_j) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ for every j .

Proof. Induction on j . Suppose the claims hold for $j - 1$. The vector $\mathbf{w}_j$ is $\mathbf{v}_j$ minus an element of $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$. If $\mathbf{w}_j$ were zero, $\mathbf{v}_j$ would lie in that span, contradicting independence. For $l < j$, $\mathbf{u}_l^T \mathbf{w}_j = \mathbf{u}_l^T \mathbf{v}_j - \sum_{i < j} (\mathbf{u}_i^T \mathbf{v}_j) \mathbf{u}_l^T \mathbf{u}_i = \mathbf{u}_l^T \mathbf{v}_j - \mathbf{u}_l^T \mathbf{v}_j = 0$, so $\mathbf{u}_j$ is orthogonal to its predecessors. Finally, $\mathbf{u}_j \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}, \mathbf{v}_j) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ and $\mathbf{v}_j \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_j)$, so the two spans agree. ■

Two consequences are used repeatedly. Every subspace has an orthonormal basis (apply Gram–Schmidt to any basis). And any orthonormal basis of a subspace $\mathcal{S}$ can be extended to an orthonormal basis of $\mathbb{R}^n$: extend it to a basis of $\mathbb{R}^n$ in any way, then apply Gram–Schmidt, which leaves the first vectors unchanged. [Section 6.10](#) shows that Gram–Schmidt, written in matrix form, is the QR factorization.

6.2.4 Direct sums and the orthogonal decomposition

For subspaces $\mathcal{S}, \mathcal{T}$, the sum $\mathcal{S} + \mathcal{T} = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in \mathcal{S}, \mathbf{v} \in \mathcal{T}\}$ is a subspace. When $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$ the sum is *direct*, written $\mathcal{S} \oplus \mathcal{T}$. Every element then decomposes as $\mathbf{u} + \mathbf{v}$ in exactly one way: if $\mathbf{u} + \mathbf{v} = \mathbf{u}' + \mathbf{v}'$ then $\mathbf{u} - \mathbf{u}' = \mathbf{v}' - \mathbf{v}$ lies in both subspaces and so is zero. When in addition $\mathcal{S} \perp \mathcal{T}$, the sum is an *orthogonal* direct sum.

Theorem 6.2.3 — Orthogonal decomposition

For every subspace $\mathcal{S} \subseteq \mathbb{R}^n$,

$$\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp, \quad \dim \mathcal{S} + \dim \mathcal{S}^\perp = n, \quad (\mathcal{S}^\perp)^\perp = \mathcal{S}.$$

Hence every $\mathbf{y} \in \mathbb{R}^n$ can be written uniquely as $\mathbf{y} = \mathbf{u} + \mathbf{v}$ with $\mathbf{u} \in \mathcal{S}$ and $\mathbf{v} \perp \mathcal{S}$.

Proof. Let $\mathbf{u}_1, \dots, \mathbf{u}_r$ be an orthonormal basis of $\mathcal{S}$, extended to an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_n$ of $\mathbb{R}^n$. A vector $\mathbf{y} = \sum_{i=1}^n a_i \mathbf{u}_i$ is orthogonal to $\mathcal{S}$ iff $a_1 = \dots = a_r = 0$, so $\mathcal{S}^\perp = \text{span}(\mathbf{u}_{r+1}, \dots, \mathbf{u}_n)$ has dimension $n - r$. Writing $\mathbf{y} = \sum_{i \leq r} a_i \mathbf{u}_i + \sum_{i > r} a_i \mathbf{u}_i$ exhibits the decomposition, and it is unique because $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$. By the same argument with the roles of the two index sets swapped, $(\mathcal{S}^\perp)^\perp = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_r) = \mathcal{S}$. ■

Combining [Theorem 6.2.3](#) with [Lemma 6.2.1](#) gives a matrix fact used in almost every later chapter.

Corollary 6.2.4

For any $n \times p$ matrix $\mathbf{A}$:

- (a) $\mathcal{N}(\mathbf{ATA}) = \mathcal{N}(\mathbf{A})$;
- (b) $\mathbb{C}(\mathbf{ATA}) = \mathbb{C}(\mathbf{AT})$ and $\mathbb{C}(\mathbf{AAT}) = \mathbb{C}(\mathbf{A})$;
- (c) $\text{rank}(\mathbf{ATA}) = \text{rank}(\mathbf{AAT}) = \text{rank}(\mathbf{A})$;
- (d) $\mathbb{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{AT})$ and $\mathbb{R}^p = \mathbb{C}(\mathbf{AT}) \oplus \mathcal{N}(\mathbf{A})$.

Proof.

- (a) If $\mathbf{Ab} = \mathbf{0}$ then $\mathbf{ATAb} = \mathbf{0}$. Conversely, $\mathbf{ATAb} = \mathbf{0}$ gives $\mathbf{bTATAb} = \|\mathbf{Ab}\|^2 = 0$, so $\mathbf{Ab} = \mathbf{0}$.
- (b) By [Lemma 6.2.1](#) applied to the symmetric matrix $\mathbf{ATA}$, and then (a) and [Lemma 6.2.1](#)

again, $\mathbb{C}(\mathbf{ATA}) = \mathcal{N}(\mathbf{ATA})^\perp = \mathcal{N}(\mathbf{A})^\perp = (\mathbb{C}(\mathbf{AT})^\perp)^\perp = \mathbb{C}(\mathbf{AT})$, using $(\mathcal{S}^\perp)^\perp = \mathcal{S}$ in the last step. The second identity is the first applied to $\mathbf{AT}$.

(c) Take dimensions in (b) and use $\text{rank}(\mathbf{AT}) = \text{rank}(\mathbf{A})$.

(d) This is [Lemma 6.2.1](#) and [Theorem 6.2.3](#). ■

Part (b) is the reason the normal equations $\mathbf{XTXb} = \mathbf{XTy}$ can always be solved: the right-hand side lies in $\mathbb{C}(\mathbf{XT}) = \mathbb{C}(\mathbf{XTX})$. We come back to this in [Section 6.4](#).

Lemma 6.2.5

For subspaces $\mathcal{S}_1, \mathcal{S}_2 \subseteq \mathbb{R}^n$,

$$(\mathcal{S}_1 + \mathcal{S}_2)^\perp = \mathcal{S}_1^\perp \cap \mathcal{S}_2^\perp \quad \text{and} \quad (\mathcal{S}_1 \cap \mathcal{S}_2)^\perp = \mathcal{S}_1^\perp + \mathcal{S}_2^\perp.$$

Proof. A vector is orthogonal to $\mathcal{S}_1 + \mathcal{S}_2$ iff it is orthogonal to every $\mathbf{u}_1 + \mathbf{u}_2$, which (taking $\mathbf{u}_2 = \mathbf{0}$ or $\mathbf{u}_1 = \mathbf{0}$) is the case iff it is orthogonal to both subspaces. For the second identity, apply the first to $\mathcal{S}_1^\perp$ and $\mathcal{S}_2^\perp$: $(\mathcal{S}_1^\perp + \mathcal{S}_2^\perp)^\perp = \mathcal{S}_1 \cap \mathcal{S}_2$, and take complements of both sides. ■

[Lemma 6.2.5](#) is what we need to describe reduced models defined by linear constraints ([Theorem 6.5.2](#)).

6.2.5 Exercises

6.2.5.1 A. Check your understanding

Exercise A1.

Show that $\mathcal{S}^\perp$ is a subspace for any set $\mathcal{S}$ (not necessarily a subspace), and that $\mathcal{S} \subseteq \mathcal{T}$ implies $\mathcal{T}^\perp \subseteq \mathcal{S}^\perp$. Show that $\mathcal{S}^\perp = \text{span}(\mathcal{S})^\perp$.

6.2.5.2 B. Practice

Exercise B1.

A one-way layout has three groups of sizes 2, 2, 1 and model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3]$. Find bases for $\mathbb{C}(\mathbf{X})$, $\mathcal{N}(\mathbf{X})$, $\mathbb{C}(\mathbf{XT})$ and $\mathbb{C}(\mathbf{X})^\perp$, and check that the dimensions agree with [Eq. 6.2.1](#) and [Theorem 6.2.3](#).

Exercise B2.

Show that $\mathbb{C}(\mathbf{AB}) \subseteq \mathbb{C}(\mathbf{A})$, and deduce $\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$. Show that $\mathbb{C}(\mathbf{AB}) = \mathbb{C}(\mathbf{A})$ whenever $\mathbf{B}$ has full row rank.

6.2.5.3 C. Going deeper

Exercise C1.

For subspaces $\mathcal{S}, \mathcal{T} \subseteq \mathbb{R}^n$, prove $\dim(\mathcal{S} + \mathcal{T}) = \dim \mathcal{S} + \dim \mathcal{T} - \dim(\mathcal{S} \cap \mathcal{T})$. Use it to show that two r -dimensional subspaces of $\mathbb{R}^n$ with $2r > n$ must share a nonzero vector.

6.3 Orthogonal projection

6.3.1 Nearest points

Theorem 6.3.1 — Projection theorem

Let $\mathcal{S}$ be a subspace of $\mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^n$. There is exactly one $\hat{\mathbf{y}} \in \mathcal{S}$ such that

$$\|\mathbf{y} - \hat{\mathbf{y}}\| \leq \|\mathbf{y} - \mathbf{u}\| \quad \text{for all } \mathbf{u} \in \mathcal{S}.$$

It is the unique $\hat{\mathbf{y}} \in \mathcal{S}$ with $\mathbf{y} - \hat{\mathbf{y}} \perp \mathcal{S}$. Moreover, for every $\mathbf{u} \in \mathcal{S}$,

$$\|\mathbf{y} - \mathbf{u}\|^2 = \|\mathbf{y} - \hat{\mathbf{y}}\|^2 + \|\hat{\mathbf{y}} - \mathbf{u}\|^2. \quad (6.3.1)$$

Proof. By Theorem 6.2.3 there is a unique decomposition $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{v}$ with $\hat{\mathbf{y}} \in \mathcal{S}$ and $\mathbf{v} \in \mathcal{S}^\perp$. For $\mathbf{u} \in \mathcal{S}$, $\mathbf{y} - \mathbf{u} = \mathbf{v} + (\hat{\mathbf{y}} - \mathbf{u})$, where $\mathbf{v} \perp \mathcal{S}$ and $\hat{\mathbf{y}} - \mathbf{u} \in \mathcal{S}$. The Pythagorean identity Eq. 6.2.3 then gives Eq. 6.3.1. The right-hand side is at least $\|\mathbf{y} - \hat{\mathbf{y}}\|^2$, with equality iff $\mathbf{u} = \hat{\mathbf{y}}$. So $\hat{\mathbf{y}}$ is the unique nearest point. If some other $\mathbf{w} \in \mathcal{S}$ also had $\mathbf{y} - \mathbf{w} \perp \mathcal{S}$, then $\mathbf{y} = \mathbf{w} + (\mathbf{y} - \mathbf{w})$ would be a second decomposition, contradicting uniqueness. ■

The theorem has two halves, and it helps to keep them apart. The *variational* half says $\hat{\mathbf{y}}$ minimizes a distance. The *orthogonality* half says the error $\mathbf{y} - \hat{\mathbf{y}}$ is perpendicular to the subspace. The first half explains why least squares is a sensible estimation principle. The second is what we compute with, because orthogonality is a set of linear equations.

6.3.2 The projection as a matrix

Write $\mathbf{P}_{\mathcal{S}}\mathbf{y}$ for the nearest point $\hat{\mathbf{y}}$. The map $\mathbf{y} \mapsto \mathbf{P}_{\mathcal{S}}\mathbf{y}$ is linear: if $\mathbf{y}_k = \hat{\mathbf{y}}_k + \mathbf{v}_k$ are the decompositions of $\mathbf{y}_1$ and $\mathbf{y}_2$, then $a\mathbf{y}_1 + b\mathbf{y}_2 = (a\hat{\mathbf{y}}_1 + b\hat{\mathbf{y}}_2) + (a\mathbf{v}_1 + b\mathbf{v}_2)$ is a decomposition of the same kind, and by uniqueness it is *the* decomposition. A linear map on $\mathbb{R}^n$ is given by a unique $n \times n$ matrix, since two matrices that agree on every vector agree on the standard basis vectors and are therefore equal.

Definition 6.3.1 — Orthogonal projection matrix

The **orthogonal projection** onto a subspace $\mathcal{S}$ is the unique $n \times n$ matrix $P_{\mathcal{S}}$ such that, for every $\mathbf{y}$, $P_{\mathcal{S}}\mathbf{y} \in \mathcal{S}$ and $\mathbf{y} - P_{\mathcal{S}}\mathbf{y} \perp \mathcal{S}$. When $\mathcal{S} = \mathbb{C}(\mathbf{X})$ for the model matrix of a linear model, we write $\mathbf{M} = P_{\mathbb{C}(\mathbf{X})}$.

The most useful formula for $P_{\mathcal{S}}$ uses an orthonormal basis.

Proposition 6.3.2

Let the columns of the $n \times r$ matrix $\mathbf{Q}$ be an orthonormal basis for $\mathcal{S}$, so that $\mathbf{Q}\mathbf{T}\mathbf{Q} = \mathbf{I}_r$. Then

$$P_{\mathcal{S}} = \mathbf{Q}\mathbf{Q}\mathbf{T} = \sum_{i=1}^r \mathbf{u}_i \mathbf{u}_i \mathbf{T},$$

where $\mathbf{u}_1, \dots, \mathbf{u}_r$ are the columns of $\mathbf{Q}$.

Proof. For any $\mathbf{y}$, $\mathbf{Q}\mathbf{Q}\mathbf{T}\mathbf{y} = \mathbf{Q}(\mathbf{Q}\mathbf{T}\mathbf{y}) \in \mathbb{C}(\mathbf{Q}) = \mathcal{S}$, and $\mathbf{Q}\mathbf{T}(\mathbf{y} - \mathbf{Q}\mathbf{Q}\mathbf{T}\mathbf{y}) = \mathbf{Q}\mathbf{T}\mathbf{y} - \mathbf{Q}\mathbf{T}\mathbf{y} = \mathbf{0}$, so the difference is orthogonal to every column of $\mathbf{Q}$ and hence to $\mathcal{S}$. By uniqueness, $\mathbf{Q}\mathbf{Q}\mathbf{T} = P_{\mathcal{S}}$. ■

Example. (Projections onto a line)

For $\mathcal{S} = \text{span}(\mathbf{v})$ with $\mathbf{v} \neq \mathbf{0}$, the single vector $\mathbf{v}/\|\mathbf{v}\|$ is an orthonormal basis, so

$$P_{\mathcal{S}} = \frac{\mathbf{v}\mathbf{v}\mathbf{T}}{\mathbf{v}\mathbf{T}\mathbf{v}}, \quad P_{\mathcal{S}}\mathbf{y} = \frac{\mathbf{v}\mathbf{T}\mathbf{y}}{\mathbf{v}\mathbf{T}\mathbf{v}} \mathbf{v}.$$

With $\mathbf{v} = \mathbf{1}$ this gives $\mathbf{P} = n^{-1}\mathbf{1}\mathbf{1}\mathbf{T}$, the matrix with every entry $1/n$, and $\mathbf{P}\mathbf{y} = \bar{y}\mathbf{1}$. *The sample mean is a projection:* fitting the intercept-only model replaces every observation by $\bar{y}$, and the residuals $y_i - \bar{y}$ form a vector in $\mathbf{1}^{\perp}$, the set of vectors whose entries sum to zero.

6.3.3 Characterizing projection matrices

Which matrices are orthogonal projections? [Proposition 6.3.2](#) shows that every projection has the form $\mathbf{Q}\mathbf{Q}\mathbf{T}$, so it is *symmetric*, and $\mathbf{Q}\mathbf{Q}\mathbf{T}\mathbf{Q}\mathbf{Q}\mathbf{T} = \mathbf{Q}\mathbf{I}\mathbf{Q}\mathbf{T} = \mathbf{Q}\mathbf{Q}\mathbf{T}$, so it is *idempotent*. The converse is the basic recognition theorem.

Theorem 6.3.3

An $n \times n$ matrix $\mathbf{P}$ is the orthogonal projection onto some subspace iff $\mathbf{P}\mathbf{T} = \mathbf{P}$ and $\mathbf{P}^2 = \mathbf{P}$. In that case the subspace is $\mathbb{C}(\mathbf{P})$, and $\mathbf{I} - \mathbf{P}$ is the orthogonal projection onto $\mathbb{C}(\mathbf{P})^{\perp}$.

Proof. Necessity was shown above. For sufficiency, let $\mathbf{P}$ be symmetric and idempotent and let $\mathbf{y} \in \mathbb{R}^n$. Certainly $\mathbf{Py} \in \mathbb{C}(\mathbf{P})$. For any $\mathbf{u} = \mathbf{Pb} \in \mathbb{C}(\mathbf{P})$,

$$\mathbf{uT}(\mathbf{y} - \mathbf{Py}) = \mathbf{bTPT}(\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{bT}(\mathbf{P} - \mathbf{P}^2)\mathbf{y} = 0,$$

so $\mathbf{y} - \mathbf{Py} \perp \mathbb{C}(\mathbf{P})$ and $\mathbf{P} = \mathbf{P}_{\mathbb{C}(\mathbf{P})}$. For the last claim, $\mathbf{I} - \mathbf{P}$ is symmetric and idempotent, since $(\mathbf{I} - \mathbf{P})^2 = \mathbf{I} - 2\mathbf{P} + \mathbf{P}^2 = \mathbf{I} - \mathbf{P}$. It sends $\mathbf{y}$ to $\mathbf{y} - \mathbf{Py} \in \mathbb{C}(\mathbf{P})^\perp$, and $\mathbf{y} - (\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{Py} \perp \mathbb{C}(\mathbf{P})^\perp$. ■

Before this theorem, checking that a matrix computes a projection meant thinking about subspaces. Afterwards it is two matrix identities. Also, a projection is determined by its range: if $\mathbf{P}_1, \mathbf{P}_2$ are symmetric idempotent with $\mathbb{C}(\mathbf{P}_1) = \mathbb{C}(\mathbf{P}_2)$, then $\mathbf{P}_1 = \mathbf{P}_2$, because both are the projection onto the same subspace.

Proposition 6.3.4 — Spectrum, rank and trace

Let $\mathbf{P}$ be the orthogonal projection onto an r -dimensional subspace $\mathcal{S}$. Then:

- (a) every eigenvalue of $\mathbf{P}$ is 0 or 1, and 1 has multiplicity r ;
- (b) $\text{rank}(\mathbf{P}) = \text{tr}(\mathbf{P}) = r$;
- (c) $\mathbf{Pv} = \mathbf{v}$ iff $\mathbf{v} \in \mathcal{S}$, and $\mathbf{Pv} = \mathbf{0}$ iff $\mathbf{v} \in \mathcal{S}^\perp$;
- (d) $\mathbf{vTPv} = \|\mathbf{Pv}\|^2 \leq \|\mathbf{v}\|^2$, so $\mathbf{P}$ and $\mathbf{I} - \mathbf{P}$ are nonnegative definite.

Proof. Write $\mathbf{P} = \mathbf{QQT}$ as in Proposition 6.3.2, and extend $\mathbf{Q}$ to an orthogonal matrix $\mathbf{U} = [\mathbf{Q}, \mathbf{Q}_\perp]$. Then $\mathbf{P} = \mathbf{U} \text{diag}(\mathbf{I}_r, \mathbf{0}) \mathbf{UT}$ is a spectral decomposition, which gives (a). Rank and trace of a symmetric matrix are the number of nonzero eigenvalues and their sum, which gives (b). Directly, $\text{tr}(\mathbf{QQT}) = \text{tr}(\mathbf{QTQ}) = \text{tr}(\mathbf{I}_r) = r$. Part (c) restates the projection theorem: the nearest point of $\mathcal{S}$ to a vector already in $\mathcal{S}$ is the vector itself, and $\mathbf{v} \perp \mathcal{S}$ iff $\mathbf{0}$ is its nearest point. For (d), $\mathbf{vTPv} = \mathbf{vTPTPv}$, and $\|\mathbf{v}\|^2 = \|\mathbf{Pv}\|^2 + \|(\mathbf{I} - \mathbf{P})\mathbf{v}\|^2$ by Pythagoras. ■

The identity $\text{rank} = \text{tr}$ gets a lot of use. The dimension of a subspace, which is the “degrees of freedom” of every sum of squares in this book, can be read off as a sum of diagonal entries without finding a basis.

6.3.4 Generalized inverses

Proposition 6.3.2 needs an orthonormal basis. In statistics the subspace usually comes as the column space of a model matrix $\mathbf{X}$ whose columns are not orthonormal and may not even be independent. To write $\mathbf{M}$ directly in terms of $\mathbf{X}$ we need a substitute for $(\mathbf{XTX})^{-1}$ that exists even when $\mathbf{XTX}$ is singular.

Definition 6.3.2 — Generalized inverse

A **generalized inverse** of an $m \times k$ matrix $\mathbf{A}$ is any $k \times m$ matrix $\mathbf{A}^-$ such that $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$.

If $\mathbf{A}$ is square and nonsingular, multiplying $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$ on both sides by $\mathbf{A}^{-1}$ gives $\mathbf{A}^- = \mathbf{A}^{-1}$, so the generalized inverse is unique and equal to the inverse. When $\mathbf{A}$ is singular there are infinitely many.

Proposition 6.3.5 — Existence

Every matrix has a generalized inverse. If $\mathbf{A}$ has rank r and is permuted so that

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$$

with $\mathbf{A}_{11}$ an $r \times r$ nonsingular block, then $\mathbf{A}^- = \begin{pmatrix} \mathbf{A}_{11}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$ (permuted back) is a generalized inverse.

Proof. A rank- r matrix has r independent rows and r independent columns, and the submatrix where they cross is nonsingular (Chapter 1). This guarantees the block form. Since the first r columns are a basis for the column space, $\begin{pmatrix} \mathbf{A}_{12} \\ \mathbf{A}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11} \\ \mathbf{A}_{21} \end{pmatrix} \mathbf{B}$ for some $\mathbf{B}$, so $\mathbf{B} = \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$ and $\mathbf{A}_{22} = \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$. Multiplying out,

$$\mathbf{A}\mathbf{A}^-\mathbf{A} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{A}_{21}\mathbf{A}_{11}^{-1} & \mathbf{0} \end{pmatrix} \mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12} \end{pmatrix} = \mathbf{A}.$$

■

Among the many generalized inverses, one is singled out by extra symmetry conditions. The **Moore–Penrose inverse** $\mathbf{A}^+$ is the unique matrix satisfying $\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}$, $\mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+$, and both $\mathbf{A}\mathbf{A}^+$ and $\mathbf{A}^+\mathbf{A}$ symmetric (Penrose 1955). Section 6.10 computes it from the singular value decomposition. For projections we will need none of these extra conditions, and that is the point of the next result.

Lemma 6.3.6

Let $\mathbf{X}$ be $n \times p$ and let $\mathbf{G}$ be any generalized inverse of $\mathbf{X}\mathbf{T}\mathbf{X}$. Then:

- (a) $\mathbf{X}\mathbf{G}\mathbf{X}\mathbf{T}\mathbf{X} = \mathbf{X}$ and $\mathbf{X}\mathbf{T}\mathbf{X}\mathbf{G}\mathbf{X}\mathbf{T} = \mathbf{X}\mathbf{T}$;
- (b) $\mathbf{X}\mathbf{G}\mathbf{X}\mathbf{T}$ does not depend on the choice of generalized inverse;
- (c) $\mathbf{X}\mathbf{G}\mathbf{X}\mathbf{T}$ is symmetric, even when $\mathbf{G}$ is not.

Proof. By Corollary 6.2.4(b), $\mathbb{C}(\mathbf{X}\mathbf{T}) = \mathbb{C}(\mathbf{X}\mathbf{T}\mathbf{X})$, so there is a $p \times n$ matrix $\mathbf{B}$ with

$\mathbf{X}\mathbf{T} = \mathbf{X}\mathbf{T}\mathbf{X}\mathbf{B}$. Transposing, $\mathbf{X} = \mathbf{B}\mathbf{T}\mathbf{X}\mathbf{T}\mathbf{X}$. Then

$$\mathbf{X}\mathbf{G}\mathbf{X}\mathbf{T}\mathbf{X} = \mathbf{B}\mathbf{T}(\mathbf{X}\mathbf{T}\mathbf{X})\mathbf{G}(\mathbf{X}\mathbf{T}\mathbf{X}) = \mathbf{B}\mathbf{T}\mathbf{X}\mathbf{T}\mathbf{X} = \mathbf{X},$$

which is the first identity in (a). The second follows the same way, using $\mathbf{X}\mathbf{T} = \mathbf{X}\mathbf{T}\mathbf{X}\mathbf{B}$ on the right. For (b) and (c),

$$\mathbf{X}\mathbf{G}\mathbf{X}\mathbf{T} = \mathbf{B}\mathbf{T}(\mathbf{X}\mathbf{T}\mathbf{X})\mathbf{G}(\mathbf{X}\mathbf{T}\mathbf{X})\mathbf{B} = \mathbf{B}\mathbf{T}\mathbf{X}\mathbf{T}\mathbf{X}\mathbf{B},$$

which involves no $\mathbf{G}$ at all, and $(\mathbf{B}\mathbf{T}\mathbf{X}\mathbf{T}\mathbf{X}\mathbf{B})\mathbf{T} = \mathbf{B}\mathbf{T}\mathbf{X}\mathbf{T}\mathbf{X}\mathbf{B}$.

■

Theorem 6.3.7

For any $n \times p$ matrix $\mathbf{X}$ and any generalized inverse $(\mathbf{X}\mathbf{T}\mathbf{X})^-$,

$$\mathbf{M} = \mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^- \mathbf{X}\mathbf{T}$$

is the orthogonal projection onto $\mathbb{C}(\mathbf{X})$. In particular $\text{rank}(\mathbf{M}) = \text{tr}(\mathbf{M}) = \text{rank}(\mathbf{X})$. If $\mathbf{X}$ has full column rank, $\mathbf{M} = \mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^{-1} \mathbf{X}\mathbf{T}$.

Proof. By Lemma 6.3.6, $\mathbf{M}$ is symmetric and $\mathbf{M}^2 = \mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^- [\mathbf{X}\mathbf{T}\mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^- \mathbf{X}\mathbf{T}] = \mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^- \mathbf{X}\mathbf{T} = \mathbf{M}$. By Theorem 6.3.3, $\mathbf{M}$ is the projection onto $\mathbb{C}(\mathbf{M})$. Clearly $\mathbb{C}(\mathbf{M}) \subseteq \mathbb{C}(\mathbf{X})$, and $\mathbf{M}\mathbf{X} = \mathbf{X}$ (again by the lemma) shows $\mathbb{C}(\mathbf{X}) \subseteq \mathbb{C}(\mathbf{M})$. The rank statement is Proposition 6.3.4(b).

■

It is worth pausing on how strong Theorem 6.3.7 is. The matrix $(\mathbf{X}\mathbf{T}\mathbf{X})^-$ can be almost anything: symmetric or not, reflexive or not, with entries of any size. The sandwich $\mathbf{X}(\cdot)\mathbf{X}\mathbf{T}$ removes all of that freedom. We will see in Section 6.4 that this is exactly the difference between coefficients and fitted values, made concrete.

Example. (The one-way layout)

In the setting of Example (A rank-deficient layout), sort the observations by group and let group k have n_k members. The model space is spanned by the orthogonal vectors $\mathbf{z}_1, \dots, \mathbf{z}_g$, so by Proposition 6.3.2 with $\mathbf{u}_k = \mathbf{z}_k / \sqrt{n_k}$,

$$\mathbf{M} = \sum_{k=1}^g \frac{\mathbf{z}_k \mathbf{z}_k^T}{n_k} = \begin{pmatrix} n_1^{-1} \mathbf{1}_{n_1} \mathbf{1}_{n_1}^T & & \\ & \ddots & \\ & & n_g^{-1} \mathbf{1}_{n_g} \mathbf{1}_{n_g}^T \end{pmatrix},$$

a block-diagonal matrix of within-group averaging operators. $\mathbf{M}\mathbf{y}$ replaces each observation by its group mean. The intercept column adds nothing to $\mathbb{C}(\mathbf{X})$, so it does not appear, and $\text{tr}(\mathbf{M}) = \sum_k n_k / n_k = g = \text{rank}(\mathbf{X})$. We got here without a generalized inverse. Theorem 6.3.7 guarantees that any generalized inverse of the singular $(g+1) \times (g+1)$ matrix $\mathbf{X}\mathbf{T}\mathbf{X}$ produces this same matrix. Section 6.4 checks this numerically.

6.3.5 Sums and products of projections

Complicated subspaces are often built from simpler ones, and their projections are built the same way.

Theorem 6.3.8

Let P_1, P_2 be orthogonal projections. Then $P_1 + P_2$ is an orthogonal projection iff $P_1 P_2 = 0$, that is, iff $\mathbb{C}(P_1) \perp \mathbb{C}(P_2)$. In that case it projects onto $\mathbb{C}(P_1) \oplus \mathbb{C}(P_2)$.

Proof. $P_1 + P_2$ is always symmetric, and $(P_1 + P_2)^2 = P_1 + P_2 + P_1 P_2 + P_2 P_1$. So it is idempotent iff $P_1 P_2 + P_2 P_1 = 0$. If $P_1 P_2 = 0$, then transposing gives $P_2 P_1 = 0$ and the condition holds. Conversely, suppose $P_1 P_2 + P_2 P_1 = 0$. Multiplying on the left by P_1 gives $P_1 P_2 + P_1 P_2 P_1 = 0$, and on the right by P_1 gives $P_1 P_2 P_1 + P_2 P_1 = 0$. Subtracting, $P_1 P_2 = P_2 P_1$, and then $2P_1 P_2 = 0$. The equivalence with orthogonality of ranges holds because $P_1 P_2 = 0$ says exactly that every vector of $\mathbb{C}(P_2)$ is sent to 0 by P_1 , that is (Proposition 6.3.4(c)), lies in $\mathbb{C}(P_1)^\perp$. Finally, $\mathbb{C}(P_1 + P_2) \subseteq \mathbb{C}(P_1) + \mathbb{C}(P_2)$, and for $u = P_1 u + P_2 u$ in that sum we have $(P_1 + P_2)u = u$, so the ranges agree. ■

By induction, if $P_1, \dots, P_k$ are projections with $P_i P_j = 0$ for $i \neq j$, their sum projects onto the orthogonal direct sum of their ranges. When the sum is I , we get an orthogonal decomposition

$$y = P_1 y + \dots + P_k y, \quad \|y\|^2 = \|P_1 y\|^2 + \dots + \|P_k y\|^2.$$

Every analysis of variance table in this book is a table of such a decomposition.

Products behave differently. $P_1 P_2$ need not be symmetric or idempotent, and it is generally not a projection. Exercise (B1) shows that it is one iff P_1 and P_2 commute, in which case it projects onto $\mathbb{C}(P_1) \cap \mathbb{C}(P_2)$. The most important instance is the nested case $\mathbb{C}(P_2) \subseteq \mathbb{C}(P_1)$, treated in Section 6.5.

6.3.6 Exercises

6.3.6.1 A. Check your understanding

Exercise A1.

Let $\mathcal{S} = \text{span}\{(1, 1, 0)\mathbf{T}, (0, 1, 1)\mathbf{T}\}$. Compute $P_{\mathcal{S}}$ in two ways: by Gram–Schmidt and Proposition 6.3.2, and by Theorem 6.3.7. Verify symmetry, idempotence and $\text{tr } P_{\mathcal{S}} = 2$. Find $\mathcal{S}^\perp$ and check that $I - P_{\mathcal{S}}$ projects onto it.

Exercise A2.

Show that a symmetric matrix whose eigenvalues all lie in $\{0, 1\}$ is an orthogonal projection. Give a non-symmetric 2×2 matrix with eigenvalues 0 and 1 that is not idempotent, and one that is idempotent.

Exercise A3.

Let $\mathbf{P}$ be an orthogonal projection and a, b real numbers. Show that the eigenvalues of $a\mathbf{P} + b\mathbf{I}$ are b and $a + b$. Show that when $b \neq 0$ and $a + b \neq 0$,

$$(a\mathbf{P} + b\mathbf{I})^{-1} = \frac{1}{b} \left(\mathbf{I} - \frac{a}{a+b} \mathbf{P} \right).$$

Use this to invert the equicorrelation matrix $(1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^T$.

Exercise A4.

Show that if $\mathbf{G}$ is a generalized inverse of the symmetric matrix $\mathbf{X}\mathbf{T}\mathbf{X}$, then so is $\mathbf{G}\mathbf{T}$, and so is $\mathbf{G}\mathbf{X}\mathbf{T}\mathbf{X}\mathbf{G}\mathbf{T}$. Which of the four Penrose conditions does the last one satisfy?

6.3.6.2 B. Practice**Exercise B1.**

Let $\mathbf{P}_1, \mathbf{P}_2$ be orthogonal projections. Show that $\mathbf{P}_1\mathbf{P}_2$ is an orthogonal projection iff $\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_2\mathbf{P}_1$, and that in this case it projects onto $\mathbb{C}(\mathbf{P}_1) \cap \mathbb{C}(\mathbf{P}_2)$. *Hint:* for the range, show $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{u}$ iff $\mathbf{u}$ lies in both subspaces.

Solution. If $\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_2\mathbf{P}_1$, then $(\mathbf{P}_1\mathbf{P}_2)^T = \mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_2$, and $(\mathbf{P}_1\mathbf{P}_2)^2 = \mathbf{P}_1\mathbf{P}_2\mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_1\mathbf{P}_2$. So the product is symmetric and idempotent. Conversely, if $\mathbf{P}_1\mathbf{P}_2$ is an orthogonal projection, it is symmetric, so $\mathbf{P}_1\mathbf{P}_2 = (\mathbf{P}_1\mathbf{P}_2)^T = \mathbf{P}_2\mathbf{P}_1$. For the range: if $\mathbf{u}$ lies in both subspaces, then $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{P}_1\mathbf{u} = \mathbf{u}$. If $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{u}$, then $\mathbf{u} \in \mathbb{C}(\mathbf{P}_1)$, and also $\mathbf{u} = \mathbf{P}_2\mathbf{P}_1\mathbf{u} \in \mathbb{C}(\mathbf{P}_2)$. A projection's range is the set of vectors it fixes, so $\mathbb{C}(\mathbf{P}_1\mathbf{P}_2) = \mathbb{C}(\mathbf{P}_1) \cap \mathbb{C}(\mathbf{P}_2)$.

Exercise B2.

Let $\mathbf{G}_0$ be a generalized inverse of $\mathbf{A}$.

- Show that $\mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{U} + \mathbf{V}(\mathbf{I} - \mathbf{A}\mathbf{G}_0)$ is a generalized inverse of $\mathbf{A}$ for all conformable $\mathbf{U}, \mathbf{V}$.
- Show that every generalized inverse $\mathbf{G}$ of $\mathbf{A}$ has this form. *Hint:* take $\mathbf{U} = \mathbf{G} - \mathbf{G}_0$ and $\mathbf{V} = \mathbf{G}_0\mathbf{A}(\mathbf{G} - \mathbf{G}_0)$, and use $\mathbf{A}(\mathbf{G} - \mathbf{G}_0)\mathbf{A} = \mathbf{0}$.
- Conclude that a matrix has a unique generalized inverse iff it is square and non-singular.

Solution. Parts a, b. For (a), multiply out: $\mathbf{A}[\mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{U} + \mathbf{V}(\mathbf{I} - \mathbf{A}\mathbf{G}_0)]\mathbf{A} = \mathbf{A}\mathbf{G}_0\mathbf{A} + (\mathbf{A} - \mathbf{A}\mathbf{G}_0\mathbf{A})\mathbf{U}\mathbf{A} + \mathbf{A}\mathbf{V}(\mathbf{A} - \mathbf{A}\mathbf{G}_0\mathbf{A}) = \mathbf{A}$. For (b), put $\mathbf{\Delta} = \mathbf{G} - \mathbf{G}_0$, so that $\mathbf{A}\mathbf{\Delta}\mathbf{A} = \mathbf{A} - \mathbf{A} = \mathbf{0}$. With $\mathbf{U} = \mathbf{\Delta}$ and $\mathbf{V} = \mathbf{G}_0\mathbf{A}\mathbf{\Delta}$,

$$(\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{\Delta} + \mathbf{G}_0\mathbf{A}\mathbf{\Delta}(\mathbf{I} - \mathbf{A}\mathbf{G}_0) = \mathbf{\Delta} - \mathbf{G}_0\mathbf{A}\mathbf{\Delta}\mathbf{A} + \mathbf{G}_0\mathbf{A}\mathbf{\Delta}\mathbf{A} - \mathbf{G}_0(\mathbf{A}\mathbf{\Delta}\mathbf{A})\mathbf{G}_0 = \mathbf{\Delta},$$

so the formula reproduces $\mathbf{G}_0 + \mathbf{\Delta} = \mathbf{G}$.

Exercise B3.

Using the singular value decomposition, show that $(\mathbf{X}\mathbf{T}\mathbf{X})^+\mathbf{X}\mathbf{T} = \mathbf{X}^+$ and $\mathbf{X}\mathbf{X}^+ = \mathbf{M}$. Show also that $\mathbf{X}^+ = (\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{X}\mathbf{T}$ when $\mathbf{X}$ has full column rank.

6.3.6.3 C. Going deeper**Exercise C1.**

Show that $\mathbf{M}$ is the only $n \times n$ matrix $\mathbf{P}$ satisfying $\mathbf{P}\mathbf{X} = \mathbf{X}$, $\mathbf{P}\mathbf{T} = \mathbf{P}$ and $\text{rank}(\mathbf{P}) = \text{rank}(\mathbf{X})$. Is the rank condition necessary? Is symmetry?

6.4 Least squares via projection

We now return to the linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, with $\mathbf{X}$ an $n \times p$ matrix of rank $r \leq p$, and $\mathbf{M}$ the orthogonal projection onto $\mathbb{C}(\mathbf{X})$. Nothing in this section uses probability. The results hold for every data vector $\mathbf{y}$ and so apply whatever the distribution of the errors.

6.4.1 The characterization of least squares estimates

Theorem 6.4.1 — Least squares is projection

A vector $\hat{\boldsymbol{\beta}}$ is a least squares estimate of $\boldsymbol{\beta}$ iff

$$\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}\mathbf{y}.$$

Consequently the fitted values $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ and residuals $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$ are the same for every least squares estimate.

Proof. Apply the projection theorem (Theorem 6.3.1) with $\mathcal{S} = \mathbb{C}(\mathbf{X})$ and $\mathbf{u} = \mathbf{X}\mathbf{b}$. By Eq. 6.3.1,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{M}\mathbf{y}\|^2 + \|\mathbf{M}\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 \quad \text{for every } \mathbf{b} \in \mathbb{R}^p.$$

The first term does not involve $\mathbf{b}$, and the second is zero iff $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y}$. Such a $\mathbf{b}$ exists because $\mathbf{M}\mathbf{y} \in \mathbb{C}(\mathbf{X})$. ■

The orthogonality half of the projection theorem turns this into linear equations. The residual $\mathbf{y} - \mathbf{X}\mathbf{b}$ is orthogonal to $\mathbb{C}(\mathbf{X})$ iff it is orthogonal to every column of $\mathbf{X}$, that is, iff $\mathbf{X}\mathbf{T}(\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$.

Theorem 6.4.2 — Normal equations

The **normal equations**

$$\mathbf{X}^T \mathbf{X} \mathbf{b} = \mathbf{X}^T \mathbf{y} \quad (6.4.1)$$

are consistent for every $\mathbf{y}$, and their solutions are exactly the least squares estimates. One solution is $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^- \mathbf{X}^T \mathbf{y}$, for any generalized inverse, and the set of all solutions is

$$\hat{\boldsymbol{\beta}} + \mathcal{N}(\mathbf{X}) = \{\hat{\boldsymbol{\beta}} + \mathbf{v} : \mathbf{X}\mathbf{v} = \mathbf{0}\}.$$

In particular, the least squares estimate is unique iff $\text{rank}(\mathbf{X}) = p$, and then $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$.

Proof. A vector $\mathbf{b}$ solves Eq. 6.4.1 iff $\mathbf{y} - \mathbf{X}\mathbf{b} \perp \mathbb{C}(\mathbf{X})$. Since $\mathbf{X}\mathbf{b} \in \mathbb{C}(\mathbf{X})$ automatically, this holds iff $\mathbf{y} = \mathbf{X}\mathbf{b} + (\mathbf{y} - \mathbf{X}\mathbf{b})$ is the orthogonal decomposition of $\mathbf{y}$, that is, iff $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y}$. By Theorem 6.4.1 this is the same as $\mathbf{b}$ being a least squares estimate. For the particular solution, $\mathbf{X}(\mathbf{X}^T \mathbf{X})^- \mathbf{X}^T \mathbf{y} = \mathbf{M}\mathbf{y}$ by Theorem 6.3.7. Finally, $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}}$ iff $\mathbf{X}(\mathbf{b} - \hat{\boldsymbol{\beta}}) = \mathbf{0}$, and $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$ iff $\text{rank}(\mathbf{X}) = p$ by Eq. 6.2.1. ■

The logic of this proof runs opposite to the usual calculus derivation, and the difference matters. Setting a gradient to zero finds stationary points. One then has to argue separately that they are minima and that one exists. Here existence, minimality and the full solution set all come from the decomposition $\mathbb{R}^n = \mathbb{C}(\mathbf{X}) \oplus \mathbb{C}(\mathbf{X})^\perp$. The consistency of the normal equations, which can look surprising when $\mathbf{X}^T \mathbf{X}$ is singular, is just the statement $\mathbf{X}^T \mathbf{y} \in \mathbb{C}(\mathbf{X}^T) = \mathbb{C}(\mathbf{X}^T \mathbf{X})$ from Corollary 6.2.4.

6.4.2 Fitted values and residuals

Collecting the consequences for $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ and $\hat{\boldsymbol{\epsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$:

Proposition 6.4.3

For any data vector $\mathbf{y}$:

- (a) $\mathbf{X}^T \hat{\boldsymbol{\epsilon}} = \mathbf{0}$ and $\hat{\mathbf{Y}}^T \hat{\boldsymbol{\epsilon}} = 0$;
- (b) if $\mathbf{1} \in \mathbb{C}(\mathbf{X})$ (for example, the model has an intercept), then $\sum_i \hat{\epsilon}_i = 0$ and the mean of the fitted values is $\bar{y}$;
- (c) $\|\mathbf{y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\epsilon}}\|^2$;
- (d) $\|\hat{\boldsymbol{\epsilon}}\|^2 = \mathbf{y}^T (\mathbf{I} - \mathbf{M}) \mathbf{y}$ and $\|\hat{\mathbf{Y}}\|^2 = \mathbf{y}^T \mathbf{M} \mathbf{y}$;
- (e) if $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ for some $\boldsymbol{\beta}$, then $\hat{\mathbf{Y}} = \mathbf{X}\boldsymbol{\beta} + \mathbf{M}\boldsymbol{\epsilon}$ and $\hat{\boldsymbol{\epsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\epsilon}$.

Proof.

- (a) is the orthogonality of $\hat{\boldsymbol{\epsilon}}$ to $\mathbb{C}(\mathbf{X})$, and $\hat{\mathbf{Y}} \in \mathbb{C}(\mathbf{X})$. For (b), $\mathbf{1} \in \mathbb{C}(\mathbf{X})$ gives $\mathbf{1}^T \hat{\boldsymbol{\epsilon}} = 0$, and then $\mathbf{1}^T \hat{\mathbf{Y}} = \mathbf{1}^T \mathbf{y}$. (c) is Pythagoras. (d) uses symmetry and idempotence: $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \mathbf{y}^T (\mathbf{I} - \mathbf{M}) \mathbf{T} (\mathbf{I} - \mathbf{M}) \mathbf{y} = \mathbf{y}^T (\mathbf{I} - \mathbf{M}) \mathbf{y}$. For (e), $\mathbf{M}\mathbf{X} = \mathbf{X}$ and $(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$.

Part (e) is simple but important. The residual vector does not depend on β at all. It is the projection of the unobservable error vector onto the $(n - r)$ -dimensional space $\mathbb{C}(\mathbf{X})^\perp$. Everything about σ^2 that the data can reveal must therefore come from $(\mathbf{I} - \mathbf{M})\varepsilon$. If $\mathbf{E}(\varepsilon) = \mathbf{0}$ and $\text{Cov}(\varepsilon) = \sigma^2 \mathbf{I}$, the expectation of a quadratic form (Chapter 2) gives

$$\mathbf{E}\|\hat{\varepsilon}\|^2 = \mathbf{E}[\varepsilon^T(\mathbf{I} - \mathbf{M})\varepsilon] = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}) = \sigma^2(n - r),$$

using Proposition 6.3.4. So $\|\hat{\varepsilon}\|^2/(n - r)$ is unbiased for σ^2 . The divisor is the dimension of the space in which the residuals live, not the number of observations and not $n - p$. Chapter 7 develops the sampling theory, and Chapter 9 develops the decomposition.

6.4.3 Coefficients are not unique; fitted values are

When $\text{rank}(\mathbf{X}) < p$, Theorem 6.4.2 gives a whole affine subspace of least squares estimates, one for each generalized inverse and more besides. Statistical software has to pick one. Some programs drop columns that are linearly dependent on earlier ones. Others impose “side conditions” such as a zero coefficient for a reference level or effects that sum to zero. Others report the minimum-norm solution. These choices look very different in the printed coefficient table. The geometry says they are all describing the same fitted vector.

Example. (Four answers, one fit)

Nine observations fall in three groups of sizes 3, 2, 4 with group means 5.700, 7.800 and 4.300. The model matrix $\mathbf{X} = [\mathbf{1}, z_1, z_2, z_3]$ has $p = 4$ columns and rank 3, so $\mathbf{X}^T\mathbf{X}$ is singular. We compute $\hat{\beta} = \mathbf{G}\mathbf{X}^T\mathbf{y}$ for four generalized inverses $\mathbf{G}$ of $\mathbf{X}^T\mathbf{X}$: the Moore–Penrose inverse; the inverse obtained by deleting the intercept row and column (Proposition 6.3.5); the one obtained by deleting the first group’s row and column; and a deliberately strange, non-symmetric one built from the general formula of Exercise (B2).

Generalized inverse	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	$\hat{\beta}_1 - \hat{\beta}_2$	$\max \mathbf{M} - \mathbf{M}_{\text{MP}} $
Moore– Penrose	4.450	1.250	3.350	-0.150	-2.100	0
drop intercept	0.000	5.700	7.800	4.300	-2.100	1.1×10^{-16}
drop group 1	5.700	0.000	2.100	-1.400	-2.100	1.1×10^{-16}
random	-9.102	14.802	16.902	13.402	-2.100	1.8×10^{-15}

The coefficient vectors disagree wildly, including in sign. Yet each produces the same projection matrix up to rounding error, and hence the same fitted values (the group means) and residuals. The difference $\hat{\beta}_1 - \hat{\beta}_2$ is the same in every row, and so is $\hat{\beta}_0 + \hat{\beta}_k$ for each k . Individual coefficients are not. The trace of $\mathbf{M}$ is 3.000, the rank of $\mathbf{X}$.

```
import numpy as np
```

```

groups = np.array([0, 0, 0, 1, 1, 2, 2, 2, 2])    # three groups, sizes 3, 2, 4
y = np.array([5.1, 6.3, 5.7, 8.2, 7.4, 4.0, 4.9, 3.8, 4.5])
n, g = len(y), 3
X = np.column_stack([np.ones(n), np.eye(g)[groups]]) # [1, indicator columns]
A = X.T @ X
print("n =", n, " p =", X.shape[1], " rank(X) =", np.linalg.matrix_rank(X))

```

```

def g_drop(j):
    """Generalized inverse obtained by deleting row and column j of A."""
    keep = [k for k in range(A.shape[0]) if k != j]
    G = np.zeros_like(A)
    G[np.ix_(keep, keep)] = np.linalg.inv(A[np.ix_(keep, keep)])
    return G

rng = np.random.default_rng(6)
G0 = np.linalg.pinv(A)                    # Moore-Penrose inverse
U, V = rng.normal(size=A.shape), rng.normal(size=A.shape)
G_random = G0 + (np.eye(4) - G0 @ A) @ U + V @ (np.eye(4) - A @ G0)

ginverses = {
    "Moore-Penrose": G0,
    "drop intercept": g_drop(0),
    "drop group 1": g_drop(1),
    "random": G_random,
}

```

```

M_ref = X @ G0 @ X.T
for name, G in ginverses.items():
    assert np.allclose(A @ G @ A, A)        # G really is a g-inverse
    beta = G @ X.T @ y
    M = X @ G @ X.T
    print(f"{name:15s} beta = {np.round(beta, 3)}",
          f" max|M - M_ref| = {np.abs(M - M_ref).max():.1e}",
          f" group1-group2 = {beta[1] - beta[2]:.3f}")

```

Which linear functions of $\hat{\beta}$ come out the same for every least squares estimate? The answer is a first look at estimability, the subject of Chapter 8.

Theorem 6.4.4

For $\lambda \in \mathbb{R}^p$, the value $\lambda^T \hat{\beta}$ is the same for every least squares estimate $\hat{\beta}$ (and every $\mathbf{y}$) iff $\lambda \in \mathbb{C}(\mathbf{X}\mathbf{T})$, that is, iff $\lambda^T \mathbf{T} = \rho^T \mathbf{X}$ for some $\rho \in \mathbb{R}^n$. In that case $\lambda^T \hat{\beta} = \rho^T \mathbf{M}\mathbf{y}$.

Proof. By Theorem 6.4.2 the least squares estimates form $\hat{\beta} + \mathcal{N}(\mathbf{X})$. So $\lambda^T \hat{\beta}$ is invariant iff $\lambda^T \mathbf{v} = 0$ for all $\mathbf{v} \in \mathcal{N}(\mathbf{X})$, that is, iff $\lambda \in \mathcal{N}(\mathbf{X})^\perp = \mathbb{C}(\mathbf{X}\mathbf{T})$ (Corollary 6.2.4). If $\lambda^T \mathbf{T} = \rho^T \mathbf{X}$ then $\lambda^T \hat{\beta} = \rho^T \mathbf{X}\hat{\beta} = \rho^T \mathbf{M}\mathbf{y}$. ■

In Example (Four answers, one fit), $\lambda = (0, 1, -1, 0)^T$ is the difference between any row of $\mathbf{X}$ from group 1, $(1, 1, 0, 0)$, and any row from group 2, $(1, 0, 1, 0)$. So $\lambda \in \mathbb{C}(\mathbf{X}\mathbf{T})$ and $\hat{\beta}_1 - \hat{\beta}_2$ is invariant. For $\lambda = (0, 1, 0, 0)^T$, note that $\rho^T \mathbf{X}$ has first entry $\sum_i \rho_i$ and remaining entries

equal to the sums of ρ_i within each group. Matching $(0, 1, 0, 0)$ would need the group sums to be $1, 0, 0$ and their total to be 0 , which is impossible. So $\hat{\beta}_1$ is not invariant.

Remark (Side conditions choose coordinates). Imposing $\beta_0 = 0$ or $\beta_1 = 0$ does not add information about the data. It picks one point from the affine set $\hat{\beta} + \mathcal{N}(\mathbf{X})$. A side condition $\mathbf{c}\mathbf{T}\beta = 0$ picks exactly one point iff $\mathbf{c} \notin \mathbb{C}(\mathbf{X}\mathbf{T})$. A condition inside $\mathbb{C}(\mathbf{X}\mathbf{T})$ would constrain the fitted values and change the model rather than just its coordinates. Chapter 8 makes this precise.

6.4.4 Exercises

6.4.4.1 A. Check your understanding

Exercise A1.

For [Example \(Three observations\)](#), write out the 3×3 matrix $\mathbf{M}$ with exact fractional entries, verify $\text{tr } \mathbf{M} = 2$, and recover the fitted values and residuals quoted in the text.

Exercise A2.

Show that the residuals sum to zero whenever $\mathbf{1} \in \mathbb{C}(\mathbf{X})$, even if no column of $\mathbf{X}$ is constant. Give an example of a model without such a column in which this happens, and one in which the residuals do not sum to zero.

6.4.4.2 B. Practice

Exercise B1.

Suppose $\mathbf{E}(\mathbf{Y}) = \mathbf{X}\beta$ and $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$. Show that $\mathbf{E}(\hat{\mathbf{Y}}) = \mathbf{X}\beta$, $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2\mathbf{M}$, $\text{Cov}(\hat{\epsilon}) = \sigma^2(\mathbf{I} - \mathbf{M})$ and $\text{Cov}(\hat{\mathbf{Y}}, \hat{\epsilon}) = \mathbf{0}$. Which of these statements survive when $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{V}$ for a general positive definite $\mathbf{V}$?

Exercise B2.

Let $\text{rank}(\mathbf{X}) = r < p$ and let $\mathbf{C}$ be a $(p - r) \times p$ matrix. Show that the constrained problem “minimize $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|$ subject to $\mathbf{C}\mathbf{b} = \mathbf{0}$ ” has a unique solution for every $\mathbf{y}$, and that its fitted values are $\mathbf{M}\mathbf{y}$, iff $\text{rank}(\begin{smallmatrix} \mathbf{X} \\ \mathbf{C} \end{smallmatrix}) = p$. Show that the condition implies $\mathbb{C}(\mathbf{C}\mathbf{T}) \cap \mathbb{C}(\mathbf{X}\mathbf{T}) = \{\mathbf{0}\}$, and interpret: the constraints must not restrict any estimable function.

6.4.4.3 C. Going deeper

Exercise C1.

Suppose $\mathbf{X}\mathbf{T}\mathbf{y} \neq \mathbf{0}$. Show that every least squares estimate $\mathbf{b}$ can be written as $\mathbf{G}\mathbf{X}\mathbf{T}\mathbf{y}$ for some generalized inverse $\mathbf{G}$ of $\mathbf{A} = \mathbf{X}\mathbf{T}\mathbf{X}$. *Hint:* with $\mathbf{c} = \mathbf{X}\mathbf{T}\mathbf{y}$, try $\mathbf{G} = \mathbf{G}_0 + (\mathbf{b} - \mathbf{G}_0\mathbf{c})\mathbf{c}\mathbf{T}/(\mathbf{c}\mathbf{T}\mathbf{c})$. What goes wrong when $\mathbf{X}\mathbf{T}\mathbf{y} = \mathbf{0}$?

6.5 Nested models and sequential projection

Statistical questions about linear models are almost always comparisons. Does a regressor matter? Are several group means equal? Is a curve needed, or will a line do? Each comparison sets a full model against a *reduced* model whose mean space is a subspace of the full one. This section studies the geometry of such nested pairs. Chapter 9 and Chapter 11 build the analysis of variance and the F test on it.

6.5.1 Two nested subspaces

Let $\mathbb{C}(\mathbf{X}_0) \subseteq \mathbb{C}(\mathbf{X})$, with ranks $r_0 \leq r$. Write $\mathbf{M}_0$ and $\mathbf{M}$ for the two projections.

Theorem 6.5.1 — Nested projections

If $\mathbb{C}(\mathbf{X}_0) \subseteq \mathbb{C}(\mathbf{X})$, then:

- (a) $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0\mathbf{M} = \mathbf{M}_0$;
- (b) $\mathbf{M} - \mathbf{M}_0$ is the orthogonal projection onto $\mathbb{C}(\mathbf{X}_0)^\perp \cap \mathbb{C}(\mathbf{X})$, the orthogonal complement of $\mathbb{C}(\mathbf{X}_0)$ within $\mathbb{C}(\mathbf{X})$;
- (c) $\text{rank}(\mathbf{M} - \mathbf{M}_0) = r - r_0$;
- (d) $(\mathbf{I} - \mathbf{M}_0) - (\mathbf{I} - \mathbf{M}) = \mathbf{M} - \mathbf{M}_0$, and $\mathbf{M}_0(\mathbf{M} - \mathbf{M}_0) = (\mathbf{M} - \mathbf{M}_0)(\mathbf{I} - \mathbf{M}) = \mathbf{0}$.

Proof.

- (a) Every column of $\mathbf{M}_0$ lies in $\mathbb{C}(\mathbf{X}_0) \subseteq \mathbb{C}(\mathbf{X})$ and is left unchanged by $\mathbf{M}$, so $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0$. Transposing gives $\mathbf{M}_0\mathbf{M} = \mathbf{M}_0$.
- (b) $\mathbf{M} - \mathbf{M}_0$ is symmetric, and by (a) $(\mathbf{M} - \mathbf{M}_0)^2 = \mathbf{M} - \mathbf{M}\mathbf{M}_0 - \mathbf{M}_0\mathbf{M} + \mathbf{M}_0 = \mathbf{M} - \mathbf{M}_0$. By Theorem 6.3.3 it projects onto its column space. If $\mathbf{v} \in \mathbb{C}(\mathbf{X})$ and $\mathbf{v} \perp \mathbb{C}(\mathbf{X}_0)$, then $(\mathbf{M} - \mathbf{M}_0)\mathbf{v} = \mathbf{v} - \mathbf{0} = \mathbf{v}$, so $\mathbf{v} \in \mathbb{C}(\mathbf{M} - \mathbf{M}_0)$. Conversely, a vector $\mathbf{v} = (\mathbf{M} - \mathbf{M}_0)\mathbf{u}$ lies in $\mathbb{C}(\mathbf{X})$ and satisfies $\mathbf{M}_0\mathbf{v} = (\mathbf{M}_0 - \mathbf{M}_0)\mathbf{u} = \mathbf{0}$.
- (c) $\text{tr}(\mathbf{M} - \mathbf{M}_0) = r - r_0$ and Proposition 6.3.4.
- (d) Direct multiplication using (a).

■

The theorem gives a three-way orthogonal decomposition of $\mathbb{R}^n$,

$$\mathbb{R}^n = \underbrace{\mathbb{C}(\mathbf{X}_0)}_{\dim r_0} \oplus \underbrace{\mathbb{C}(\mathbf{X}_0)^\perp \cap \mathbb{C}(\mathbf{X})}_{\dim r - r_0} \oplus \underbrace{\mathbb{C}(\mathbf{X})^\perp}_{\dim n - r},$$

and correspondingly $\mathbf{I} = \mathbf{M}_0 + (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$, with the three terms mutually orthogonal projections. Applying the decomposition to $\mathbf{y}$ and using Pythagoras,

$$\|\mathbf{y}\|^2 = \|\mathbf{M}_0\mathbf{y}\|^2 + \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 + \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2. \quad (6.5.1)$$

The middle term has a second reading that is the key to model comparison. Since $(\mathbf{I} - \mathbf{M}_0) = (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$ is an orthogonal sum,

$$\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \|(\mathbf{I} - \mathbf{M}_0)\mathbf{y}\|^2 - \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \text{SSE}_0 - \text{SSE}. \quad (6.5.2)$$

The *drop in residual sum of squares* from the reduced to the full model is the squared length of a projection onto an $(r - r_0)$ -dimensional space. That is why it has $r - r_0$ degrees of freedom, and why it is independent of SSE under normality: the two lie in orthogonal subspaces (Chapter 4). It also shows that SSE can never increase when the model space grows, with equality iff $\mathbf{M}\mathbf{y} = \mathbf{M}_0\mathbf{y}$.

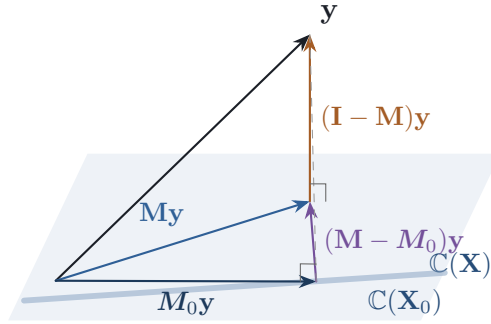


Figure 6.5.1. Nested projections. The fitted vector of the reduced model, $\mathbf{M}_0\mathbf{y}$, is also the projection of $\mathbf{M}\mathbf{y}$ onto $\mathbb{C}(\mathbf{X}_0)$. The three coloured pieces $\mathbf{M}_0\mathbf{y}$, $(\mathbf{M} - \mathbf{M}_0)\mathbf{y}$ and $(\mathbf{I} - \mathbf{M})\mathbf{y}$ are mutually orthogonal. The dashed segment is the reduced model's residual, whose squared length is the sum of the squared lengths of the two segments meeting at $\mathbf{M}\mathbf{y}$.

Part (a) also says $\mathbf{M}_0\mathbf{y} = \mathbf{M}_0(\mathbf{M}\mathbf{y})$. *Fitting the reduced model to the data gives the same result as fitting it to the full model's fitted values.* Projection can be done in stages (Figure 6.5.1).

Example. (Intercept inside simple regression)

Let $\mathbf{X}_0 = \mathbf{1}$ and $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ with $\mathbf{x}$ not constant. Then $\mathbb{C}(\mathbf{X}_0)^\perp \cap \mathbb{C}(\mathbf{X})$ is spanned by the centred regressor $\mathbf{x} - \bar{x}\mathbf{1} = (\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^T)\mathbf{x}$, which lies in $\mathbb{C}(\mathbf{X})$ and is orthogonal to $\mathbf{1}$. By Example (Projections onto a line),

$$\mathbf{M} - \mathbf{M}_0 = \frac{(\mathbf{x} - \bar{x}\mathbf{1})(\mathbf{x} - \bar{x}\mathbf{1})^T}{S_{xx}}, \quad S_{xx} = \sum_i (x_i - \bar{x})^2,$$

and so

$$\mathbf{M}\mathbf{y} = \bar{y}\mathbf{1} + \frac{S_{xy}}{S_{xx}}(\mathbf{x} - \bar{x}\mathbf{1}), \quad S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}).$$

Reading off coordinates, the slope is S_{xy}/S_{xx} and the intercept is $\bar{y} - \bar{x} S_{xy}/S_{xx}$. These are the familiar formulas, obtained without solving any equations, and the decomposition Eq. 6.5.1 becomes

$$\sum_i y_i^2 = n\bar{y}^2 + \frac{S_{xy}^2}{S_{xx}} + \text{SSE}.$$

6.5.2 Chains of subspaces

The argument extends to any chain $\mathbb{C}(\mathbf{X}_1) \subseteq \mathbb{C}(\mathbf{X}_2) \subseteq \dots \subseteq \mathbb{C}(\mathbf{X}_k)$ with projections $\mathbf{M}_1, \dots, \mathbf{M}_k$. Setting $\mathbf{M}_0 = \mathbf{0}$ and $\mathbf{M}_{k+1} = \mathbf{I}$, the differences $\mathbf{M}_j - \mathbf{M}_{j-1}$ for $j = 1, \dots, k+1$ are mutually

orthogonal projections summing to $\mathbf{I}$. The corresponding lengths

$$\|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{y}\|^2 = \text{SSE}_{j-1} - \text{SSE}_j$$

are the **sequential sums of squares** of Chapter 9. Their values depend on the *order* in which the subspaces are nested. A regressor added early can take credit that it would not get if added last. The Frisch–Waugh–Lovell theorem of [Section 6.6](#) describes what the regressor added last actually contributes.

Example. (Order matters)

For the state data of [Example \(Poverty and murder rates\)](#), enter the three regressors after the intercept in two different orders. The listing computes each squared length $\|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{y}\|^2$ directly from the projections.

Order A		Order B	
poverty	102.49	single-parent	162.84
single-parent	80.04	urban	3.59
urban	0.27	poverty	16.37
residual	101.96	residual	101.96

Entered first, poverty accounts for 102.49 of the corrected total sum of squares 284.76. Entered last, it accounts for only 16.37. The residual sum of squares, and the total explained by all three regressors together (182.80), do not depend on the order, because the final subspace does not. The individual pieces do depend on the order, because the intermediate subspaces differ. Poverty and single parenthood are correlated, so whichever enters first takes the variation they share. The piece for the *last* regressor is the only one with an order-free meaning. By [Theorem 6.6.2](#), it is the squared length of the projection of $\mathbf{y}$ onto the part of that regressor orthogonal to all the others.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
cols = {"poverty": data["poverty"], "single": data["single"], "urban": data["urban"]}

def projection(Z):
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T
    # fine for n = 50; never do this for large n

def sequential_ss(y, columns):
    """Squared lengths  $\|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{y}\|^2$  as columns enter one at a time."""
    n = len(y)
    Z = np.ones((n, 1))
    M_prev = projection(Z)
    pieces = {"mean": y @ M_prev @ y}
    for name, x in columns:
        Z = np.column_stack([Z, x])
        M = projection(Z)
        pieces[name] = y @ (M - M_prev) @ y
        M_prev = M
```

```

pieces["residual"] = y @ (np.eye(n) - M_prev) @ y
return pieces

order_a = sequential_ss(y, [(k, cols[k]) for k in ["poverty", "single", "urban"]])
order_b = sequential_ss(y, [(k, cols[k]) for k in ["single", "urban", "poverty"]])
for name in ["poverty", "single", "urban", "residual"]:
    print(f"{name:9s}  order A: {order_a[name]:8.2f}  order B: {order_b[name]:8.2f}
")

```

6.5.3 Two factors and when the pieces are orthogonal

A sum of projections is a projection exactly when the pieces are orthogonal ([Theorem 6.3.8](#)). In designed experiments, orthogonality comes from balance, and when it fails, it fails in ways that later chapters have to face.

Example. (Additive two-factor model, one observation per cell)

Let factor A have a levels and factor B have b levels, with one observation y_{ij} for each of the $n = ab$ combinations. Let P_A project onto vectors that depend only on the A level, so $P_A \mathbf{y}$ has entries $\bar{y}_{i\cdot}$, and define P_B similarly. Let $P_0 = n^{-1} \mathbf{1}\mathbf{1}^T$. Both $\mathbb{C}(P_A)$ and $\mathbb{C}(P_B)$ contain $\mathbf{1}$, so $P_A P_0 = P_B P_0 = P_0$ by [Theorem 6.5.1](#).

The crucial computation is $P_A P_B$. Averaging over j with i fixed a vector that depends only on j gives the overall average of that vector. So for every $\mathbf{y}$ the entries of $P_A(P_B \mathbf{y})$ are all equal to $b^{-1} \sum_j \bar{y}_{\cdot j} = \bar{y}_{\cdot\cdot}$. Therefore

$$P_A P_B = P_0,$$

and so $(P_A - P_0)(P_B - P_0) = P_0 - P_0 - P_0 + P_0 = \mathbf{0}$. The two *centred* factor spaces are orthogonal. By [Theorem 6.3.8](#), the additive model space has projection $P_0 + (P_A - P_0) + (P_B - P_0) = P_A + P_B - P_0$, and

$$\mathbf{I} = P_0 + (P_A - P_0) + (P_B - P_0) + (\mathbf{I} - P_A - P_B + P_0)$$

is an orthogonal decomposition with ranks 1, $a-1$, $b-1$ and $(a-1)(b-1)$. The residual projection sends y_{ij} to $y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$.

The key step used the fact that every A level appears with every B level equally often. If cells are missing or unequally replicated, $P_A P_B \neq P_0$, the centred factor spaces are no longer orthogonal, and the sum of squares for A depends on whether B was fitted first. Chapter 17 is about that situation.

6.5.4 Reduced models defined by constraints

Reduced models often come as *constraints* on the parameters of the full model, for example $\beta_2 = \beta_3$ or $\beta_4 = 0$, rather than as a smaller model matrix. In mean space, a constraint $\mathbf{A}\mathbf{T}\boldsymbol{\beta} = \mathbf{0}$ on an estimable function $\mathbf{A}\mathbf{T} = \mathbf{P}\mathbf{T}\mathbf{X}$ restricts $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ to the subspace

$$\mathcal{S}_0 = \mathbb{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{P}\mathbf{T}) = \{\boldsymbol{\mu} \in \mathbb{C}(\mathbf{X}) : \mathbf{P}\mathbf{T}\boldsymbol{\mu} = \mathbf{0}\}.$$

To test the constraint we need the piece of $\mathbb{C}(\mathbf{X})$ that $\mathcal{S}_0$ removes.

Theorem 6.5.2

Let $\mathbf{M}$ project onto $\mathbb{C}(\mathbf{X})$ and let $\mathbf{P}$ be an $n \times q$ matrix. If $\mathcal{S}_0 = \mathbb{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{P}\mathbf{T})$, then

$$\mathcal{S}_0^\perp \cap \mathbb{C}(\mathbf{X}) = \mathbb{C}(\mathbf{M}\mathbf{P}).$$

Hence the projection onto $\mathcal{S}_0$ is $\mathbf{M} - \mathbf{P}_{\mathbb{C}(\mathbf{M}\mathbf{P})}$, and $\dim \mathcal{S}_0 = r - \text{rank}(\mathbf{M}\mathbf{P})$.

Proof. By Lemma 6.2.5 and Lemma 6.2.1, $\mathcal{S}_0^\perp = \mathbb{C}(\mathbf{X})^\perp + \mathcal{N}(\mathbf{P}\mathbf{T})^\perp = \mathbb{C}(\mathbf{X})^\perp + \mathbb{C}(\mathbf{P})$. If $\mathbf{v} \in \mathcal{S}_0^\perp \cap \mathbb{C}(\mathbf{X})$, write $\mathbf{v} = \mathbf{w} + \mathbf{P}\mathbf{a}$ with $\mathbf{w} \perp \mathbb{C}(\mathbf{X})$. Applying $\mathbf{M}$ gives $\mathbf{v} = \mathbf{M}\mathbf{v} = \mathbf{M}\mathbf{P}\mathbf{a} \in \mathbb{C}(\mathbf{M}\mathbf{P})$. Conversely, $\mathbf{M}\mathbf{P}\mathbf{a}$ lies in $\mathbb{C}(\mathbf{X})$ and equals $\mathbf{P}\mathbf{a} - (\mathbf{I} - \mathbf{M})\mathbf{P}\mathbf{a} \in \mathbb{C}(\mathbf{P}) + \mathbb{C}(\mathbf{X})^\perp = \mathcal{S}_0^\perp$. The remaining claims follow from Theorem 6.5.1 with the roles $\mathbb{C}(\mathbf{X}_0) = \mathcal{S}_0$. ■

The subspace $\mathbb{C}(\mathbf{M}\mathbf{P})$ is the *test space* of the hypothesis. Its dimension $\text{rank}(\mathbf{M}\mathbf{P})$ counts how many independent restrictions the constraint really places on the mean. It can be smaller than q if some rows of the constraint are redundant. Chapter 11 uses exactly this space for the numerator of the F statistic.

6.5.5 Exercises

6.5.5.1 A. Check your understanding

Exercise A1.

Construct the projections $\mathbf{P}_A$, $\mathbf{P}_B$ and $\mathbf{P}_0$ of Example (Additive two-factor model, one observation per cell) for $a = 3$, $b = 2$ and verify $\mathbf{P}_A\mathbf{P}_B = \mathbf{P}_0$ numerically. Delete one observation and recompute. Show that the centred factor spaces are no longer orthogonal, and find the largest entry of $(\mathbf{P}_A - \mathbf{P}_0)(\mathbf{P}_B - \mathbf{P}_0)$.

6.5.5.2 B. Practice

Exercise B1.

With $\mathbb{C}(\mathbf{X}_0) \subseteq \mathbb{C}(\mathbf{X})$, show that $\text{SSE}_0 - \text{SSE} = \|\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_0\|^2$, the squared distance between the two fitted vectors.

Solution. $\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_0 = (\mathbf{M} - \mathbf{M}_0)\mathbf{y}$, and $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \text{SSE}_0 - \text{SSE}$ by Eq. 6.5.2.

Exercise B2.

In the model $\mathbf{E}(\mathbf{Y}) = \beta_0\mathbf{1} + \beta_1\mathbf{x}_1 + \beta_2\mathbf{x}_2 + \beta_3\mathbf{x}_3$ with full-rank $\mathbf{X}$, the hypothesis $\beta_2 = \beta_3$ defines a reduced model with model matrix $\mathbf{X}_0 = [\mathbf{1}, \mathbf{x}_1, \mathbf{x}_2 + \mathbf{x}_3]$. Show that the test space $\mathbb{C}(\mathbf{X}_0)^\perp \cap \mathbb{C}(\mathbf{X})$ is spanned by $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2$, and that $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2 = -(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_3$.

Verify that this agrees with [Theorem 6.5.2](#) for a suitable $\mathbf{P}$.

Solution. The two model spaces differ in dimension by one, so the test space is a line. The vector $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2$ lies in $\mathbb{C}(\mathbf{X})$ and is orthogonal to $\mathbb{C}(\mathbf{X}_0)$. It is nonzero, because $\mathbf{x}_2 \in \mathbb{C}(\mathbf{X}_0)$ would make the columns of $\mathbf{X}$ dependent. Since $\mathbf{x}_2 + \mathbf{x}_3 \in \mathbb{C}(\mathbf{X}_0)$, $(\mathbf{I} - \mathbf{M}_0)(\mathbf{x}_2 + \mathbf{x}_3) = \mathbf{0}$. For [Theorem 6.5.2](#), write the constraint as $\boldsymbol{\lambda}\mathbf{T}\boldsymbol{\beta} = 0$ with $\boldsymbol{\lambda} = (0, 0, 1, -1)\mathbf{T}$, and take $\mathbf{P} = \mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\boldsymbol{\lambda}$, so that $\mathbf{P}\mathbf{T}\mathbf{X} = \boldsymbol{\lambda}\mathbf{T}$. Then $\mathbf{M}\mathbf{P} = \mathbf{P}$. Each column of $\mathbf{X}_0$ is $\mathbf{X}\mathbf{k}$ with $\boldsymbol{\lambda}\mathbf{T}\mathbf{k} = 0$, so $\mathbf{P} \perp \mathbb{C}(\mathbf{X}_0)$ and $\mathbf{P}$ spans the same line.

6.5.5.3 C. Going deeper

Exercise C1.

Extend [Example \(Additive two-factor model, one observation per cell\)](#) to $m \geq 2$ observations in every cell and a model with interaction. Find five mutually orthogonal projections summing to $\mathbf{I}$, corresponding to the grand mean, factor A , factor B , the interaction, and within-cell error, and find their ranks. Describe the action of each on $\mathbf{y}$ in terms of cell, row, column and grand means.

6.6 The Frisch–Waugh–Lovell theorem

A coefficient in a multiple regression is often described as the effect of its regressor “holding the other regressors fixed”. In observational data nobody held anything fixed, so the phrase needs a precise meaning. The Frisch–Waugh–Lovell (FWL) theorem supplies one. A coefficient in a multiple regression equals the slope of a simple regression in which the other regressors have been *projected out* of both the response and the regressor of interest.

6.6.1 Statement and proof

Partition the model matrix and coefficients as

$$\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2], \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix},$$

with $\mathbf{X}_1$ of size $n \times p_1$ and $\mathbf{X}_2$ of size $n \times p_2$, and assume $\mathbf{X}$ has full column rank. Let $\mathbf{M}_2$ project onto $\mathbb{C}(\mathbf{X}_2)$, and write

$$\tilde{\mathbf{X}}_1 = (\mathbf{I} - \mathbf{M}_2)\mathbf{X}_1, \quad \tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_2)\mathbf{y}$$

for the parts of $\mathbf{X}_1$ and $\mathbf{y}$ that $\mathbf{X}_2$ cannot explain linearly. Each column of $\tilde{\mathbf{X}}_1$ is the residual vector from regressing the corresponding column of $\mathbf{X}_1$ on $\mathbf{X}_2$.

Lemma 6.6.1 — Orthogonalizing a partitioned model space

Under the assumptions above:

- (a) $\mathbb{C}(\mathbf{X}) = \mathbb{C}(\mathbf{X}_2) \oplus \mathbb{C}(\tilde{\mathbf{X}}_1)$, and the two summands are orthogonal;
- (b) $\tilde{\mathbf{X}}_1$ has full column rank;

$$(c) \mathbf{M} = \mathbf{M}_2 + \tilde{\mathbf{M}}_1, \text{ where } \tilde{\mathbf{M}}_1 = \tilde{\mathbf{X}}_1(\tilde{\mathbf{X}}_1\mathbf{T}\tilde{\mathbf{X}}_1)^{-1}\tilde{\mathbf{X}}_1\mathbf{T}.$$

Proof.

- (a) The columns of $\tilde{\mathbf{X}}_1 = \mathbf{X}_1 - \mathbf{M}_2\mathbf{X}_1$ lie in $\mathbb{C}(\mathbf{X})$ and are orthogonal to $\mathbb{C}(\mathbf{X}_2)$. Conversely, for any $\mathbf{b} = (\mathbf{b}_1\mathbf{T}, \mathbf{b}_2\mathbf{T})\mathbf{T}$,

$$\mathbf{X}\mathbf{b} = \mathbf{X}_1\mathbf{b}_1 + \mathbf{X}_2\mathbf{b}_2 = \tilde{\mathbf{X}}_1\mathbf{b}_1 + (\mathbf{M}_2\mathbf{X}_1\mathbf{b}_1 + \mathbf{X}_2\mathbf{b}_2),$$

with the second term in $\mathbb{C}(\mathbf{X}_2)$. So $\mathbb{C}(\mathbf{X}) \subseteq \mathbb{C}(\tilde{\mathbf{X}}_1) + \mathbb{C}(\mathbf{X}_2)$.

- (b) If $\tilde{\mathbf{X}}_1\mathbf{a} = \mathbf{0}$, then $\mathbf{X}_1\mathbf{a} = \mathbf{M}_2\mathbf{X}_1\mathbf{a} = \mathbf{X}_2\mathbf{c}$ for some $\mathbf{c}$, so $\mathbf{X}(\mathbf{a}\mathbf{T}, -\mathbf{c}\mathbf{T})\mathbf{T} = \mathbf{0}$. Full column rank of $\mathbf{X}$ forces $\mathbf{a} = \mathbf{0}$.
- (c) Combine (a) with [Theorem 6.3.8](#) and [Theorem 6.3.7](#). ■

Theorem 6.6.2 — Frisch–Waugh–Lovell

Let $\hat{\beta} = (\hat{\beta}_1\mathbf{T}, \hat{\beta}_2\mathbf{T})\mathbf{T}$ be the least squares estimate in the full model. Then:

- (a) $\hat{\beta}_1 = (\tilde{\mathbf{X}}_1\mathbf{T}\tilde{\mathbf{X}}_1)^{-1}\tilde{\mathbf{X}}_1\mathbf{T}\tilde{\mathbf{y}} = (\tilde{\mathbf{X}}_1\mathbf{T}\tilde{\mathbf{X}}_1)^{-1}\tilde{\mathbf{X}}_1\mathbf{T}\mathbf{y}$. So $\hat{\beta}_1$ is the coefficient from regressing $\tilde{\mathbf{y}}$ (or $\mathbf{y}$ itself) on $\tilde{\mathbf{X}}_1$;
- (b) the residual vector from regressing $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$ equals the residual vector $(\mathbf{I} - \mathbf{M})\mathbf{y}$ of the full model;
- (c) $\hat{\beta}_2 = (\mathbf{X}_2\mathbf{T}\mathbf{X}_2)^{-1}\mathbf{X}_2\mathbf{T}(\mathbf{y} - \mathbf{X}_1\hat{\beta}_1)$.

Proof. Let $\tilde{\mathbf{b}} = (\tilde{\mathbf{X}}_1\mathbf{T}\tilde{\mathbf{X}}_1)^{-1}\tilde{\mathbf{X}}_1\mathbf{T}\mathbf{y}$. Since $\mathbf{I} - \mathbf{M}_2$ is symmetric and idempotent, $\tilde{\mathbf{X}}_1\mathbf{T}\mathbf{y} = \mathbf{X}_1\mathbf{T}(\mathbf{I} - \mathbf{M}_2)\mathbf{y} = \tilde{\mathbf{X}}_1\mathbf{T}\tilde{\mathbf{y}}$, so the two expressions in (a) agree. By [Lemma 6.6.1\(c\)](#),

$$\mathbf{M}\mathbf{y} = \mathbf{M}_2\mathbf{y} + \tilde{\mathbf{X}}_1\tilde{\mathbf{b}} = \mathbf{M}_2\mathbf{y} + \mathbf{X}_1\tilde{\mathbf{b}} - \mathbf{M}_2\mathbf{X}_1\tilde{\mathbf{b}} = \mathbf{X}_1\tilde{\mathbf{b}} + \mathbf{M}_2(\mathbf{y} - \mathbf{X}_1\tilde{\mathbf{b}}).$$

The last term lies in $\mathbb{C}(\mathbf{X}_2)$, so it equals $\mathbf{X}_2\mathbf{c}$ with $\mathbf{c} = (\mathbf{X}_2\mathbf{T}\mathbf{X}_2)^{-1}\mathbf{X}_2\mathbf{T}(\mathbf{y} - \mathbf{X}_1\tilde{\mathbf{b}})$. Thus $\mathbf{M}\mathbf{y} = \mathbf{X}_1\tilde{\mathbf{b}} + \mathbf{X}_2\mathbf{c}$. Because $\mathbf{X}$ has full column rank, the coefficient vector representing $\mathbf{M}\mathbf{y}$ is unique, so $\hat{\beta}_1 = \tilde{\mathbf{b}}$ and $\hat{\beta}_2 = \mathbf{c}$. This proves (a) and (c). For (b), $\mathbf{y} - \mathbf{M}\mathbf{y} = (\mathbf{I} - \mathbf{M}_2)\mathbf{y} - \tilde{\mathbf{M}}_1\mathbf{y} = \tilde{\mathbf{y}} - \tilde{\mathbf{X}}_1\tilde{\mathbf{b}}$, which is the residual of the regression of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$, because $\tilde{\mathbf{X}}_1\tilde{\mathbf{b}} = \tilde{\mathbf{M}}_1\mathbf{y} = \tilde{\mathbf{M}}_1\tilde{\mathbf{y}}$. ■

The proof uses nothing beyond the orthogonal decomposition of the model space. Part (a) contains a small surprise. Residualizing $\mathbf{y}$ is optional, because the part of $\mathbf{y}$ lying in $\mathbb{C}(\mathbf{X}_2)$ is orthogonal to $\tilde{\mathbf{X}}_1$ and drops out of $\tilde{\mathbf{X}}_1\mathbf{T}\mathbf{y}$ anyway. Residualizing the regressor of interest is essential.

Idea. In a multiple regression, $\hat{\beta}_j$ measures how $\mathbf{y}$ moves with *the part of $\mathbf{x}_j$ that is orthogonal to the other regressors*. A regressor that is almost a linear combination of the others has a tiny orthogonal part, so its coefficient rests on very little information. That is collinearity, seen geometrically.

6.6.2 Consequences

Variances. The FWL formula writes $\hat{\beta}_1$ as a linear function of $\mathbf{y}$. If $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, then

$$\text{Cov}(\hat{\beta}_1) = \sigma^2 (\tilde{\mathbf{X}}_1 \mathbf{T} \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1 \mathbf{T} \tilde{\mathbf{X}}_1 (\tilde{\mathbf{X}}_1 \mathbf{T} \tilde{\mathbf{X}}_1)^{-1} = \sigma^2 (\tilde{\mathbf{X}}_1 \mathbf{T} \tilde{\mathbf{X}}_1)^{-1}.$$

Comparing with $\text{Cov}(\hat{\beta}) = \sigma^2 (\mathbf{X} \mathbf{T} \mathbf{X})^{-1}$ (Chapter 5) identifies the leading block of the inverse of a partitioned matrix,

$$[(\mathbf{X} \mathbf{T} \mathbf{X})^{-1}]_{11} = (\mathbf{X}_1 \mathbf{T} (\mathbf{I} - \mathbf{M}_2) \mathbf{X}_1)^{-1} = (\mathbf{X}_1 \mathbf{T} \mathbf{X}_1 - \mathbf{X}_1 \mathbf{T} \mathbf{X}_2 (\mathbf{X}_2 \mathbf{T} \mathbf{X}_2)^{-1} \mathbf{X}_2 \mathbf{T} \mathbf{X}_1)^{-1}, \quad (6.6.1)$$

a Schur-complement identity that [Chapter 1](#) proves by algebra. For a single regressor $\mathbf{x}_j$, with all other columns (including the intercept) in $\mathbf{X}_2$,

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\|(\mathbf{I} - \mathbf{M}_{(j)}) \mathbf{x}_j\|^2} = \frac{\sigma^2}{S_{jj} (1 - R_j^2)}, \quad (6.6.2)$$

where $S_{jj} = \sum_i (x_{ij} - \bar{x}_j)^2$ and R_j^2 is the coefficient of determination from regressing $\mathbf{x}_j$ on the other regressors. The second equality is [Exercise \(B1\)](#). The factor $1/(1 - R_j^2)$ is the *variance inflation factor* of Chapter 26.

Centring. Take $\mathbf{X}_2 = \mathbf{1}$. Then $\mathbf{M}_2 = n^{-1} \mathbf{1} \mathbf{1} \mathbf{T}$ and $\tilde{\mathbf{X}}_1, \tilde{\mathbf{y}}$ are the centred regressors and response. FWL says the slopes in a model with an intercept are the coefficients of a no-intercept regression of centred $\mathbf{y}$ on centred regressors, and part (c) gives the intercept $\bar{y} - \bar{\mathbf{x}} \mathbf{T} \hat{\beta}_1$. [Example \(Intercept inside simple regression\)](#) was the case $p_1 = 1$.

Detrending and deseasonalizing. The original application was economic time series. Frisch and Waugh (1933) showed that regressing on a time trend together with other variables gives the same slope as first detrending every series. Lovell (1963) proved the analogous statement for seasonal dummy variables, and noted a trap: if only the *response* is seasonally adjusted and the regressors are not, the result is generally wrong. Part (a) of the theorem shows why. What must be residualized is the regressor.

A warning about standard errors. If the residualized regression of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$ is fitted with ordinary software, the coefficient and residuals are right, but the software divides the residual sum of squares by $n - p_1$ instead of $n - p_1 - p_2$. It does not know that p_2 dimensions were used up in forming $\tilde{\mathbf{y}}$. The reported standard error is too small by the factor $\sqrt{(n - p_1)/(n - p)}$.

Example. (Poverty and murder rates)

For the 50 US states in 2009, regress the murder rate (per 100,000) on the poverty rate, the percentage of single-parent households, and the percentage of the population living in urban areas, with an intercept. The coefficient of poverty is 0.2538. On its own, a simple regression of murder rate on poverty has slope 0.4706, nearly twice as large. Poverty and single parenthood are correlated, and the simple regression gives poverty credit for part of their shared association.

[Figure 6.6.1](#) shows both regressions. Panel (b) is an **added-variable plot**: the residual of murder rate after regression on the other two regressors and the intercept, against the corresponding residual of poverty. By FWL its least squares slope is exactly the multiple-regression coefficient. The listing confirms the equality to machine precision.

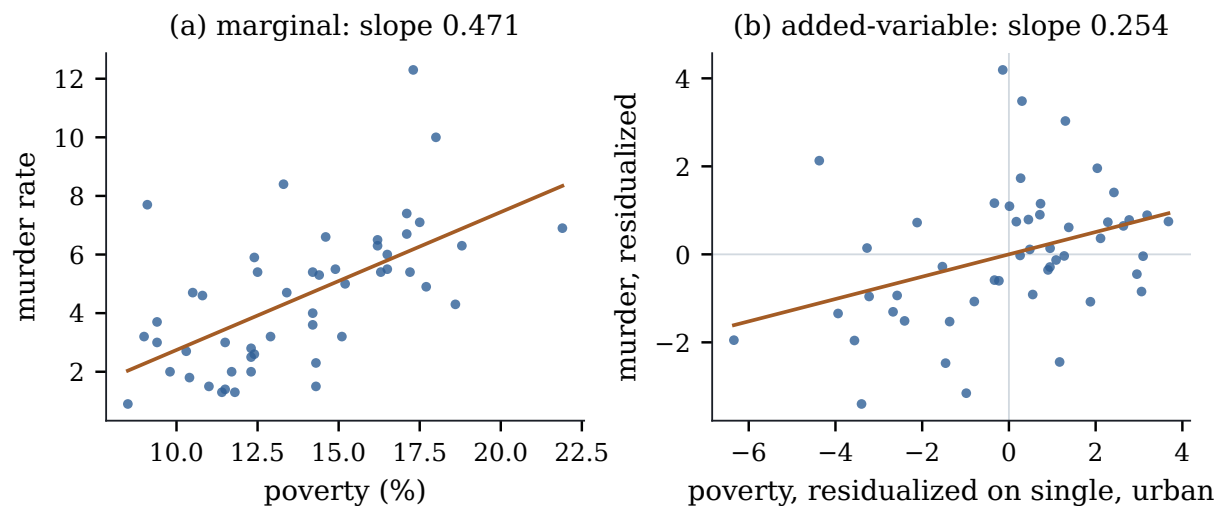


Figure 6.6.1: Murder rate and poverty for 50 US states. (a) The marginal relationship. (b) The added-variable plot: both variables residualized on the single-parent percentage, urban percentage and intercept. The slope in (b) is the multiple-regression coefficient of poverty (Theorem 6.6.2).

The standard error of the coefficient in the full model is 0.0934. Fitting the residualized regression naively gives 0.0905, smaller by the factor 1.0321, which is $\sqrt{(n-1)/(n-4)}$ as predicted.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data
data = data.drop(index="District of Columbia")  # an extreme point; see the
                                                # exercises of Section 6.6

y = data["murder"].to_numpy()
x1 = data["poverty"].to_numpy()  # the coefficient we care
                                # about
X2 = np.column_stack([np.ones(len(y)), data["single"], data["urban"]])
X = np.column_stack([x1, X2])

def resid(v, Z):
    """Residual of v after projecting onto C(Z)."""
    coef, *_ = np.linalg.lstsq(Z, v, rcond=None)
    return v - Z @ coef

beta_full, *_ = np.linalg.lstsq(X, y, rcond=None)  # regress y on everything

y_tilde = resid(y, X2)  # (I - M2) y
x_tilde = resid(x1, X2)  # (I - M2) x1
beta_fwl = (x_tilde @ y_tilde) / (x_tilde @ x_tilde)  # simple regression, no
                                                        # intercept

print(f"full regression: {beta_full[0]:.6f}")
print(f"FWL regression: {beta_fwl:.6f}")
```

Remark (What FWL does not say). FWL is an algebraic identity about fitted coefficients. It says what the number $\hat{\beta}_j$ is: an association with the unexplained part of $\mathbf{x}_j$. It does not say that changing x_j in the world, with the other regressors held fixed, would

change the response by $\hat{\beta}_j$. That is a causal claim, and it needs assumptions about how the data arose. Chapter 25 takes up this question, including cases where adding a regressor to $\mathbf{X}_2$ makes the coefficient *less* interpretable rather than more.

6.6.3 Fixed effects are a projection

One of the most widely used estimators in applied statistics is the FWL theorem in disguise. Suppose units $k = 1, \dots, K$ are observed repeatedly: firms over years, patients over visits, schools over cohorts. The analyst wants the effect of time-varying regressors $\mathbf{X}_1$ while allowing each unit its own intercept:

$$\mathbf{E}(\mathbf{Y}) = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{D}\boldsymbol{\alpha},$$

where $\mathbf{D}$ is the $n \times K$ matrix of unit indicators. Fitting this directly means a regression with K extra columns, and K may be in the thousands.

FWL with $\mathbf{X}_2 = \mathbf{D}$ removes that burden. The projection onto $\mathbb{C}(\mathbf{D})$ replaces each entry by its unit's average (Example (The one-way layout)), so $\mathbf{I} - \mathbf{M}_D$ subtracts unit means. Therefore $\hat{\beta}_1$ is the least squares coefficient from regressing the *within-unit deviations* $y_{kt} - \bar{y}_k$ on $\mathbf{x}_{kt} - \bar{\mathbf{x}}_k$, with no indicators at all. This is the **within estimator** of panel-data econometrics. Its standard error needs the degrees-of-freedom correction described above, with $p_2 = K$, which is large.

Example. (Grunfeld's investment data)

Grunfeld's classic panel follows 11 US firms for 20 years, giving $n = 220$ observations of gross investment, market value and capital stock. With a separate intercept for each firm, the coefficients of value and capital are 0.1101 and 0.3100. The same numbers come from demeaning every variable within firm and fitting two columns. Ignoring the firms altogether (one common intercept) gives 0.1145 and 0.2275 instead. That comparison uses variation *between* firms, of the sort "large firms invest more", which the within estimator deliberately discards.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
df = sm.datasets.grunfeld.load_pandas().data
y = df["invest"].to_numpy()
X1 = df[["value", "capital"]].to_numpy()

# (i) least squares with one indicator column per firm ("dummy variables")
D = pd.get_dummies(df["firm"]).to_numpy(dtype=float)
beta_dummies, *_ = np.linalg.lstsq(np.column_stack([X1, D]), y, rcond=None)

# (ii) FWL: projecting onto C(D) replaces each value by its firm mean,
#         so (I - M_D) subtracts firm means -- the "within" transformation
def demean_by(v, groups):
    v = pd.DataFrame(v)
    return (v - v.groupby(groups.to_numpy()).transform("mean")).to_numpy()

X1_within = demean_by(X1, df["firm"])
y_within = demean_by(y, df["firm"]).ravel()
beta_within, *_ = np.linalg.lstsq(X1_within, y_within, rcond=None)

print("with firm indicators:", beta_dummies[:2])
```

```
print("within transformation:", beta_within)
```

Whether discarding between-unit variation is wise depends on why units differ. That question is taken up for random-effects alternatives in Chapter 32 and Chapter 33.

Added-variable plots are among the most useful diagnostics for multiple regression. They show the evidence for a single coefficient in two dimensions, together with any points that dominate it. They reappear in Chapter 20.

6.6.4 Exercises

6.6.4.1 A. Check your understanding

Exercise A1.

Refit the model of [Example \(Poverty and murder rates\)](#) with the District of Columbia included. How does the poverty coefficient change? Draw the added-variable plot and locate the District. Relate what you see to its leverage ([Example \(High-leverage states\)](#)).

6.6.4.2 B. Practice

Exercise B1.

Let $\mathbf{x}_j$ be a column of $\mathbf{X}$, let $\mathbf{X}_{(j)}$ be $\mathbf{X}$ with that column removed, and assume $\mathbf{1} \in \mathbb{C}(\mathbf{X}_{(j)})$. Show that $\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2 = S_{jj}(1 - R_j^2)$, where R_j^2 is the R^2 from regressing $\mathbf{x}_j$ on $\mathbf{X}_{(j)}$. Deduce [Eq. 6.6.2](#).

Solution. Since $\mathbf{1} \in \mathbb{C}(\mathbf{X}_{(j)})$, the vector $(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$ is the residual from regressing $\mathbf{x}_j$ on $\mathbf{X}_{(j)}$. Its squared length is that regression's residual sum of squares, which equals $\text{SST}(1 - R_j^2)$ with $\text{SST} = S_{jj}$ by the definition of R^2 . Substituting into the variance formula from FWL gives [Eq. 6.6.2](#).

Exercise B2.

In [Theorem 6.6.2](#), regressing $\mathbf{y}$ (not $\tilde{\mathbf{y}}$) on $\tilde{\mathbf{X}}_1$ gives the coefficient $\hat{\beta}_1$. Show that its residual vector is $(\mathbf{I} - \mathbf{M})\mathbf{y} + \mathbf{M}_2\mathbf{y}$, so the residuals are generally wrong even though the coefficient is right.

Solution. The fitted vector of the regression of $\mathbf{y}$ on $\tilde{\mathbf{X}}_1$ is $\tilde{\mathbf{M}}_1\mathbf{y}$. By [Lemma 6.6.1](#), $\mathbf{y} - \tilde{\mathbf{M}}_1\mathbf{y} = \mathbf{y} - (\mathbf{M} - \mathbf{M}_2)\mathbf{y} = (\mathbf{I} - \mathbf{M})\mathbf{y} + \mathbf{M}_2\mathbf{y}$. The extra term $\mathbf{M}_2\mathbf{y}$ is zero only if $\mathbf{y} \perp \mathbb{C}(\mathbf{X}_2)$.

Exercise B3.

For a single regressor $\mathbf{x}_j$, the *partial correlation* of $\mathbf{y}$ and $\mathbf{x}_j$ given the other regressors is the sample correlation of $\tilde{\mathbf{y}}$ and $\tilde{\mathbf{x}}_j$ (both residualized on $\mathbf{X}_{(j)}$, which contains $\mathbf{1}$). Let $t = \hat{\beta}_j/\text{se}(\hat{\beta}_j)$ with $\text{se}(\hat{\beta}_j)^2 = \text{SSE}/[(n - p)\|\tilde{\mathbf{x}}_j\|^2]$. Show that the squared partial

correlation equals $t^2/(t^2 + n - p)$.

Solution. Write $\tilde{\mathbf{x}} = \tilde{\mathbf{x}}_j$ and $\tilde{\mathbf{y}} = \tilde{\mathbf{y}}$. Both have mean zero because $\mathbf{1} \in \mathbb{C}(\mathbf{X}_{(j)})$, so their correlation is $r = \tilde{\mathbf{x}}\mathbf{T}\tilde{\mathbf{y}}/(\|\tilde{\mathbf{x}}\|\|\tilde{\mathbf{y}}\|)$. By FWL, $\hat{\beta}_j = \tilde{\mathbf{x}}\mathbf{T}\tilde{\mathbf{y}}/\|\tilde{\mathbf{x}}\|^2$, and the full-model residual sum of squares equals that of the residualized regression: $\text{SSE} = \|\tilde{\mathbf{y}}\|^2 - (\tilde{\mathbf{x}}\mathbf{T}\tilde{\mathbf{y}})^2/\|\tilde{\mathbf{x}}\|^2 = \|\tilde{\mathbf{y}}\|^2(1 - r^2)$. Hence

$$t^2 = \frac{\hat{\beta}_j^2 (n - p) \|\tilde{\mathbf{x}}\|^2}{\text{SSE}} = \frac{(n - p) r^2}{1 - r^2},$$

and solving gives $r^2 = t^2/(t^2 + n - p)$.

6.6.4.3 C. Going deeper

Exercise C1.

Drop the full-rank assumption in [Theorem 6.6.2](#). Show that $\mathbf{b}_1$ solves the normal equations of the regression of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$ iff there exists $\mathbf{b}_2$ such that $(\mathbf{b}_1\mathbf{T}, \mathbf{b}_2\mathbf{T})\mathbf{T}$ is a least squares estimate in the full model.

6.7 Reparameterization, centring and R^2

6.7.1 Same subspace, same fit

The key idea of [Section 6.1](#) now becomes a theorem.

Theorem 6.7.1 — Invariance under reparameterization

Let $\mathbf{X}$ ($n \times p$) and $\mathbf{W}$ ($n \times q$) satisfy $\mathbb{C}(\mathbf{X}) = \mathbb{C}(\mathbf{W})$. Then the two linear models $\mathbf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\mathbf{E}(\mathbf{Y}) = \mathbf{W}\boldsymbol{\gamma}$ have the same projection $\mathbf{M}$, the same fitted values and residuals for every $\mathbf{y}$, and the same residual sum of squares. There are a $p \times q$ matrix $\mathbf{T}$ with $\mathbf{W} = \mathbf{X}\mathbf{T}$ and a $q \times p$ matrix $\mathbf{S}$ with $\mathbf{X} = \mathbf{W}\mathbf{S}$. If $\hat{\boldsymbol{\gamma}}$ is a least squares estimate for $\mathbf{W}$, then $\mathbf{T}\hat{\boldsymbol{\gamma}}$ is one for $\mathbf{X}$, and $\mathbf{S}\hat{\boldsymbol{\beta}}$ is one for $\mathbf{W}$ whenever $\hat{\boldsymbol{\beta}}$ is one for $\mathbf{X}$. If both matrices have full column rank, then $p = q$, $\mathbf{T} = \mathbf{S}^{-1}$, and $\hat{\boldsymbol{\gamma}} = \mathbf{S}\hat{\boldsymbol{\beta}}$.

Proof. A projection is determined by its range ([Theorem 6.3.3](#)), so both models have projection $\mathbf{M}$, and fitted values and residuals follow from [Theorem 6.4.1](#). Each column of $\mathbf{W}$ lies in $\mathbb{C}(\mathbf{X})$, so $\mathbf{W} = \mathbf{X}\mathbf{T}$ for some $\mathbf{T}$, and likewise $\mathbf{X} = \mathbf{W}\mathbf{S}$. If $\mathbf{W}\hat{\boldsymbol{\gamma}} = \mathbf{M}\mathbf{y}$, then $\mathbf{X}(\mathbf{T}\hat{\boldsymbol{\gamma}}) = \mathbf{M}\mathbf{y}$, so $\mathbf{T}\hat{\boldsymbol{\gamma}}$ is a least squares estimate for $\mathbf{X}$. The other direction is symmetric. With full column rank, $\mathbf{X} = \mathbf{X}\mathbf{T}\mathbf{S}$ gives $\mathbf{X}(\mathbf{I} - \mathbf{T}\mathbf{S}) = \mathbf{0}$ and so $\mathbf{T}\mathbf{S} = \mathbf{I}$, and similarly $\mathbf{S}\mathbf{T} = \mathbf{I}$. Uniqueness of least squares estimates then gives $\hat{\boldsymbol{\gamma}} = \mathbf{S}\hat{\boldsymbol{\beta}}$. ■

Many operations that change the printed coefficient table leave the fit exactly unchanged:

- **Changing units** of a regressor (metres to centimetres, dollars to thousands of dollars)

rescales one column: $\mathbf{W} = \mathbf{X}\mathbf{D}$ with $\mathbf{D}$ diagonal. The coefficient is divided by the scale factor, and nothing else changes.

- **Centring** regressors, $\mathbf{x}_j \mapsto \mathbf{x}_j - \bar{x}_j \mathbf{1}$, when the model has an intercept. The slopes are unchanged, and the intercept becomes the fitted value at the mean of the regressors instead of at zero, which is often far outside the data.
- **Recoding a factor**. Reference-level dummies, sum-to-zero effect coding, and cell-means coding without an intercept all span the space of vectors that are constant within levels.
- **Replacing polynomial terms by orthogonal polynomials** (Chapter 42), which span the same space with much better numerical behaviour (Section 6.10).

Operations that *do* change the fit are exactly those that change $\mathbb{C}(\mathbf{X})$. Examples are dropping the intercept when no other combination of columns produces $\mathbf{1}$, replacing a regressor by its logarithm, and adding an interaction.

6.7.2 Centring as a projection

Suppose $\mathbf{1} \in \mathbb{C}(\mathbf{X})$. By Theorem 6.5.1 with $\mathbf{X}_0 = \mathbf{1}$, the fitted vector splits as

$$\mathbf{M}\mathbf{y} = \bar{y}\mathbf{1} + (\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^T)\mathbf{y},$$

and the second projection is onto $\mathbb{C}(\mathbf{X}) \cap \mathbf{1}^\perp$, the vectors in the model space with entries summing to zero. By Lemma 6.6.1 with $\mathbf{X}_2 = \mathbf{1}$, that subspace is spanned by the centred columns $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^T)\mathbf{X}$. Applying Pythagoras to the orthogonal pieces $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$ and $\hat{\mathbf{e}}$ of $\mathbf{y} - \bar{y}\mathbf{1}$ gives the *corrected* decomposition

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{SST}} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\text{SSR}} + \underbrace{\sum_i (y_i - \hat{y}_i)^2}_{\text{SSE}}, \quad (6.7.1)$$

with dimensions $n - 1 = (r - 1) + (n - r)$. Chapter 9 builds the analysis of variance on this identity.

6.7.3 R^2 is a squared cosine

The **coefficient of determination** of a model with $\mathbf{1} \in \mathbb{C}(\mathbf{X})$ is

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}.$$

It depends on the model only through $\mathbb{C}(\mathbf{X})$, so by Theorem 6.7.1 it is invariant under every reparameterization. Its geometric meaning is simple.

Theorem 6.7.2

Assume $\mathbf{1} \in \mathbb{C}(\mathbf{X})$ and $\mathbf{y} \notin \text{span}(\mathbf{1})$. Let θ be the angle between the centred response $\mathbf{y} - \bar{y}\mathbf{1}$ and the centred fitted vector $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$. Then

$$R^2 = \cos^2 \theta = \text{corr}(\mathbf{y}, \hat{\mathbf{Y}})^2,$$

where corr is the sample correlation. For simple regression on one regressor $\mathbf{x}$, $R^2 =$

inside $\mathbf{1}^\perp$: $R^2 = \cos^2 \theta = 0.617$

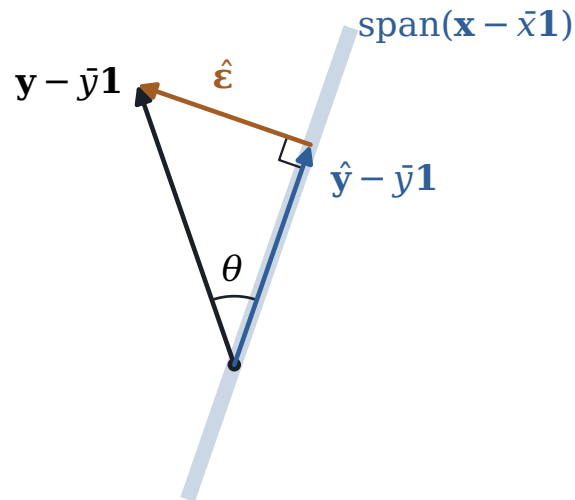


Figure 6.7.1: R^2 as a squared cosine, for $n = 3$ and one regressor. Centred vectors lie in the plane $\mathbf{1}^\perp$, so the whole configuration is two-dimensional. The centred fitted vector is the projection of the centred response onto the line spanned by the centred regressor.

$\text{corr}(\mathbf{y}, \mathbf{x})^2$.

Proof. Write $\mathbf{u} = \mathbf{y} - \bar{y}\mathbf{1}$ and $\mathbf{v} = \hat{\mathbf{Y}} - \bar{y}\mathbf{1}$. Then $\mathbf{u} = \mathbf{v} + \hat{\boldsymbol{\varepsilon}}$ with $\mathbf{v} \perp \hat{\boldsymbol{\varepsilon}}$, so $\mathbf{u}^T \mathbf{v} = \|\mathbf{v}\|^2$ and

$$\cos^2 \theta = \frac{(\mathbf{u}^T \mathbf{v})^2}{\|\mathbf{u}\|^2 \|\mathbf{v}\|^2} = \frac{\|\mathbf{v}\|^2}{\|\mathbf{u}\|^2} = \frac{\text{SSR}}{\text{SST}}.$$

(If $\mathbf{v} = \mathbf{0}$ the angle is undefined, but $R^2 = 0$ and the correlation is taken to be zero.) Since $\hat{\mathbf{Y}}$ has mean $\bar{y}$ (Proposition 6.4.3(b)), $\cos \theta$ is the sample correlation of $\mathbf{y}$ and $\hat{\mathbf{Y}}$. In simple regression $\mathbf{v} = (S_{xy}/S_{xx})(\mathbf{x} - \bar{x}\mathbf{1})$ by Example (Intercept inside simple regression), a scalar multiple of the centred regressor. Rescaling a vector by a nonzero constant does not change a squared cosine, so $\cos^2 \theta$ is the squared correlation of $\mathbf{y}$ and $\mathbf{x}$. When $S_{xy} = 0$ both sides are zero. ■

So R^2 measures how closely the model space lines up with the centred data. Figure 6.7.1 draws the case $n = 3$. For the three-regressor model of Example (Poverty and murder rates), $R^2 = 0.6419$, meaning the centred response makes an angle of 36.8° with the model space. The simple regression on poverty alone has $R^2 = 0.3599$, the square of the correlation 0.5999.

```
import numpy as np
import statsmodels.api as sm

def r_squared_as_cosine(y, X):
    """X must contain the intercept column. Returns (R^2 from sums of squares, cos^2
    of angle)."""
    beta, *_ = np.linalg.lstsq(X, y, rcond=None)
    y_hat = X @ beta
    yc, yhc = y - y.mean(), y_hat - y.mean()           # centre both vectors
```

```

r2 = 1 - np.sum((y - y_hat) ** 2) / np.sum(yc ** 2)
cos = (yc @ yhc) / (np.linalg.norm(yc) * np.linalg.norm(yhc))
return r2, cos ** 2

```

Three consequences follow directly from the geometry.

1. R^2 never decreases when the model space grows. If $\mathbb{C}(\mathbf{X}_0) \subseteq \mathbb{C}(\mathbf{X})$ and both contain $\mathbf{1}$, then $\text{SSE}_0 - \text{SSE} = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 \geq 0$ by Eq. 6.5.2. Adding a regressor of pure noise increases R^2 almost surely. This is why R^2 cannot be used for model selection (Chapter 29).
2. $R^2 = 1$ once the model space contains $\mathbf{y}$, which always happens when $\text{rank}(\mathbf{X}) = n$. A model with as many free parameters as observations interpolates, and its R^2 says nothing.
3. Without an intercept, the definition changes. If $\mathbf{1} \notin \mathbb{C}(\mathbf{X})$, the corrected decomposition Eq. 6.7.1 fails, because $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$ and $\hat{\mathbf{e}}$ need not be orthogonal. Software then usually reports the *uncentred* $R_0^2 = \|\hat{\mathbf{Y}}\|^2 / \|\mathbf{y}\|^2$, the squared cosine of the angle between $\mathbf{y}$ and $\mathbb{C}(\mathbf{X})$ itself. It is typically much larger and cannot be compared with the centred R^2 of a model with an intercept (Exercise (B1)).

6.7.4 Exercises

6.7.4.1 A. Check your understanding

Exercise A1.

A factor with three levels is coded with an intercept and indicators for levels 2 and 3. Write down the matrices $\mathbf{T}$ and $\mathbf{S}$ of Theorem 6.7.1 that convert to the coding in which level 3, rather than level 1, is the omitted baseline, and to sum-to-zero effect coding. Express each new coefficient in terms of the old ones.

Exercise A2.

Show that, for a model with $\mathbf{1} \in \mathbb{C}(\mathbf{X})$, R^2 is unchanged when $\mathbf{y}$ is replaced by $a\mathbf{1} + b\mathbf{y}$ with $b \neq 0$, and when any column of $\mathbf{X}$ is replaced by an affine function of itself. Is it unchanged when a regressor is replaced by its logarithm?

6.7.4.2 B. Practice

Exercise B1.

Show that if $\mathbf{1} \in \mathbb{C}(\mathbf{X})$ then $R_0^2 = \|\hat{\mathbf{Y}}\|^2 / \|\mathbf{y}\|^2 = (n\bar{y}^2 + \text{SSR}) / (n\bar{y}^2 + \text{SST}) \geq R^2$. Construct a data set and a model without an intercept for which $R_0^2 > 0.95$ but the model's residual sum of squares exceeds that of the intercept-only model.

Solution. With $\mathbf{1} \in \mathbb{C}(\mathbf{X})$, the fitted vector splits orthogonally as $\bar{y}\mathbf{1} + (\hat{\mathbf{Y}} - \bar{y}\mathbf{1})$, so $\|\hat{\mathbf{Y}}\|^2 = n\bar{y}^2 + \text{SSR}$, and likewise $\|\mathbf{y}\|^2 = n\bar{y}^2 + \text{SST}$. For $a \geq 0$ and $0 \leq \text{SSR} \leq \text{SST}$, $(a + \text{SSR}) / (a + \text{SST}) \geq \text{SSR} / \text{SST}$. For the example, take responses near 100 with small

noise and a regressor unrelated to them, and fit through the origin. The fitted line captures the large mean, so R_0^2 is near 1, but its residuals exceed those around $\bar{y}$.

6.7.4.3 C. Going deeper

Exercise C1.

Suppose $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$ and $\mathbf{1} \in \mathbb{C}(\mathbf{X})$ with $\text{rank}(\mathbf{X}) = r$. Using the results of Chapter 4, show that $R^2 \sim \text{Beta}((r-1)/2, (n-r)/2)$, so that $\mathbf{E}(R^2) = (r-1)/(n-1)$ even though no regressor is related to the response. Use this to motivate the adjusted $R^2 = 1 - (1 - R^2)(n-1)/(n-r)$.

6.8 Leverage as geometry

The projection $\mathbf{M}$ is an $n \times n$ matrix whose entries have a direct interpretation. Since $\hat{y}_i = \sum_j m_{ij}y_j$, the entry m_{ij} is the weight observation j receives in the fitted value for observation i . So

$$m_{ij} = \frac{\partial \hat{y}_i}{\partial y_j}.$$

The diagonal entries measure how strongly each observation pulls the fitted surface towards itself.

Definition 6.8.1 — Leverage

The **leverage** of observation i is $h_{ii} = m_{ii}$, the i th diagonal entry of $\mathbf{M}$. In this context $\mathbf{M}$ is often called the **hat matrix** and written $\mathbf{H}$, because it “puts the hat on” $\mathbf{y}$.

Leverage is a property of the design, not of the response. It depends on $\mathbf{X}$ not at all, and on $\mathbf{X}$ only through $\mathbb{C}(\mathbf{X})$, so it is unchanged by reparameterization.

Proposition 6.8.1 — Properties of leverage

Let $\text{rank}(\mathbf{X}) = r$, and let $\mathbf{e}_i$ denote the i th standard basis vector. Then:

- (a) $h_{ii} = \|\mathbf{M}\mathbf{e}_i\|^2 = \sum_j m_{ij}^2$, the squared length of the projection of $\mathbf{e}_i$ onto $\mathbb{C}(\mathbf{X})$;
- (b) $0 \leq h_{ii} \leq 1$, and $\sum_i h_{ii} = r$, so the average leverage is r/n ;
- (c) if $\mathbf{1} \in \mathbb{C}(\mathbf{X})$, then $h_{ii} \geq 1/n$;
- (d) $h_{ii} = 1$ iff $\mathbf{e}_i \in \mathbb{C}(\mathbf{X})$. In that case $\hat{y}_i = y_i$ for every $\mathbf{y}$, the i th residual is always zero, and $m_{ij} = 0$ for $j \neq i$;
- (e) if $\mathbf{X} = \mathbb{Q}\mathbb{R}$ with $\mathbb{Q}$ having orthonormal columns spanning $\mathbb{C}(\mathbf{X})$, then h_{ii} is the squared length of the i th row of $\mathbb{Q}$.

Proof.

- (a) $h_{ii} = \mathbf{e}_i^T \mathbf{M} \mathbf{e}_i = \mathbf{e}_i^T \mathbf{M} \mathbf{T} \mathbf{T}^T \mathbf{M} \mathbf{e}_i$, and $\mathbf{M} \mathbf{e}_i$ is the i th column of $\mathbf{M}$.

- (b) By [Proposition 6.3.4\(d\)](#), $0 \leq \|\mathbf{M}\mathbf{e}_i\|^2 \leq \|\mathbf{e}_i\|^2 = 1$. The sum is $\text{tr}(\mathbf{M}) = r$.
- (c) With $\mathbf{M}_0 = n^{-1}\mathbf{1}\mathbf{1}^\top$, the difference $\mathbf{M} - \mathbf{M}_0$ is a projection ([Theorem 6.5.1](#)), hence nonnegative definite, so $h_{ii} \geq (\mathbf{M}_0)_{ii} = 1/n$.
- (d) Equality holds in $\|\mathbf{M}\mathbf{e}_i\| \leq \|\mathbf{e}_i\|$ iff $(\mathbf{I} - \mathbf{M})\mathbf{e}_i = \mathbf{0}$, that is, iff $\mathbf{e}_i \in \mathbb{C}(\mathbf{X})$. Then $\mathbf{M}\mathbf{e}_i = \mathbf{e}_i$, so column i of $\mathbf{M}$ is $\mathbf{e}_i$, and by symmetry so is row i . Hence $\hat{y}_i = y_i$.
- (e) $\mathbf{M} = \mathbf{Q}\mathbf{Q}^\top$, so $m_{ii} = \sum_k q_{ik}^2$.

■

Part (d) describes the extreme case. An observation with leverage one has a coordinate direction of its own in the model space. This happens, for example, when it is the only member of a group in a one-way layout, or when a column of $\mathbf{X}$ is an indicator for that single observation. The model then fits the observation exactly, whatever its value, and the data give no check on it. High leverage is a weaker form of the same thing. If h_{ii} is close to one, then $\hat{y}_i \approx y_i$, and a gross error in y_i barely shows up in its residual.

This shows up in the residual variances. If $\text{Cov}(\varepsilon) = \sigma^2\mathbf{I}$, then by [Proposition 6.4.3\(e\)](#)

$$\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii}),$$

so residuals at high-leverage points are *less* variable than others, even though those are exactly the points where an error would do the most damage. Chapter 20 standardizes residuals to correct for this.

6.8.1 Leverage as distance from the centre

When the model has an intercept, leverage has a second description in variable space: distance from the centroid of the regressors. Write $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$, let $\bar{\mathbf{x}}$ be the vector of column means of $\mathbf{X}_1$, and let $\tilde{\mathbf{X}}_1 = (\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{X}_1$ be the centred regressors, assumed to have full column rank.

Proposition 6.8.2

Under these assumptions,

$$h_{ii} = \frac{1}{n} + (\mathbf{x}_{(i)} - \bar{\mathbf{x}})^\top (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} (\mathbf{x}_{(i)} - \bar{\mathbf{x}}),$$

where $\mathbf{x}_{(i)}$ is the i th row of $\mathbf{X}_1$, written as a column vector.

Proof. By [Lemma 6.6.1](#) with $\mathbf{X}_2 = \mathbf{1}$, $\mathbf{M} = n^{-1}\mathbf{1}\mathbf{1}^\top + \tilde{\mathbf{X}}_1(\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1}\tilde{\mathbf{X}}_1^\top$. The i th row of $\tilde{\mathbf{X}}_1$ is $(\mathbf{x}_{(i)} - \bar{\mathbf{x}})^\top$.

■

The quadratic form is $(n-1)^{-1}$ times the squared Mahalanobis distance of $\mathbf{x}_{(i)}$ from the centroid, measured in the metric of the sample covariance matrix of the regressors. Leverage is therefore high for points that are unusual *jointly*, even when no single coordinate is extreme. A state with moderate poverty and moderate single parenthood, but an unusual *combination* of the two, can have high leverage. For simple regression the formula reduces to $h_{ii} = n^{-1} + (x_i - \bar{x})^2/S_{xx}$.

Example. (High-leverage states)

Return to the state data, now with all 51 jurisdictions including the District of Columbia, and the model of [Example \(Poverty and murder rates\)](#). The average leverage is $p/n = 0.078$, and the smallest is 0.021, just above the bound $1/n$. The three largest are

Jurisdiction	h_{ii}	multiple of average
District of Columbia	0.506	6.5
Mississippi	0.212	2.7
Alaska	0.177	2.3

The District of Columbia is entirely urban, with by far the highest proportion of single-parent households. It sits at a corner of the regressor cloud that no state comes near, and its leverage is about half the maximum possible. It was omitted from [Example \(Poverty and murder rates\)](#). [Exercise \(A1\)](#) asks what happens to the poverty coefficient when it is put back.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statelcrime.load_pandas().data           # District of Columbia
kept here
X = np.column_stack([np.ones(len(data)), data["poverty"], data["single"], data["
urban"]])

Q, _ = np.linalg.qr(X)                                     # orthonormal basis for C(X)
h = np.sum(Q ** 2, axis=1)                                  # h_ii = m_ii = ||row i of Q||^2
n, p = X.shape
print("sum of leverages:", h.sum(), " (= p =", p, ")")
print("range:", h.min(), h.max(), " bounds:", 1 / n, 1)
top = np.argsort(h)[::-1][:3]
for i in top:
    print(f"{data.index[i]:22s} h = {h[i]:.3f}    ({h[i] / (p / n):.1f} x average)")
```

The listing computes leverages from a QR factorization rather than by forming $\mathbf{M}$. For large n the $n \times n$ matrix $\mathbf{M}$ should never be formed. [Section 6.10](#) explains why.

6.8.2 Exercises

6.8.2.1 A. Check your understanding

Exercise A1.

For simple regression, derive $h_{ii} = n^{-1} + (x_i - \bar{x})^2 / S_{xx}$ directly from [Example \(Intercept inside simple regression\)](#), and show that the largest leverage belongs to the observation with x_i farthest from $\bar{x}$.

6.8.2.2 B. Practice

Exercise B1.

Show that $m_{ij}^2 \leq m_{ii}(1 - m_{ii})$ for $i \neq j$, and hence $|m_{ij}| \leq 1/2$. When is the bound attained?

Solution. From Proposition 6.8.1(a), $m_{ii} = \sum_k m_{ik}^2 \geq m_{ii}^2 + m_{ij}^2$. So $m_{ij}^2 \leq m_{ii}(1 - m_{ii}) \leq 1/4$. Equality needs $m_{ii} = 1/2$ and m_{ij} the only other nonzero entry of row i . The intercept-only model with $n = 2$, where every entry of $\mathbf{M}$ is $1/2$, attains it.

Exercise B2.

Show that adding a column to $\mathbf{X}$ cannot decrease any leverage. Show that the leverages in a one-way layout are $1/n_k$ for members of group k , and explain why an observation forming a group on its own has leverage one.

6.9 Other inner products

Everything so far has used the Euclidean inner product $\mathbf{u}^T \mathbf{v}$. That choice treats the n coordinates of observation space symmetrically: an error of a given size counts the same in every observation, and errors in different observations are unrelated. When observations have different variances or are correlated, Euclidean distance is the wrong way to measure how far $\mathbf{y}$ is from the model space. This section shows that the whole theory carries over to any inner product, and that the result is generalized least squares.

6.9.1 Projection in a general inner product

Let $\mathbf{A}$ be an $n \times n$ symmetric positive definite matrix, and define

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{A}} = \mathbf{u}^T \mathbf{A} \mathbf{v}, \quad \|\mathbf{u}\|_{\mathbf{A}} = (\mathbf{u}^T \mathbf{A} \mathbf{u})^{1/2}.$$

This is an inner product: bilinear, symmetric, and positive on nonzero vectors. Say $\mathbf{u}$ and $\mathbf{v}$ are $\mathbf{A}$ -orthogonal if $\mathbf{u}^T \mathbf{A} \mathbf{v} = 0$. Nothing in the proofs of Proposition 6.2.2, Theorem 6.2.3 or Theorem 6.3.1 used anything about $\mathbf{u}^T \mathbf{v}$ except these three properties. So those results hold verbatim with $\langle \cdot, \cdot \rangle_{\mathbf{A}}$ in place of the Euclidean inner product. In particular, every $\mathbf{y}$ has a unique $\mathbf{A}$ -nearest point in a subspace $\mathcal{S}$, characterized by $\mathbf{A}$ -orthogonality of the error to $\mathcal{S}$. The map sending $\mathbf{y}$ to it is linear, and we write it $\mathbf{P}_{\mathcal{S}, \mathbf{A}}$.

Theorem 6.9.1

Let $\mathbf{A}$ be positive definite and $\mathbf{X}$ any $n \times p$ matrix. The $\mathbf{A}$ -orthogonal projection onto $\mathbb{C}(\mathbf{X})$ is

$$\mathbf{P}_{\mathbf{X}, \mathbf{A}} = \mathbf{X}(\mathbf{X}^T \mathbf{A} \mathbf{X})^{-} \mathbf{X}^T \mathbf{A},$$

for any generalized inverse. It satisfies $\mathbf{P}_{\mathbf{X}, \mathbf{A}}^2 = \mathbf{P}_{\mathbf{X}, \mathbf{A}}$ and $\mathbb{C}(\mathbf{P}_{\mathbf{X}, \mathbf{A}}) = \mathbb{C}(\mathbf{X})$, and the matrix $\mathbf{A} \mathbf{P}_{\mathbf{X}, \mathbf{A}}$ is symmetric. The matrix $\mathbf{P}_{\mathbf{X}, \mathbf{A}}$ itself is symmetric iff $\mathbf{A} \mathbb{C}(\mathbf{X}) \subseteq \mathbb{C}(\mathbf{X})$,

and in general it is not.

Proof. Let $\mathbf{A} = \mathbf{L}\mathbf{L}^T$ with $\mathbf{L}$ nonsingular (for instance the Cholesky factor), and put $\mathbf{X}_* = \mathbf{L}^T\mathbf{X}$ and $\mathbf{y}_* = \mathbf{L}^T\mathbf{y}$. Then $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|_{\mathbf{A}}^2 = \|\mathbf{y}_* - \mathbf{X}_*\mathbf{b}\|^2$, so $\mathbf{b}$ minimizes the $\mathbf{A}$ -distance iff it is an ordinary least squares estimate for $(\mathbf{y}_*, \mathbf{X}_*)$. By [Theorem 6.3.7](#) and [Theorem 6.4.2](#), one such estimate is $(\mathbf{X}_*^T\mathbf{X}_*)^{-1}\mathbf{X}_*^T\mathbf{y}_* = (\mathbf{X}^T\mathbf{A}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{A}\mathbf{y}$, and $\mathbf{X}_*(\mathbf{X}_*^T\mathbf{X}_*)^{-1}\mathbf{X}_*^T$ is invariant to the choice of generalized inverse. Multiplying on the left by $\mathbf{L}^{-T}$ and on the right by $\mathbf{L}^T$ shows that $\mathbf{P}_{\mathbf{X},\mathbf{A}} = \mathbf{L}^{-T}\mathbf{M}_*\mathbf{L}^T$, where $\mathbf{M}_*$ projects onto $\mathbb{C}(\mathbf{X}_*)$. Idempotence and $\mathbb{C}(\mathbf{P}_{\mathbf{X},\mathbf{A}}) = \mathbf{L}^{-T}\mathbb{C}(\mathbf{X}_*) = \mathbb{C}(\mathbf{X})$ follow, and $\mathbf{A}\mathbf{P}_{\mathbf{X},\mathbf{A}} = \mathbf{L}\mathbf{M}_*\mathbf{L}^T$ is symmetric. For the last claim, see [Exercise \(B1\)](#). ■

The proof also gives the general principle: *$\mathbf{A}$ -geometry is Euclidean geometry after the change of variables $\mathbf{y} \mapsto \mathbf{L}^T\mathbf{y}$.* Any fact about orthogonal projections has an $\mathbf{A}$ -version, obtained by transporting it through $\mathbf{L}$.

Symmetry is the Euclidean signature of an orthogonal projection. The $\mathbf{A}$ -version of [Theorem 6.3.3](#) replaces it with self-adjointness.

Proposition 6.9.2

An idempotent matrix $\mathbf{P}$ is the $\mathbf{A}$ -orthogonal projection onto $\mathbb{C}(\mathbf{P})$ iff $\mathbf{A}\mathbf{P}$ is symmetric, that is, iff $\langle \mathbf{P}\mathbf{u}, \mathbf{v} \rangle_{\mathbf{A}} = \langle \mathbf{u}, \mathbf{P}\mathbf{v} \rangle_{\mathbf{A}}$ for all $\mathbf{u}, \mathbf{v}$.

Proof. For idempotent $\mathbf{P}$, the residual $(\mathbf{I} - \mathbf{P})\mathbf{y}$ is $\mathbf{A}$ -orthogonal to $\mathbb{C}(\mathbf{P})$ for every $\mathbf{y}$ iff $\mathbf{P}^T\mathbf{A}(\mathbf{I} - \mathbf{P}) = \mathbf{0}$, that is, iff $\mathbf{P}^T\mathbf{A} = \mathbf{P}^T\mathbf{A}\mathbf{P}$. If $\mathbf{A}\mathbf{P}$ is symmetric, then $\mathbf{P}^T\mathbf{A} = \mathbf{A}\mathbf{P}$, and $\mathbf{P}^T\mathbf{A}\mathbf{P} = \mathbf{A}\mathbf{P}\mathbf{P} = \mathbf{A}\mathbf{P} = \mathbf{P}^T\mathbf{A}$. Conversely, $\mathbf{P}^T\mathbf{A} = \mathbf{P}^T\mathbf{A}\mathbf{P}$ has a symmetric right-hand side, so $\mathbf{P}^T\mathbf{A}$ and hence its transpose $\mathbf{A}\mathbf{P}$ are symmetric. ■

6.9.2 Generalized least squares

Suppose $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{V}$ with $\mathbf{V}$ known and positive definite. Observations with large variance should count for less, and correlated observations partly repeat each other. The natural distance is the one that makes the errors look uncorrelated: $\|\cdot\|_{\mathbf{V}^{-1}}$. If $\mathbf{V} = \mathbf{L}\mathbf{L}^T$, then $\mathbf{L}^{-1}\boldsymbol{\varepsilon}$ has covariance $\sigma^2\mathbf{I}$, and $\|\mathbf{u}\|_{\mathbf{V}^{-1}} = \|\mathbf{L}^{-1}\mathbf{u}\|$.

Definition 6.9.1 — Generalized least squares

A **generalized least squares** (GLS) estimate minimizes $(\mathbf{y} - \mathbf{X}\mathbf{b})^T\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\mathbf{b})$. Equivalently, $\mathbf{X}\hat{\boldsymbol{\beta}}_{\text{GLS}} = \mathbf{P}_{\mathbf{X},\mathbf{V}^{-1}}\mathbf{y}$, and $\hat{\boldsymbol{\beta}}_{\text{GLS}}$ solves the generalized normal equations $\mathbf{X}^T\mathbf{V}^{-1}\mathbf{X}\mathbf{b} = \mathbf{X}^T\mathbf{V}^{-1}\mathbf{y}$.

[Theorem 6.9.1](#) with $\mathbf{A} = \mathbf{V}^{-1}$ is the whole of the algebra. Computationally, GLS is ordinary least squares applied to the *whitened* data $(\mathbf{L}^{-1}\mathbf{y}, \mathbf{L}^{-1}\mathbf{X})$. Aitken (1935) showed that GLS gives

the best linear unbiased estimator under this covariance, which is the Gauss–Markov theorem transported through $\mathbf{L}$. Chapter 31 proves this, and handles singular $\mathbf{V}$ and the more realistic case where $\mathbf{V}$ must be estimated.

When is the ordinary least squares fit already the GLS fit? Geometrically, the two projections onto $\mathbb{C}(\mathbf{X})$ coincide iff the Euclidean and the $\mathbf{V}^{-1}$ notions of orthogonality agree on this particular subspace.

Theorem 6.9.3 — Kruskal

For positive definite $\mathbf{V}$, $\mathbf{M}\mathbf{y} = \mathbf{P}_{\mathbf{X}, \mathbf{V}^{-1}}\mathbf{y}$ for every $\mathbf{y}$ iff $\mathbb{C}(\mathbf{V}\mathbf{X}) \subseteq \mathbb{C}(\mathbf{X})$.

Proof. The two projections have the same range. They agree iff they have the same null space, since an idempotent matrix is determined by its range and null space (Proposition 6.9.4). The null space of $\mathbf{M}$ is $\mathbb{C}(\mathbf{X})^\perp = \mathcal{N}(\mathbf{X}\mathbf{T})$. The null space of $\mathbf{P}_{\mathbf{X}, \mathbf{V}^{-1}}$ is the set of $\mathbf{u}$ with $\mathbf{X}\mathbf{T}\mathbf{V}^{-1}\mathbf{u} = \mathbf{0}$, that is, $\mathbf{V}\mathcal{N}(\mathbf{X}\mathbf{T})$. So the projections agree iff $\mathbf{V}\mathcal{N}(\mathbf{X}\mathbf{T}) = \mathcal{N}(\mathbf{X}\mathbf{T})$, and since $\mathbf{V}$ is nonsingular and the spaces have equal dimension, iff $\mathbf{V}\mathcal{N}(\mathbf{X}\mathbf{T}) \subseteq \mathcal{N}(\mathbf{X}\mathbf{T})$. By symmetry of $\mathbf{V}$, this says $\mathbf{X}\mathbf{T}\mathbf{V}\mathbf{u} = \mathbf{0}$ whenever $\mathbf{X}\mathbf{T}\mathbf{u} = \mathbf{0}$, that is, $\mathbb{C}(\mathbf{V}\mathbf{X}) \perp \mathbb{C}(\mathbf{X})^\perp$, which is $\mathbb{C}(\mathbf{V}\mathbf{X}) \subseteq \mathbb{C}(\mathbf{X})$. ■

The condition holds, for example, for equicorrelated errors $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^T$ whenever $\mathbf{1} \in \mathbb{C}(\mathbf{X})$ (Exercise A1). It also underlies the analysis of balanced split-plot designs in Chapter 33. The result is due to Kruskal (1968), and the coordinate-free argument is his.

Example. (Autocorrelated errors)

Take eight equally spaced observations with a linear trend and error covariance $v_{st} = 0.7^{|s-t|}$. The $\mathbf{V}^{-1}$ -projection onto $\mathbb{C}(\mathbf{X})$ is idempotent and self-adjoint for $\langle \cdot, \cdot \rangle_{\mathbf{V}^{-1}}$ but not symmetric: its largest asymmetry $\max|p_{st} - p_{ts}|$ is 0.444. The ordinary and generalized slope estimates are 0.648 and 0.650. The listing checks each claim, including that the GLS fit is ordinary least squares on whitened data.

```
import numpy as np
rng = np.random.default_rng(9)
n = 8
X = np.column_stack([np.ones(n), np.arange(n, dtype=float)])
y = X @ np.array([1.0, 0.5]) + rng.normal(size=n)
t = np.arange(n)
V = 0.7 ** np.abs(t[:, None] - t[None, :])           # AR(1)-type covariance

Vinv = np.linalg.inv(V)
P_V = X @ np.linalg.solve(X.T @ Vinv @ X, X.T @ Vinv) # V^{-1}-orthogonal
               projection onto C(X)

print("idempotent:", np.allclose(P_V @ P_V, P_V))
print("symmetric: ", np.allclose(P_V, P_V.T))           # False: oblique in
               the usual geometry
print("self-adjoint for <u,v> = u'V^{-1}v:", np.allclose(Vinv @ P_V, (Vinv @ P_V).T)
)
```

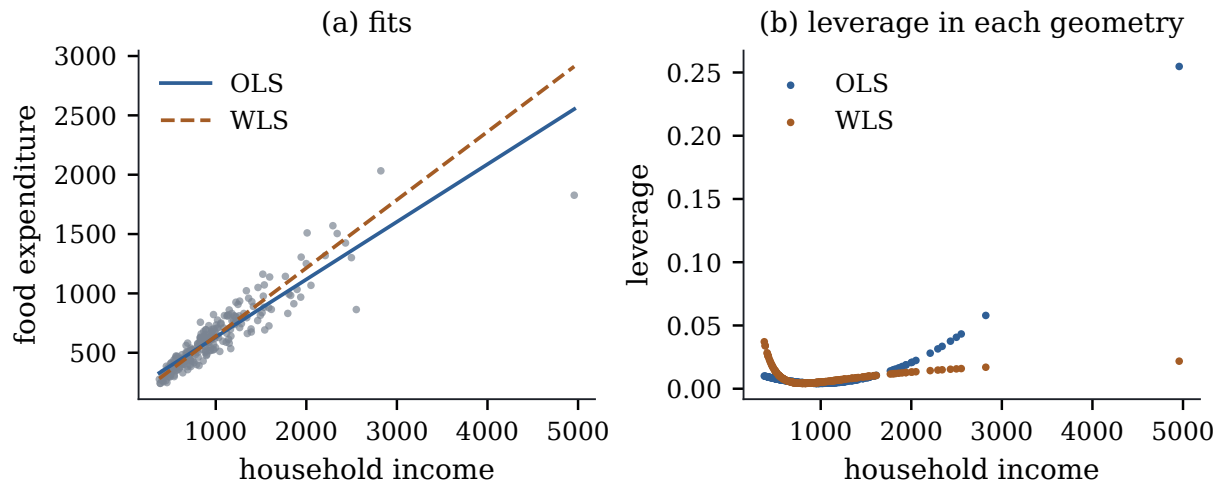


Figure 6.9.1: Engel's data. (a) Ordinary and weighted least squares fits, with weights proportional to $1/\text{income}^2$. (b) Leverages under each geometry. The weighted projection spreads influence away from the high-variance, high-income households.

```
# GLS = ordinary least squares after whitening with a square root of  $V^{-1}$ 
L = np.linalg.cholesky(V) #  $V = L L'$ 
Xw, yw = np.linalg.solve(L, X), np.linalg.solve(L, y)
beta_whitened, *_ = np.linalg.lstsq(Xw, yw, rcond=None)
beta_gls = np.linalg.solve(X.T @ Vinv @ X, X.T @ Vinv @ y)
print("GLS via P_V equals OLS on whitened data:", np.allclose(beta_gls,
    beta_whitened))
```

6.9.3 Weighted least squares

The simplest non-Euclidean geometry is a diagonal one. If the errors are uncorrelated with variances σ^2/w_i , then $\mathbf{V}^{-1} = \mathbf{W} = \text{diag}(w_1, \dots, w_n)$, and GLS minimizes $\sum_i w_i (y_i - \mathbf{x}_{(i)} \mathbf{T} \mathbf{b})^2$. Observations believed to be noisier get less weight in the distance. Whitening multiplies row i of $\mathbf{X}$ and $\mathbf{y}$ by $\sqrt{w_i}$. The fitted values are $\mathbf{P}_{\mathbf{X}, \mathbf{W}} \mathbf{y}$, and the natural leverages are the diagonal entries of $\mathbf{P}_{\mathbf{X}, \mathbf{W}}$. These equal the diagonal of the orthogonal projection onto $\mathbb{C}(\mathbf{W}^{1/2} \mathbf{X})$ and still sum to $\text{rank}(\mathbf{X})$ (Exercise (B3)).

Example. (Engel's food expenditure data)

Engel's 1857 survey of 235 Belgian working-class households records annual income and food expenditure. The spread of expenditure grows with income, as Figure 6.9.1(a) shows. Weighting by $w_i = 1/\text{income}_i^2$ (standard deviation proportional to income) changes the fitted slope from 0.4852 to 0.5740. The change in geometry shows up more clearly in the leverages, panel (b). Under ordinary least squares the richest households have the most leverage, with a maximum of 0.255. In the weighted geometry their large variance shrinks their influence, and no household has leverage above 0.037. Whether this is the *right* geometry depends on whether the variance model holds. Chapter 21 discusses how to check it, and how to get valid standard errors when it fails.

```
import numpy as np
import statsmodels.api as sm
```

```

df = sm.datasets.engel.load_pandas().data
x, y = df["income"].to_numpy(), df["foodexp"].to_numpy()
X = np.column_stack([np.ones(len(x)), x])

w = 1.0 / x**2                                # working model: sd proportional to
income
sw = np.sqrt(w)
beta_ols, *_ = np.linalg.lstsq(X, y, rcond=None)
beta_wls, *_ = np.linalg.lstsq(X * sw[:, None], y * sw, rcond=None) # OLS on (W
~1/2 X, W~1/2 y)

# leverages in the weighted geometry: squared row norms of an orthonormal basis of C
(W~1/2 X)
Qw, _ = np.linalg.qr(X * sw[:, None])
h_wls = np.sum(Qw**2, axis=1)
Q, _ = np.linalg.qr(X)
h_ols = np.sum(Q**2, axis=1)
print("OLS:", beta_ols, " WLS:", beta_wls)
print("max leverage OLS %.3f, WLS %.3f" % (h_ols.max(), h_wls.max()))

```

6.9.4 Oblique projections

The matrices $P_{X,A}$ are idempotent without being symmetric. In the Euclidean geometry of $\mathbb{R}^n$ they are **oblique projections**. Idempotent matrices in general have a clean description.

Proposition 6.9.4

Let P be idempotent. Then $\mathbb{R}^n = \mathbb{C}(P) \oplus \mathcal{N}(P)$, and P maps each $y = u + v$ ($u \in \mathbb{C}(P)$, $v \in \mathcal{N}(P)$) to u . Conversely, for any decomposition $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{T}$ there is exactly one idempotent P with $\mathbb{C}(P) = \mathcal{S}$ and $\mathcal{N}(P) = \mathcal{T}$, the *projection onto $\mathcal{S}$ along $\mathcal{T}$* . It is an orthogonal projection iff $\mathcal{T} = \mathcal{S}^\perp$, that is, iff P is symmetric.

Proof. Write $y = Py + (I - P)y$. The first term lies in $\mathbb{C}(P)$, and $P(I - P)y = 0$ puts the second in $\mathcal{N}(P)$. If $u = Pw \in \mathcal{N}(P)$ then $u = P^2w = Pu = 0$, so the sum is direct. For $u \in \mathbb{C}(P)$, $Pu = u$ (write $u = Pw$), so $P(u + v) = u$. Conversely, given $\mathcal{S} \oplus \mathcal{T}$, the map $u + v \mapsto u$ is well defined and linear by uniqueness of the decomposition, and it is idempotent with the required range and null space. It is unique because its values are forced on all of $\mathbb{R}^n$. If $\mathcal{T} = \mathcal{S}^\perp$, the map is the orthogonal projection, which is symmetric. If P is symmetric, then $\mathcal{N}(P) = \mathcal{N}(PT) = \mathbb{C}(P)^\perp$ by Lemma 6.2.1. ■

Figure 6.9.2 shows the difference. The nearest-point property belongs to the orthogonal projection alone. An oblique projection can move a vector *farther* away: its operator norm exceeds one unless it is orthogonal (Exercise (B2)). The GLS projection is still a nearest-point map, but in the V^{-1} norm, not the Euclidean one. Which geometry is “right” is not a mathematical question. It is decided by the covariance of the errors.

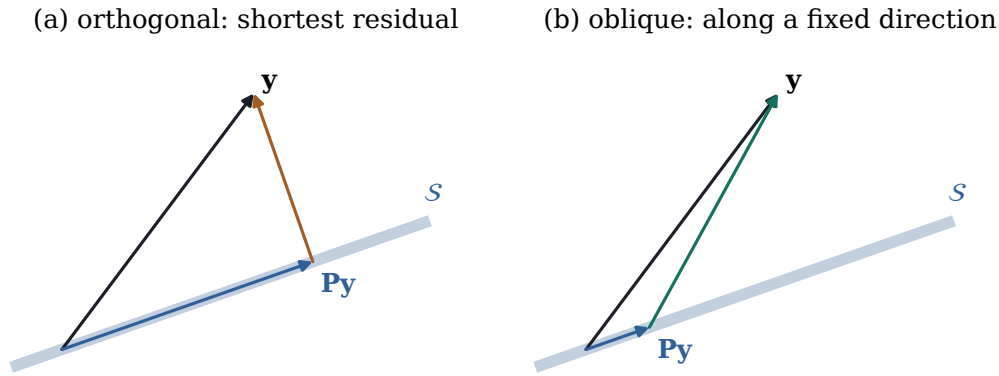


Figure 6.9.2: Projections onto a line S in $\mathbb{R}^2$. (a) The orthogonal projection: the residual is perpendicular to S , and Py is the nearest point. (b) An oblique projection moves y along a fixed direction that is not perpendicular to S . The result is idempotent but is not the nearest point, and the residual is longer.

6.9.5 Exercises

6.9.5.1 A. Check your understanding

Exercise A1.

Let $V = (1 - \rho)I + \rho \mathbf{1}\mathbf{1}^T$ with $-1/(n-1) < \rho < 1$. Show that V is positive definite and that $\mathbb{C}(VX) \subseteq \mathbb{C}(X)$ whenever $\mathbf{1} \in \mathbb{C}(X)$. Conclude that ordinary least squares is generalized least squares for equicorrelated errors in any model with an intercept.

Solution. $V\mathbf{1} = (1 + (n-1)\rho)\mathbf{1}$ and $Vu = (1 - \rho)u$ for $u \perp \mathbf{1}$. So the eigenvalues are $1 + (n-1)\rho$ and $1 - \rho$, both positive exactly when $-1/(n-1) < \rho < 1$. Then $VX = (1 - \rho)X + \rho \mathbf{1}(\mathbf{1}^T X)$, whose columns lie in $\mathbb{C}(X)$ when $\mathbf{1}$ does. [Theorem 6.9.3](#) applies.

6.9.5.2 B. Practice

Exercise B1.

Complete the proof of [Theorem 6.9.1](#). Show that $P_{X,A}$ is symmetric iff it equals M , and deduce from [Theorem 6.9.3](#) (with $V = A^{-1}$) that this happens iff $A\mathbb{C}(X) \subseteq \mathbb{C}(X)$.

Exercise B2.

Let $P \neq 0$ be idempotent. Show that the operator norm $\|P\|_2 = \max_{\|u\|=1} \|Pu\|$ is at least 1, with equality iff P is symmetric. *Hint:* if P is not symmetric, find $u \in \mathbb{C}(P)$ and $v \in \mathcal{N}(P)$ with $u^T v \neq 0$, and consider $u + tv$ for small t .

Solution. For $u \in \mathbb{C}(P)$, $Pu = u$, so $\|P\|_2 \geq 1$. If P is symmetric, $\|Pu\| \leq \|u\|$ by [Proposition 6.3.4\(d\)](#). Suppose P is not symmetric. Then $\mathcal{N}(P) \neq \mathbb{C}(P)^\perp$ by [Proposition 6.9.4](#). The two spaces have the same dimension, so some $v \in \mathcal{N}(P)$ is not orthogonal to $\mathbb{C}(P)$. Choose a unit $u \in \mathbb{C}(P)$ with $u^T v > 0$. For $t > 0$, $P(u - tv) = u$, while

$\|\mathbf{u} - t\mathbf{v}\|^2 = 1 - 2t\mathbf{u}^T\mathbf{v} + t^2\|\mathbf{v}\|^2 < 1$ for small t . So $\mathbf{P}$ stretches $\mathbf{u} - t\mathbf{v}$, and $\|\mathbf{P}\|_2 > 1$.

Exercise B3.

For weighted least squares, $\mathbf{V} = \text{diag}(1/w_1, \dots, 1/w_n)$ with $w_i > 0$. Show that the diagonal entries of $\mathbf{P}_{\mathbf{X}, \mathbf{V}^{-1}}$ equal those of the orthogonal projection onto $\mathbb{C}(\mathbf{W}^{1/2}\mathbf{X})$, where $\mathbf{W} = \text{diag}(w_i)$, and that they sum to $\text{rank}(\mathbf{X})$. Interpret them as leverages.

6.10 Computation

The formulas $\mathbf{M} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ and $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$ are the right way to *reason* about least squares and the wrong way to *compute* it. There are two problems. $\mathbf{M}$ is $n \times n$, so forming it costs $O(n^2)$ memory for a quantity that is never needed in full. And forming $\mathbf{X}^T\mathbf{X}$ squares the condition number of the problem, which can throw away half of the available floating-point digits before any equation is solved. This section shows how the geometry itself suggests the fix, which is to build an orthonormal basis for $\mathbb{C}(\mathbf{X})$. Chapter 10 develops the numerical linear algebra in full.

6.10.1 QR: Gram–Schmidt as a factorization

Apply Gram–Schmidt (Proposition 6.2.2) to the columns $\mathbf{x}_1, \dots, \mathbf{x}_p$ of a full-column-rank $\mathbf{X}$. Each $\mathbf{x}_j$ is a combination of $\mathbf{u}_1, \dots, \mathbf{u}_j$ with coefficients $r_{ij} = \mathbf{u}_i^T\mathbf{x}_j$ for $i < j$ and $r_{jj} = \|\mathbf{w}_j\| > 0$. In matrix form,

$$\mathbf{X} = \mathbf{Q}\mathbf{R},$$

where $\mathbf{Q} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$ is $n \times p$ with $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_p$, and $\mathbf{R}$ is $p \times p$ upper triangular with positive diagonal. This is the **thin QR factorization**. Since $\mathbb{C}(\mathbf{Q}) = \mathbb{C}(\mathbf{X})$, Proposition 6.3.2 gives $\mathbf{M} = \mathbf{Q}\mathbf{Q}^T$ at once, and the rest of least squares follows:

$$\hat{\mathbf{Y}} = \mathbf{Q}(\mathbf{Q}^T\mathbf{y}), \quad \mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{Q}^T\mathbf{y}, \quad \text{SSE} = \|\mathbf{y}\|^2 - \|\mathbf{Q}^T\mathbf{y}\|^2, \quad h_{ii} = \sum_k q_{ik}^2. \quad (6.10.1)$$

The coefficient equation holds because the normal equations become $\mathbf{R}^T\mathbf{R}\mathbf{b} = \mathbf{R}^T\mathbf{Q}^T\mathbf{y}$, and $\mathbf{R}^T$ is nonsingular. It is solved by back substitution, with no matrix inverse. Nothing of size $n \times n$ is ever formed.

In floating point, Gram–Schmidt as written in Proposition 6.2.2 gradually loses orthogonality among the computed $\mathbf{u}_j$. Production software computes the same factorization with Householder reflections, which are backward stable (Golub and Van Loan 2013; Higham 2002). The geometry does not change: $\mathbf{Q}$ is an orthonormal basis for the model space.

6.10.2 Conditioning

The sensitivity of least squares to rounding is governed by the **condition number**

$$\kappa(\mathbf{X}) = \frac{\sigma_{\max}(\mathbf{X})}{\sigma_{\min}(\mathbf{X})},$$

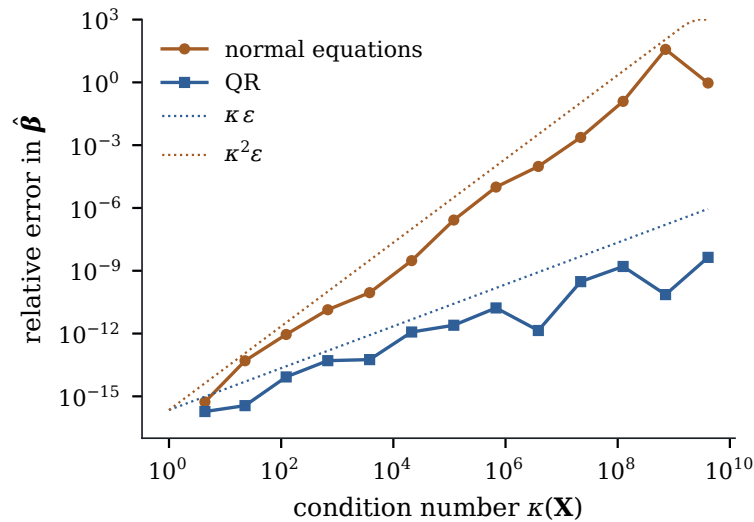


Figure 6.10.1: Accuracy of least squares coefficients for polynomial regressions of degree 1 to 13, against the condition number of the model matrix. The error of the normal equations tracks $\kappa^2\epsilon$ until it reaches order one. The error of QR tracks $\kappa\epsilon$.

the ratio of the largest and smallest singular values of $\mathbf{X}$ (Chapter 1). Geometrically, $\kappa(\mathbf{X})$ is large when some combination of columns of unit coefficient norm is nearly zero. The columns are then close to linearly dependent, and the coordinates of a point in $\mathbb{C}(\mathbf{X})$ are poorly determined, even though the point itself is not. The singular values of $\mathbf{X}^T\mathbf{X}$ are the squares of those of $\mathbf{X}$, so

$$\kappa(\mathbf{X}^T\mathbf{X}) = \kappa(\mathbf{X})^2.$$

A useful rule of thumb from numerical analysis says that solving a linear system with condition number κ in double precision loses about $\log_{10} \kappa$ of the roughly 16 available significant digits. The normal equations solve a system whose matrix is $\mathbf{X}^T\mathbf{X}$, so they lose about $2\log_{10} \kappa(\mathbf{X})$ digits. QR solves the triangular system $\mathbf{R}\mathbf{b} = \mathbf{Q}^T\mathbf{y}$, and $\kappa(\mathbf{R}) = \kappa(\mathbf{X})$ because $\mathbf{Q}$ has orthonormal columns, so it loses about $\log_{10} \kappa(\mathbf{X})$. The precise perturbation theory has one more subtlety. When the residual is large, the least squares *problem* itself has a sensitivity term proportional to $\kappa(\mathbf{X})^2\|\hat{\epsilon}\|$, and no algorithm can avoid it (Higham 2002, ch. 20). QR’s advantage is largest when the model fits well.

Example. (Polynomial regression)

The columns $1, t, t^2, \dots, t^d$ evaluated at 200 equally spaced points in $[0, 1]$ become nearly dependent as d grows. To isolate the numerical error, the listing sets $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$ exactly for a random $\boldsymbol{\beta}$, so that the true least squares solution is $\boldsymbol{\beta}$ itself. For $d = 8$ the condition number is 6.8×10^5 . The normal equations return $\boldsymbol{\beta}$ with relative error 1.0×10^{-5} , while QR achieves 1.7×10^{-11} . At $d = 13$, with $\kappa(\mathbf{X}) = 4.1 \times 10^9$, the normal-equation answer has relative error 9.3×10^{-1} and not one correct digit, while QR still gets 4.4×10^{-9} . Figure 6.10.1 plots the whole sequence against the reference lines $\kappa\epsilon$ and $\kappa^2\epsilon$, where $\epsilon \approx 2.2 \times 10^{-16}$ is machine precision.

```
import numpy as np
import statsmodels.api as sm

def fit_normal_equations(X, y):
    return np.linalg.solve(X.T @ X, X.T @ y)
```

```

def fit_qr(X, y):
    Q, R = np.linalg.qr(X)          # X = QR, Q has orthonormal columns
    return np.linalg.solve(R, Q.T @ y) # triangular system R b = Q'y

rng = np.random.default_rng(10)
n = 200
t = np.linspace(0, 1, n)
rows = []

for degree in range(1, 14):
    X = np.vander(t, degree + 1, increasing=True) # 1, t, t^2, ..., t^degree
    beta = rng.normal(size=degree + 1)
    y = X @ beta # exact fit: the true answer is beta
    kappa = np.linalg.cond(X)
    err_ne = np.linalg.norm(fit_normal_equations(X, y) - beta) / np.linalg.norm(beta)
    err_qr = np.linalg.norm(fit_qr(X, y) - beta) / np.linalg.norm(beta)
    rows.append((degree, kappa, err_ne, err_qr))
    print(f"degree {degree:2d} cond(X) = {kappa:9.2e} normal eq. {err_ne:8.1e} QR {err_qr:8.1e}")

```

The polynomial example is extreme by design, but ill-conditioning is common in real data. The macroeconomic series of Longley's classic test problem (in `statsmodels.datasets.longley`) give a model matrix with $\kappa(\mathbf{X}) \approx 4.86 \times 10^9$. Then $\kappa(\mathbf{X}\mathbf{T}\mathbf{X}) \approx 2.36 \times 10^{19}$, which is larger than $1/\epsilon$. The rule of thumb predicts about 6 correct digits from QR and about 0 from the normal equations. Part of that condition number is only a matter of units: one column is a year, another is population in thousands. Rescaling columns ([Theorem 6.7.1](#)) changes κ without changing the fit ([Exercise \(B1\)](#)). The part that remains after rescaling reflects genuine near-dependence among the regressors, the subject of Chapter 26.

Two practical lessons follow. First, never compute $(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}$ just to obtain $\hat{\beta}$. Solve with QR, or call a least squares routine that does. Second, reparameterizations that leave $\mathbb{C}(\mathbf{X})$ unchanged, such as centring, scaling and orthogonal polynomials, are statistically neutral but can be numerically decisive.

6.10.3 Rank deficiency and the singular value decomposition

QR with positive diagonal needs full column rank. For rank-deficient or nearly rank-deficient $\mathbf{X}$, the most informative tool is the **singular value decomposition**

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}\mathbf{T} = \mathbf{U}_r\mathbf{D}_r\mathbf{V}_r\mathbf{T},$$

where $\mathbf{U}$ ($n \times n$) and $\mathbf{V}$ ($p \times p$) are orthogonal, $\mathbf{D}$ is $n \times p$ with nonnegative diagonal entries $\sigma_1 \geq \dots \geq \sigma_r > 0$, and the subscript r keeps only the first r columns. Then:

- $\mathbf{U}_r$ is an orthonormal basis for $\mathbb{C}(\mathbf{X})$, so $\mathbf{M} = \mathbf{U}_r\mathbf{U}_r\mathbf{T}$;
- the last $p - r$ columns of $\mathbf{V}$ form an orthonormal basis for $\mathcal{N}(\mathbf{X})$;
- the Moore–Penrose inverse is $\mathbf{X}^+ = \mathbf{V}_r\mathbf{D}_r^{-1}\mathbf{U}_r\mathbf{T}$, and $\mathbf{M} = \mathbf{X}\mathbf{X}^+$.

Proposition 6.10.1 — Minimum-norm least squares

$\hat{\beta}^+ = \mathbf{X}^+ \mathbf{y}$ is a least squares estimate, and it has the smallest Euclidean norm among all least squares estimates.

Proof. $\mathbf{X}\mathbf{X}^+ \mathbf{y} = \mathbf{U}_r \mathbf{D}_r \mathbf{V}_r \mathbf{T} \mathbf{V}_r \mathbf{D}_r^{-1} \mathbf{U}_r \mathbf{T} \mathbf{y} = \mathbf{U}_r \mathbf{U}_r \mathbf{T} \mathbf{y} = \mathbf{M} \mathbf{y}$, so $\hat{\beta}^+$ is a least squares estimate. It lies in $\mathbb{C}(\mathbf{V}_r) = \mathbb{C}(\mathbf{X}\mathbf{T}) = \mathcal{N}(\mathbf{X})^\perp$. Every other least squares estimate is $\hat{\beta}^+ + \mathbf{v}$ with $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ (Theorem 6.4.2), and $\|\hat{\beta}^+ + \mathbf{v}\|^2 = \|\hat{\beta}^+\|^2 + \|\mathbf{v}\|^2$. ■

In Example (Four answers, one fit), the Moore–Penrose row is this minimum-norm solution. In floating point, “ $\sigma_j > 0$ ” becomes “ σ_j larger than a tolerance”, and choosing the tolerance decides the numerical rank. That choice is a statistical decision in disguise. Chapter 10 discusses it, together with pivoted QR, which gives most of the SVD’s rank-revealing ability at lower cost.

6.10.4 Exercises**6.10.4.1 A. Check your understanding****Exercise A1.**

Prove the four identities in Eq. 6.10.1.

6.10.4.2 B. Practice**Exercise B1.**

Compute $\kappa(\mathbf{X})$ for the Longley data (with intercept) as given, after scaling every non-intercept column to unit Euclidean norm, and after centring and scaling. Confirm that the fitted values agree to the precision you expect in all three cases. Which part of the original condition number is “just units”?

Exercise B2.

Implement classical Gram–Schmidt as in Proposition 6.2.2, and the *modified* version that subtracts each projection from the working vector before computing the next inner product. For the degree-10 polynomial design of Example (Polynomial regression), compare $\|\mathbf{Q}\mathbf{T}\mathbf{Q} - \mathbf{I}\|$ for the two and for `numpy.linalg.qr`.

6.10.4.3 C. Going deeper**Exercise C1.**

Let $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ with $\bar{x} = c$ and $S_{xx} = s^2 n$. Find $\kappa(\mathbf{X})$ as a function of c , s and n ,

and show that it grows without bound as $c \rightarrow \infty$ with s fixed. Show that centring the regressor, a reparameterization, reduces the condition number to $\max(1, s)/\min(1, s)$.

Solution. Write $\mathbf{x} = c\mathbf{1} + \mathbf{d}$ with $\mathbf{d} \perp \mathbf{1}$ and $\|\mathbf{d}\|^2 = ns^2$. Then $\mathbf{X}\mathbf{T}\mathbf{X} = n \begin{pmatrix} 1 & c \\ c & c^2 + s^2 \end{pmatrix}$. The matrix in parentheses has trace $T = 1 + c^2 + s^2$ and determinant s^2 , so its eigenvalues are $\lambda_{\pm} = (T \pm \sqrt{T^2 - 4s^2})/2$, with $\lambda_+\lambda_- = s^2$ and $\lambda_+ \geq T/2$. Hence

$$\kappa(\mathbf{X})^2 = \frac{\lambda_+}{\lambda_-} = \frac{\lambda_+^2}{s^2} \geq \frac{T^2}{4s^2} \rightarrow \infty \quad \text{as } c \rightarrow \infty.$$

After centring, $\mathbf{X} = [\mathbf{1}, \mathbf{d}]$ has orthogonal columns of lengths $\sqrt{n}$ and $s\sqrt{n}$, so its singular values are these lengths and $\kappa = \max(1, s)/\min(1, s)$.

6.11 Projection in the population: best linear prediction

So far the geometry has lived in observation space $\mathbb{R}^n$, and every statement was about one fixed data vector. There is a parallel geometry one level up, in which the vectors are *random variables* and the inner product is an expectation. In that geometry, least squares estimates something even when the linear model is false, namely the population projection of Y onto a span of regressors. The conditional expectation turns out to be a projection too. This section sets up the correspondence. Chapter 14 develops prediction theory from it, and Chapter 19 uses it to study what least squares estimates under misspecification.

6.11.1 Random variables as vectors

Consider random variables with finite second moments on a common probability space. They form a vector space, and

$$\langle U, V \rangle = \mathbf{E}(UV), \quad \|U\| = (\mathbf{E}U^2)^{1/2}$$

is an inner product, provided we identify variables that are equal with probability one, so that $\|U\| = 0$ forces $U = 0$. This space is usually written L^2 . Two random variables are orthogonal when $\mathbf{E}(UV) = 0$. For variables with mean zero, this is the same as being uncorrelated. The Cauchy–Schwarz inequality Eq. 6.2.2 becomes the statement that correlations lie in $[-1, 1]$. The *cosine* of the angle between two centred random variables is their correlation.

For a random k -vector $\mathbf{X}$ with finite second moments, the set $\mathcal{L} = \{a + \mathbf{b}\mathbf{T}\mathbf{X} : a \in \mathbb{R}, \mathbf{b} \in \mathbb{R}^k\}$ is a subspace of dimension at most $k + 1$. The proofs of Proposition 6.2.2, Theorem 6.2.3 and Theorem 6.3.1 used only the axioms of an inner product and a finite spanning set, so they apply unchanged. Every Y with $\mathbf{E}Y^2 < \infty$ has a unique nearest point in $\mathcal{L}$, characterized by orthogonality of the error to $\mathcal{L}$.

Definition 6.11.1 — Best linear predictor

The **best linear predictor** of Y given $\mathbf{X}$, written $L(Y \mid \mathbf{X})$, is the projection of Y

onto $\mathcal{L}$. It is the $a + \mathbf{b}^T \mathbf{X}$ that minimizes the mean squared error $\mathbf{E}(Y - a - \mathbf{b}^T \mathbf{X})^2$.

Theorem 6.11.1

Let $\boldsymbol{\mu}_X = \mathbf{E}\mathbf{X}$, $\mu_Y = \mathbf{E}Y$, $\boldsymbol{\Sigma}_{XX} = \text{Cov}(\mathbf{X})$ and $\boldsymbol{\sigma}_{XY} = \text{Cov}(\mathbf{X}, Y)$, and suppose $\boldsymbol{\Sigma}_{XX}$ is positive definite. Then

$$L(Y | \mathbf{X}) = \alpha^* + \boldsymbol{\beta}^{*\top} \mathbf{X}, \quad \boldsymbol{\beta}^* = \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}, \quad \alpha^* = \mu_Y - \boldsymbol{\beta}^{*\top} \boldsymbol{\mu}_X.$$

The prediction error $U = Y - L(Y | \mathbf{X})$ has mean zero and is uncorrelated with every component of $\mathbf{X}$. Moreover

$$\text{Var}(Y) = \text{Var}(L(Y | \mathbf{X})) + \mathbf{E}U^2, \quad \frac{\text{Var}(L(Y | \mathbf{X}))}{\text{Var}(Y)} = \frac{\boldsymbol{\sigma}_{XY}^T \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}}{\text{Var}(Y)}. \quad (6.11.1)$$

Proof. Orthogonality of $U = Y - a - \mathbf{b}^T \mathbf{X}$ to the spanning set $\{1, X_1, \dots, X_k\}$ of $\mathcal{L}$ gives the *population normal equations*

$$\mathbf{E}(U) = 0, \quad \mathbf{E}(\mathbf{X}U) = \mathbf{0}.$$

The first gives $a = \mu_Y - \mathbf{b}^T \boldsymbol{\mu}_X$. Substituting into the second and subtracting $\boldsymbol{\mu}_X \mathbf{E}(U) = \mathbf{0}$ gives $\text{Cov}(\mathbf{X}, Y) - \text{Cov}(\mathbf{X})\mathbf{b} = \mathbf{0}$, so $\mathbf{b} = \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}$. By the projection theorem the solution is the unique minimizer. Since U has mean zero and is uncorrelated with $\mathbf{X}$, it is uncorrelated with $L(Y | \mathbf{X})$, so the variances add. Finally, $\text{Var}(\boldsymbol{\beta}^{*\top} \mathbf{X}) = \boldsymbol{\beta}^{*\top} \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta}^* = \boldsymbol{\sigma}_{XY}^T \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}$. ■

The ratio in Eq. 6.11.1 is the population *squared multiple correlation*, the population counterpart of R^2 . By the same cosine argument as Theorem 6.7.2, it is the squared correlation between Y and $L(Y | \mathbf{X})$.

6.11.2 Conditional expectation is also a projection

The linear span $\mathcal{L}$ is one subspace of L^2 . A much larger one is $\mathcal{G} = \{g(\mathbf{X}) : \mathbf{E}g(\mathbf{X})^2 < \infty\}$, all square-integrable functions of $\mathbf{X}$. It is infinite-dimensional, but it is closed, and the projection theorem still holds in it. We accept that fact from measure-theoretic probability and only use its conclusion.

Proposition 6.11.2

Let $m(\mathbf{X}) = \mathbf{E}(Y | \mathbf{X})$ with $\mathbf{E}Y^2 < \infty$. Then:

- (a) $m(\mathbf{X})$ is the projection of Y onto $\mathcal{G}$. It minimizes $\mathbf{E}(Y - g(\mathbf{X}))^2$ over g , and $\mathbf{E}[(Y - m(\mathbf{X}))g(\mathbf{X})] = 0$ for all $g(\mathbf{X}) \in \mathcal{G}$;
- (b) $L(Y | \mathbf{X}) = L(m(\mathbf{X}) | \mathbf{X})$: the best linear predictor of Y equals the best linear predictor of the regression function;

- (c) $\mathbf{E}(Y - L(Y | \mathbf{X}))^2 = \mathbf{E}(Y - m(\mathbf{X}))^2 + \mathbf{E}(m(\mathbf{X}) - L(Y | \mathbf{X}))^2$;
 (d) if m is affine, $m(\mathbf{X}) = L(Y | \mathbf{X})$.

Proof.

- (a) The orthogonality property follows from the tower rule: $\mathbf{E}[(Y - m(\mathbf{X}))g(\mathbf{X})] = \mathbf{E}[g(\mathbf{X}) \mathbf{E}(Y - m(\mathbf{X}) | \mathbf{X})] = 0$. Orthogonality to the subspace characterizes the nearest point, by the argument of Theorem 6.3.1, which needs only Pythagoras.
 (b) This is Theorem 6.5.1(a) in L^2 . Since $\mathcal{L} \subseteq \mathcal{G}$, projecting onto $\mathcal{L}$ can be done in two stages.
 (c) $Y - L = (Y - m) + (m - L)$, and $m - L \in \mathcal{G}$ is orthogonal to $Y - m$ by (a).
 (d) If $m \in \mathcal{L}$, the second term in (c) is zero. ■

Part (c) is the population version of Figure 6.5.1. The error of the best linear predictor splits into two orthogonal parts. One is irreducible noise, which no function of $\mathbf{X}$ can remove. The other is *approximation error*, the price of insisting on linearity. Chapters in Part IX reduce the second part by enlarging $\mathcal{L}$ towards $\mathcal{G}$.

6.11.3 Least squares estimates the population projection

Here is the connection with the rest of the chapter. Given data $(\mathbf{x}_{(i)}, y_i)$, put probability $1/n$ on each row. For this *empirical distribution*, the L^2 inner product of two variables is $n^{-1} \sum_i u_i v_i = n^{-1} \mathbf{u}^T \mathbf{v}$, which is the Euclidean inner product of observation space up to the factor $1/n$. So the best linear predictor under the empirical distribution is the least squares fit, and the population normal equations become the normal equations Eq. 6.4.1. Observation-space geometry is L^2 geometry for the empirical distribution.

As the sample grows, the empirical distribution approaches the population distribution, and the projections converge with it.

Proposition 6.11.3 — Consistency for the projection

Let $(\mathbf{X}_i, Y_i)$, $i = 1, 2, \dots$, be independent copies of $(\mathbf{X}, Y)$ with finite second moments and Σ_{XX} positive definite. Let $(\hat{\alpha}_n, \hat{\beta}_n)$ be the least squares coefficients from the first n observations, with an intercept. Then $(\hat{\alpha}_n, \hat{\beta}_n) \rightarrow (\alpha^*, \beta^*)$ with probability one. No assumption about the form of $\mathbf{E}(Y | \mathbf{X})$, the distribution of the errors, or their variance is needed.

Proof. For large n , $\hat{\beta}_n = \mathbf{S}_{XX}^{-1} \mathbf{s}_{XY}$, where $\mathbf{S}_{XX}$ and $\mathbf{s}_{XY}$ are sample covariances with divisor n (Section 6.6, centring). By the strong law of large numbers the sample means, second moments and cross-moments converge almost surely to their population values. So $\mathbf{S}_{XX} \rightarrow \Sigma_{XX}$ and $\mathbf{s}_{XY} \rightarrow \sigma_{XY}$. Matrix inversion is continuous at the nonsingular Σ_{XX} , so eventually $\mathbf{S}_{XX}$ is invertible and $\hat{\beta}_n \rightarrow \beta^*$. The intercept follows the same way.

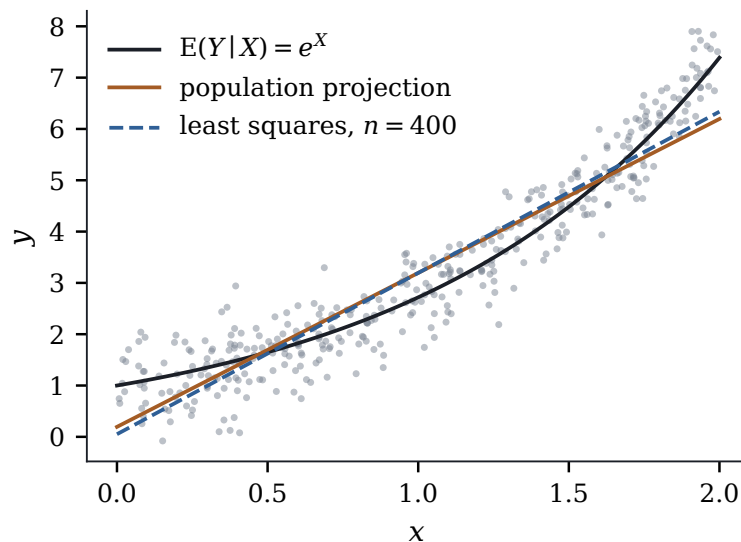


Figure 6.11.1: When the regression function is curved, least squares estimates its best linear approximation in the L^2 sense, the population projection. The approximation error is largest where e^x bends away from the line.

Example. (A curved mean)

Let $X \sim \text{Uniform}(0, 2)$ and $Y = e^X + \varepsilon$ with independent $\varepsilon \sim N(0, 0.25)$. The regression function is not linear, but [Theorem 6.11.1](#) still identifies a best line. A short calculation gives $\text{Cov}(X, Y) = 1$ and $\text{Var}(X) = 1/3$, so the population projection has slope 3.000 and intercept 0.195. The listing fits least squares lines to samples of increasing size. The estimates settle on these values, not on any feature of e^x . [Figure 6.11.1](#) shows the regression function, the population projection, and a least squares fit.

```
import numpy as np
rng = np.random.default_rng(611)

# X ~ Uniform(0, 2), E(Y | X) = exp(X), Var(Y | X) = 0.25.
# Population projection of Y onto span{1, X}: slope Cov(X, Y)/Var(X), intercept E Y
# - slope * E X.
EX, VarX = 1.0, 4.0 / 12.0
EY = (np.exp(2) - 1) / 2
EXY = (np.exp(2) + 1) / 2
slope_pop = (EXY - EX * EY) / VarX
intercept_pop = EY - slope_pop * EX

# E exp(X)
# E X exp(X)

for n in [20, 200, 2000, 20000]:
    x = rng.uniform(0, 2, n)
    y = np.exp(x) + rng.normal(0, 0.5, n)
    b, *_ = np.linalg.lstsq(np.column_stack([np.ones(n), x]), y, rcond=None)
    print(f"n = {n:6d}    intercept {b[0]:.3f}    slope {b[1]:.3f}")
print(f"population    intercept {intercept_pop:.3f}    slope {slope_pop:.3f}")
```

Remark (What changes under misspecification). [Proposition 6.11.3](#) is reassuring about the *target*. A least squares line always estimates a well-defined population quantity. It says nothing about the *uncertainty* of the estimate. When $\mathbf{E}(Y \mid \mathbf{X})$ is not linear, or the variance is not constant, the approximation error $m(\mathbf{X}) - L(Y \mid \mathbf{X})$ behaves like extra noise that depends on $\mathbf{X}$. The textbook formula $\sigma^2(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}$ for the covariance of $\hat{\boldsymbol{\beta}}$ is then wrong, even in large samples. The “sandwich” covariance estimators of Chapter 21 are designed for exactly this situation.

6.12 Summary and Notes

6.12.1 Summary

Idea.

1. Least squares is the nearest-point problem in observation space. The fitted vector $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ is the orthogonal projection of $\mathbf{y}$ onto $\mathbb{C}(\mathbf{X})$, and the residual is orthogonal to $\mathbb{C}(\mathbf{X})$ ([Theorem 6.4.1](#)).
2. Orthogonal projections are exactly the symmetric idempotent matrices. Their rank equals their trace ([Theorem 6.3.3](#) and [Proposition 6.3.4](#)).
3. $\mathbf{M} = \mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^{-}\mathbf{X}\mathbf{T}$ for *every* generalized inverse ([Theorem 6.3.7](#)). Fitted values, residuals and every function of them are unique, whatever the rank. Coefficients are unique only when $\text{rank}(\mathbf{X}) = p$, and $\boldsymbol{\lambda}\mathbf{T}\hat{\boldsymbol{\beta}}$ is invariant iff $\boldsymbol{\lambda} \in \mathbb{C}(\mathbf{X}\mathbf{T})$ ([Theorem 6.4.2](#) and [Theorem 6.4.4](#)).
4. Nested model spaces give orthogonal decompositions $\mathbf{I} = \mathbf{M}_0 + (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$. The drop in residual sum of squares is $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2$, with $r - r_0$ degrees of freedom ([Theorem 6.5.1](#)).
5. A multiple-regression coefficient is the slope on the part of its regressor orthogonal to the others. Fixed-effects estimators and added-variable plots are applications ([Theorem 6.6.2](#)).
6. Anything that depends only on $\mathbb{C}(\mathbf{X})$ is invariant to reparameterization. R^2 is the squared cosine between centred data and centred fit ([Theorem 6.7.1](#) and [Theorem 6.7.2](#)).
7. Leverage $h_{ii} = m_{ii}$ is the squared length of the projection of a coordinate vector. It lies in $[1/n, 1]$ with an intercept and averages r/n ([Proposition 6.8.1](#)).
8. Replacing the inner product by $\mathbf{u}\mathbf{T}\mathbf{V}^{-1}\mathbf{v}$ turns least squares into generalized least squares. The projection is then idempotent and $\mathbf{V}^{-1}$ -self-adjoint, but not symmetric ([Theorem 6.9.1](#) and [Theorem 6.9.3](#)).
9. Compute projections with orthonormal bases (QR, SVD), never by inverting $\mathbf{X}\mathbf{T}\mathbf{X}$, which squares the condition number ([Section 6.10](#)).
10. In the population, least squares estimates the L^2 projection of Y onto linear functions of $\mathbf{X}$, whether or not the regression function is linear ([Theorem 6.11.1](#) and [Proposition 6.11.3](#)).

6.12.2 Notes and sources

The geometric tradition. Seeing least squares as projection is as old as least squares itself, but it became the organizing principle of linear-model theory only in the twentieth century. Herr (1980) traces the history, from Fisher’s geometric derivations of sampling distributions through the coordinate-free treatments of the 1960s. Book-length geometric developments at different levels include Saville and Wood (1991), written for a first course, and Wickens (1995), for multivariate methods. Christensen (2020) is the most sustained vector-space treatment of linear model theory. Its chapter 2 and appendices A–B cover the material of [Section 6.2](#), [Section 6.3](#) and [Section 6.4](#) for the rank-deficient case throughout, and influenced this chapter’s choice to state results for general rank from the start. Seber and Lee (2003, appendix B) gives a compact account of projections, including the constraint-space result that appears here as [Theorem 6.5.2](#). Agresti (2015, chapter 2) develops projection matrices through worked null-model, one-way and two-way examples, and is a gentle companion to [Section 6.5](#). For the underlying linear algebra, Halmos (1958) remains a model of coordinate-free exposition.

Generalized inverses. The four-condition inverse is due to Penrose (1955). Its statistical uses, and the theory of general $\mathbf{A}^-$ including the characterization in [Exercise \(B2\)](#), are developed by Rao (1973). The invariance of $\mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^-\mathbf{X}\mathbf{T}$ ([Lemma 6.3.6](#)) is the fact that makes generalized inverses harmless in linear models. Every quantity of statistical interest is sandwiched in this way.

Frisch–Waugh–Lovell. The theorem is due to Frisch and Waugh (1933) for time trends and Lovell (1963) for seasonal dummies and general partitions. It is also known under Lovell’s name alone, and in econometrics as the “partialling-out” result. It underlies added-variable plots, double/debiased machine-learning estimators, and the fixed-effects (“within”) estimator for panel data.

Leverage. Hoaglin and Welsch (1978) introduced the name “hat matrix” into common use and established the basic properties in [Proposition 6.8.1](#). Diagnostic use of leverage is taken up in Chapter 20.

Other inner products. Generalized least squares is due to Aitken (1935). The coordinate-free condition for equality of ordinary and generalized least squares ([Theorem 6.9.3](#)) is from Kruskal (1968). Christensen (2020, chapter 10) develops the consequences for the general Gauss–Markov model, which Chapter 31 follows.

Computation. Golub and Van Loan (2013) is the standard reference for QR, the SVD and their perturbation theory. Higham (2002) gives the precise statement of the conditioning of least squares problems, including the $\kappa^2\|\hat{\epsilon}\|$ term mentioned in [Section 6.10](#). Seber and Lee (2003, chapter 11) surveys algorithms from a statistician’s perspective, and Chapter 10 builds on it.

6.12.3 References

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